

SUPPORTING INFORMATION

Enzymatic Synthesis of Kynurenic Acid-linked Dipeptides with C-terminal Amide

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SUPPLEMENTARY METHODS

1. Strains and general techniques for DNA manipulation.

Aspergillus nidulans FGSC A1145 was obtained from the Fungal Genetics Stock Center (<http://www.fgsc.net/>) and was maintained on Czapek-Dox (CD) agar. *Saccharomyces cerevisiae* strain JHY686 (*prb1Δ pep4Δ ADH2p-npgA-ACS1t lys2Δ0*) was used for *in vivo* homologous recombination to construct the *A. nidulans* overexpression plasmids. Genomic DNA from *Aspergillus asperescens* NRRL 4770 was prepared using LETS isolation buffer (10 mM Tris-HCl, pH 8.0, 20 mM EDTA, 0.5% SDS, 0.1 M LiCl). KOD One™ PCR master Mix (Toyobo) or Q5® High-Fidelity DNA Polymerase (New England Biolab) was used for PCR. *Escherichia coli* One Shot® TOP10 was used for plasmid propagation. NEBuilder HiFi DNA Assembly Cloning Kit was used for gene clone following the protocol (New England Biolab). *E. coli* BL21(DE3) (Sangon Biotech) was used for protein expression. DNA restriction enzymes (New England Biolab) were used according to the protocol provided by the manufacturer. cDNA was synthesized with the PrimeScript™ II 1st Strand cDNA Synthesis Kit (Takara). DNA was synthesized by Sangon Biotech (Shanghai) company. DNA sequencing was provided by Tsingke Biotechnology company. The gene-specific primers are listed in **Table S1**.

2. Spectroscopic analysis.

NMR spectra were obtained with a Bruker AV500 spectrometer with a 5 mm dual cryoprobe (¹H 500 MHz, ¹³C 125 MHz) or Bruker AV600 spectrometer with a 5 mm dual cryoprobe (¹H 600 MHz, ¹³C 150 MHz). High-resolution electrospray ionization mass spectrometry (HRESIMS) was performed on a Waters Xevo G2-XS-Q-TOF (Manchester, UK). Optical rotations were measured on a Rudolph Research Analytical Autopol III Automatic Polarimeter. The *in vivo* and *in vitro* assay of **all HPLC figures** was measured on Agilent Technologies 1200 Series HPLC (ZORBAX Eclipse XDB-C18, 4.6×150 mm, 5 μm) with a linear gradient of CH₃CN-H₂O (from 1:19 to 19:1, v/v, 30 min) supplemented with 0.1% (v/v) formic acid. The *in vivo* and *in vitro* assay of **all MS figures** was measured on SHIMADZU LC/MS 2050 using a column (Shim-pack GWS-C18, 4.6×250 mm, 5 μm) with a linear gradient of CH₃CN-H₂O (from 1:19 to 19:1, v/v, 30 min) supplemented with 0.1% (v/v) formic acid.

3. Heterologous expression of *kaz* in *A. nidulans*.

To construct plasmid pJM01004, *kazB* was amplified by PCR using primers R-KazB-F/ R-KazB(AmyB5)-R, *kazE* was amplified by PCR using primers R-KazE(AmyB5)-F/R- KazE-R, *AmyB5* was amplified by PCR using primers R-AmyB5(KazB)-F/R-AmyB5(KazE)-R. The overlapping DNA fragments and PacI/SwaI-digested pYTR expression vector were transformed into yeast for homologous recombination. The correct colonies checked by PCR were combined, and subjected to yeast miniprep to get small amount of plasmids. The plasmids obtained from yeast miniprep were introduced into *Escherichia coli* One Shot® TOP10 by electroporation. Single *E. coli* colony was cultured and minipreped to obtain pJM01004, which was confirmed by restriction enzymes digestion. The same protocol was used to generate other plasmids, the plasmids used for the heterologous expression are illustrated in **Figure S2**.

For the preparation of the protoplasts of *A. nidulans* A1145, fresh spores of *A. nidulans* were inoculated into 25 mL liquid CD medium (1.25 g/L yeast extract) supplemented with 10 mM uridine, 5 mM uracil, 0.5 μg/mL pyridoxine and 2.5 μg/mL riboflavin in a 250 mL flask and germinated at 37 °C and 250 rpm for approximately 5 h for germination. The culture was harvest by centrifugation at 4000 rpm, 5 min, and the mycelia was washed with Osmotic Medium. Washed mycelia were transfered into a 125 mL erlenmeyer flask

containing 30 mg of Lysing Enzymes from Trichoderma and 20 mg of Yatalase in 10 mL of Osmotic Medium. The mixture was shaken at 80 rpm for 10 hours at 28 °C. The cells were poured into a sterile 30 mL Corex tube and overlaid with 10 mL of Trapping Buffer. The content was centrifuged at 4000 rpm for 15 min at 4 °C. The protoplasts were removed from the interface with a pipet and transferred to a sterile 15 mL falcon tube. Two volumes of STC buffer was added and centrifuged at 4000 rpm for 8 min to harvest the protoplasts. The supernatant was removed and the protoplasts were resuspended in STC buffer.

For heterologous expression of genes in *A. nidulans* 1145. The plasmids containing the relative genes were transformed into fresh protoplast, followed by incubation on ice for 1 hour, mix well with PEG solution (10 * volume of the protoplast) before transforming to CD-Sorbitol plate, followed by incubation at 37 °C for 4-5 days. The resulting transformants were used for fermentation check.

CD medium (1 L): 10 g of Glucose, 50 mL of 20x Nitrate salts, 1 mL of Trace elements, 5 g of Yeast extract

Osmotic medium: 1.2 M of MgSO_4 and 10 mM of NaPB (pH 6.5).

Trapping buffer: Sorbitol (0.6 M) and Tris-HCl (0.1 M, pH 7.0).

STC buffer: Sorbitol (1.2 M), CaCl_2 (10 mM) and Tris-HCl (10 mM, pH 7.5).

PEG solution: 60% PEG 4000 (W/V), CaCl_2 (50 mM) and Tris-HCl (10 mM, pH 7.5).

20x Nitrate salts (1 L): NaNO_3 (120 g), KCl (10.4 g), $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ (10.4 g) and KH_2PO_4 (30.4 g).

Trace elements(0.1 L): $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ (2.2 g), H_3BO_3 (1.1 g), $\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$ (0.5 g), $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ (0.16 g), $\text{CoCl}_2 \cdot 5\text{H}_2\text{O}$ (0.16 g), $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (0.16 g), $(\text{NH}_4)_6\text{Mo}_7\text{O}_{24} \cdot 4\text{H}_2\text{O}$ (0.11 g), pH 6.5.

CD-Sorbitol (1 L): 10 g of Glucose, Sorbitol (218.6 g), 50 mL of 20x Nitrate salts, 1 mL of Trace elements, Agar (20 g).

CD-ST: Starch (20 g), N-Z-Amine (20 g), 50 mL of 20x Nitrate salts, 1 mL of Trace elements and 20 g of Argar.

4. Chemical analysis and compound isolation.

Small scale: To detect the metabolites of the transformants, the transformants were grown in petri dish with 6 mL solid CD-ST medium at 28 °C for 5 days, then take 0.8 mL of solid medium for extraction by adding 1 mL of methanol and ultrasonic for 30 min, centrifuge to separate the solid and organic phase, subject 10 μL methanol into HPLC for analysis.

Large scale: For **1** and **2**, 6 L of solid CD-ST medium containing pJM01001, pJM01004 and pJM01006 was used for fermentation. After 5 days of incubation at 28 °C, the culture was extracted three times with ethyl acetate, followed by removal of the organic solvent by evaporation. After three extractions of petroleum ether, the crude extract was extracted four times using ethyl acetate and subjected to Silica gel column using a gradient elution with DCM-MeOH (from 400:1 to 3:1, v/v). The fractions containing **1** were combined and evaporated. Fractions was further separated by semi-preparative HPLC (MeOH- H_2O , 1:9, v/v, 60 min) to afford **1** (2 mg/L). **2** in the aqueous was subjected to gel column using a mixture elution with 50% MeOH- H_2O . The fractions containing **2** were combined and evaporated. Fractions was further separated by semi-preparative HPLC (MeOH- H_2O , 1:9, v/v, 70 min) to afford **2** (0.2 mg/L).

For **3**, 6 L of CD-ST medium containing pJM01001 and pJM01002 was used for fermentation. After 5 days of incubation at 28 °C, the culture was extracted three times with ethyl acetate, followed by removal of the organic solvent by evaporation. After three extractions of petroleum ether, the crude extract was extracted four times using ethyl acetate and subjected to Silica gel column using a gradient elution with DCM-MeOH (from 400:1 to 3:1, v/v). The fraction containing **3** were combined and evaporated. This fraction was further separated by semi-preparative HPLC (MeOH- H_2O , 9:41, v/v, 30 min) to afford **3** (3.0 mg/L).

For **5**, 6 L of CD-ST medium containing pJM01001 and pJM01004 was used for fermentation. After 5 days

of incubation at 28 °C, the culture was extracted three times with ethyl acetate, followed by removal of the organic solvent by evaporation. After three extractions of petroleum ether, the crude extract was extracted four times using ethyl acetate and subjected to Silica gel column using a gradient elution with DCM-MeOH (from 400:1 to 3:1, v/v). The fraction containing **5** were combined and evaporated. This fraction was further separated by semi-preparative HPLC (MeOH-H₂O, 9:41, v/v, 50 min) to afford **5** (3.0 mg/L).

For **6**, 6 L of CD-ST medium containing pJM01001, pJM01004 and pJM01006 was used for fermentation. After 3 days of incubation at 28 °C, the culture was extracted three times with ethyl acetate, followed by removal of the organic solvent by evaporation. After three extractions of petroleum ether, the crude extract was extracted four times using ethyl acetate and subjected to Silica gel column using a gradient elution with DCM-MeOH (from 400:1 to 3:1, v/v). The fraction containing **6** were combined and evaporated. This fraction was further separated by semi-preparative HPLC (MeOH-H₂O, 9:41, v/v, 40 min) to afford **6** (0.8 mg/L).

For **7**, 6 L of CD-ST medium containing pJM01001, pJM01002 and pJM01006 was used for fermentation. After 3 days of incubation at 28 °C, the culture was extracted three times with ethyl acetate, followed by removal of the organic solvent by evaporation. After three extractions of petroleum ether, the crude extract was extracted four times using ethyl acetate and subjected to Silica gel column using a gradient elution with DCM-MeOH (from 400:1 to 3:1, v/v). The fraction containing **7** were combined and evaporated. This fraction was further separated by semi-preparative HPLC (MeOH-H₂O, 9:41, v/v, 20 min) to afford **7** (0.8 mg/L).

For **8** and **9**, 6 L of CD-ST medium containing pJM01001, pJM01002 and pJM01006 was used for fermentation. After 5 days of incubation at 28 °C, the culture was extracted three times with ethyl acetate, followed by removal of the organic solvent by evaporation. After three extractions of petroleum ether, the crude extract was extracted four times using ethyl acetate and subjected to Silica gel column using a gradient elution with DCM-MeOH (from 400:1 to 3:1, v/v). The fractions containing **8** were combined and evaporated. Fractions was further separated by semi-preparative HPLC (MeOH-H₂O, 1:9, v/v, 55 min) to afford **8** (2 mg/L). **9** in the aqueous was subjected to gel column using a mixture elution with 50% MeOH-H₂O. The fractions containing **9** were combined and evaporated. Fractions was further separated by semi-preparative HPLC (MeOH-H₂O, 1:9, v/v, 65 min) to afford **9** (0.3 mg/L).

For **10**, 6 L of CD-ST medium (added 0.1 g/L **7-F Trp**) containing pJM01001, pJM01002 and pJM01006 was used for fermentation. After 5 days of incubation at 28 °C, the culture was extracted three times with ethyl acetate, followed by removal of the organic solvent by evaporation. After three extractions of petroleum ether, the crude extract was extracted four times using ethyl acetate and subjected to Silica gel column using a gradient elution with DCM-MeOH (from 400:1 to 3:1, v/v). The fractions containing **10** were combined and evaporated. Fractions was further separated by semi-preparative HPLC (MeOH-H₂O, 1:9, v/v, 60 min) to afford **10** (2 mg/L).

For **11** and **12**, 6 L of CD-ST medium (added 0.1 g/L **N-MT Gly**) containing pJM01001 and pJM01006 was used for fermentation. After 5 days of incubation at 28 °C, the culture was extracted three times with ethyl acetate, followed by removal of the organic solvent by evaporation. After three extractions of petroleum ether, the crude extract was extracted four times using ethyl acetate and subjected to Silica gel column using a gradient elution with DCM-MeOH (from 400:1 to 3:1, v/v). The fractions containing **11** were combined and evaporated. Fractions was further separated by semi-preparative HPLC (MeOH-H₂O, 1:9, v/v, 50 min) to afford **11** (0.5 mg/L). **12** in the aqueous was subjected to gel column using a mixture elution with 50% MeOH-H₂O. The fractions containing **12** were combined and evaporated. Fractions was further separated by semi-preparative HPLC (MeOH-H₂O, 1:9, v/v, 55 min) to afford **12** (0.3 mg/L).

For **13** and **14**, 6 L of CD-ST medium (added 0.1 g/L **N-MT Gly**) containing pJM01001, pJM01003 and pJM01006 was used for fermentation. After 5 days of incubation at 28 °C, the culture was extracted three times with ethyl acetate, followed by removal of the organic solvent by evaporation. After three extractions of

petroleum ether, the crude extract was extracted four times using ethyl acetate and subjected to Silica gel column using a gradient elution with DCM-MeOH (from 400:1 to 3:1, v/v). The fractions containing **13** were combined and evaporated. Fractions was further separated by semi-preparative HPLC (MeOH-H₂O, 1:9, v/v, 55 min) to afford **13** (2 mg/L). **14** in the aqueous was subjected to gel column using a mixture elution with 50% MeOH-H₂O. The fractions containing **14** were combined and evaporated. Fractions was further separated by semi-preparative HPLC (MeOH-H₂O, 1:9, v/v, 60 min) to afford **14** (0.2 mg/L).

For **35**, 6 L of CD-ST medium (added 0.1 g/L **6-F Trp**) containing pJM01001, pJM01002 and pJM01006 was used for fermentation. After 5 days of incubation at 28 °C, the culture was extracted three times with ethyl acetate, followed by removal of the organic solvent by evaporation. After three extractions of petroleum ether, the crude extract was extracted four times using ethyl acetate and subjected to Silica gel column using a gradient elution with DCM-MeOH (from 400:1 to 3:1, v/v). The fractions containing **35** were combined and evaporated. Fractions was further separated by semi-preparative HPLC (MeOH-H₂O, 1:9, v/v, 65 min) to afford **35** (0.3 mg/L).

For **38**, 6 L of CD-ST medium (added 0.1 g/L **6-Cl Trp**) containing pJM01001, pJM01004 and pJM01006 was used for fermentation. After 5 days of incubation at 28 °C, the culture was extracted three times with ethyl acetate, followed by removal of the organic solvent by evaporation. After three extractions of petroleum ether, the crude extract was extracted four times using ethyl acetate and subjected to Silica gel column using a gradient elution with DCM-MeOH (from 400:1 to 3:1, v/v). The fractions containing **38** were combined and evaporated. Fractions was further separated by semi-preparative HPLC (MeOH-H₂O, 9:41, v/v, 30 min) to afford **38** (1.3 mg/L).

For **40**, 6 L of CD-ST medium (added 0.1 g/L **7-F Trp** + 0.1 g/L **N-MT Gly**) containing pJM01001 and pJM01006 was used for fermentation. After 5 days of incubation at 28 °C, the culture was extracted three times with ethyl acetate, followed by removal of the organic solvent by evaporation. After three extractions of petroleum ether, the crude extract was extracted four times using ethyl acetate and subjected to Silica gel column using a gradient elution with DCM-MeOH (from 400:1 to 3:1, v/v). The fractions containing **40** were combined and evaporated. Fractions was further separated by semi-preparative HPLC (MeOH-H₂O, 1:9, v/v, 55 min) to afford **40** (1.2 mg/L).

5. Preparation of **4**.

4 mg Kynurenine was dissolved in H₂O to prepare a 10 mM stock solution. Preparation for **4** was carried out in 50 mM sodium phosphate buffer containing 10 µM of KazE, 2 mM of SAM and 2 mM of **KYN** in 500 µL of solution at pH 7.4. The reaction was incubated at 30 °C for overnight and quenched by the addition of 2 mL of methanol. The organic layer was evaporated, further separated by semi-preparative HPLC (MeOH-H₂O, 1:4, v/v, 30 min) to afford **4**.

6. Gene cloning, protein expression and purification.

The gene encoding KazE was amplified by PCR from cDNA extracted from *A. nidulans* expression the whole *kaz* gene cluster. The sequences of primers are presented in **Table S1**. The PCR product of target genes and pET-28a (+) vector (digest by NdeI/BamHI for N-his tag or digest by NcoI/HindIII for C-his tag) are assembled using NEBuilder HiFi DNA Assembly Cloning kit according to the manufacturer's instructions. The resulting recombinant plasmids were confirmed by sequencing. The resulting recombinant plasmids listed in **Table S2**, were transferred into *E. coli* BL21 (DE3) for overexpression. Cultures (2 × 1 L) in LB medium supplemented with 50 µg/mL of kanamycin were grown at 37 °C to an OD₆₀₀ of 0.6 ~ 0.8 and protein expression was induced by the addition of isopropyl-β-D-thiogalactopyranoside (IPTG) to a final concentration of 100 µM,

followed by further incubation for 20 hours at 16 °C. The cultures were centrifuged for 20 min at 4,500 rpm at 4 °C, and then resuspended the cell pellet in 50 mL of lysis buffer (50 mM Tris-HCl, 500 mM NaCl, 10 mM imidazole, 10% glycerol, pH 7.5). The enzymes were purified with Ni-NTA Beads according to the manufacturer's manual (Smart-Lifesciences). The purified protein was concentrated using an Amicon Ultra-15 Centrifugal Filter (Ultracel® - 30KD) and stored at -80 °C in buffer (50 mM Tris-HCl, 50 mM NaCl, 10% glycerol, pH 7.5). The protein concentrations were calculated by measuring the ultraviolet absorption at A₂₈₀. Amidases and KAT were expressed and purified using the same protocol as KazE.

7. *In vitro* characterization of KazE.

Assay for KazE activity was carried out in 50 mM sodium phosphate buffer containing 5 µM of KazE, 2 mM of SAM and i) containing 2 mM of **KYN**, ii) containing 2 mM of **KYNA**, iii) containing 2 mM of **3**, iv) containing 2 mM of **7**, v) containing 2 mM of **8**, vi) containing 2 mM of **9** in 50 µL of solution at pH 7.4. The reaction was incubated at 30 °C for 1h and quenched by the addition of 200 µL of methanol, analyzed on an Agilent Technologies 1200 Series HPLC. Control reaction was carried out using standard substrate and buffer.

8. *In vitro* characterization of amidases.

Assay for Amidases activity was carried out in 50 mM sodium phosphate buffer containing *E. coli* lysate and 1 mM of **8** or **1** in 50 µL of solution at pH 7.4. The reaction was incubated at 30 °C for overnight and quenched by the addition of 200 µL of methanol, analyzed on an Agilent Technologies 1200 Series HPLC. Control reaction was carried out using denatured Amidases.

Assay for purified Amidases activity was carried out in 50 mM sodium phosphate buffer containing 20 µM Amidases and 1 mM of **8** or **1** in 50 µL of solution at pH 7.4. The reaction was incubated at 30 °C for overnight and quenched by the addition of 200 µL of methanol, analyzed on an Agilent Technologies 1200 Series HPLC. Control reaction was carried out using denatured Amidases.

9. *In vitro* assays of KazC with microsomes purified from *A.nidulans*.

The transformant spores were cultured in 100 mL CD-ST solid medium at 28 °C for 2.5 days. The culture broth was removed agar to harvest mycelia. The mycelia were further powdered in liquid nitrogen. The powder was suspended in 20-30 mL TES B buffer (50 mM Tris-HCl, 1 mM EDTA, 0.6 M sorbitol, pH 6.5). All subsequent steps were performed at 4 °C or on ice. The combined lysate was then centrifuged at 1,2000 rpm for 30 min at 4 °C to pellet large cell debris. The supernatant (around 20-30 mL) was removed from the pellet and the pellet was discarded. Microsomes were precipitated by the addition of NaCl to a final volume of 150 mM and polyethylene glycol PEG-4000 to a final concentration of 0.1 g/mL. After an incubation of about 1 h on ice with periodic mixing, microsomal protein fractions were collected by centrifugation at 1,2000 rpm for 10 min at 4 °C. The supernatant was discarded, and the microsomal pellet was resuspended in a certain volume of ice-cold Tris-HCl buffer.

Assay for KazC activity was carried out in 50 mM Tris-HCl buffer containing extracted KazC microsome from *A.nidulans*, 0.2 mM NADPH, 0.2 mM FAD and i) containing 0.2 mM of **3**, ii) containing 0.2 mM of **5**, iii) containing 0.2 mM of **7**, iv) containing 0.2 mM of **6** in 50 µL of solution at pH 8.5. The reaction was incubated at 20 °C in shaker for overnight and quenched by the addition of 200 µL of methanol, analyzed on a SHIMADZU LC/MS 2050. Control reaction was carried out using denatured KazC and UPR.

10. Evaluation of blood-brain barrier permeability by PAMPA.

In separate centrifuge tubes, prepare 500 µM test compound or control solutions by combining 25 µL of

10 mM compound in DMSO with 475 μ L PBS. For equilibrium standards (200 μ M), mix 80 μ L of the 500 μ M solution with 120 μ L PBS. Prepare a blank control using 5 μ L DMSO and 245 μ L PBS. Reserve standards and blank for subsequent analysis.

Add 300 μ L PBS to each well of an acceptor plate. Carefully pipette 5 μ L of 4% lecithin in dodecane onto the membranes of a donor plate. Apply 200 μ L of each 500 μ M test compound or control in duplicate to the donor plate. Assemble the system by placing the donor plate into the acceptor plate. Incubate at room temperature or 37 $^{\circ}$ C for 16–24 hours.

After incubation, carefully separate the plates and collect the acceptor solutions. Analyze 100 μ L aliquots of acceptor samples, equilibrium standards, and blank control by HPLC. The blank confirms absence of DMSO interference. Typical absorbance peaks for controls are 280 nm (high permeability) and 270 nm (medium/low permeability).

11. Theory and calculation details.

Conformational searches were performed using the CREST program^[1] to identify the lowest-energy conformers for quantum mechanical calculations. All quantum mechanical calculations were performed using the Gaussian software package^[2] on the CAMS Biomedical HPC Platform. Structures were optimized at the M06-2X^[3]/6-31+G(d,p) level of theory using the SMD^[4] solvent model for water. Frequency calculations were used to confirm the presence of local minima (no imaginary frequencies) and transition states (one imaginary frequency). All transition states were further verified by IRC calculations. Thermochemistry corrections to obtain enthalpies and free energies were computed at 298.15 K and 1 atm; Zero-point vibrational energies (ZPVEs) were scaled by 0.9670^[5]. Low-frequency modes below 100 cm^{-1} were treated with the Truhlar quasi-harmonic approach as implemented in Shermo^[6,7] software package. To obtain more accurate energetics, single-point energy calculations were performed on the optimized structures at the M06-2X/aug-cc-pVTZ level of theory using the SMD solvent model for water. Visualizations of all computed structures were created using the CYLview20^[8] software package.

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SUPPLEMENTARY TABLES

Table S1. Primers used in this study.

Primers	SEQUENCE 5'-3'
P-KazA-F	CTTCTCTGAACAATAAACCCACAGAAAGGCATTTTAAATATGACTAGAACAGATCGTAT
P-KazA-R	GGAGTGATGAGACCCAACAACCATGATACCAGGGGATTTCCACGCAGTGAGTAACCAAT
R-KazB-F	AACCATTACCCCGCCACATAGACACATCTAAACATTAATATGCACGACATAGCTACAAT
R-KazB-R	TCTGCTAAAGGGTATCATCGAAAGGGAGTCATCCAATTTAGGGGAAATATGGTGCTTGC
R-KazE-F	AACCATTACCCCGCCACATAGACACATCTAAACATTAATATGTCGCAATCCGTTAACGC
R-KazE-R	TCTGCTAAAGGGTATCATCGAAAGGGAGTCATCCAATTTACAGGAAAGGCTGCTGTTCT
R-KazB (AmyB5)-R	CTCGTTCGGCACCTTTAATCAGGGGAAATATGGTGCTTGC
R-AmyB5 (KazB)-F	GCAAGCACCATATTTCCCCTGATTAAAGGTGCCGAACGAG
R-AmyB5 (KazE)-R	GCGTTAACGGATTGCGACATAAATGCCTTCTGTGGGGTTT
R-KazE (AmyB5)-F	AAACCCACAGAAAGGCATTTATGTCGCAATCCGTTAACGC
U-KazC-F	TGAGCTTCATCCCCAGCATCATTACACCTCAGCATTAATATGTCCGTAGCTGTAAGGAT
U-KazC-R	AACACAGTGGAGGACATACCCGTAATTTTCTGGGCATTTGAAGCTCTTTGGTACAGCCAT
U-KazC (AmyB5)-R	CTCGTTCGGCACCTTTAATCGAAGCTCTTTGGTACAGCCA
U-AmyB5 (KazC)-F	TGGCTGTACCAAAGAGCTTCGATTAAAGGTGCCGAACGAG
U-AmyB5 (KazD)-R	ATATGGTGAGGTTGCAGCATAAATGCCTTCTGTGGGGTTT
U-KazD (AmyB5)-F	AAACCCACAGAAAGGCATTTATGTCGCAACCTCACCATAT
U-KazD-R	AACACAGTGGAGGACATACCCGTAATTTTCTGGGCATTTTCAGACACAACAATCAGGTTT
U-KazD (gpdA)-R	CCAAATCAATTCACCGGAGTCAGACACAACAATCAGGTTT
U-gpdA (KazD)-F	GAACCTGATTGTTGTGTCTGACTCCGGTGAATTGATTGG
U-gpdA (KazF)-R	GCAGACAGTGAAAAGACCATTTGTTTAGATGTGTCTATGTG
U-KazF(gpdA)-F	CACATAGACACATCTAAACAATGGTCTTTTCACTGTCTGC
U-KazF-R	AACACAGTGGAGGACATACCCGTAATTTTCTGGGCATTTGTGCGATAGTGTTCCATCT
R-KazC-F	AACCATTACCCCGCCACATAGACACATCTAAACATTAATATGTCCGTAGCTGTAAGGAT
R-KazC-R	TCTGCTAAAGGGTATCATCGAAAGGGAGTCATCCAATTTGAAGCTCTTTGGTACAGCCAT
Amidase 1-pet28a-F	TGCCGCGCGGCAGCCATATGGGCAGCCTTCAACCATGGGA
Amidase 1-pet28a-R	CGGAGCTCGAATTCGGATCCCTAACTCCGCCAAATCCGCT
Amidase 2-pet28a-F	TGCCGCGCGGCAGCCATATGCCTCAATCCTGGGAAGAACT
Amidase 2-pet28a-R	CGGAGCTCGAATTCGGATCCCTATGGAGTCACCACATTTT
Amidase 3-pet28a-F	TGCCGCGCGGCAGCCATATGTCTTCCACCTGGGAAGCCCA
Amidase 3-pet28a-R	CGGAGCTCGAATTCGGATCCCTATACGTTTGTAGTGTCTT
Amidase 4-pet28a-F	TGCCGCGCGGCAGCCATATGAATTTCAAATATCTCTTTGG
Amidase 4-pet28a-R	CGGAGCTCGAATTCGGATCCTCACTGAGCCTCCTGTTGAA
Amidase 5-pet28a-F	TGCCGCGCGGCAGCCATATGGCGGAAGAAACCTGGCAAGC
Amidase 5-pet28a-R	CGGAGCTCGAATTCGGATCCTCAGACTCGAATGTCCCTAT
Amidase 6-pet28a-F	TGCCGCGCGGCAGCCATATGGACTGGCAGCAGAAGGCTCA
Amidase 6-pet28a-R	CGGAGCTCGAATTCGGATCCCTACTTTCCATCGGCCAAGA
Amidase 7-pet28a-F	TGCCGCGCGGCAGCCATATGACGGAACCTCACTGGAAATC
Amidase 7-pet28a-R	CGGAGCTCGAATTCGGATCCCTAACGCCCAAAAAGCGTAT

Amidase 8-pet28a-F	TGCCGCGCGGCAGCCATATGGAGTACTCTTCGATCGCCCA
Amidase 8-pet28a-R	CGGAGCTCGAATTCGGATCCCTACAGCTTTGACCTGTCAG
KAT-pet28a-F	TGCCGCGCGGCAGCCATATGTCGGCCACCACAGCTCGTTC
KAT-pet28a-R	TGGTGGTGGTGGTGCTCGAGTTACCGCACGATATACTTCT

Table S2. Plasmids used in this study.

Plasmid	Description	Aim
pJM01001	pYTP: amyB+kazA	<i>A. nidulans</i> overexpression
pJM01002	pYTR: gpdA+kazB	<i>A. nidulans</i> overexpression
pJM01003	pYTR: gpdA+kazE	<i>A. nidulans</i> overexpression
pJM01004	pYTR: gpdA+kazB+amyB+kazE	<i>A. nidulans</i> overexpression
pJM01005	pYTU: glaA+kazC	<i>A. nidulans</i> overexpression
pJM01006	pYTU: glaA+kazC+amyB+kazD	<i>A. nidulans</i> overexpression
pJM01007	pYTU: glaA+kazC+amyB+kazD+gpdA+kazF	<i>A. nidulans</i> overexpression
pJM01008	pYTR: gpdA+kazC	<i>A. nidulans</i> microsome
pJM01101	pET28a: NcoI/HindIII, C-histag-KazE	<i>E. coli</i> overexpression
pJM01102	pET28a: NdeI/BamHI, N-histag-Amidase 1	<i>E. coli</i> overexpression
pJM01103	pET28a: NdeI/BamHI, N-histag-Amidase 2	<i>E. coli</i> overexpression
pJM01104	pET28a: NdeI/BamHI, N-histag-Amidase 3	<i>E. coli</i> overexpression
pJM01105	pET28a: NdeI/BamHI, N-histag-Amidase 4	<i>E. coli</i> overexpression
pJM01106	pET28a: NdeI/BamHI, N-histag-Amidase 5	<i>E. coli</i> overexpression
pJM01107	pET28a: NdeI/BamHI, N-histag-Amidase 6	<i>E. coli</i> overexpression
pJM01108	pET28a: NdeI/BamHI, N-histag-Amidase 7	<i>E. coli</i> overexpression
pJM01109	pET28a: NdeI/BamHI, N-histag-Amidase 8	<i>E. coli</i> overexpression
pJM01110	pET28a: NdeI/XhoI, N-histag-KAT	<i>E. coli</i> overexpression

Table S3. Deduced functions of the ORFs in *kaz* gene cluster.

A. asperescens NRRL 4770



Gens	Proteins	Size (aa)	Proposed function	Identities to homologues	Protein homologues
<i>kazA</i>	KazA	1751	Nonribosomal Peptide Synthetase	88%	Nonribosomal Peptide Synthetase AacA
<i>kazB</i>	KazB	355	Methyltransferase	86%	N-methyltransferase AacB
<i>kazC</i>	KazC	515	Cytochrome P450 monooxygenase	89%	Cytochrome P450 monooxygenase AacC
<i>kazD</i>	KazD	486	Indoleamine 2,3 dioxygenase	92%	L-saccharopine oxidase AacD
<i>kazE</i>	KazE	405	Methyltransferase	35%	O-methyltransferase XanE
<i>kazF</i>	KazF	504	Cytochrome P450 monooxygenase	36%	Cytochrome P450 monooxygenase AstJ

Note: The name *kaz* was assigned in this work, with K representing kynurenic acid (KYNA), A indicating C-terminal amide formation, and Z serving as an arbitrary identifier.

Table S4. The hydrolyzation of **8** and **1** by amidases from *A.nidulans*.

Enzyme/Substrate	8	1
Amidase 1	×	×
Amidase 2	√	×
Amidase 3	×	√
Amidase 4	×	×
Amidase 5	×	×
Amidase 6	×	×
Amidase 7	×	×
Amidase 8	×	√

“×” indicates the substrate cannot be hydrolyzed by the corresponding enzyme, and “√” indicates the substrate can be hydrolyzed.

Table S5. Feeding experiments with substituted tryptophans.

A)

Substrate/Product	DKP (Ala)	N-MT DKP (Ala)	DKP (Gly)	N-MT DKP (Gly)
Trp	++++++	++++++	+++++	+++++
4-F Trp	+++	+++	-	-
5-F Trp	++	+	-	-
6-F Trp	++++	+++	-	-
7-F Trp	+++++	++	+++	+
4-Cl Trp	+	-	-	-
6-Cl Trp	++	++	+++	+
7-Cl Trp	+++	-	+++	-
5-Br Trp	-	-	-	-
6-Br Trp	+	-	-	-
5-OH Trp	-	-	-	-

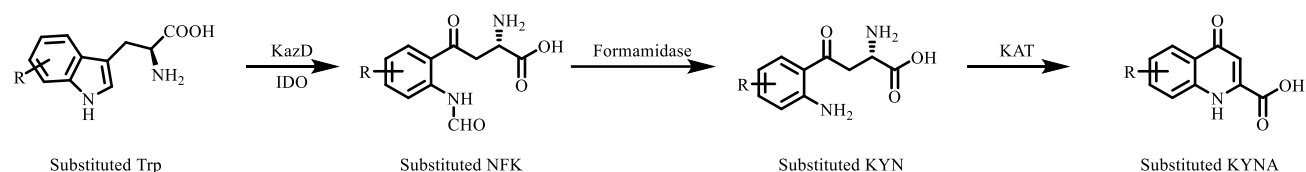
B)

Substrate/Product	DPP (Ala)	N-MT DPP (Ala)	DPP (Gly)	N-MT DPP (Gly)
Trp	++++++	++++++	+++++	+++++
4-F Trp	-	-	-	-
5-F Trp	-	-	-	-
6-F Trp	+++	+++	-	-
7-F Trp	++	-	+++	-
4-Cl Trp	-	-	-	-
6-Cl Trp	+++	+++	++	-
7-Cl Trp	++	-	+++	-
5-Br Trp	-	-	-	-
6-Br Trp	-	-	-	-
5-OH Trp	-	-	-	-

A) Production of diketopiperazine (DKP) products. Feeding various substituted tryptophans (0.2 g/L) to *A. nidulans* A1145 expressing *kazAB* for DKP (Ala) production, *kazABE* for N-MT DKP (Ala) production; Feeding various substituted tryptophans (0.2 g/L) and N-methylglycine (0.2 g/L) to *A. nidulans* A1145 expressing *kazA* for DKP(Gly) production, *kazAE* for N-MT DKP(Gly) production; B) Production of dipeptide (DPP) products. Feeding various substituted tryptophans (0.2 g/L) to *A. nidulans* A1145 expressing *kazABCD* for DPP (Ala) production, *kazABCDE* for N-MT DPP (Ala) production; Feeding various substituted tryptophans (0.2 g/L) and N-methylglycine (0.2 g/L) to *A. nidulans* A1145 expressing *kazACD* for DPP (Gly) production, *kazACDE* for N-MT DPP(Gly) production. The number of “+” denotes the relative peak height of the UV-detected (254nm) products. “-” denotes no expected product formed.

Feeding 5 Cl-substituted tryptophan did not yield corresponding metabolites, however, experimental data confirmed it can be taken up by the cells, suggesting potential metabolism via unknown alternative pathways. Thus, it has not been included in the preceding discussion.

Table S6. The recognition of substituted kynurenines by KazD or endogenous IDOs.



A)

	F-substituted Trp	F-substituted NFK	F-substituted KYN	F-substituted KYNA
vector-4F	√	-	-	√
KazD-4F	√	-	-	√
vector-5F	√	-	-	-
KazD-5F	√	-	-	-
vector-6F	-	-	-	√
KazD-6F	-	-	-	√
vector-7F	√	-	-	√
KazD-7F	√	-	-	√

B)

	Cl-substituted Trp	Cl -substituted NFK	Cl -substituted KYN	Cl-substituted KYNA
vector-4 Cl	-	-	-	√
KazD-4 Cl	-	-	-	√
vector-6 Cl	-	-	-	√
KazD-6 Cl	-	-	-	√
vector-7 Cl	√	-	-	√
KazD-7 Cl	√	-	-	√

C)

	Br-substituted Trp	Br -substituted NFK	Br -substituted KYN	Br -substituted KYNA
vector-5 Br	-	-	-	-
KazD-5 Br	-	-	-	-
vector-6 Br	-	-	-	-
KazD-6 Br	-	-	-	-

D)

	OH-substituted Trp	OH -substituted NFK	OH -substituted KYN	OH -substituted KYNA
vector-5 OH	-	-	-	-
KazD-5 OH	-	-	-	-

Detection of kynurenine pathway metabolites produced after feeding (A) F-substituted tryptophans; (B) Cl-substituted tryptophans; (C) Br-substituted tryptophans; (D) OH-substituted tryptophan. The reaction was performed by feeding 0.2 g/L substituted tryptophans to *A. nidulans* A1145 expressing empty vector or KazD. The number of “√” denotes expected product formed. “-” denotes no expected product formed.

Table S7. Summary of BBB-PAMPA permeability analysis of compounds.

Compound ID	Test Concentration (μM)	Incubation Time (hour)	Mean Pe (nm/s)	Mean %Solution Recovery	CNS +/- CNS-*
Nadolol	10.0	4	0.00289	93.1	CNS-
Carbamazepine	10.0	4	67.8	99.0	CNS+
KYNA	10.0	4	<0.185	<94.0	CNS-
1	10.0	4	0.0828	100.0	CNS-
2	10.0	4	<0.185	<107.2	CNS-
8	10.0	4	0.0348	85.0	CNS-
9	10.0	4	<0.185	<104.4	CNS-
10	10.0	4	0.0670	103.7	CNS-
11	10.0	4	<0.185	<98.0	CNS-
12	10.0	4	<0.185	<87.6	CNS-
13	10.0	4	0.0475	93.6	CNS-
14	10.0	4	<0.185	<91.9	CNS-

The signal responses of some test compounds in acceptor samples were undetectable. For the convenience of calculation, 1/300 of the measured peak area ratio (PAR) of the relevant AD samples were used as the PAR values of these test compounds in acceptor samples instead.

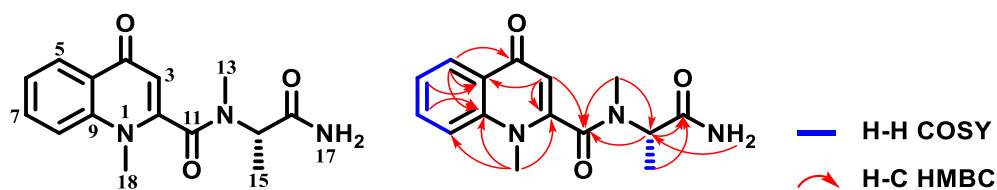
Classification:

CNS-: Pe <15.0nm/s

CNS+: Pe ≥ 15.0 nm/s

*BBB-PAMPA model may overestimate the BBB permeation because there are no efflux transporters expressed and the passive diffusion plays a dominant role.

Table S8. Spectroscopic data of **1**.



1a: 1b: 1c: 1d=1:1:1:2*

Position	1a in (500 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c	1b in (500 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c
1				
2		148.7		148.8
3	6.03 (s, 1H)	106.4	6.06 (s, 1H)	106.3
4		175.9		176.0
5	8.17 (m, 1H)	125.4	8.18 (m, 1H)	125.4
6	7.46 (m, 1H)	124.0	7.43 (m, 1H)	124.0
7	7.80 (m, 1H)	132.6	7.80 (m, 1H)	132.7
8	7.78 (m, 1H)	117.3	7.78 (m, 1H)	117.2
9		141.0		141.1
10		126.5		126.5
11		164.0		164.0
12				
13	2.97 (s, 3H)	28.7	2.84 (s, 3H)	32.6
14	4.27 (q, 1H, $J = 7.1$)	56.8	4.93 (q, 1H, $J = 7.2$)	52.4
15	1.38 (d, 3H, $J = 7.1$)	15.9	1.39 (d, 3H, $J = 7.2$)	14.1
16		171.7		171.6
17	7.26, 7.35 (NH ₂)		7.21, 7.54 (NH ₂)	
18	3.65 (s, 3H)	36.5	3.63 (s, 3H)	36.6
Position	1c in (500 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c	1d in (500 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c
1				
2		148.9		149.2
3	6.17 (s, 1H)	106.7	6.04 (s, 1H)	106.2
4		176.0		176.1
5	8.18 (m, 1H)	125.4	8.20 (m, 1H)	125.4
6	7.47 (m, 1H)	124.0	7.44 (m, 1H)	124.0
7	7.80 (m, 1H)	132.6	7.80 (m, 1H)	132.6
8	7.78 (m, 1H)	117.3	7.78 (m, 1H)	117.3
9		140.8		140.9
10		126.5		126.5
11		164.2		164.3
12				
13	2.98 (s, 3H)	28.6	2.88 (s, 3H)	32.5
14	4.36 (q, 1H, $J = 7.0$)	56.7	4.89 (q, 1H, $J = 7.4$)	51.9
15	1.42 (d, 3H, $J = 7.0$)	15.6	1.45 (d, 3H, $J = 7.5$)	14.6

16		171.8		172.6
17	7.29,7.38 (NH ₂)		7.20,7.56 (NH ₂)	
18	3.67 (s, 3H)	36.1	3.76 (s, 3H)	36.2

In dms_o-d₆, 500 MHz for ¹H NMR and 125 MHz for ¹³C NMR; Chemical shifts are reported in ppm.

The ¹H NMR, ¹³C NMR and 2D NMR spectra are provided in **Figures S24-28**.

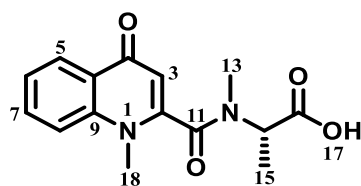
HRMS (ESI, M+H⁺) calculated for C₁₅H₁₈N₃O₃⁺ 288.1348; found 288.1358.

UV (MeCN : H₂O = 30:70) λ_{max}: 335nm.

[α]_D^{20.0} -18.57° (c 0.56, MeOH).

*Semi-preparative separation attempts of the four compounds were unsuccessful. According to the literature^{*1}, because the existent of group 16C(=O)-CH(Me)-N(Me)-C(=O)11, free rotation about the C-C single bond becomes restricted, resulting in a pair of atropisomers. Additionally, resonance occurs between the C=O and nitrogen atoms in the amide group, which imparts a certain double-bond character to the C-N bond. As a result, the C-N bond exhibits restricted rotation. This resonance interconversion between the keto and enol forms generates a nearly planar enol-like structure, leading to the nonequivalence of the two substituents on the nitrogen atom.

Table S9. Spectroscopic data of **2**.



2a: 2b: 2c: 2d=0.8:1:1:2

Position	2a in (600 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	2b in (600 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)
1		
2		
3	5.93 (s, 1H)	5.96 (s, 1H)
4		
5	8.20 (m, 1H)	8.20 (m, 1H)
6	7.47 (m, 1H)	7.47 (m, 1H)
7	7.80 (m, 1H)	7.80 (m, 1H)
8	7.80 (m, 1H)	7.80 (m, 1H)
9		
10		
11		
12		
13	2.96 (s, 3H)	2.90 (s, 3H)
14	4.42 (q, 1H, $J = 7.1$)	4.72 (q, 1H, $J = 7.3$)
15	1.38 (d, 3H, $J = 7.1$)	1.35 (d, 3H, $J = 6.9$)
16		
17	OH*	OH*
18	3.69 (s, 3H)	3.66 (s, 3H)
Position	2c in (600 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	2d in (600 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)
1		
2		
3	6.05 (s, 1H)	6.05 (s, 1H)
4		
5	8.20 (m, 1H)	8.20 (m, 1H)
6	7.47 (m, 1H)	7.47 (m, 1H)
7	7.80 (m, 1H)	7.80 (m, 1H)
8	7.80 (m, 1H)	7.80 (m, 1H)
9		
10		
11		
12		
13	2.92 (s, 3H)	2.87 (s, 3H)
14	4.38 (q, 1H, $J = 7.4$)	4.99 (q, 1H, $J = 7.3$)
15	1.46 (d, 3H, $J = 7.0$)	1.45 (d, 3H, $J = 7.5$)

16		
17	OH*	OH*
18	3.72 (s, 3H)	3.76 (s, 3H)

In dms o - d_6 , 600 MHz for ^1H NMR and 150 MHz for ^{13}C NMR; Chemical shifts are reported in ppm.

The ^1H NMR spectra are provided in **Figures S29-30**.

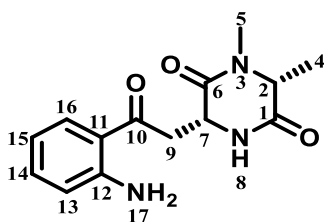
HRMS (ESI, $\text{M}+\text{H}^+$) calculated for $\text{C}_{15}\text{H}_{17}\text{N}_2\text{O}_4^+$ 289.1188; found 289.1196.

UV (MeCN : H_2O = 35:65) λ_{max} : 335nm.

$[\alpha]_{\text{D}}^{20.0}$ -28.00° (c 0.07 MeOH).

*The NMR signal of OH group was not detected. As only a small quantity of **2** was obtained, its structure was elucidated primarily by comparing its ^1H NMR with those of **1**, further supported by molecular weight from mass spectrometry.

Table S10. Spectroscopic data of **3**.



Position	3 in (500 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c	Reported 3* in (400 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c
1		167.5		167.4
2	3.96 (q, 1H, $J = 7.0$)	56.8	3.96 (q, 1H, $J = 7.0$)	56.7
3				
4	1.52 (d, 3H, $J = 7.0$)	17.5	1.52 (d, 3H, $J = 7.0$)	17.5
5	2.85 (s, 3H)	31.6	2.85 (s, 3H)	31.5
6		165.6		165.5
7	4.28 (m, 1H)	51.0	4.28 (m, 1H)	51.0
8	8.05 (d, NH, $J = 2.5$)		8.02 (NH)	
9	3.40 (d, 2H, $J = 5.0$)	42.5	3.40 (d, 2H, $J = 5.0$)	42.5
10		198.1		198.1
11		116.3		116.3
12		151.3		151.2
13	6.76 (dd, 1H, $J = 8.4, 1.2$)	117.0	6.76 (d, 1H, $J = 8.3$)	117.0
14	7.25 (ddd, 1H, $J = 8.4, 6.8, 1.5$)	134.5	7.25 (m, 1H)	134.4
15	6.53 (ddd, 1H, $J = 8.2, 6.9, 1.2$)	114.5	6.54 (t, 1H, $J = 7.2$)	114.5
16	7.73 (dd, 1H, $J = 8.3, 1.6$)	131.3	7.73 (d, 1H, $J = 7.3$)	131.2
17	7.22 (s, NH ₂)		7.20 (NH ₂)	

In dmsO- d_6 , 500 MHz for ^1H NMR and 125 MHz for ^{13}C NMR; Chemical shifts are reported in ppm.

The ^1H NMR and ^{13}C NMR spectra are provided in **Figures S31-32**.

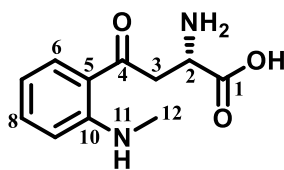
HRMS (ESI, $\text{M}+\text{H}^+$) calculated for $\text{C}_{14}\text{H}_{18}\text{N}_3\text{O}_3^+$ 276.1348; found 276.1343.

UV (MeCN : $\text{H}_2\text{O} = 35:65$) λ_{max} : 228 nm, 258nm and 364nm.

$[\alpha]_{\text{D}}^{20.0}$ -62.00° (c 0.20, MeOH).

*Kong D, Wang X, Liu L. Biosynthesis of a Novel Diketopiperazine Aspkyncin Incorporating a Kynurenine Unit from *Aspergillus aculeatus*. *J. Fungi*. **2025**, 11, 171.

Table S11. Spectroscopic data of **4**.



Position	4 in (500 MHz DMSO- <i>d</i> ₆), δ_H (mult., J_{H-H} in Hz)	δ_c	Reported 4* in (500 MHz DMSO- <i>d</i> ₆), δ_H (mult., J_{H-H} in Hz)	δ_c
1		169.3		169.6
2	3.62 (m, 1H)	49.9	3.62 (m, 1H)	49.9
3	3.29** (m, 1H) 3.54 (dd, 1H, $J = 3.3, 18.0$)	40.2	3.29 (m, 1H) 3.54 (dd, 1H, $J = 3.3, 18.0$)	40.2
4		199.5		199.5
5		116.7		116.7
6	7.79 (d, 1H, $J = 8.1$)	131.8	7.79 (dd, 1H, $J = 1.6, 8.1$)	131.7
7	6.61 (dd, 1H, $J = 7.5, 7.5$)	114.1	6.60 (m, 1H)	114.0
8	7.42 (dd, 1H, $J = 7.8, 7.8$)	135.3	7.42 (m, 1H)	135.2
9	6.75 (d, 1H, $J = 8.5$)	111.4	6.74 (m, 1H)	111.3
10		151.4		151.3
11	8.61 (q, NH, $J = 5.1$)		8.60 (NH)	
12	2.85 (d, 3H, $J = 4.9$)	29.1	2.86 (d, 3H, $J = 5.0$)	29.1

In dmsO-*d*₆, 500 MHz for ¹H NMR and 125 MHz for ¹³C NMR; Chemical shifts are reported in ppm.

The ¹H NMR and ¹³C NMR spectra are provided in **Figures S33-34**.

HRMS (ESI, M+H⁺) calculated for C₁₁H₁₅N₂O₃⁺ 223.1083; found 223.1079.

UV (MeCN : H₂O = 30:70) λ_{max} : 232 nm, 262nm and 390nm.

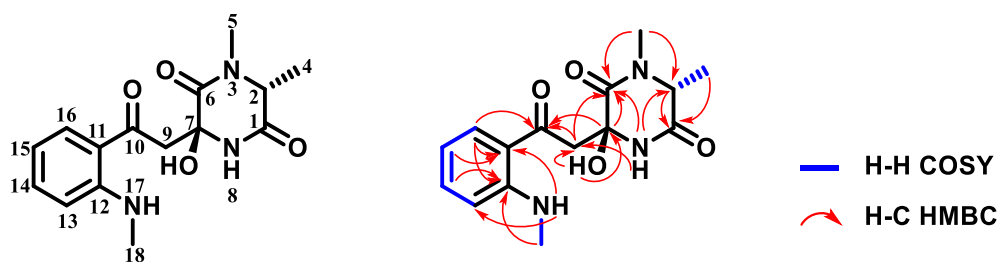
$[\alpha]_D^{20.0}$ 8.00° (*c* 0.10, MeOH).

*Liu M, Ohashi M, Tang Y. Engineered Biosynthesis of Fungal 4-Quinolone Natural Products. *Org Lett.* **2020**, 22, 6637-6641.

**The signal in the region δ 3.2-3.5 ppm were obscured by the intense water resonance.

$$[\alpha]_{\text{D}}^{20.0} -90.72^{\circ} (c\ 0.28, \text{MeOH}).$$

Table S13: Spectroscopic data of **6**.



Position	6 in (600 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c
1		167.4
2	4.02 (q, 1H, $J = 7.0$)	56.7
3		
4	1.56 (d, 3H, $J = 7.0$)	17.3
5	2.86 (s, 3H)	31.5
6		165.9
7	6.51 (s, OH) *	79.3
8	8.57 (s, NH)	
9	3.45 (d, 1H, $J = 17.3$) 3.68 (d, 1H, $J = 17.4$)	47.1
10		198.1
11		116.3
12		151.1
13	6.74 (dd, 1H, $J = 8.5, 1.2$)	117.1
14	7.24 (ddd, 1H, $J = 8.4, 6.9, 1.5$)	134.4
15	6.55 (ddd, 1H, $J = 8.2, 6.9, 1.2$)	114.6
16	7.66 (dd, 1H, $J = 8.3, 1.5$)	130.9
17	7.13 (s, NH ₂)	

In dmsO- d_6 , 600 MHz for ^1H NMR and 150 MHz for ^{13}C NMR; Chemical shifts are reported in ppm.

The ^1H NMR, ^{13}C NMR and 2D NMR spectra are provided in **Figures S40-44**.

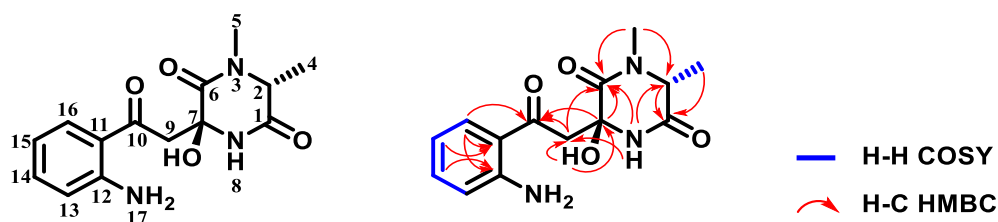
HRMS (ESI, $[\text{M}+\text{H}-\text{H}_2\text{O}]^+$) calculated for $\text{C}_{15}\text{H}_{18}\text{N}_3\text{O}_3^+$ 288.1303; found 288.1355.

UV (MeCN : $\text{H}_2\text{O} = 40:60$) λ_{max} : 232, 263 and 390nm.

$[\alpha]_{\text{D}}^{20.0}$ 50.63° (c 0.32, MeOH).

*The chirality of OH group is deduced by the consistence with direction of hydrogen atom released by P450 monooxygenase.

Table S14: Spectroscopic data of **7**.



Position	7 in (600 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c
1		167.5
2	4.02 (q, 1H, $J = 6.9$)	56.6
3		
4	1.56 (d, 3H, $J = 7.0$)	17.4
5	2.86 (s, 3H)	31.5
6		165.9
7	6.55 (s, OH) *	79.3
8	8.62 (s, NH)	
9	3.47 (d, 1H, $J = 17.4$) 3.71 (d, 1H, $J = 17.4$)	47.0
10		198.2
11		116.5
12		151.4
13	6.73 (dd, 1H, $J = 8.7, 1.1$)	111.5
14	7.41 (ddd, 1H, $J = 8.6, 7.0, 1.6$)	135.3
15	6.61 (ddd, 1H, $J = 8.2, 7.0, 1.2$)	114.0
16	7.75 (dd, 1H, $J = 8.2, 1.6$)	131.7
17	8.50 (q, NH, $J = 5.1$)	
18	2.82 (d, 3H, $J = 5.0$)	29.0

In dmsO- d_6 , 600 MHz for ^1H NMR and 150 MHz for ^{13}C NMR; Chemical shifts are reported in ppm.

The ^1H NMR, ^{13}C NMR and 2D NMR spectra are provided in **Figures S45-49**.

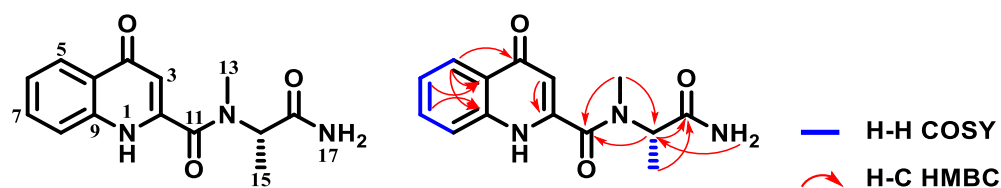
HRMS (ESI, $[\text{M}+\text{H}-\text{H}_2\text{O}]^+$) calculated for $\text{C}_{14}\text{H}_{16}\text{N}_3\text{O}_3^+$ 274.1147; found 274.1179.

UV (MeCN : $\text{H}_2\text{O} = 30:70$) λ_{max} : 230, 259 and 366nm.

$[\alpha]_{\text{D}}^{20.0}$ 35.92° (c 1.03, MeOH).

*The chirality of OH group is deduced by the consistence with direction of hydrogen atom released by P450 monooxygenase.

Table S15. Spectroscopic data of **8**.



8a: 8b=0.9:1

Position	8a in (500 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c	8b in (500MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c
1	11.93 (NH)		11.93 (NH)	
2		146.4*		146.4*
3	6.09 (s, 1H)	105.8	6.22 (s, 1H)	107.0
4		177.1*		177.3*
5	8.08 (d, 1H, $J = 8.2$)	124.7	8.08 (d, 1H, $J = 8.1$)	124.7
6	7.35 (m, 1H)	123.3	7.35 (m, 1H)	123.7
7	7.69 (m, 1H)	132.0	7.68 (m, 1H)	132.2
8	7.67 (d, 1H, $J = 8.6$)	118.9	7.60 (d, 1H, $J = 8.4$)	119.4
9		140.4		140.4
10		125.2		125.2
11		164.9		165.1
12				
13	2.92 (s, 3H)	28.8	2.94 (s, 3H)	33.1
14	4.39 (q, 1H, $J = 7.0$)	56.7	4.91 (q, 1H, $J = 7.2$)	52.4
15	1.36 (d, 3H, $J = 6.9$)	15.7	1.38 (d, 3H, $J = 7.3$)	14.3
16		171.9		172.1
17	7.28,7.47 (NH ₂)		7.23,7.47 (NH ₂)	

In dmsO- d_6 , 500 MHz for ^1H NMR and 125 MHz for ^{13}C NMR; Chemical shifts are reported in ppm.

The ^1H NMR, ^{13}C NMR and 2D NMR spectra are provided in **Figures S50-54**.

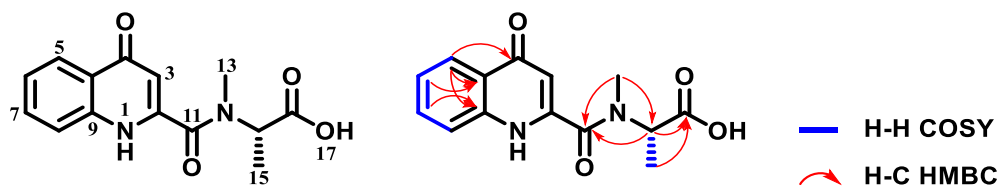
HRMS (ESI, $\text{M}+\text{H}^+$) calculated for $\text{C}_{14}\text{H}_{16}\text{N}_3\text{O}_3^+$ 274.1192; found 274.1183.

UV (MeCN : $\text{H}_2\text{O} = 30:70$) λ_{max} : 236 nm and 330nm.

$[\alpha]_{\text{D}}^{20.0}$ -36° (c 0.20, MeOH).

*The signal was only observed by HMBC correlation. According to the literature^{*2}, there is a tautomeric equilibrium between 4-quinolinone and 4-hydroxyquinoline, so it is hard to detect the NMR signal of position C2 and C4. Additionally, the NMR signal of carbon atom adjacent to a quadrupolar nucleus (such as nitrogen) typically is hard to be detected.

Table S16. Spectroscopic data of **9**.



9a: 9b=0.9:1

Position	9a in (500 MHz DMSO- <i>d</i> ₆), δ_H (mult., J_{H-H} in Hz)	δ_c	9b in (500 MHz DMSO- <i>d</i> ₆), δ_H (mult., J_{H-H} in Hz)	δ_c
1	12.35 (NH)		12.13 (NH)	
2		144.0		144.0
3	6.08 (s, 1H)	105.9	6.08 (s, 1H)	107.1
4		176.7		176.7
5	8.08 (m, 1H)	124.8	8.07 (m, 1H)	124.8
6	7.37 (m, 1H)	123.7	7.35 (m, 1H)	123.7
7	7.69 (m, 1H)	132.1	7.68 (m, 1H)	132.3
8	7.67 (d, 1H, $J = 8.4$)	118.8	7.62 (d, 1H, $J = 8.4$)	118.8
9		140.0		140.0
10		125.4		125.4
11		164.6		164.6
12				
13	2.90 (s, 3H)	28.2	2.98 (s, 3H)	33.8
14	4.60 (q, 1H, $J = 7.0$)	56.6	4.92 (q, 1H, $J = 7.3$)	52.7
15	1.39 (d, 3H, $J = 7.1$)	15.2	1.45 (d, 3H, $J = 7.3$)	14.0
16		172.2		172.3
17	12.96 (OH)		12.96 (OH)	

In dmso-*d*₆, 500 MHz for ¹H NMR and 125 MHz for ¹³C NMR; Chemical shifts are reported in ppm.

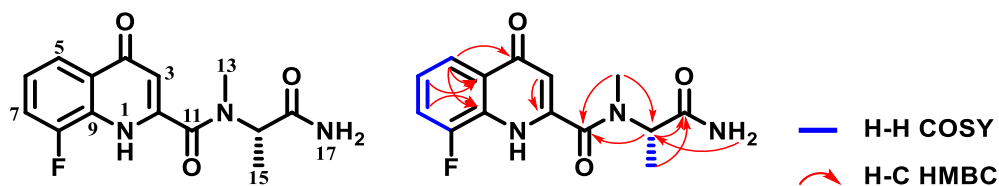
The ¹H NMR, ¹³C NMR and 2D NMR spectra are provided in **Figures S55-59**.

HRMS (ESI, M+H⁺) calculated for C₁₄H₁₅N₂O₄⁺ 275.1032; found 275.1050.

UV (MeCN : H₂O = 30:70) $\lambda_{\max}$: 239 nm and 330nm.

$[\alpha]_D^{20.0}$ -39.2° (*c* 0.07, MeOH).

Table S17: Spectroscopic data of **10**.



10a: 10b=0.9:1.3

Position	10a in (500 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c	10b in (500MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c
1	12.14 (NH)*		12.14 (NH)*	
2		Not observed**		Not observed**
3	6.48 (s, 1H)	105.7	6.48 (s, 1H)	105.7
4		Not observed**		Not observed**
5	7.93 (d, 1H, $J = 8.2$)	119.5	7.93 (d, 1H, $J = 8.2$)	119.5
6	7.38 (m, 1H)	123.8	7.38 (m, 1H)	123.8
7	7.61 (m, 1H)	116.2	7.55 (m, 1H)	115.2
8		Not observed**		Not observed**
9		136.1		136.1
10		130.4		130.4
11		165.6***		166.7***
12				
13	2.85 (s, 3H)	32.5	2.87 (s, 3H)	28.4
14	4.98 (q, 1H, $J = 7.3$)	56.1	4.42 (m, 1H)	51.9
15	1.35 (d, 3H, $J = 7.3$)	14.9	1.26 (d, 3H, $J = 7.0$)	14.0
16		172.2		172.0
17	7.24, 7.43 (NH ₂)		7.30 (NH ₂)	

In dmso- d_6 , 500 MHz for ^1H NMR and 125 MHz for ^{13}C NMR; Chemical shifts are reported in ppm.

The ^1H NMR, ^{13}C NMR, 2D NMR and ^{19}F NMR spectra are provided in **Figures S60-64**.

HRMS (ESI, $\text{M}+\text{H}^+$) calculated for $\text{C}_{14}\text{H}_{15}\text{N}_3\text{O}_3\text{F}^+$ 292.1097; found 292.1106

UV (MeCN : $\text{H}_2\text{O} = 30:70$) λ_{max} : 332nm.

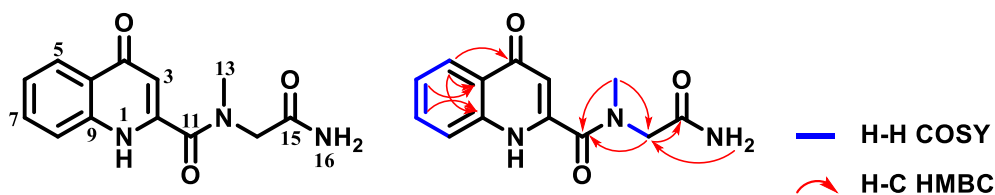
$[\alpha]_{\text{D}}^{20.0}$ -23.53° (c 0.34, MeOH).

* The NMR signal of NH group requires amplification to be seen.

Like **8, there is a tautomeric equilibrium between 4-quinolinone and 4-hydroxyquinoline, so it may can't observe the NMR signal of position C2 and C4 if a small quantity of the compound was isolated.

***The signal was only observed by HMBC correlation. The NMR signal of carbon atom adjacent to a quadrupolar nucleus (such as nitrogen or fluorine atom) typically is hard to be detected.

Table S18. Spectroscopic data of **11**.



11a: 11b=1:1.2

Position	11a in (500MHz DMSO- <i>d</i> ₆), δ_H (mult., J_{H-H} in Hz)	δ_c	11b in (500MHz DMSO- <i>d</i> ₆), δ_H (mult., J_{H-H} in Hz)	δ_c
1	12.05 (NH)		12.05 (NH)	
2		Not observed*		Not observed*
3	6.19 (s, 1H)	106.4	6.11 (s, 1H)	106.4
4		Not observed*		Not observed*
5	8.09 (d, 1H, $J = 8.1$)	124.6	8.07 (d, 1H, $J = 8.1$)	124.6
6	7.38 (m, 1H)	123.7	7.35 (m, 1H)	123.7
7	7.70 (m, 1H)	132.1	7.72 (m, 1H)	132.1
8	7.65 (m, 1H)	119.0	7.67 (m, 1H)	118.9
9		140.3**		140.3**
10		125.1**		125.1**
11		165.0**		165.5**
12				
13	3.05 (s, 3H)	38.1	2.99 (s, 3H)	34.2
14	4.06 (s, 2H)	49.8	4.01 (s, 2H)	53.0
15		169.3		169.5
16	7.25, 7.56 (NH ₂)		7.24, 7.52 (NH ₂)	

In dmsO-*d*₆, 500 MHz for ¹H NMR and 125 MHz for ¹³C NMR; Chemical shifts are reported in ppm.

The ¹H NMR, ¹³C NMR and 2D NMR spectra are provided in **Figures S65-69**.

HRMS (ESI, M+H⁺) calculated for C₁₃H₁₄N₃O₃⁺ 260.1035; found 260.1036.

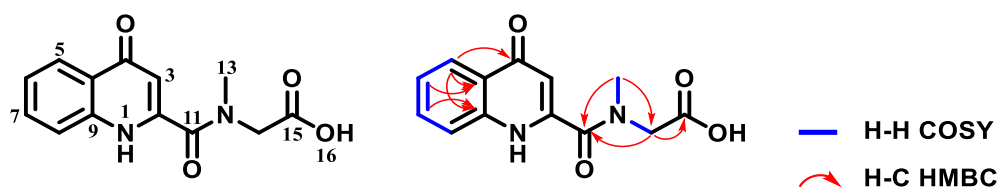
UV (MeCN : H₂O = 25:75) λ_{max} : 317nm.

$[\alpha]_D^{20.0}$ -1.74° (*c* 0.23, MeOH).

*Like **8**, there is a tautomeric equilibrium between 4-quinolinone and 4-hydroxyquinoline, so it may can't observe the NMR signal of position C2 and C4 if a small quantity of the compound was isolated.

**The signal was only observed by HMBC correlation.

Table S19: Spectroscopic data of **12**.



12a: 12b=1:1

Position	12a in (500 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c	12b in (500 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c
1	12.59 (NH)		12.28 (NH)	
2		143.6		143.5
3	6.09 (s, 1H)	106.1	6.09 (s, 1H)	107.2
4		176.7		176.8
5	8.06 (d, 1H, $J = 8.1$)	124.5	8.09 (d, 1H, $J = 8.1$)	124.7
6	7.36 (m, 1H)	123.9	7.35 (m, 1H)	123.9
7	7.69 (m, 1H)	132.0	7.67 (m, 1H)	132.2
8	7.66 (m, 1H)	118.9	7.64 (m, 1H)	119.7*
9		141.0*		140.1*
10		125.1*		125.1*
11		165.2		164.7
12				
13	3.02 (s, 3H)	37.9	3.08 (s, 3H)	34.3
14	4.18 (s, 2H)	52.8	4.13 (s, 2H)	48.8
15		169.9		170.7
16	12.98 (OH)		12.98 (OH)	

In dmsO- d_6 , 500 MHz for ^1H NMR and 125 MHz for ^{13}C NMR; Chemical shifts are reported in ppm.

The ^1H NMR, ^{13}C NMR and 2D NMR spectra are provided in **Figures S70-74**.

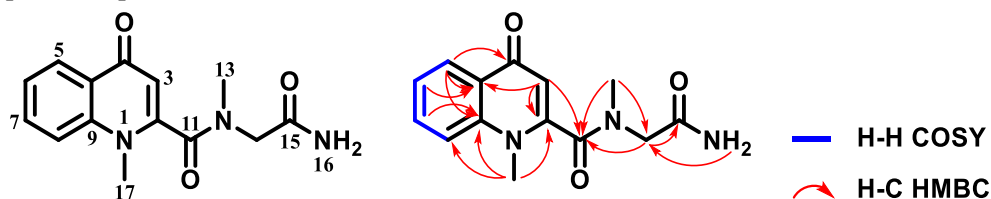
HRMS (ESI, $\text{M}+\text{H}^+$) calculated for $\text{C}_{13}\text{H}_{13}\text{N}_2\text{O}_4^+$ 261.0875; found 261.0880.

UV (MeCN : $\text{H}_2\text{O} = 25:75$) λ_{max} : 317nm.

(similar to **11**, **12**, **13**, **14** and **40** has no chirality, so the specific rotation data of them will not be provided)

*The signal was only observed by HMBC correlation.

Table S20: Spectroscopic data of **13**.



13a: 13b=0.9:1.7

Position	13a in (500 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c	13b in (500 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c
1				
2		148.3		148.9
3	6.03 (s, 1H)	106.8	6.04 (s, 1H)	106.4
4		175.9		176.0
5	8.17 (d, 1H, $J = 7.8$)	125.4	8.19 (d, 1H, $J = 8.0$)	125.4
6	7.46 (m, 1H)	123.9	7.46 (m, 1H)	124.0
7	7.80 (m, 1H)	132.7	7.80 (m, 1H)	132.6
8	7.79 (m, 1H)	117.2	7.79 (m, 1H)	117.3
9		141.1		140.9
10		126.4		126.5
11		164.4		164.2
12				
13	3.01 (s, 3H)	33.6	2.94 (s, 3H)	37.4
14	3.93 (d, 1H, $J = 17.4$) 3.99 (d, 1H, $J = 17.4$)	52.8	3.83 (d, 1H, $J = 16.2$) 4.35 (d, 1H, $J = 16.2$)	49.0
15		169.1		169.1
16	7.21, 7.46 (NH ₂)		7.21, 7.61 (NH ₂)	
17	3.65 (s, 3H)	36.6	3.81 (s, 3H)	36.2

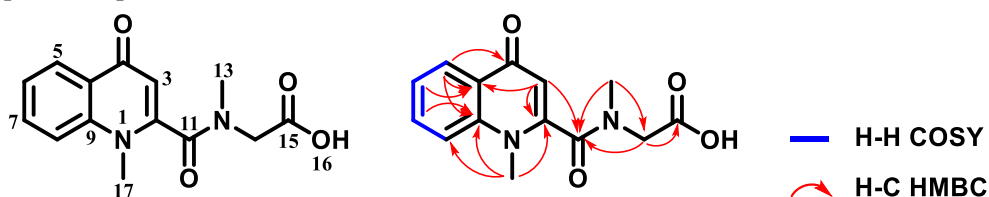
In dms o - d_6 , 500 MHz for ^1H NMR and 125 MHz for ^{13}C NMR; Chemical shifts are reported in ppm.

The ^1H NMR, ^{13}C NMR and 2D NMR spectra are provided in **Figures S75-79**.

HRMS (ESI, $\text{M}+\text{H}^+$) calculated for $\text{C}_{14}\text{H}_{16}\text{N}_3\text{O}_3^+$ 274.1192; found 274.1183.

UV (MeCN : H₂O = 25:75) λ_{max} : 336nm.

Table S21: Spectroscopic data of **14**.



14a: 14b=0.8:1

Position	14a in (600 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c	14b in (600 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c
1				
2		149.0		149.3
3	6.00 (s, 1H)	106.7	6.00 (s, 1H)	106.3
4		176.3		176.1
5	8.17 (d, 1H, $J = 7.9$)	125.4	8.20 (d, 1H, $J = 8.0$)	125.4
6	7.46 (m, 1H)	123.9	7.43 (m, 1H)	124.0
7	7.80 (m, 1H)	132.6	7.80 (m, 1H)	132.7
8	7.77 (m, 1H)	117.2	7.77 (m, 1H)	117.3
9		141.1		140.9
10		126.4		126.4
11		164.0		163.9
12				
13	3.01 (s, 3H)	33.5	2.93 (s, 3H)	36.2
14	3.74(m, 2H)	52.4*	4.29 (d, 2H, $J = 16.3$)	48.6*
15		168.7*		168.7*
16	OH**		OH**	
17	3.67 (s, 3H)	36.4	3.83 (s, 3H)	37.1

In dmso- d_6 , 600 MHz for ^1H NMR and 150 MHz for ^{13}C NMR; Chemical shifts are reported in ppm.

The ^1H NMR, ^{13}C NMR and 2D NMR spectra are provided in **Figures S80-82**.

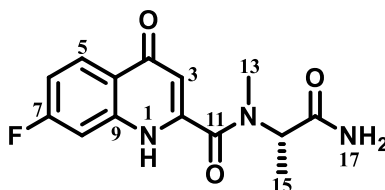
HRMS (ESI, $\text{M}+\text{H}^+$) calculated for $\text{C}_{14}\text{H}_{15}\text{N}_2\text{O}_4^+$ 275.1032; found 275.1050.

UV (MeCN : H_2O = 30:70) λ_{max} : 336nm.

*The signal was only observed by HMBC correlation.

**The NMR signal of OH group was not detected.

Table S22: Spectroscopic data of **35**.



35a: 35b=1:1.1

Position	35a in (500 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c	35b in (500MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c
1	12.26 (NH)*		12.26 (NH)*	
2		Not observed**		Not observed**
3	6.13 (s, 1H)	107.3	6.27 (s, 1H)	106.1
4		Not observed**		Not observed**
5	8.13 (d, 1H, $J = 8.0$)	128.0	8.14 (d, 1H, $J = 7.8$)	128.0
6	7.22 (m, 1H)	112.7	7.22 (m, 1H)	112.9
7		164.7		165.0
8	7.44 (m, 1H)	104.3	7.44 (m, 1H)	104.3
9		Not observed**		Not observed**
10		123.6		122.1
11		163.1		163.1
12				
13	2.92 (s, 3H)	28.8	2.95 (s, 3H)	33.2
14	4.90 (q, 1H, $J = 7.1$)	56.7	4.41 (q, 1H, $J = 6.9$)	52.5
15	1.37 (d, 3H, $J = 7.2$)	15.7	1.37 (d, 3H, $J = 7.2$)	14.3
16		171.8		172.1
17	7.24,7.41 (NH ₂)		7.20,7.35 (NH ₂)	

In dmso- d_6 , 500 MHz for ^1H NMR and 125 MHz for ^{13}C NMR; Chemical shifts are reported in ppm.

The ^1H NMR, ^{13}C NMR and ^{19}F NMR spectra are provided in **Figures S83-85**.

HRMS (ESI, $\text{M}+\text{H}^+$) calculated for $\text{C}_{14}\text{H}_{15}\text{N}_3\text{O}_3\text{F}^+$ 292.1097; found 292.1143.

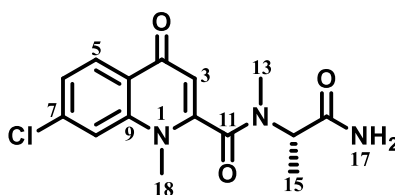
UV (MeCN : $\text{H}_2\text{O} = 30:70$) λ_{max} : 332nm.

$[\alpha]_{\text{D}}^{20.0}$ -26.00° (c 0.20, MeOH).

* The NMR signal of NH group requires amplification to be seen.

Like **8, there is a tautomeric equilibrium between 4-quinolinone and 4-hydroxyquinoline, so it may can't observe the NMR signal of position C2 and C4 if a small quantity of the compound was isolated.

Table S23: Spectroscopic data of **38**.



38a: 38b: 38c: 38d=1:1:1:2

Position	38a in (500 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c	38b in (500 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c
1				
2		149.1		149.2
3	6.06 (s, 1H)	107.0	6.09 (s, 1H)	106.9
4		175.3		175.4
5	8.15 (m, 1H)	127.9	8.16 (m, 1H)	127.5
6	7.49 (m, 1H)	124.3	7.46 (m, 1H)	124.2
7		137.5		137.5
8	7.90 (m, 1H)	117.0	7.90 (m, 1H)	117.0
9		142.0		142.0
10		125.1		125.0
11		163.6		163.6
12				
13	2.96 (s, 3H)	28.7	2.84 (s, 3H)	32.6
14	4.28 (q, 1H, $J = 7.0$)	56.7	4.91 (q, 1H, $J = 7.2$)	52.5
15	1.37 (d, 3H, $J = 6.9$)	15.8	1.38 (d, 3H, $J = 7.1$)	14.0
16		171.6		171.5
17	7.26, 7.32 (NH ₂)		7.18, 7.51 (NH ₂)	
18	3.64 (s, 3H)	36.7	3.62 (s, 3H)	36.8
Position	38c in (500 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c	38d in (500 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c
1				
2		149.3		149.6
3	6.19 (s, 1H)	107.3	6.07 (s, 1H)	106.8
4		175.4		175.4
5	8.16 (m, 1H)	129.4	8.18 (m, 1H)	127.5
6	7.50 (m, 1H)	124.2	7.47 (m, 1H)	124.2
7		137.5		137.5
8	7.90 (m, 1H)	117.0	7.88 (m, 1H)	117.0
9		141.8		141.8
10		126.1		125.0
11		163.9		163.9
12				
13	2.97 (s, 3H)	28.6	2.88 (s, 3H)	32.4
14	4.36 (q, 1H, $J = 7.0$)	56.6	4.88 (q, 1H, $J = 7.4$)	51.9
15	1.41 (d, 3H, $J = 7.1$)	15.4	1.45 (d, 3H, $J = 7.4$)	14.6

16		171.7		172.5
17	7.27,7.35 (NH ₂)		7.17,7.53 (NH ₂)	
18	3.66 (s, 3H)	36.2	3.74 (s, 3H)	36.4

In dms-*d*₆, 500 MHz for ¹H NMR and 125 MHz for ¹³C NMR; Chemical shifts are reported in ppm.

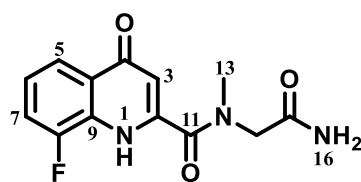
The ¹H NMR and ¹³C NMR spectra are provided in **Figures S86-87**.

HRMS (ESI, M+H⁺) calculated for C₁₅H₁₇N₃O₃Cl⁺ 322.0958; found 322.0977.

UV (MeCN : H₂O = 35:65) λ_{max}: 335nm.

[α]_D^{20.0} -20.8° (c 0.50, MeOH).

Table S24: Spectroscopic data of **40**.



40a: 40b=1:1.3

Position	40a in (500 MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c	40b in (500MHz DMSO- d_6), δ_H (mult., J_{H-H} in Hz)	δ_c
1	Not observed		Not observed	
2		Not observed*		Not observed*
3	6.55 (s, 1H)	105.8	6.55 (s, 1H)	105.8
4		Not observed*		Not observed*
5	7.94 (dd, 1H, $J = 8.2, 1.3$)	119.4	7.92 (dd, 1H, $J = 8.3, 1.3$)	119.3
6	7.40 (m, 1H)	125.3	7.40 (m, 1H)	124.1
7	7.59 (m, 1H)	115.4	7.56 (m, 1H)	115.4
8		Not observed**		Not observed**
9		Not observed**		Not observed**
10		Not observed**		Not observed**
11		168.6		166.4
12				
13	2.98 (s, 3H)	34.1	2.98 (s, 3H)	37.6
14	4.04 (s, 2H)	49.8	4.01 (s, 2H)	53.0
15		169.5		169.9
16	7.20, 7.48 (NH ₂)		7.14, 7.53 (NH ₂)	

In dmsO- d_6 , 500 MHz for ^1H NMR and 125 MHz for ^{13}C NMR; Chemical shifts are reported in ppm.

The ^1H NMR, ^{13}C NMR and ^{19}F NMR spectra are provided in **Figures S88-90**.

HRMS (ESI, $\text{M}+\text{H}^+$) calculated for $\text{C}_{13}\text{H}_{14}\text{N}_3\text{O}_3\text{F}^+$ 278.0941; found 278.0930.

UV (MeCN : H₂O = 25:75) λ_{max} : 332nm.

*Like **8**, there is a tautomeric equilibrium between 4-quinolinone and 4-hydroxyquinoline, so it may can't observe the NMR signal of position C2 and C4 if a small quantity of the compound was isolated.

**Due to the influence of fluorine substitution, the ^{13}C signal of the relevant quaternary carbon cannot be observed.

SUPPLEMENTARY FIGURES

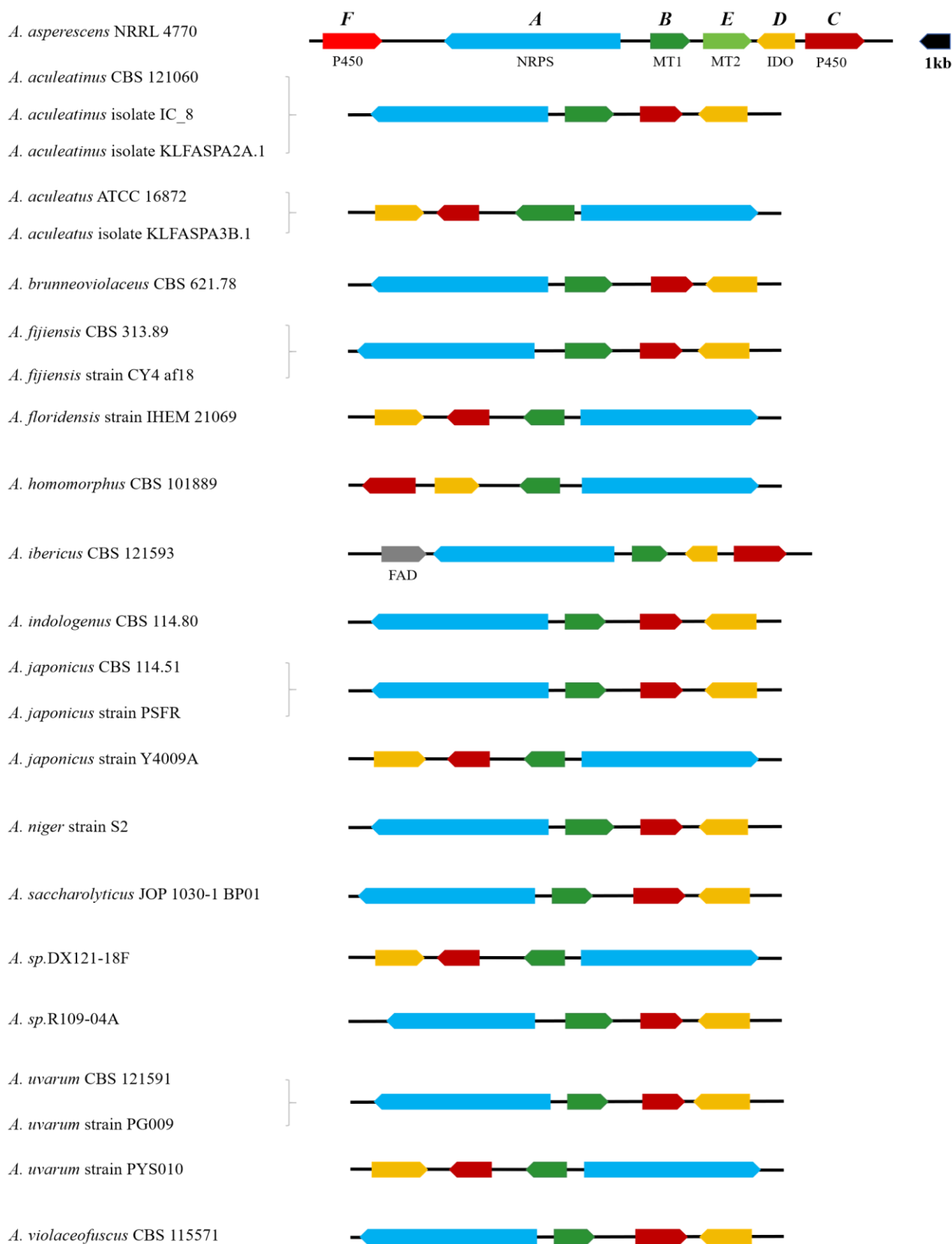


Figure S1. Homologous gene clusters of *kaz* in fungi.

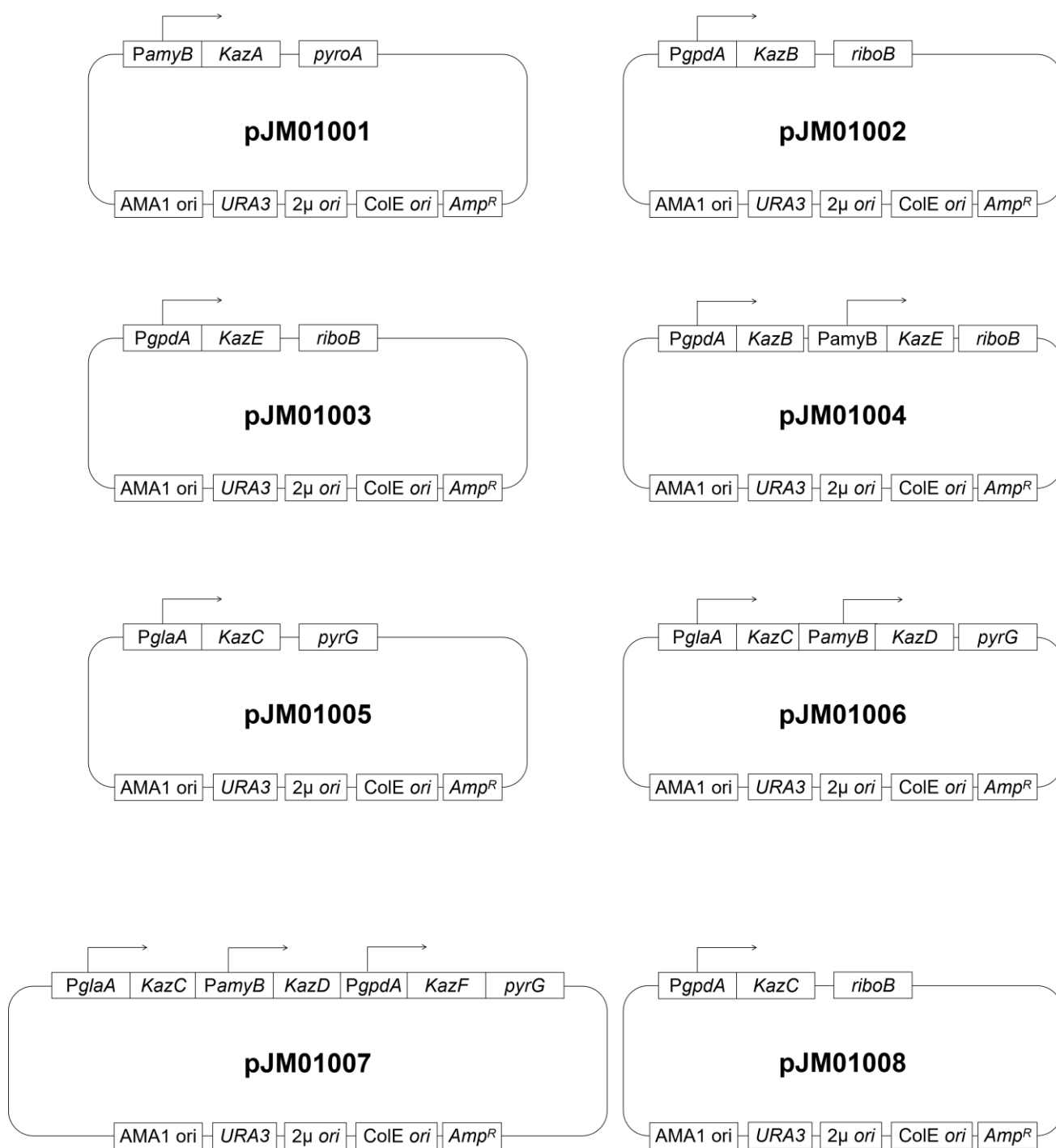


Figure S2. Plasmids for heterologous expression of *kaz* in *A. nidulans* A1145.

In *A. nidulans* ($\lambda=254\text{nm}$)

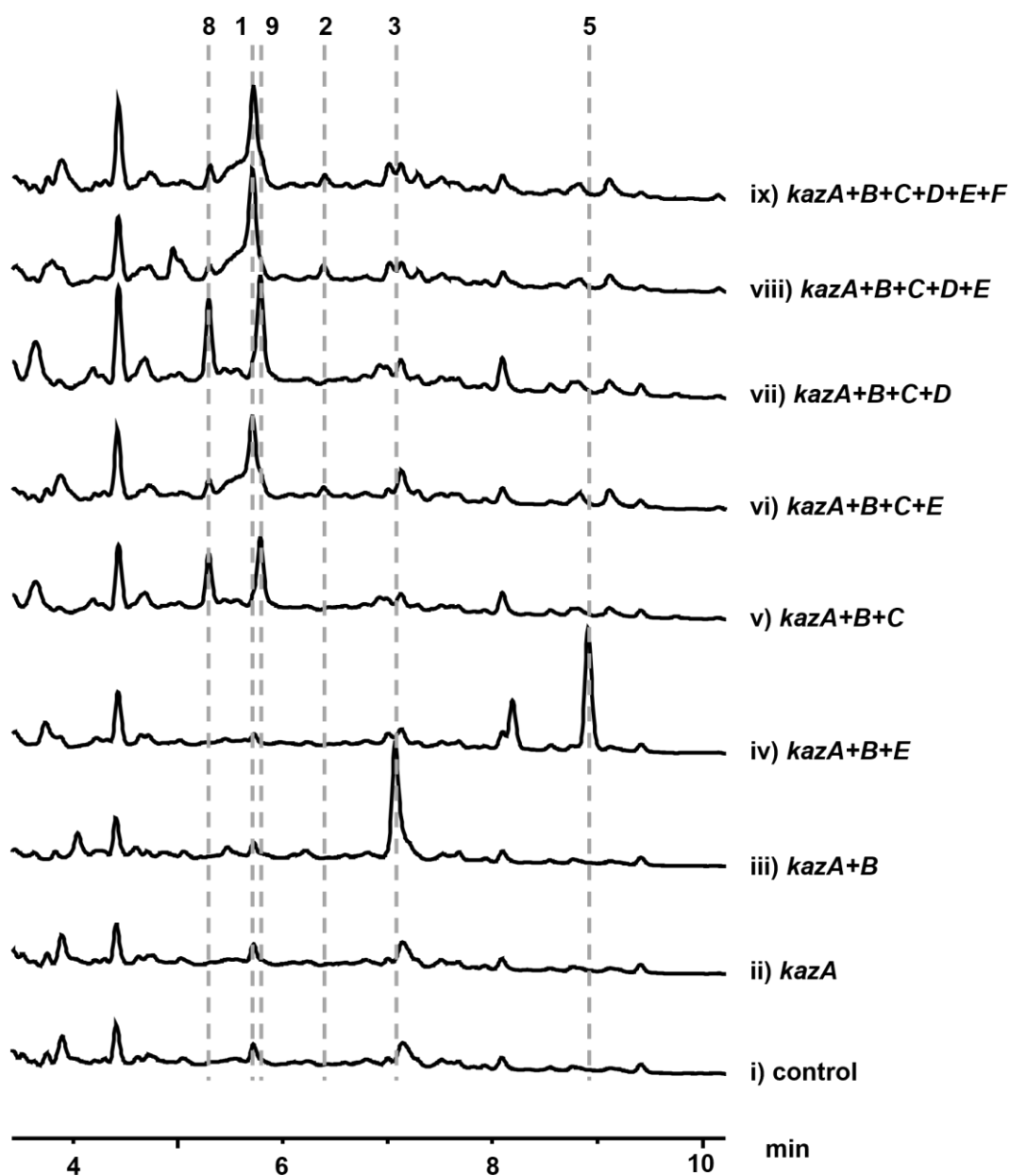


Figure S3. The HPLC profile of heterologous expression of *kazA-F* in *A. nidulans* A1145. Introduction of *kazD* can increase the production of **1**, **2**, **8** and **9** slightly. No new trace was observed by adding *kazF* to *kazABCDE* expression strain. The samples were analyzed on an Agilent Technologies 1200 Series HPLC.

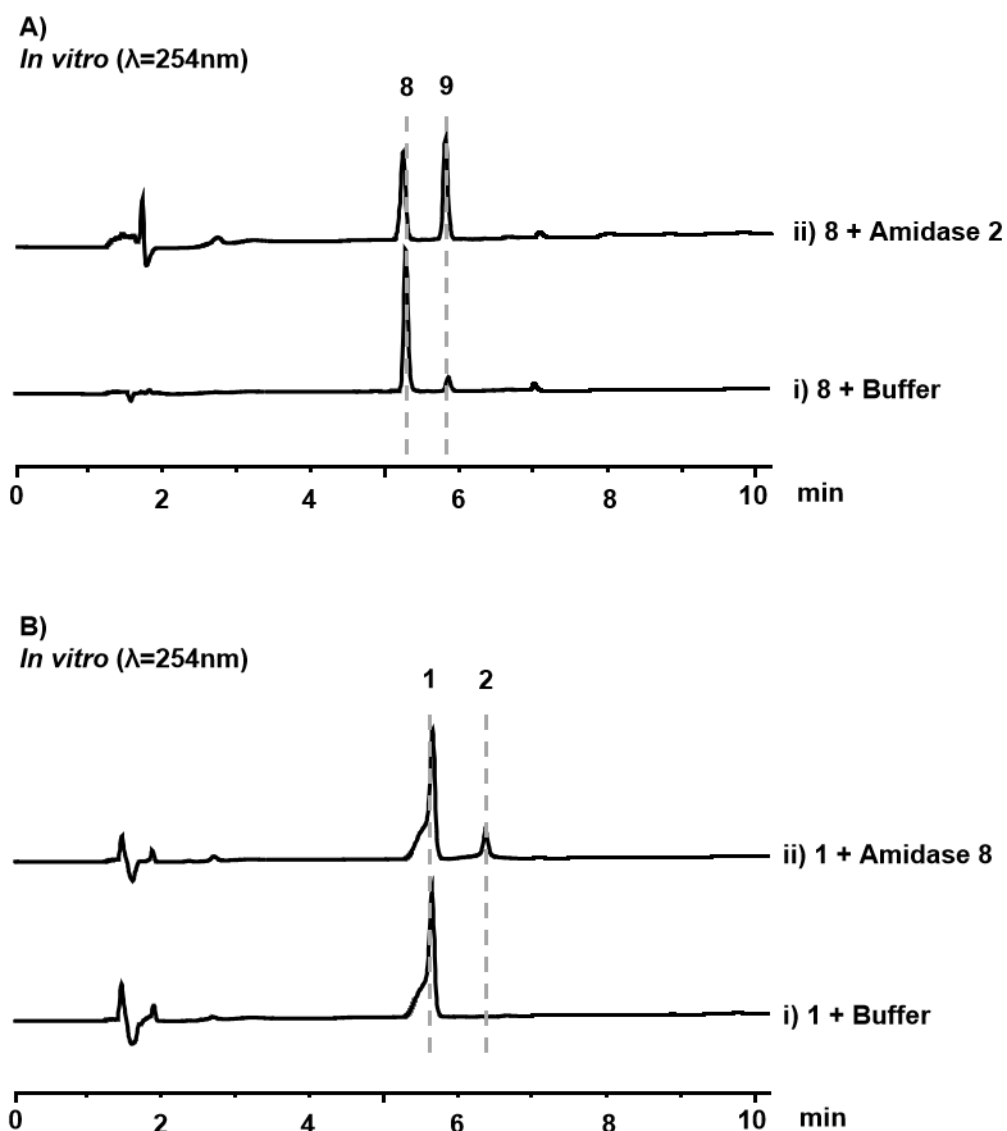
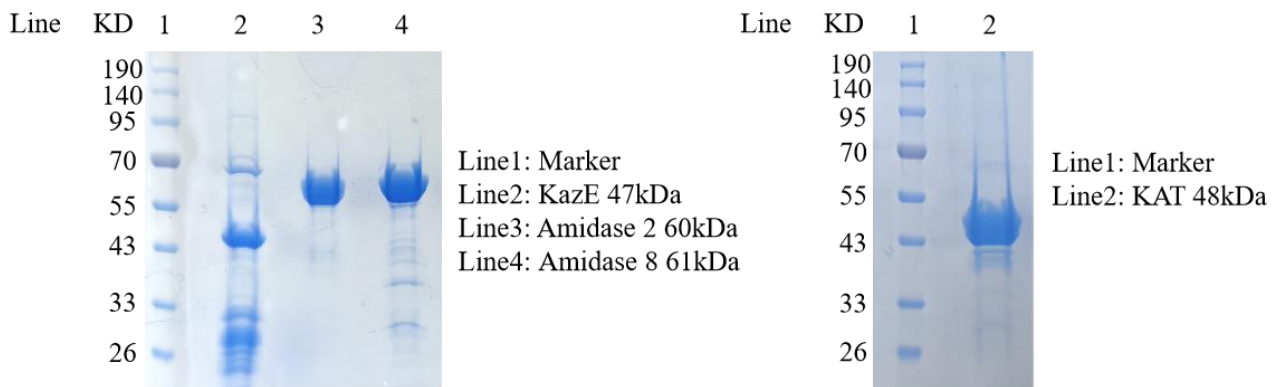


Figure S4. *In vitro* characterization of amidase 2 and 8. Assay for amidases 1-8 activity of **8** or **1** can be found in **Table S4**. A) The conversion of **8** to **9** by Amidase 2. B) The conversion of **1** to **2** by Amidase 8. i) Control reaction using buffer without enzyme. ii) The amidase catalyzed reaction was carried out in 50 mM sodium phosphate buffer containing 20 μM amidases and 1 mM of **8** or **1** in 50 μL of solution at pH 7.4. The reaction was incubated at 30 $^{\circ}\text{C}$ for overnight and quenched by the addition of 200 μL of methanol, analyzed on an Agilent Technologies 1200 Series HPLC.

A)



B)

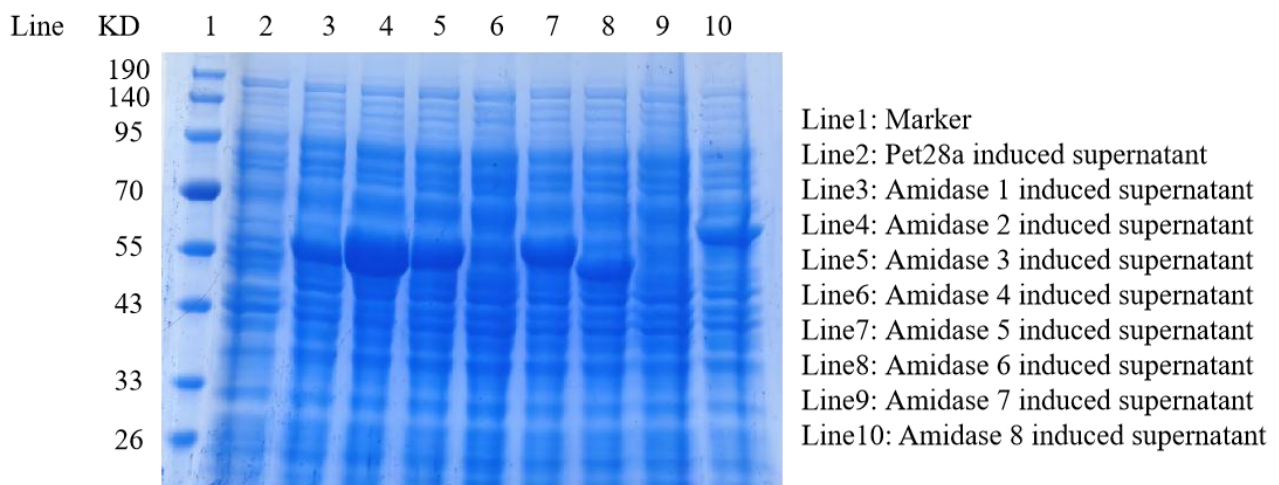
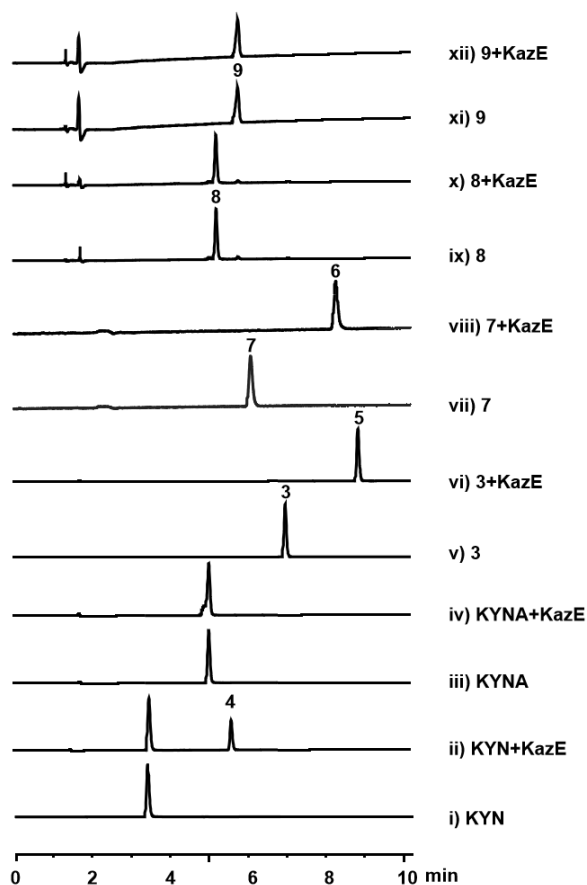


Figure S5. SDS-PAGE analysis of wild-type KazE, KAT and amidases 1-8. A) Purified wild-type KazE, amidase 2, amidase 8 and KAT. B) Supernatant of wild-type amidases 1-8.

A)

In vitro ($\lambda=360\text{nm}$)

B)

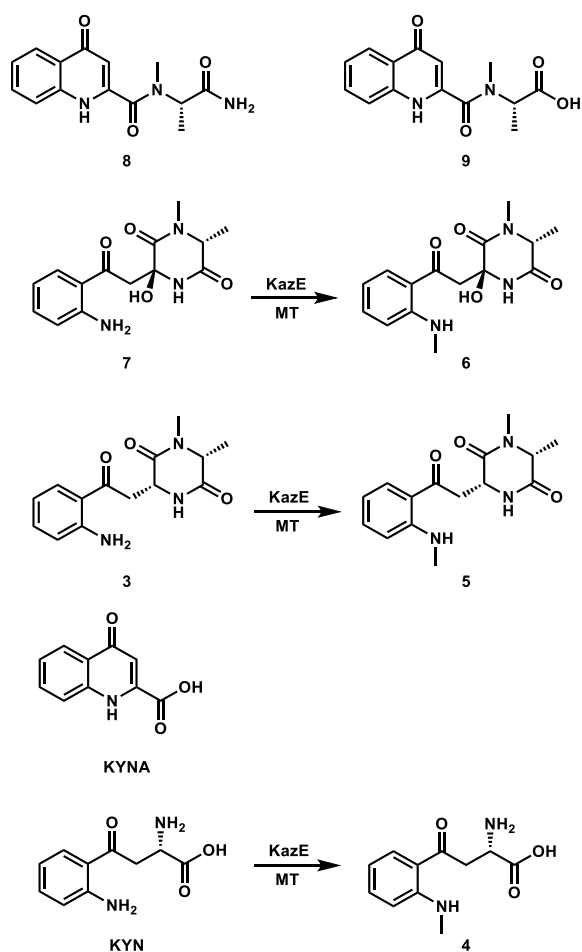


Figure S6. *In vitro* characterization of KazE. A) The reaction was carried out in 50 mM sodium phosphate buffer (pH 7.4) containing 5 μM of KazE, 2 mM of SAM and i) containing 2 mM of **KYN**, ii) containing 2 mM of **KYNA**, iii) containing 2 mM of **3**, iv) containing 2 mM of **7**, v) containing 2 mM of **8**, vi) containing 2 mM of **9** in 50 μL of solution. Reactions were incubated at 30°C for 1h, then quenched by the addition of 200 μL of methanol, analyzed on an Agilent Technologies 1200 Series HPLC. Control reaction was carried out using standard substrate and buffer. B) Structures involved in the reaction.

In *A. nidulans* ($\lambda=360\text{nm}$)

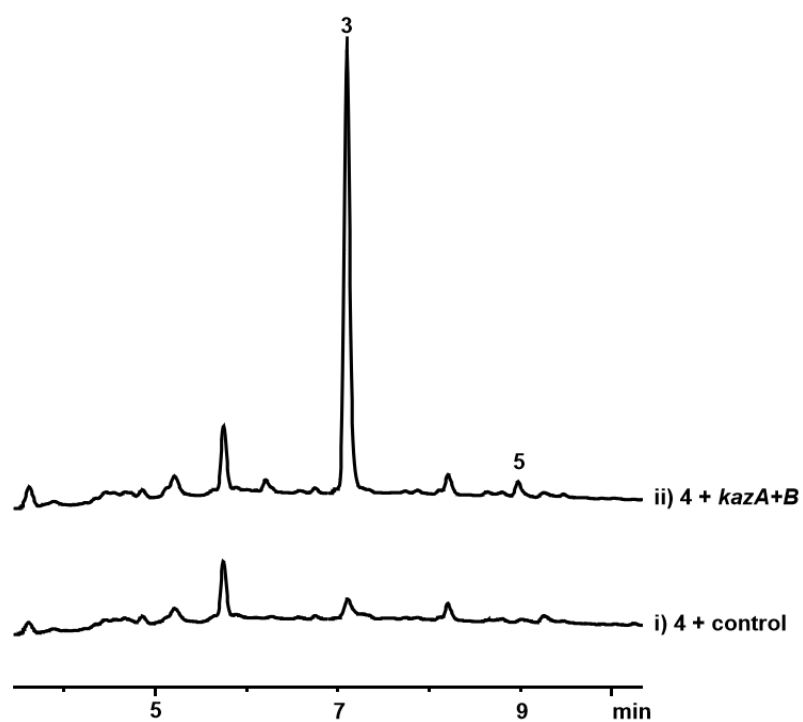


Figure S7. The recognition of **4** by *kaz*. Feeding 0.2 g/L **4** in to *A. nidulans* A1145 expressing *kazAB* for **5** production, analyzed on an Agilent Technologies 1200 Series HPLC.

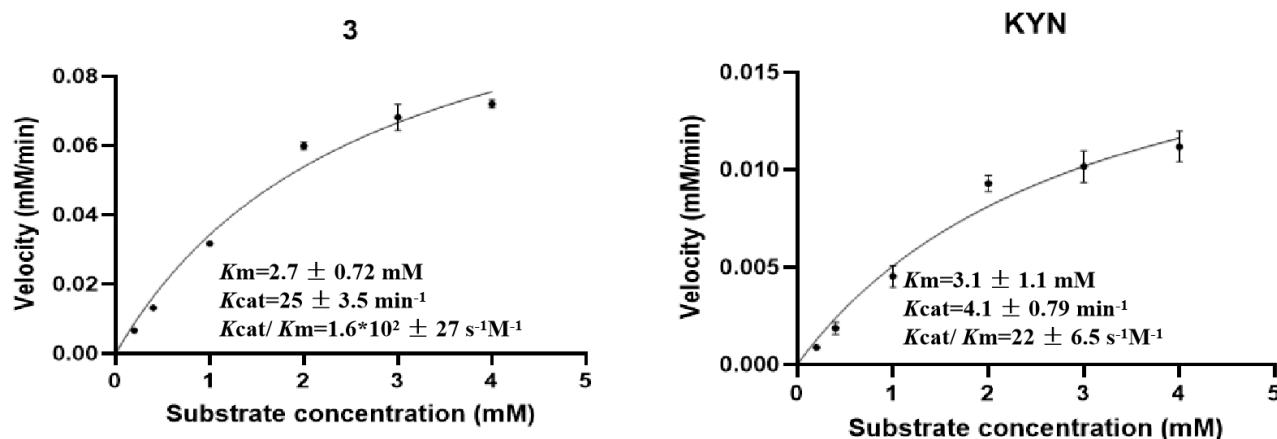


Figure S8. The kinetic analysis of KazE. Determination of K_m and K_{cat} values for KazE. Michaelis-Menten plots calculated using **3** or **KYN** as the substrate. The experiment was performed in duplicate and the values represent their average.

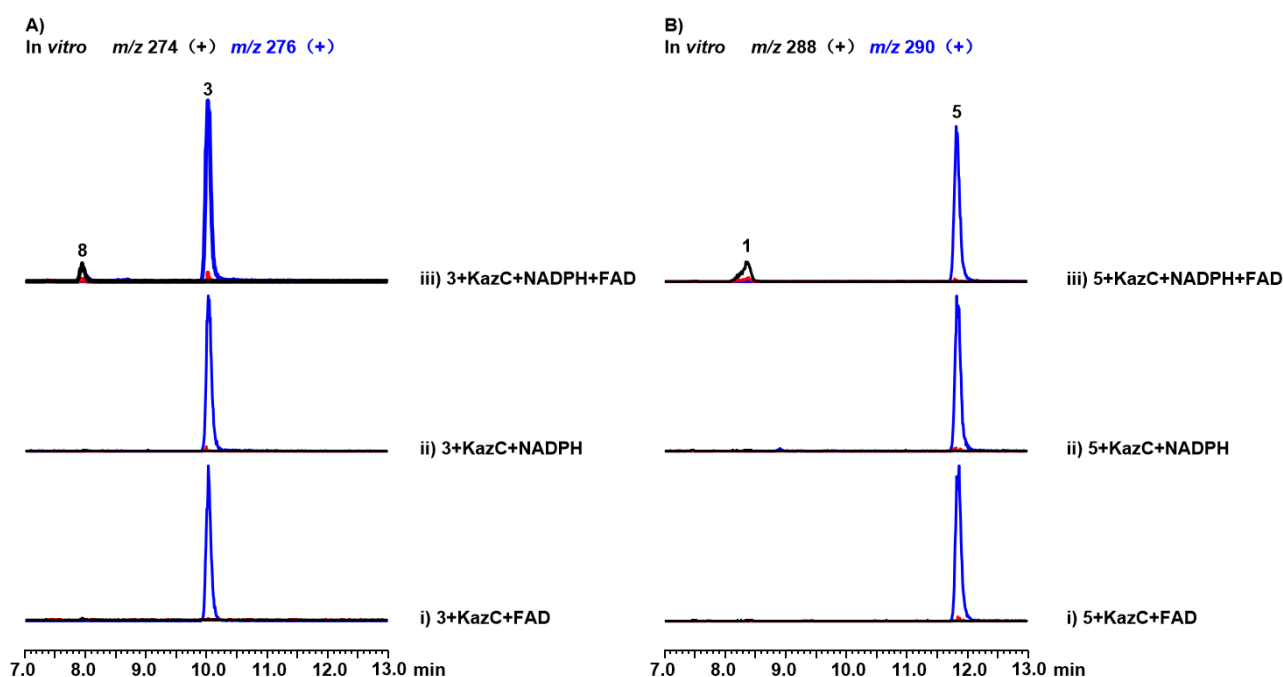


Figure S9. The hydroxylation of **3** and **5** by KazC is FAD dependent. A) The LC-MS profile of **3** with KazC microsome in the presence of i) FAD, ii) NADPH, iii) NADPH+FAD. B) The LC-MS profile of **5** with KazC microsome in the presence of i) FAD, ii) NADPH, iii) NADPH+FAD. Reactions were incubated at 28°C for overnight, then quenched by the addition of 200 μL of methanol, analyzed on a SHIMADZU LC/MS 2050.

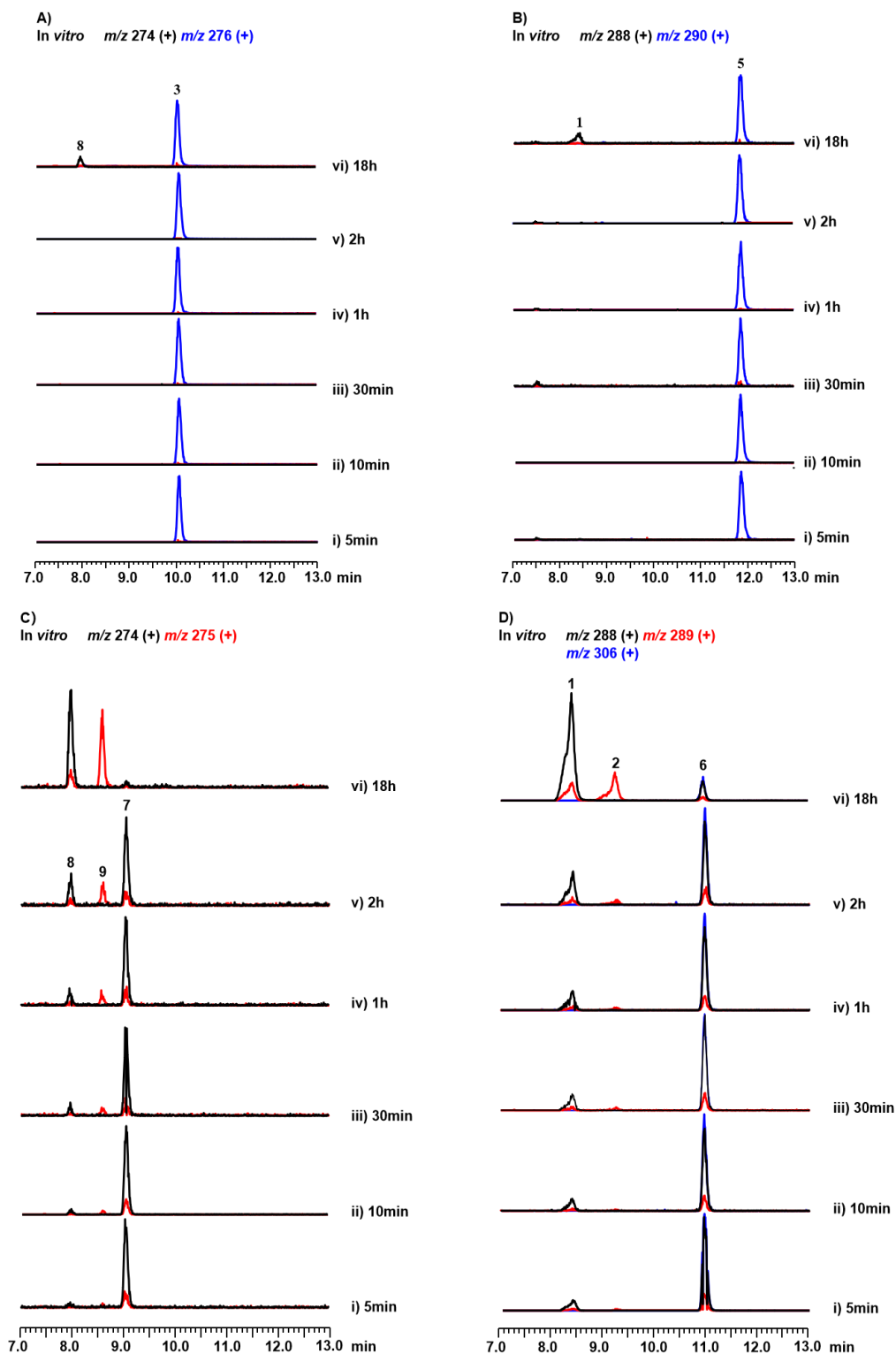


Figure S10. Stability tests of **3**, **5**, **6** and **7** in Tris-HCl of pH 8.5 with KazC microsomes. A) **3** B) **5** C) **7** D) **6** in 50mM Tris-HCl buffer of pH 8.5, detected at 5 min, 10 min, 15 min, 30 min, 1 h, 2 h and 18h. Reactions were incubated at 28°C, then quenched by the addition of 200 μ L of methanol, analyzed on a SHIMADZU LC/MS 2050. Extracted ion chromatograms (EICs) at m/z 274 contain signals from both **7** (MW 291), detected as the dehydrated ion $[M+H-H_2O]^+$, and **8** (MW 273), detected as $[M+H]^+$. m/z 288 contain signals from both **6** (MW 305), detected as the dehydrated ion $[M+H-H_2O]^+$, and **1** (MW 287), detected as $[M+H]^+$.

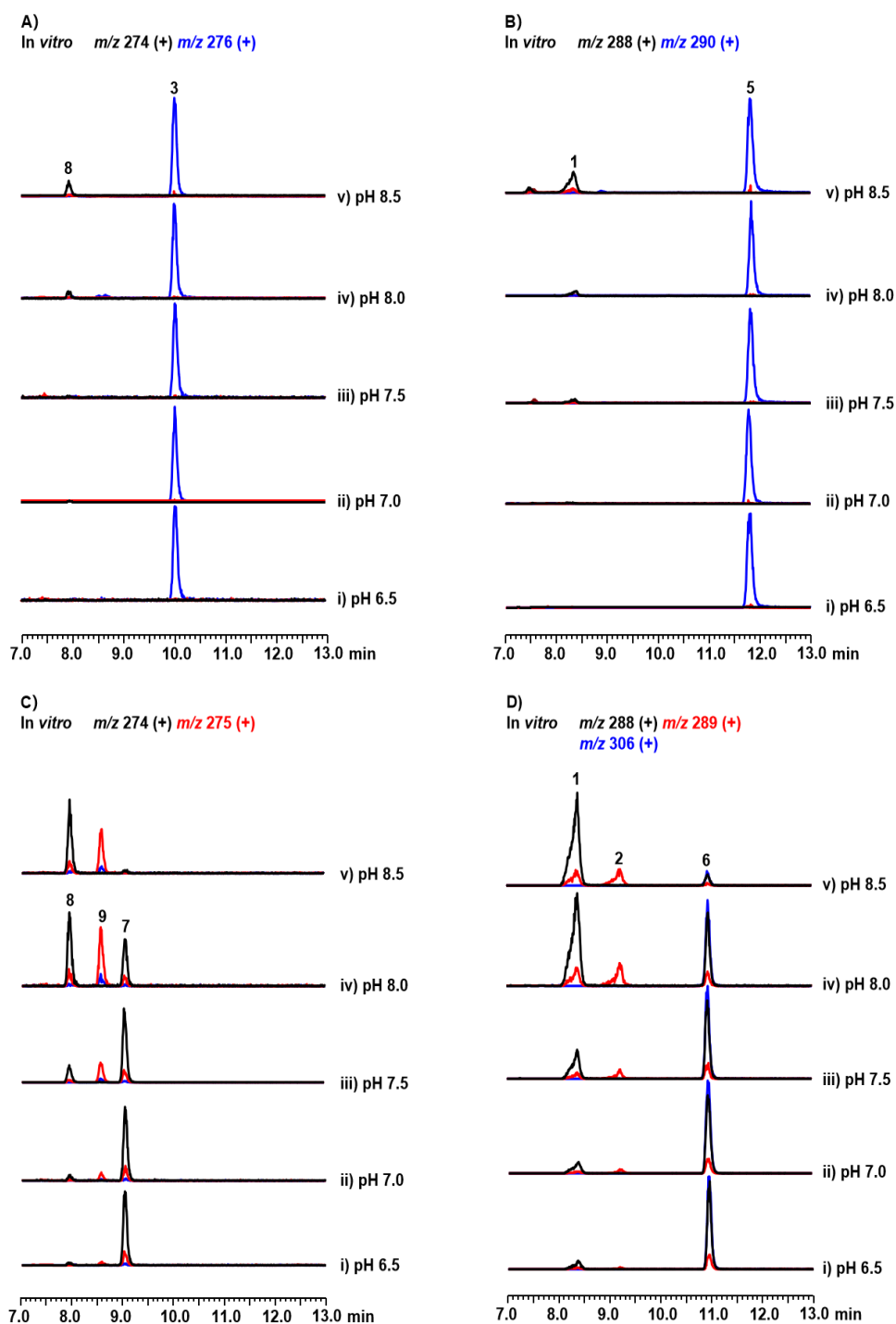


Figure S11. *In vitro* assay of **3**, **5**, **6** and **7** with KazC microsome at different pH. (A) LC-MS analysis of transformation of **3** to **8** with KazC microsome at different pH. (B) LC-MS analysis of transformation of **5** to **1** with KazC microsome at different pH. (C) LC-MS analysis of transformation of **7** to **8** and **9** with KazC microsome at different pH. (D) LC-MS analysis of transformation of **6** to **1** and **2** with KazC microsome at different pH. Reactions were incubated at 28°C for overnight (about 17 h), then quenched by the addition of 200 μ L of methanol, analyzed on a SHIMADZU LC/MS 2050. Extracted ion chromatograms (EICs) at m/z 274 contain signals from both **7** (MW 291), detected as the dehydrated ion $[M+H-H_2O]^+$, and **8** (MW 273), detected as $[M+H]^+$. m/z 288 contain signals from both **6** (MW 305), detected as the dehydrated ion $[M+H-H_2O]^+$, and **1** (MW 287), detected as $[M+H]^+$.

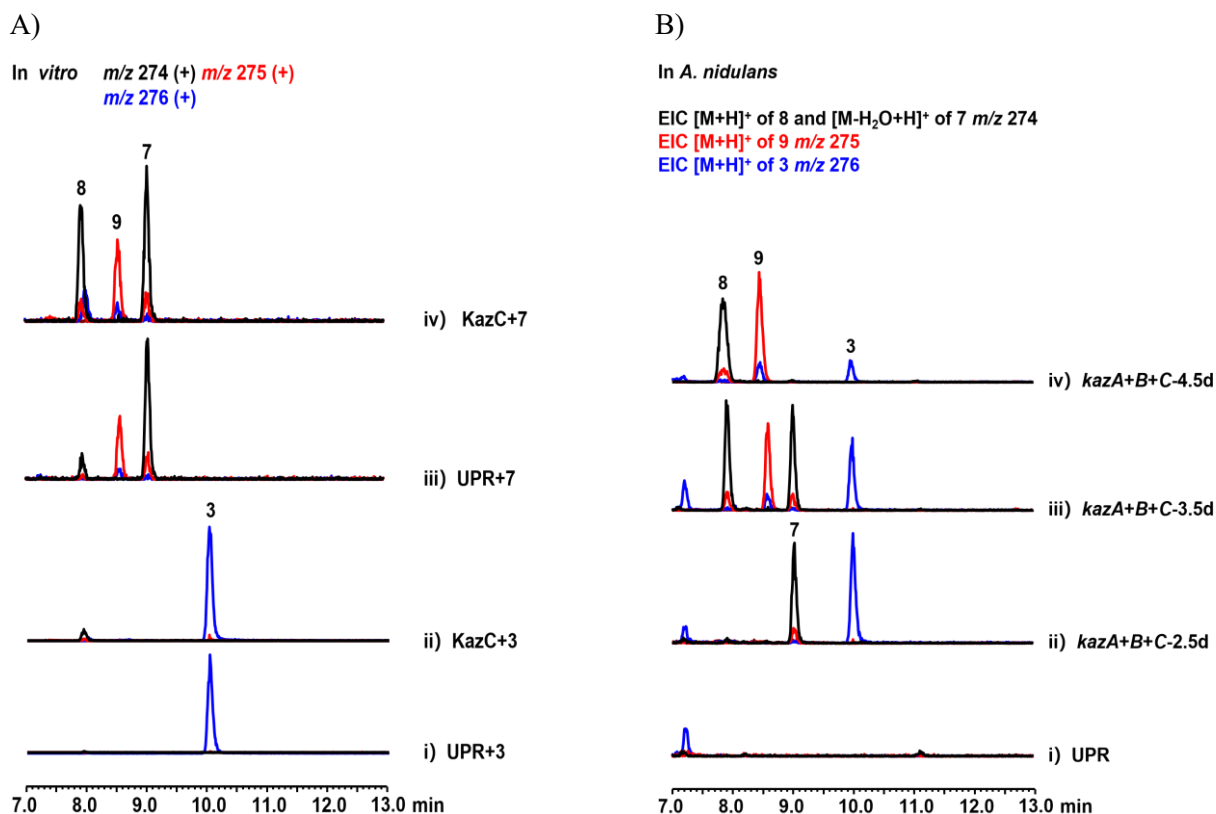


Figure S12. LC-MS analysis of transformation of **3** to **8** and **9** *in vivo* and *in vitro*. A) LC-MS analysis of transformation of **3** to **8** and **9** using KazC microsome. Reactions were incubated at 28°C for overnight, then quenched by the addition of 200 μ L of methanol, analyzed on a SHIMADZU LC/MS 2050. Extracted ion chromatograms (EICs) at m/z 274 contain signals from both **7** (MW 291), detected as the dehydrated ion $[M+H-H_2O]^+$, and **8** (MW 273), detected as $[M+H]^+$. B) LC-MS analysis of transformation of **3** to **8** and **9** *in vivo*. m/z 274 contain signals from both **7** (MW 291), detected as the dehydrated ion $[M+H-H_2O]^+$, and **8** (MW 273), detected as $[M+H]^+$. *A. nidulans* A1145 containing the empty vector was used as control. UPR represents the cell lysis of control strain.

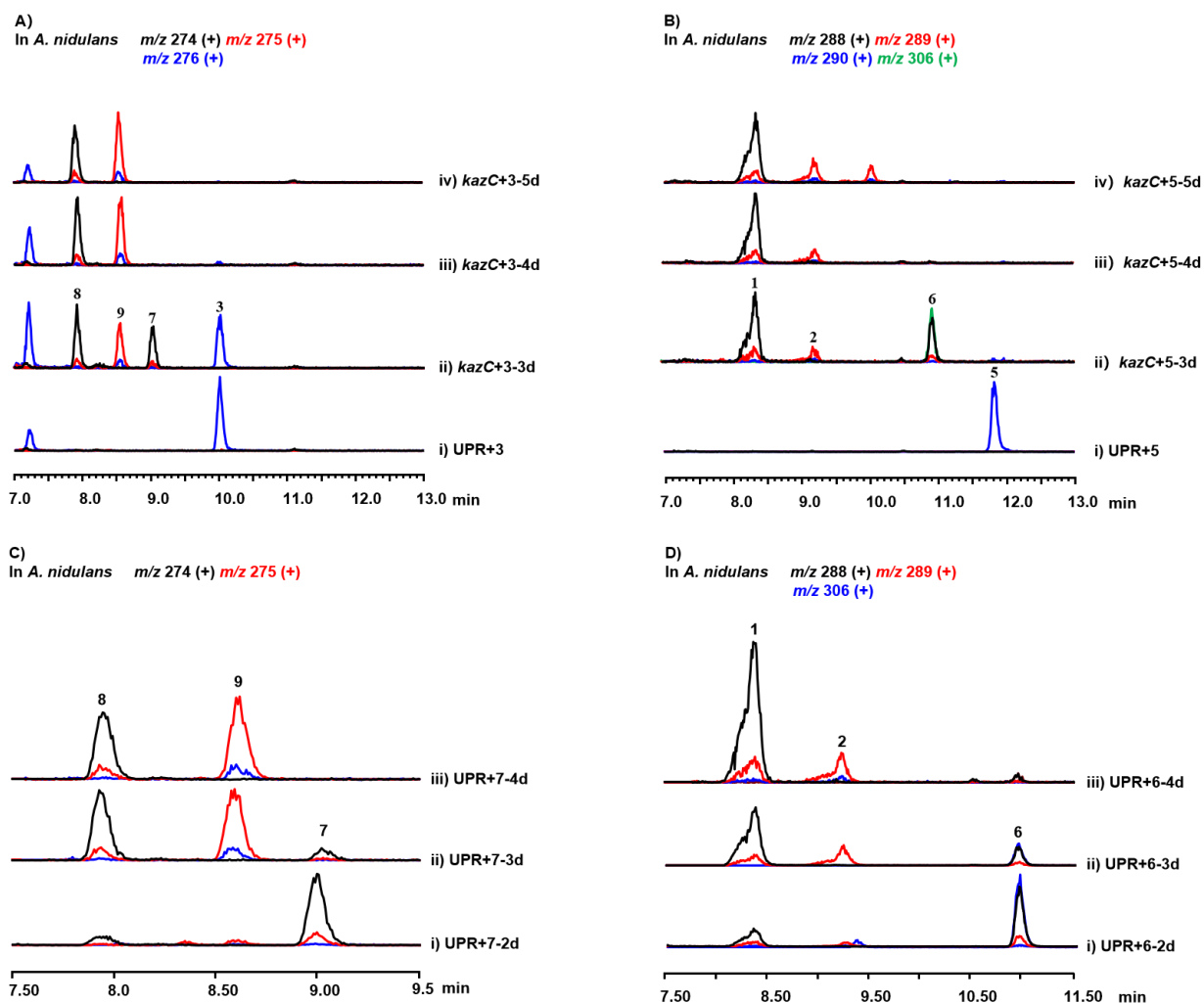


Figure S13. Feeding experiments of **3**, **5**, **6** and **7**. A) Feeding **3** of *kazC* and UPR in *A.nidulans*, detecting products for 3-5 days. B) Feeding **5** of *kazC* and UPR in *A.nidulans*, detecting products for 3-5 days. C) Feeding **6** of UPR in *A.nidulans*, detecting products for 2-4 days. D) Feeding **7** of UPR in *A.nidulans*, detecting products for 2-4 days. Samples were analyzed on a SHIMADZU LC/MS 2050. Extracted ion chromatograms (EICs) at m/z 274 contain signals from both **7** (MW 291), detected as the dehydrated ion $[M+H-H_2O]^+$, and **8** (MW 273), detected as $[M+H]^+$. m/z 288 contain signals from both **6** (MW 305), detected as the dehydrated ion $[M+H-H_2O]^+$, and **1** (MW 287), detected as $[M+H]^+$.

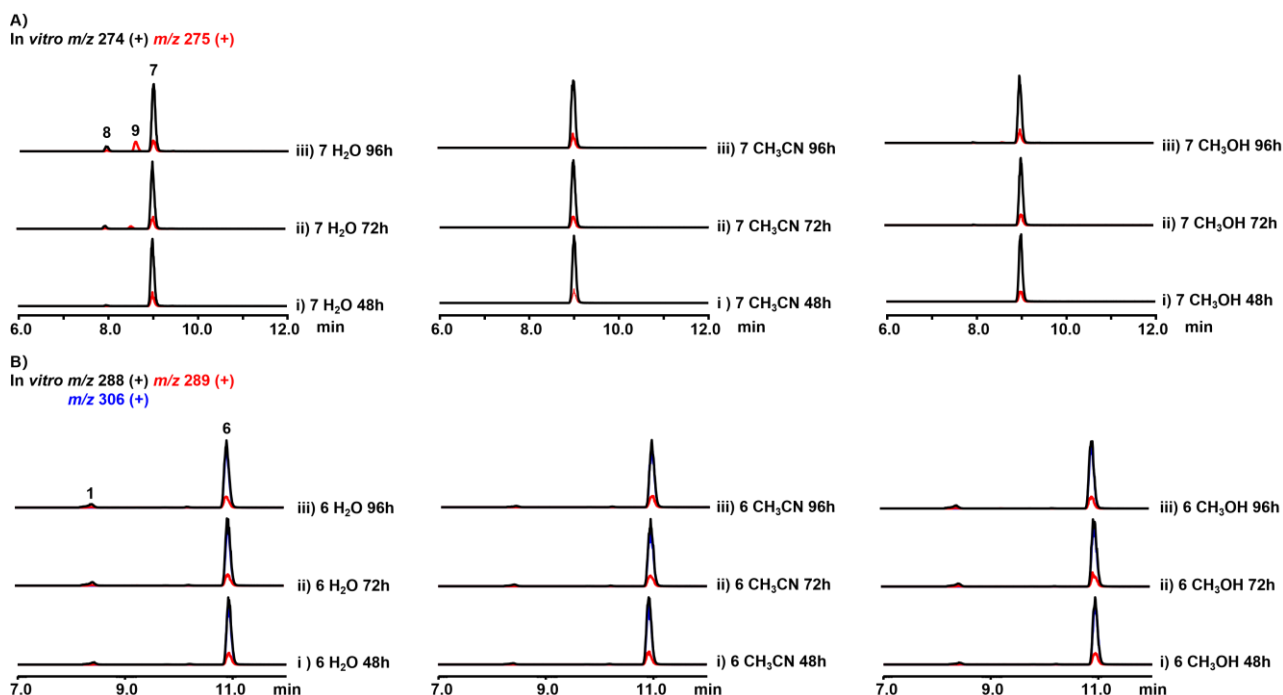


Figure S14. Stability assays of **6** and **7** in protic and aprotic solvents. A) 0.2 mM **7** B) 0.2 mM **6** in H₂O, CH₃CN and CH₃OH, detected at 48 h, 72 h and 96 h. Reactions were incubated at 28°C, analyzed on a SHIMADZU LC/MS 2050. Extracted ion chromatograms (EICs) at m/z 274 contain signals from both **7** (MW 291), detected as the dehydrated ion $[M+H-H_2O]^+$, and **8** (MW 273), detected as $[M+H]^+$. m/z 288 contain signals from both **6** (MW 305), detected as the dehydrated ion $[M+H-H_2O]^+$, and **1** (MW 287), detected as $[M+H]^+$.

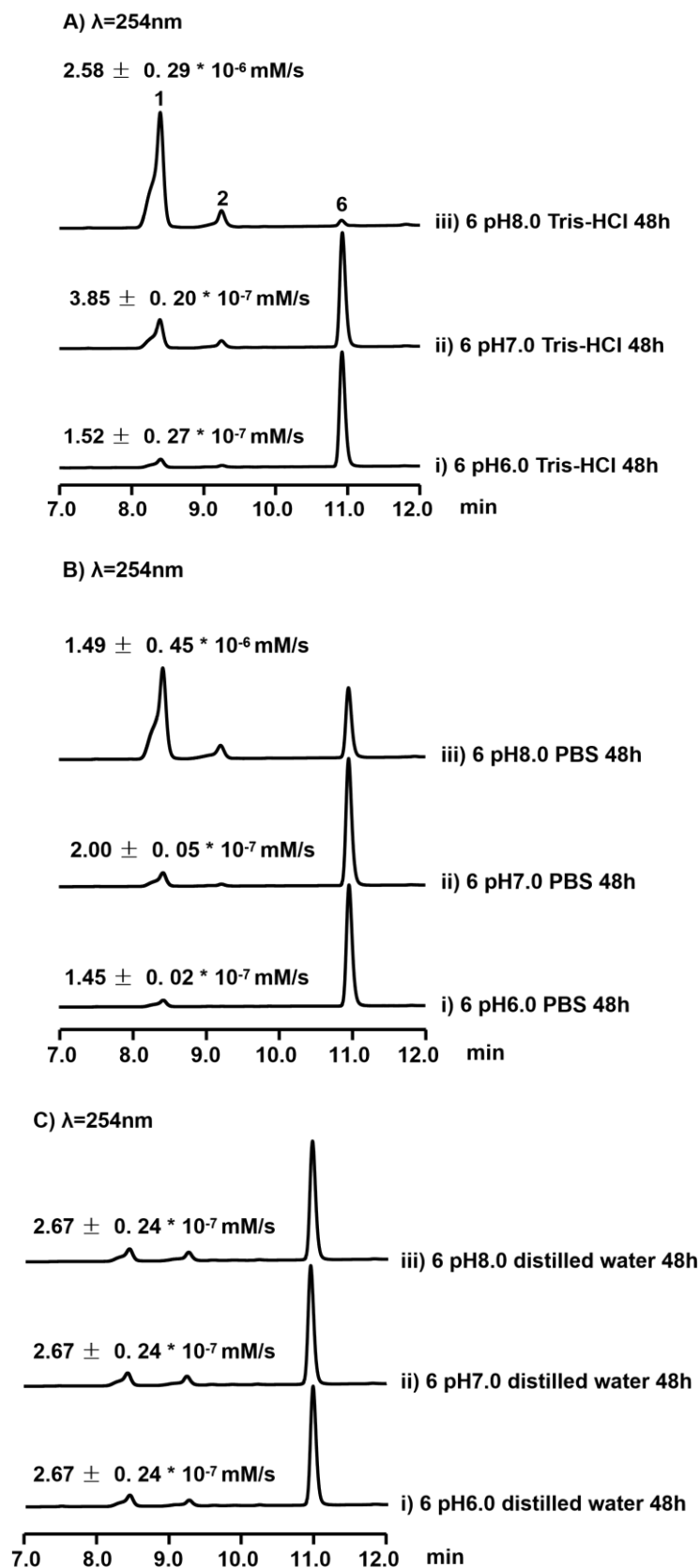
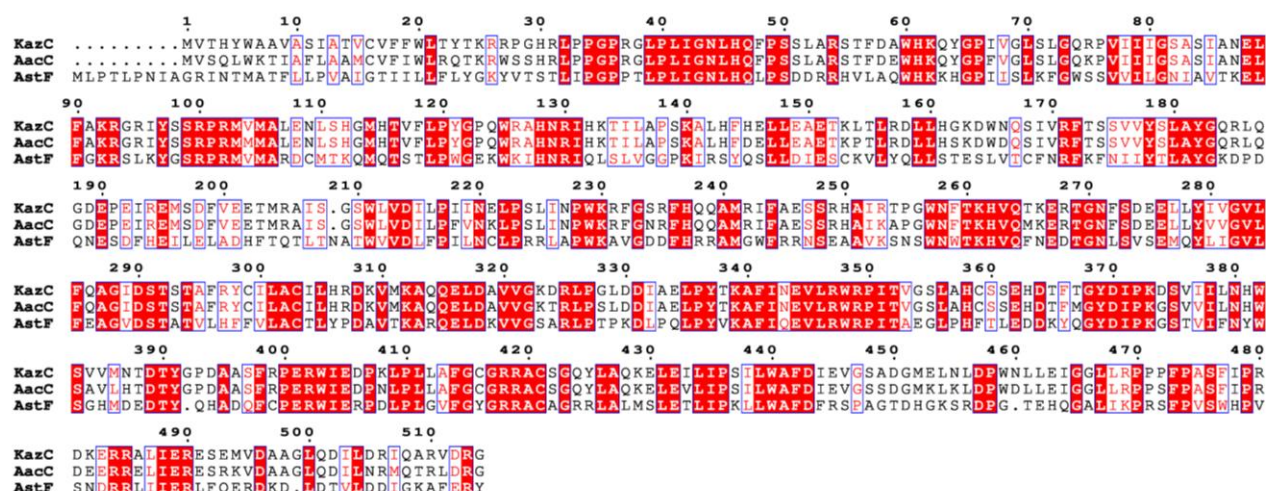


Figure S15. Transformation of **6** to **1** and **2** at pH 6.0, 7.0, 8.0 in Tris-HCl buffer (the pH was adjusted at 28°C in a water bath), PBS buffer and distilled water (adjusted with HCl or NaOH), detected at 48 h. Reactions were incubated at 28°C, analyzed on a SHIMADZU LC/MS 2050.

A)



B)

TMHMM result

```
# WEBSEQUENCE Length: 514
# WEBSEQUENCE Number of predicted TMHs: 1
# WEBSEQUENCE Exp number of AAs in TMHs: 21.66217
# WEBSEQUENCE Exp number, first 60 AAs: 20.05193
# WEBSEQUENCE Total prob of N-in: 0.91178
# WEBSEQUENCE POSSIBLE N-term signal sequence
WEBSEQUENCE TMHMM2.0 inside 1 4
WEBSEQUENCE TMHMM2.0 TMhelix 5 24
WEBSEQUENCE TMHMM2.0 outside 25 514
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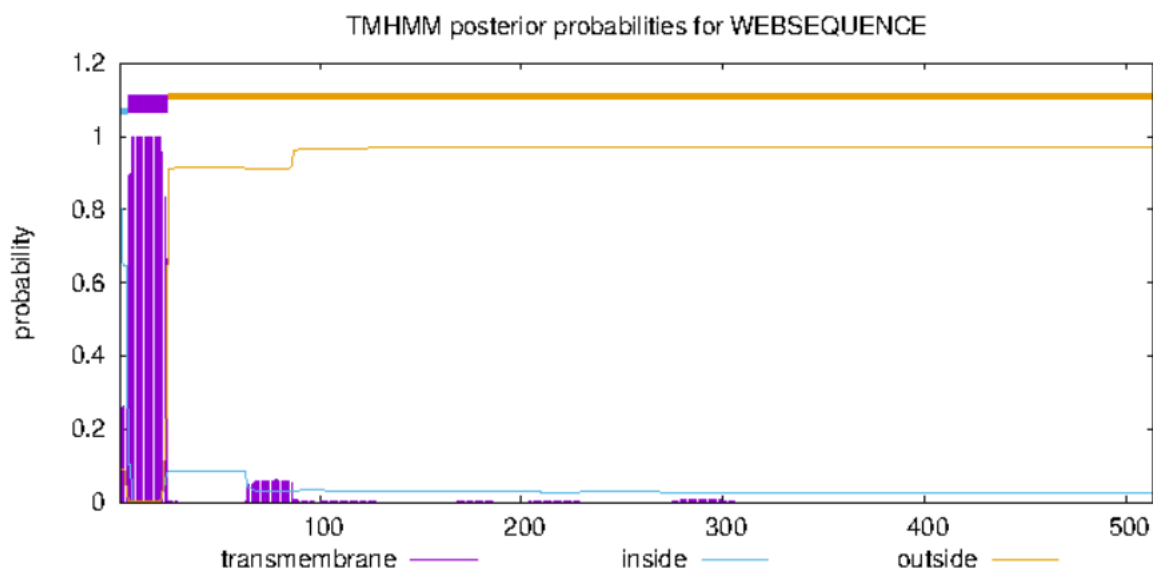
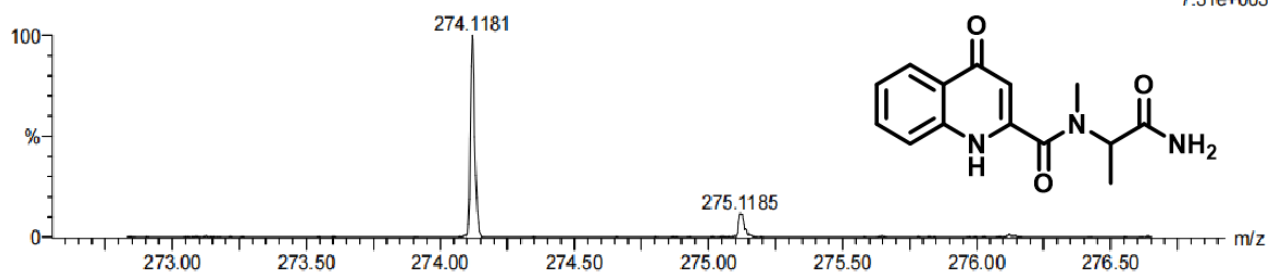


Figure S17. Sequence alignment and TM analysis* of KazC. A) Amino acid sequence alignment of KazC, AacC and AstF. B) TM analysis of KazC indicates that KazC is a membrane protein.

*(TMHMM 2.0-DTU Health Tech - Bioinformatic Services)

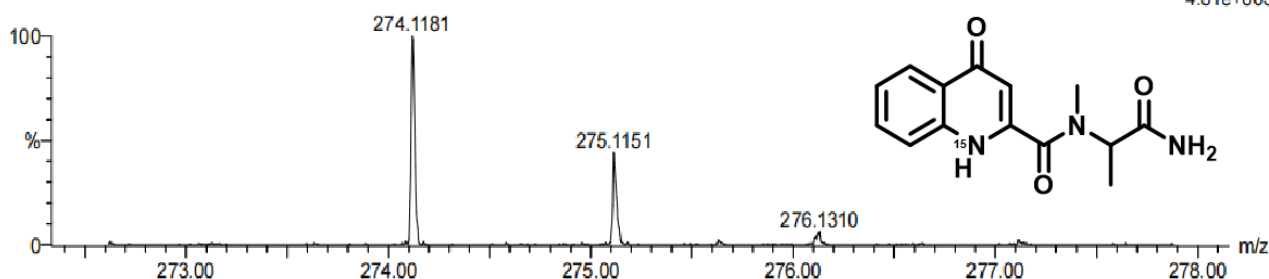
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7.31e+003



Minimum: -1.5
Maximum: 5.0 10.0 50.0

Mass	Calc. Mass	mDa	PPM	DBE	Formula
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	274.1151	3.0	10.9	4.5	C9 H16 N5 O5

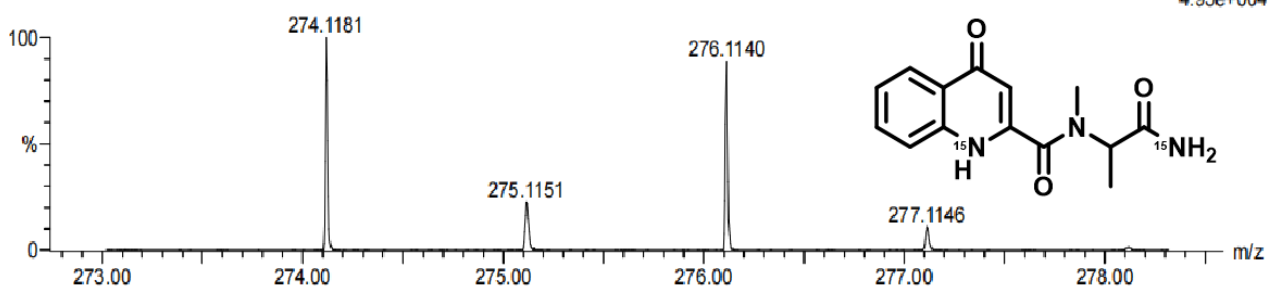
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4.81e+003



Minimum: -1.5
Maximum: 5.0 10.0 50.0

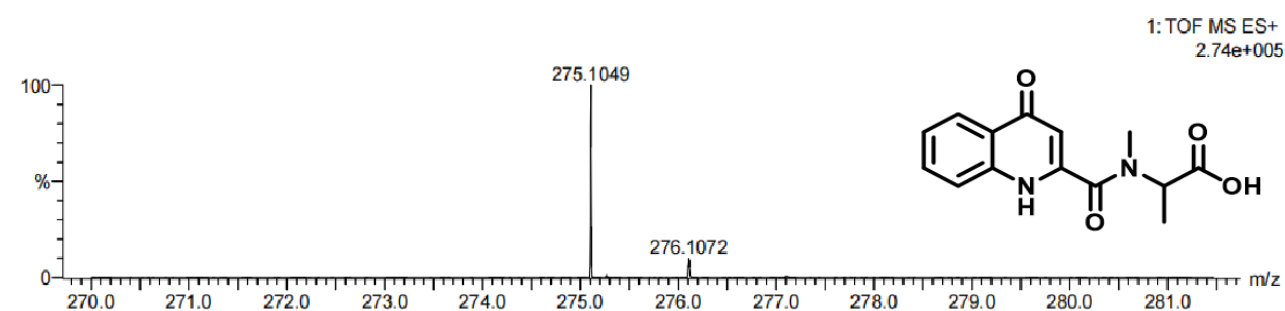
Mass	Calc. Mass	mDa	PPM	DBE	Formula
275.1151	275.1162	-1.1	-4.0	8.5	C14 H16 14N2 15N O3

1: TOF MS ES+
4.95e+004

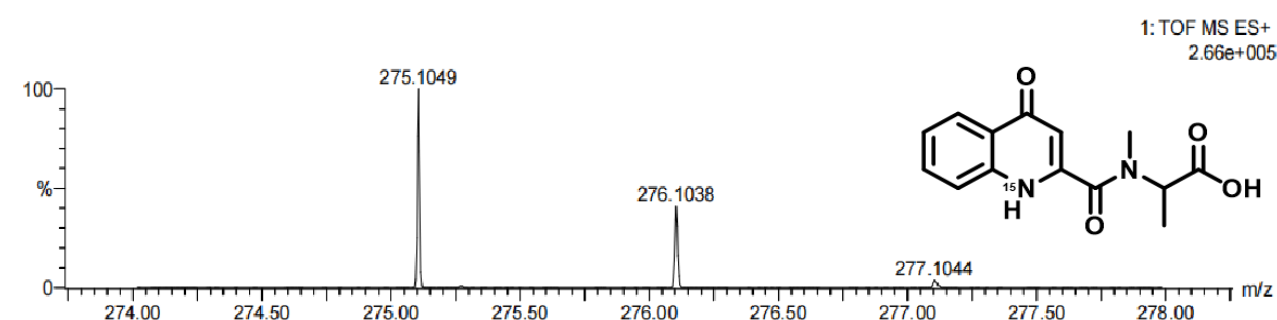


Minimum: -1.5
Maximum: 5.0 10.0 50.0

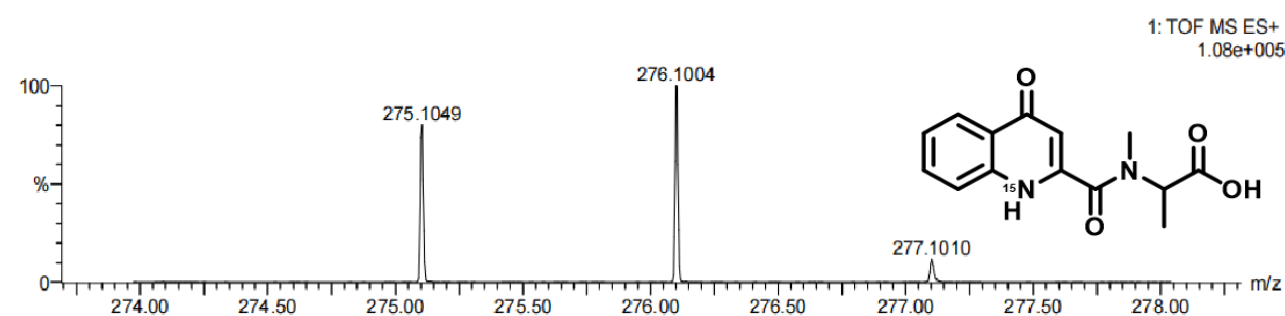
Mass	Calc. Mass	mDa	PPM	DBE	Formula
276.1140	276.1132	0.8	2.9	8.5	C14 H16 14N 15N2 O3



Minimum:				-1.5	
Maximum:	5.0	10.0		50.0	
Mass	Calc. Mass	mDa	PPM	DBE	Formula
275.1049	275.1032	1.7	6.2	8.5	C ₁₄ H ₁₅ N ₂ O ₄
	275.1072	-2.3	-8.4	12.5	C ₁₉ H ₁₅ O ₂



Minimum:				-1.5	
Maximum:	5.0	10.0		50.0	
Mass	Calc. Mass	mDa	PPM	DBE	Formula
276.1038	276.1002	3.6	13.0	8.5	C ₁₄ H ₁₅ ¹⁴ N ¹⁵ N O ₄



Minimum:				-1.5	
Maximum:	5.0	10.0		50.0	
Mass	Calc. Mass	mDa	PPM	DBE	Formula
276.1004	276.1002	0.2	0.7	8.5	C ₁₄ H ₁₅ ¹⁴ N ¹⁵ N O ₄

Figure S18. Isotope feeding experiments with ¹⁵N labelled tryptophans. Feeding 0.2 g/L ¹⁵N labelled tryptophans in to *A. nidulans* A1145 expressing *kazABCD* for **8** and **9** production. Analysis the fermentation extract by high-resolution mass spectrometry. A mass shift of +1 Da was observed in **8** and **9** upon feeding with L-Trp-1-¹⁵N, whereas L-Trp-¹⁵N₂ labeled assay results in a mass increase of +2 Da in **8**, and a mass increase of +1 Da in **9**.

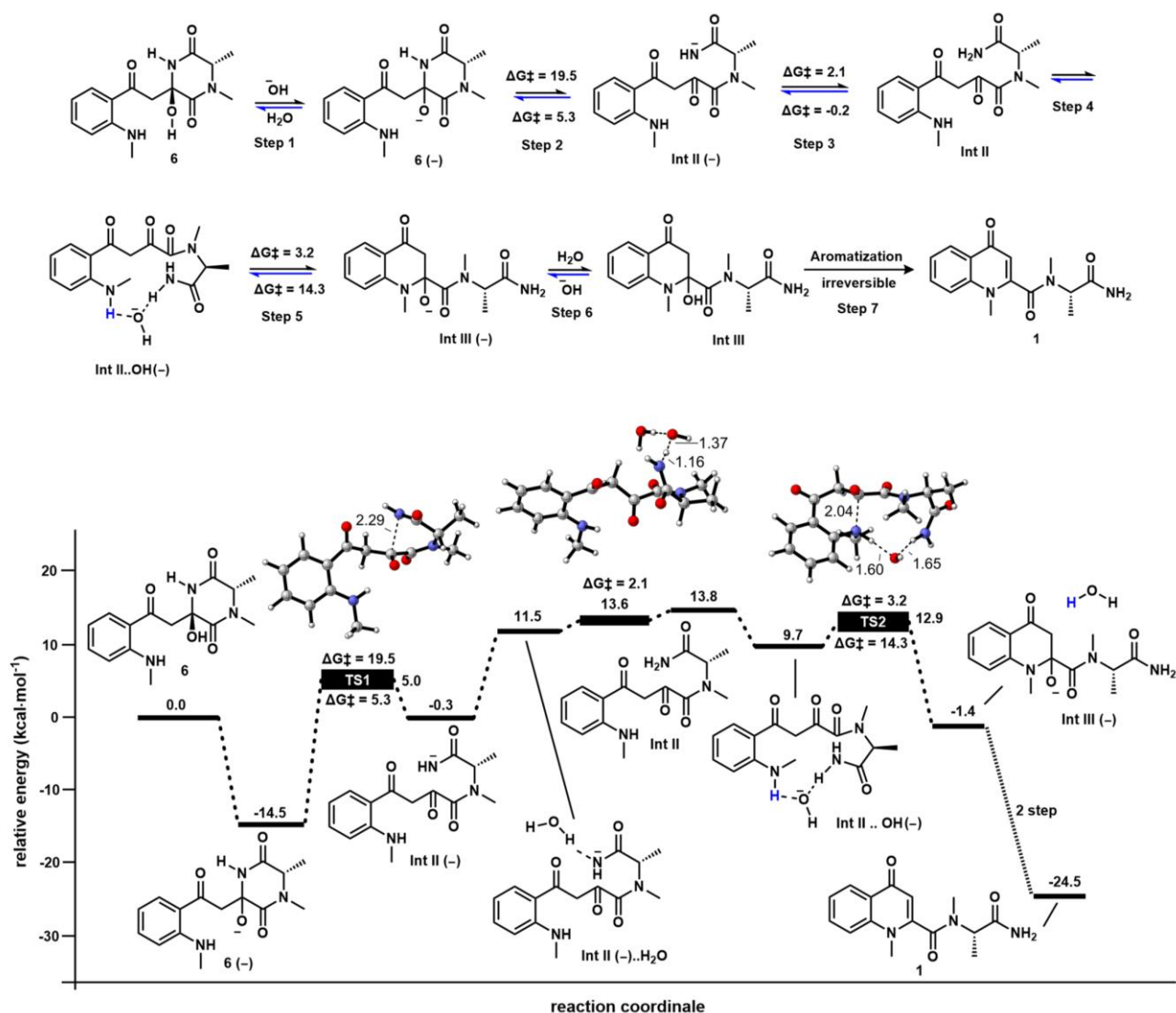


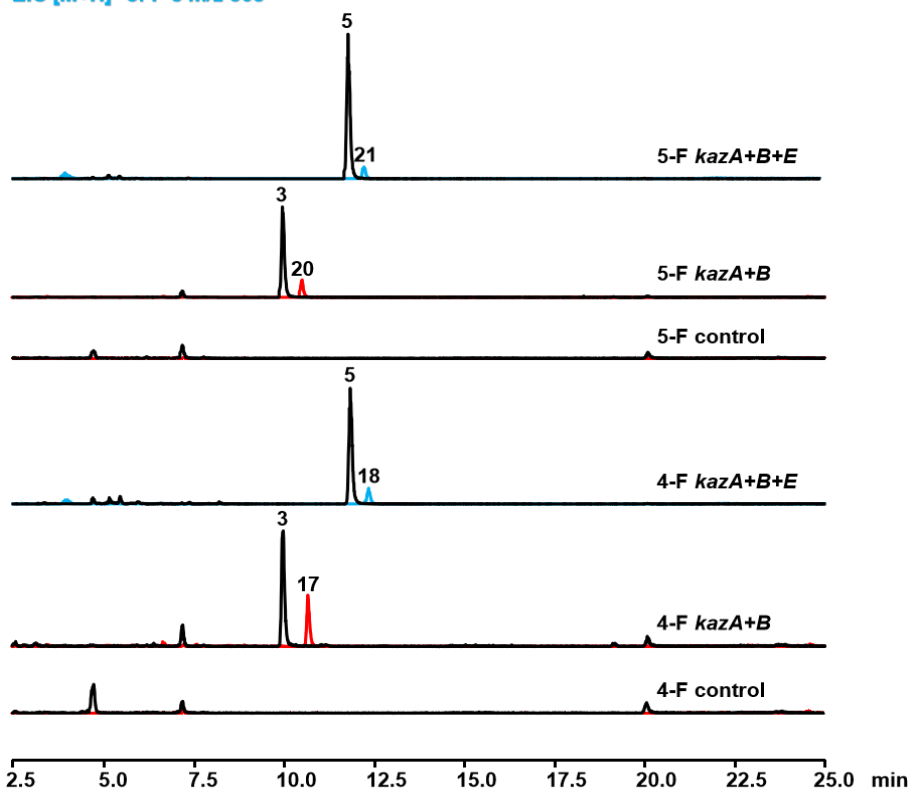
Figure S19: Reaction free-energy profile from 6 to 1. Step 1: Compound **6** undergoes deprotonation by a hydroxide ion (OH^-) to yield the oxyanion intermediate **6(-)** and a water molecule. Step 2: **6(-)** undergoes C–N bond cleavage to afford the amide nitrogen anion intermediate **Int II(-)**. Step 3: **Int II(-)** forms an explicit complex with water (**Int II(-)·H₂O**) and subsequently abstracts a proton to generate **Int II** and a hydroxide ion. Step 4: The hydroxide ion interacts with **Int II** to form a stable hydrogen-bonded complex **Int II·OH(-)**. Step 5: The amide nitrogen in **Int II·OH(-)** performs a nucleophilic attack on the carbonyl carbon, leading to cyclization and dehydration to yield **Int III(-)**. Finally, **Int III(-)** is converted to product **1** through an aromatization process. Energies are Gibbs free energies (kcal/mol, 298 K) relative to a reference state of separated **6** + OH^- + H_2O .

* Dey J, O'Donoghue AC, More O'Ferrall RA. Equilibrium constants for dehydration of water adducts of aromatic carbon–carbon double bonds. *J. Am. Chem. Soc.*, **2002**, 124, 8561–8574.

A)
In *A. nidulans*

EIC [M+H]⁺ of F-3 m/z 294

EIC [M+H]⁺ of F-5 m/z 308



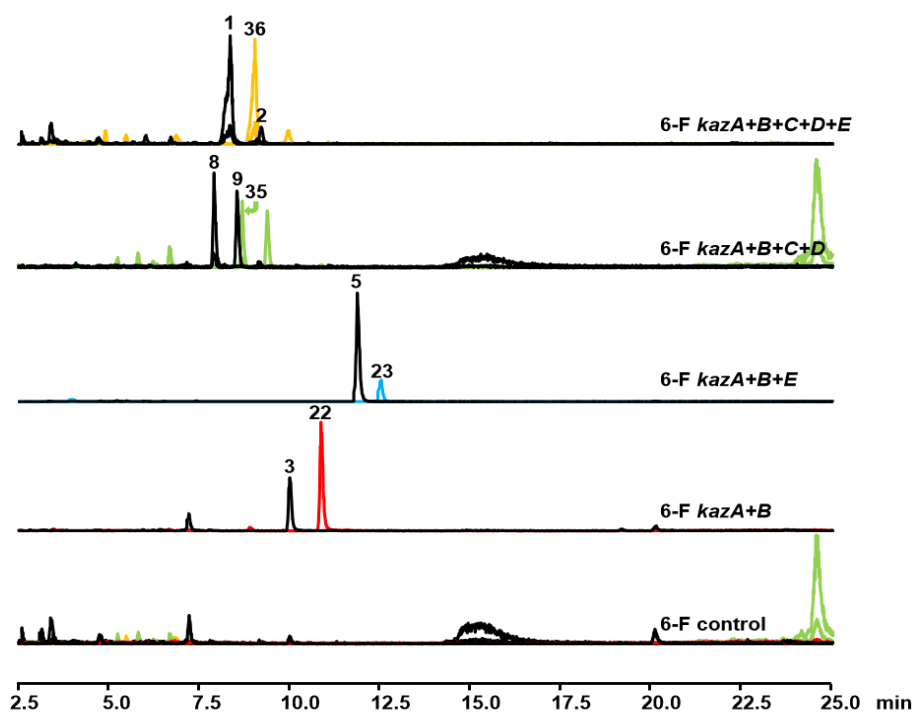
B)
In *A. nidulans*

EIC [M+H]⁺ of F-3 m/z 294

EIC [M+H]⁺ of F-5 m/z 308

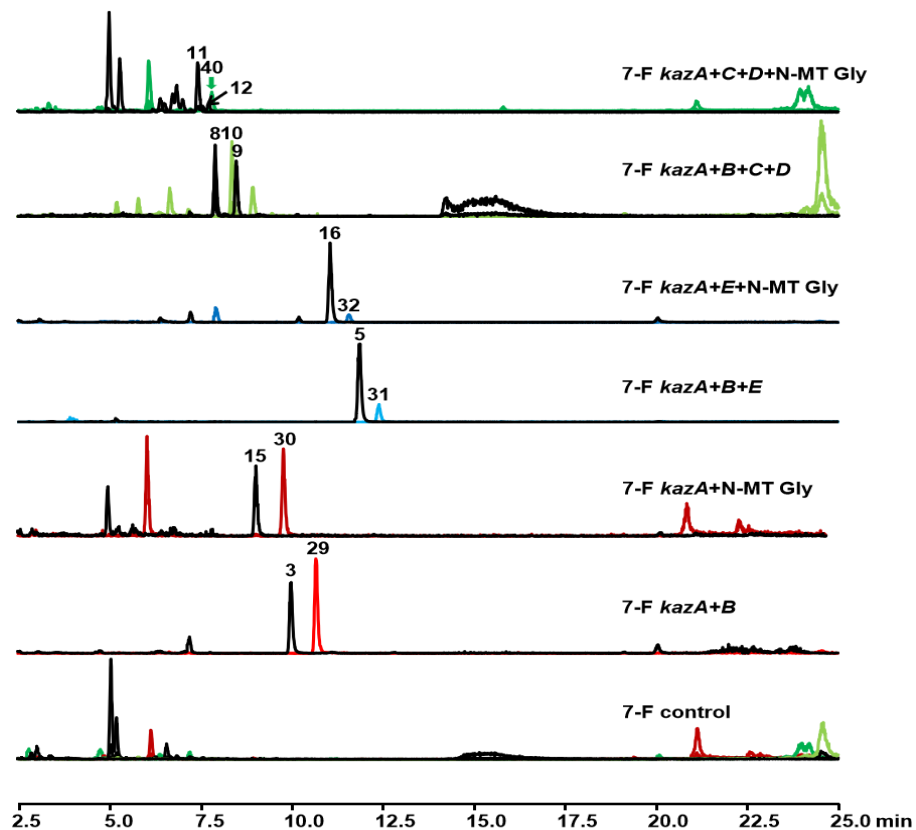
EIC [M+H]⁺ of F-8 m/z 292 [M+H]⁺ of F-9 m/z 293

EIC [M+H]⁺ of F-1 m/z 306 [M+H]⁺ of F-2 m/z 307



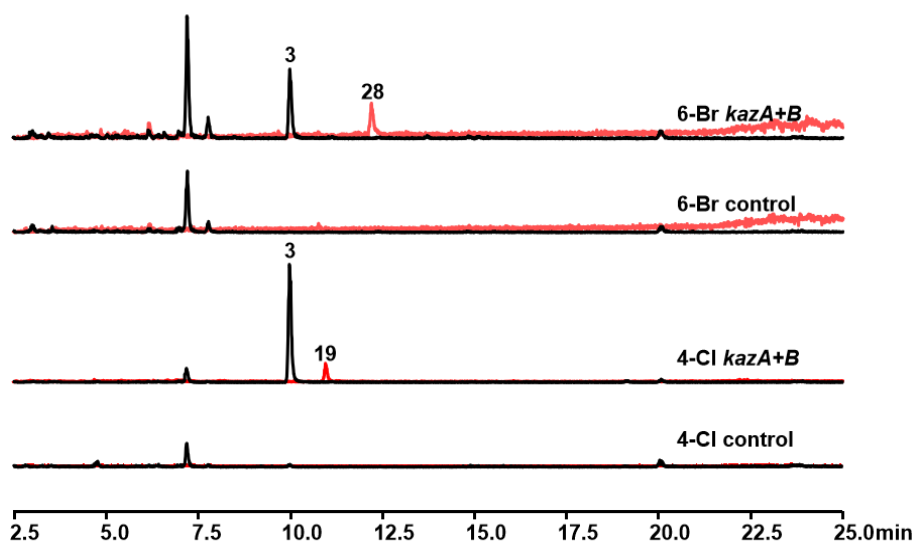
C)
In *A. nidulans*

EIC [M+H]⁺ of F-3 m/z 294
 EIC [M+H]⁺ of F-15 m/z 280
 EIC [M+H]⁺ of F-5 m/z 308
 EIC [M+H]⁺ of F-16 m/z 294
 EIC [M+H]⁺ of F-8 m/z 292 [M+H]⁺ of F-9 m/z 293
 EIC [M+H]⁺ of F-11 m/z 278 [M+H]⁺ of F-12 m/z 279



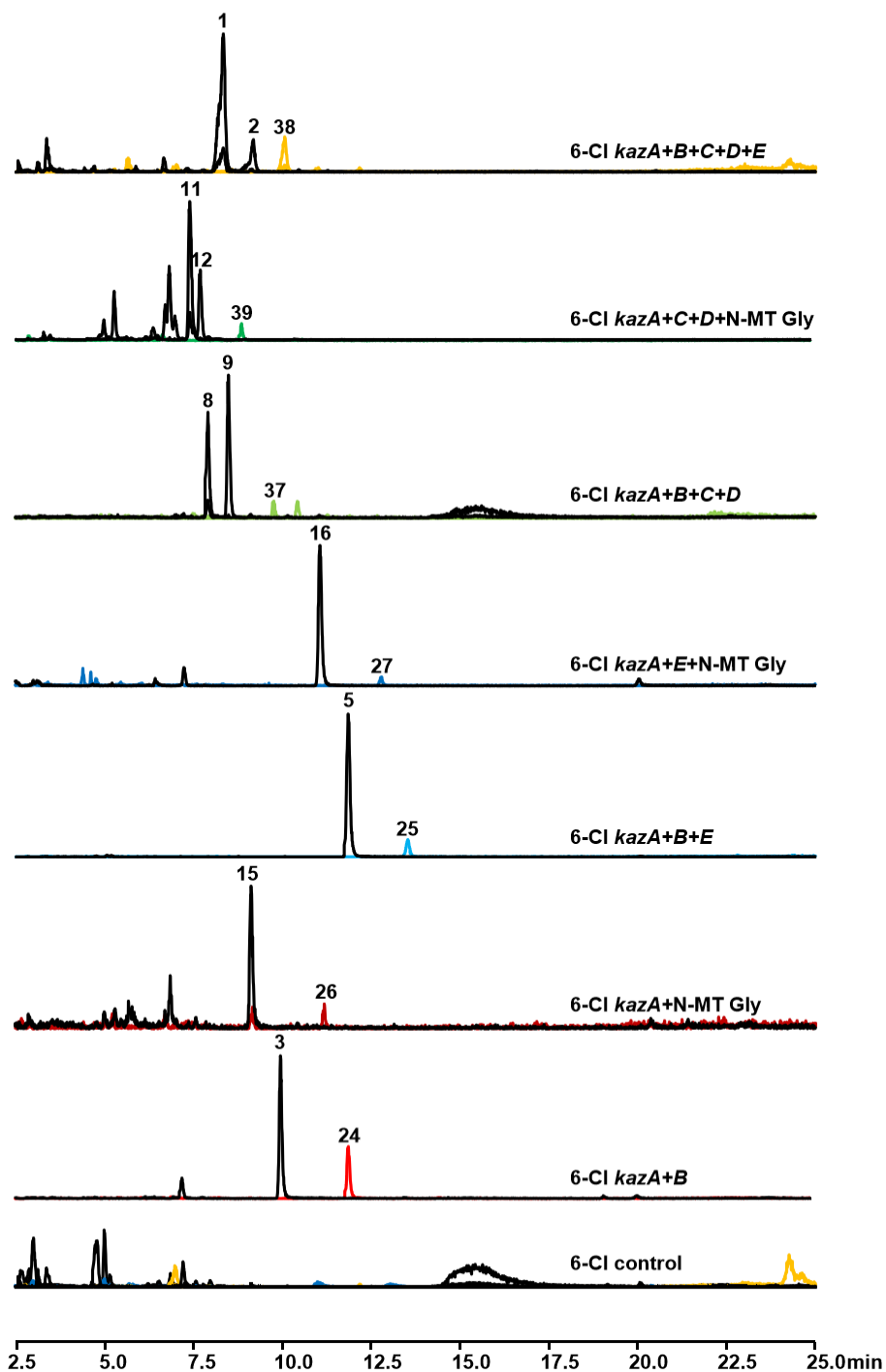
D)
In *A. nidulans*

EIC [M+H]⁺ of Cl-3 m/z 310
 EIC [M+H]⁺ of Br-3 m/z 354



E)
In *A. nidulans*

EIC [M+H]⁺ of Cl-3 m/z 310
 EIC [M+H]⁺ of Cl-15 m/z 296
 EIC [M+H]⁺ of Cl-5 m/z 324
 EIC [M+H]⁺ of Cl-16 m/z 310
 EIC [M+H]⁺ of Cl-8 m/z 308 [M+H]⁺ of Cl-9 m/z 309
 EIC [M+H]⁺ of Cl-11 m/z 294 [M+H]⁺ of Cl-12 m/z 295
 EIC [M+H]⁺ of Cl-1 m/z 322 [M+H]⁺ of Cl-2 m/z 323



F)
In *A. nidulans*

EIC [M+H]⁺ of Cl-3 m/z 310
EIC [M+H]⁺ of Cl-15 m/z 296
EIC [M+H]⁺ of Cl-8 m/z 308 [M+H]⁺ of Cl-9 m/z 309
EIC [M+H]⁺ of Cl-11 m/z 294 [M+H]⁺ of Cl-12 m/z 295

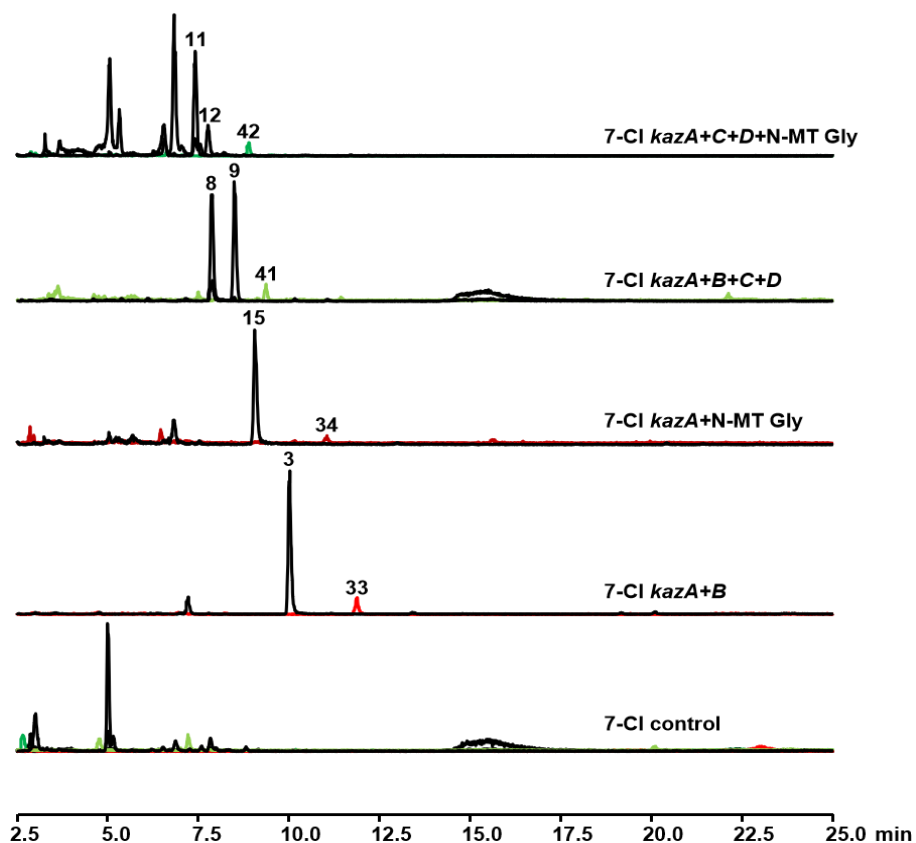
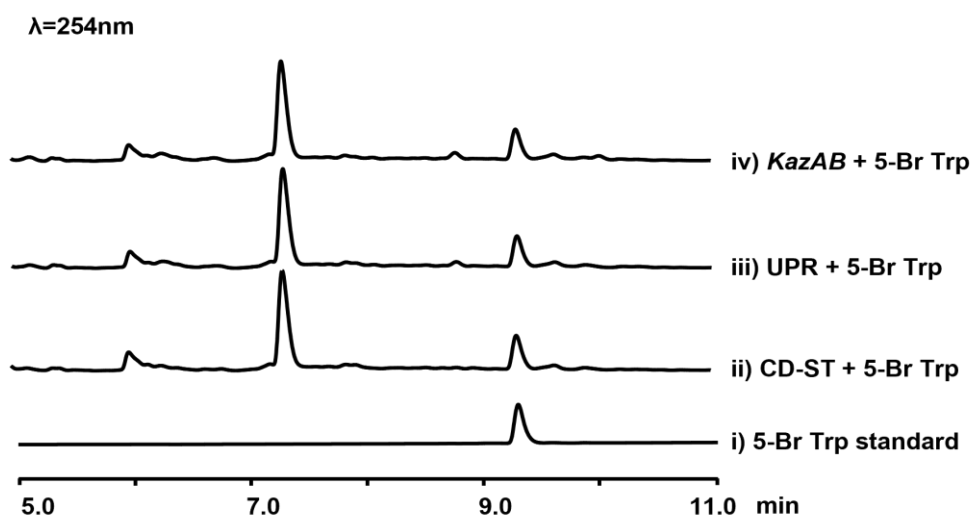


Figure S20. LC-MS analysis of feeding experiments with substituted tryptophans. The result can be found in **Table S5**. A) The products profile of feeding 4-F and 5-F tryptophans. B) The products profile of feeding 6-F tryptophans. C) The products profile of feeding 7-F tryptophans. D) The products profile of feeding 4-Cl and 6-Br tryptophans. E) The products profile of feeding 6-Cl tryptophans. F) The products profile of feeding 7-Cl tryptophans.

A)



B)

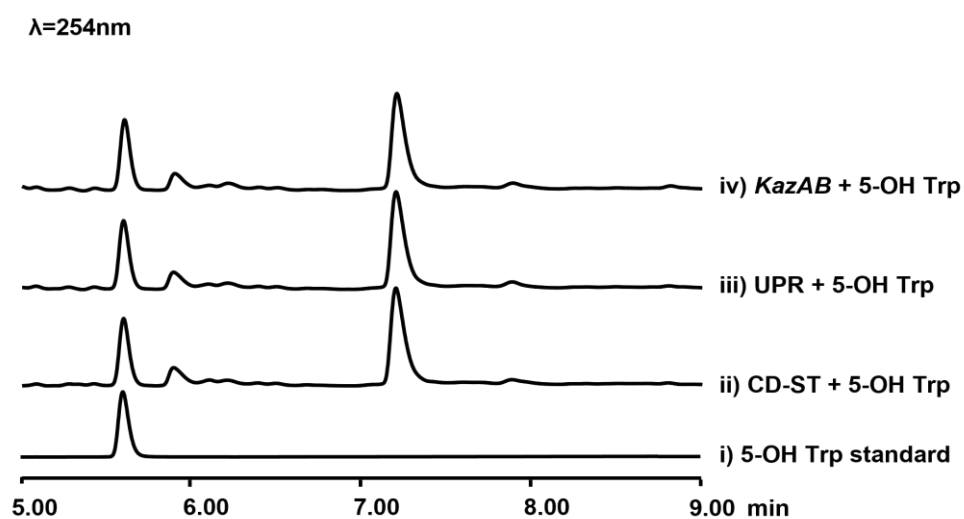


Figure S21. 5-Substitued tryptophan is not taken up by *A.nidulans*. A) Feeding 5-Br tryptophan into CD-ST liquid medium, UPR control and *KazAB* respectively. B) Feeding 5-OH tryptophan into CD-ST liquid medium, UPR control and *KazAB* respectively.

In *A. nidulans* ($\lambda=254\text{nm}$)

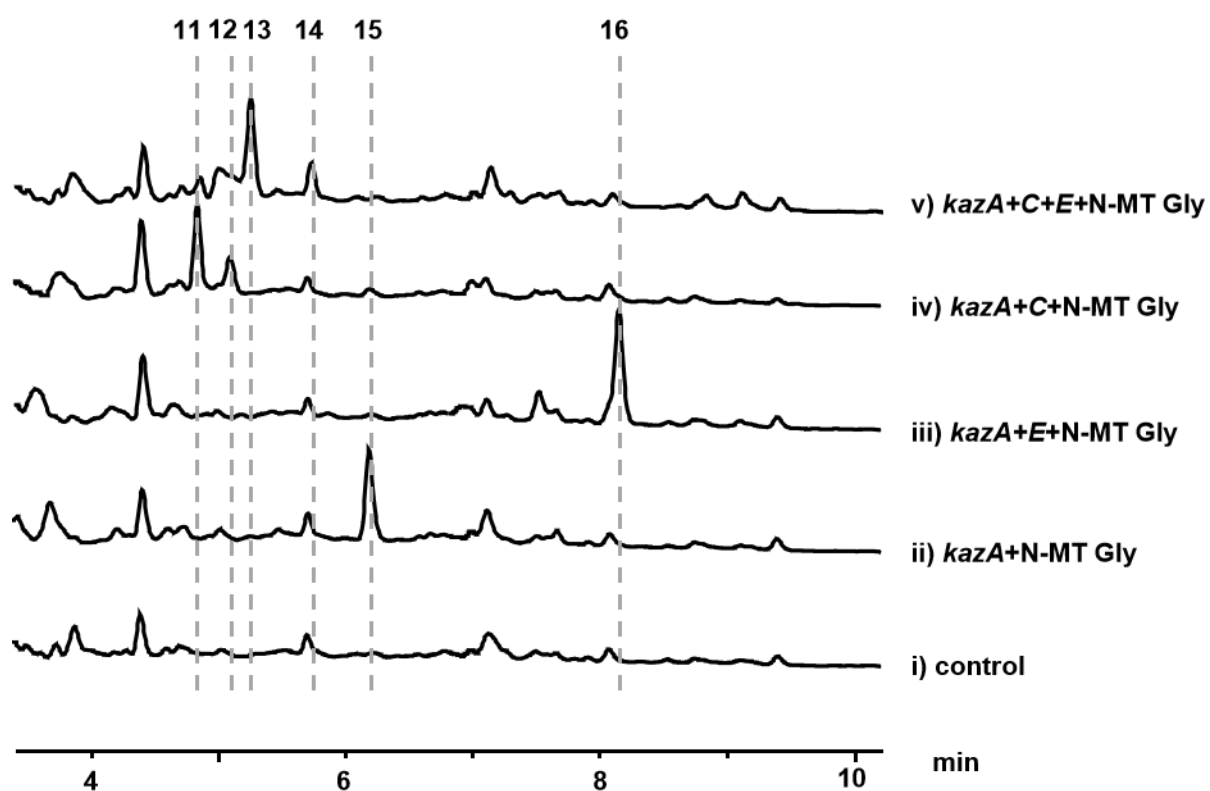
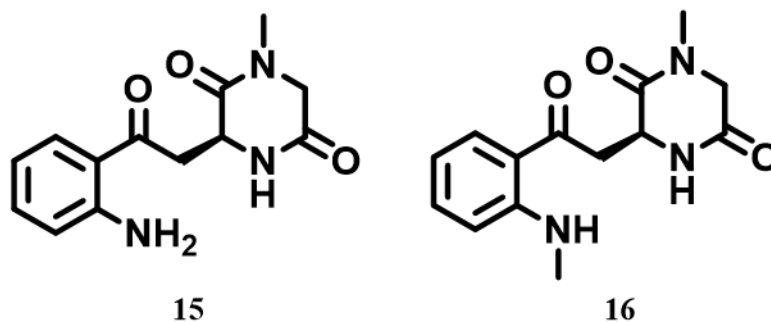
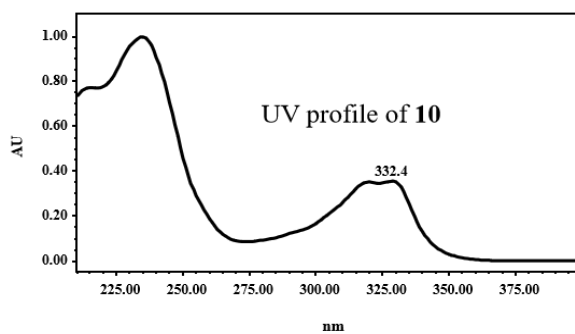
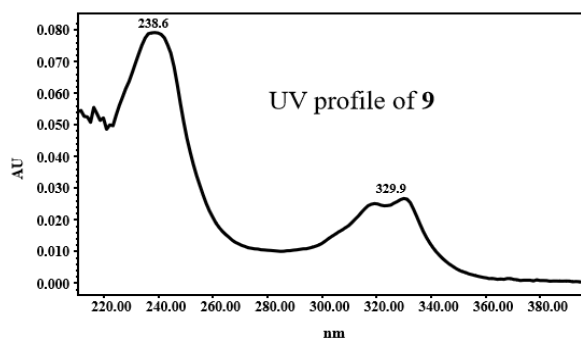
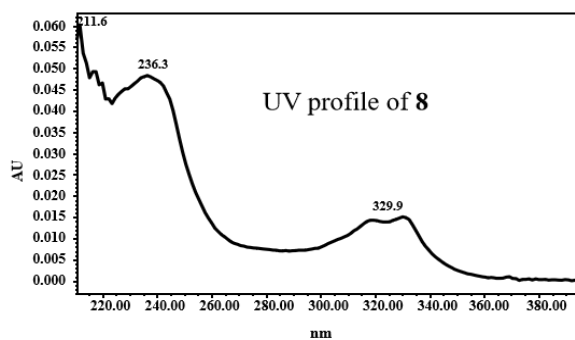
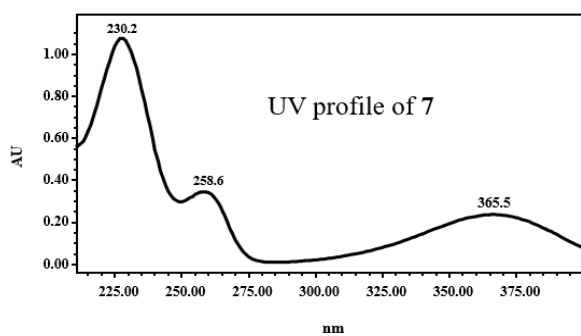
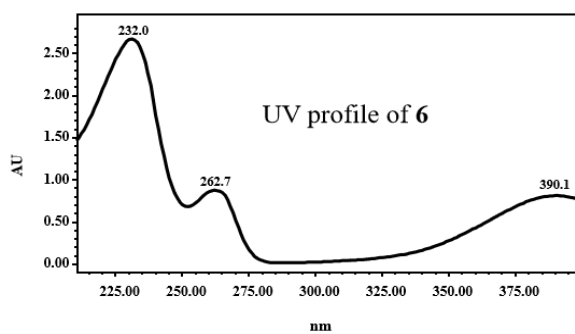
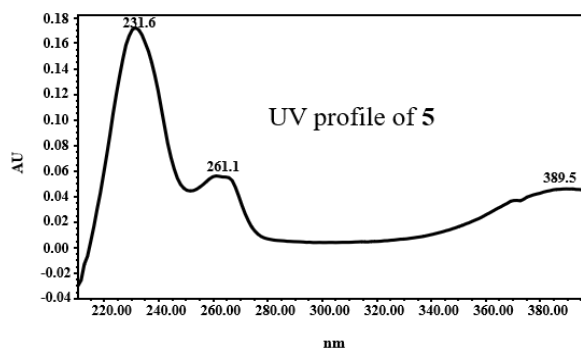
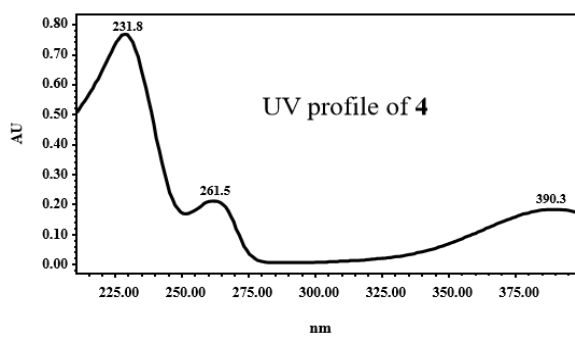
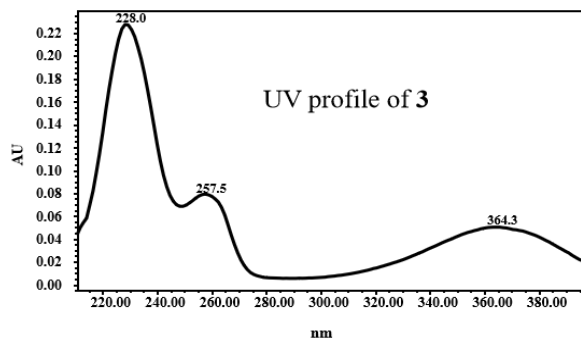
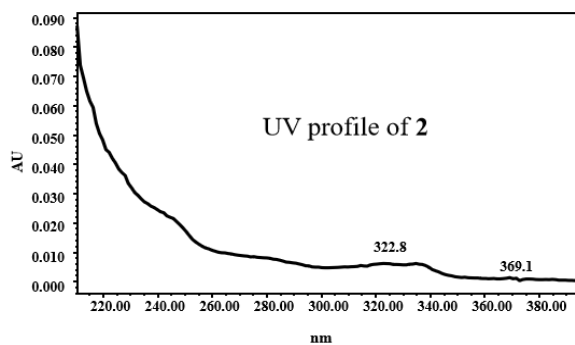
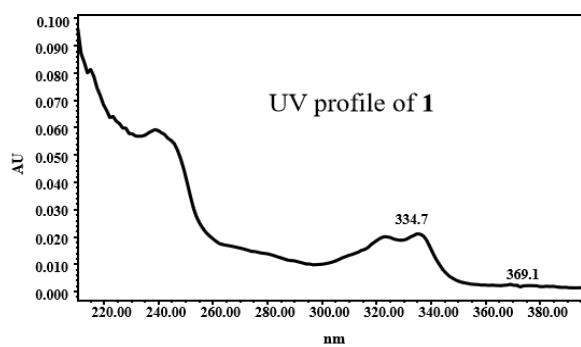


Figure S22. Production of 11-16 by feeding N-MT Gly. The proposed structures of 15 and 16 are as shown above.



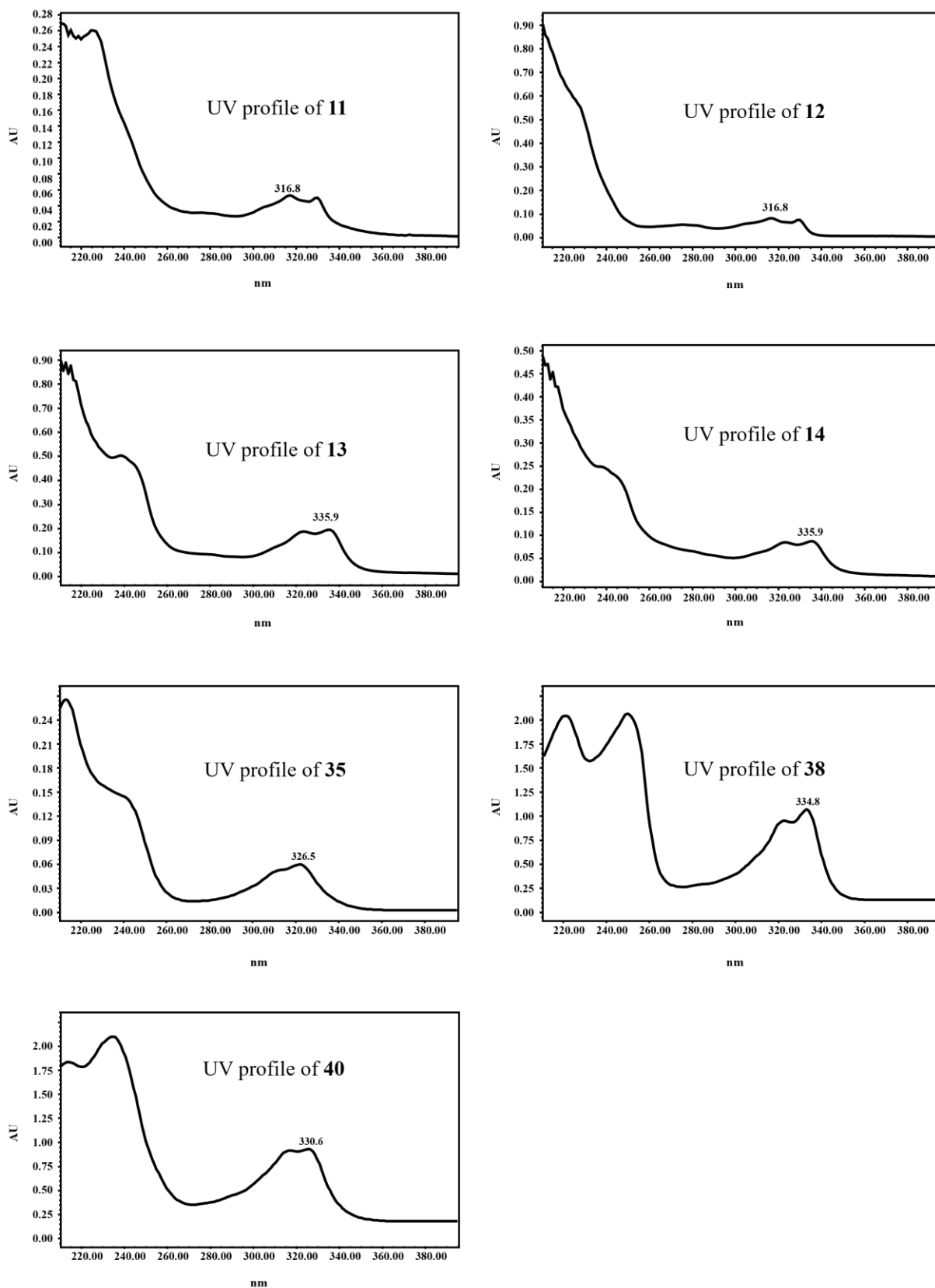
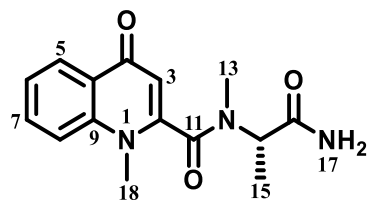


Figure S23. UV profiles of **1-14, 35, 38** and **40**. **4, 6, 7, 10, 35, 38** and **40** are detected on an Agilent Technologies 1200 Series HPLC, others are detected on a Waters 2996 HPLC.



1

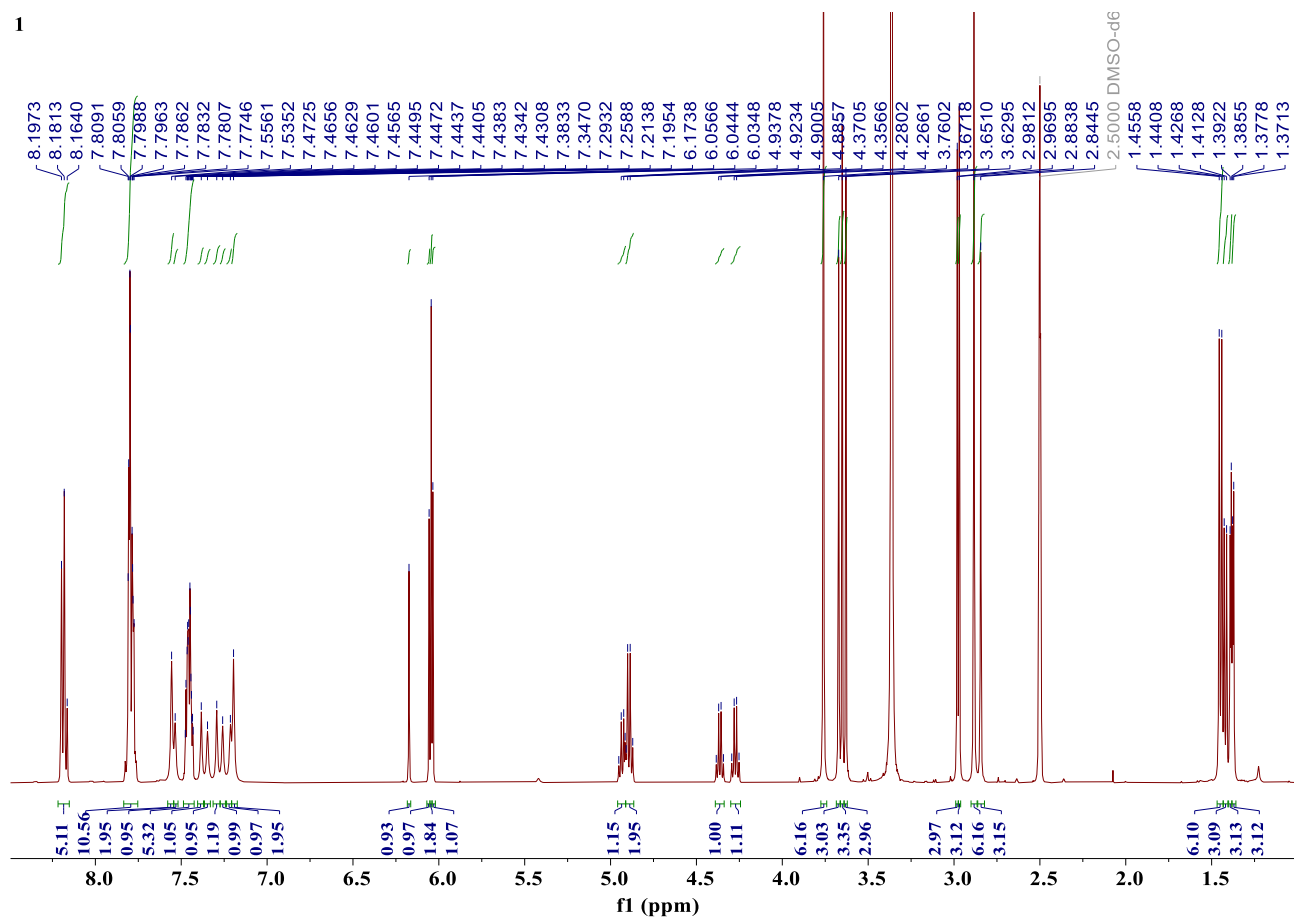


Figure S24: ¹H NMR of 1. DMSO- *d*₆, 500 MHz for ¹H NMR.

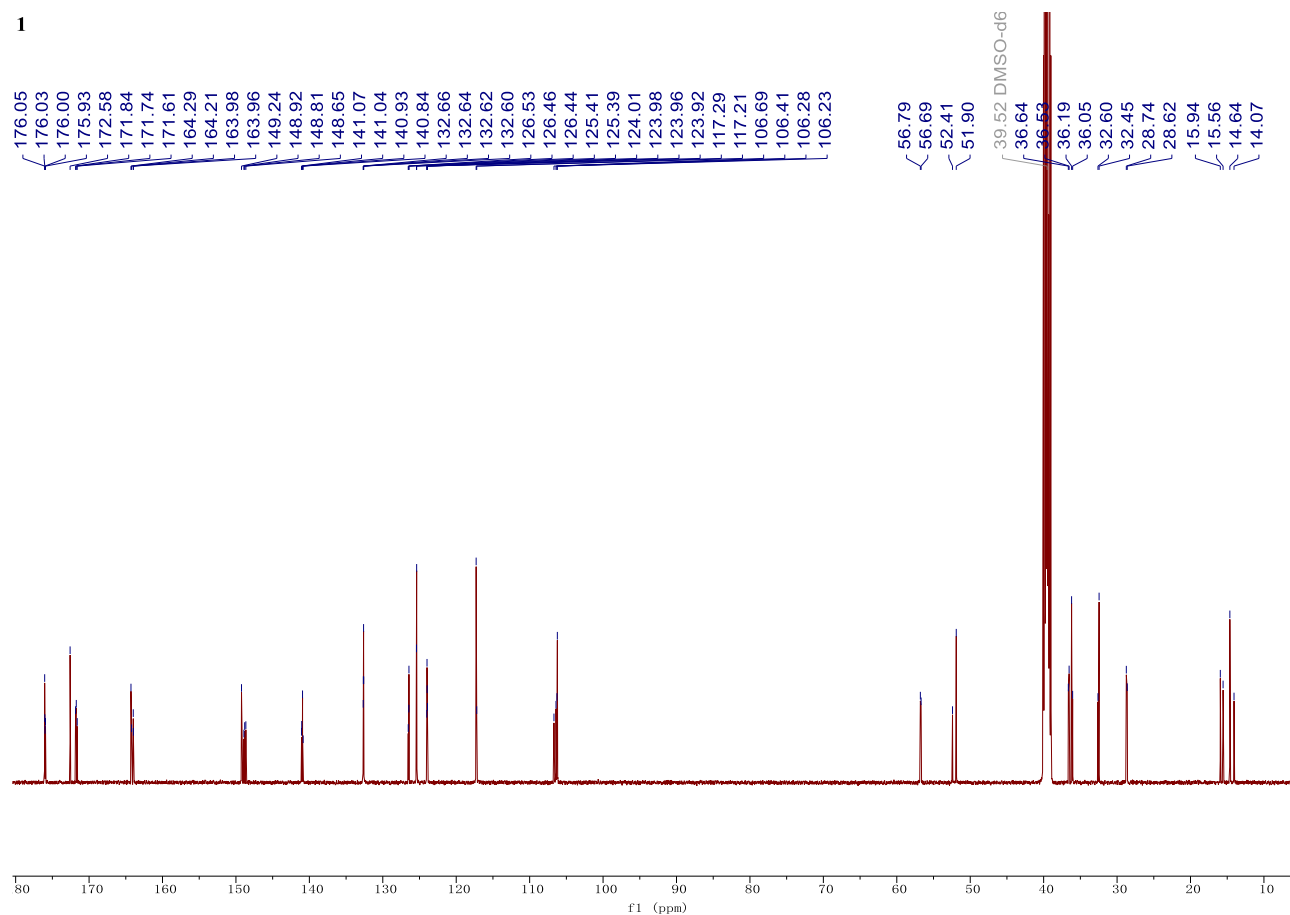
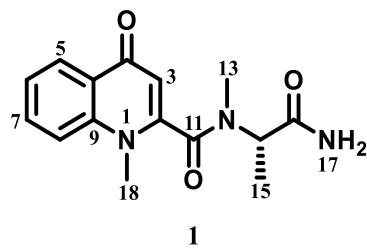


Figure S25: ^{13}C NMR of **1**. DMSO- d_6 , 125 MHz for ^{13}C NMR.

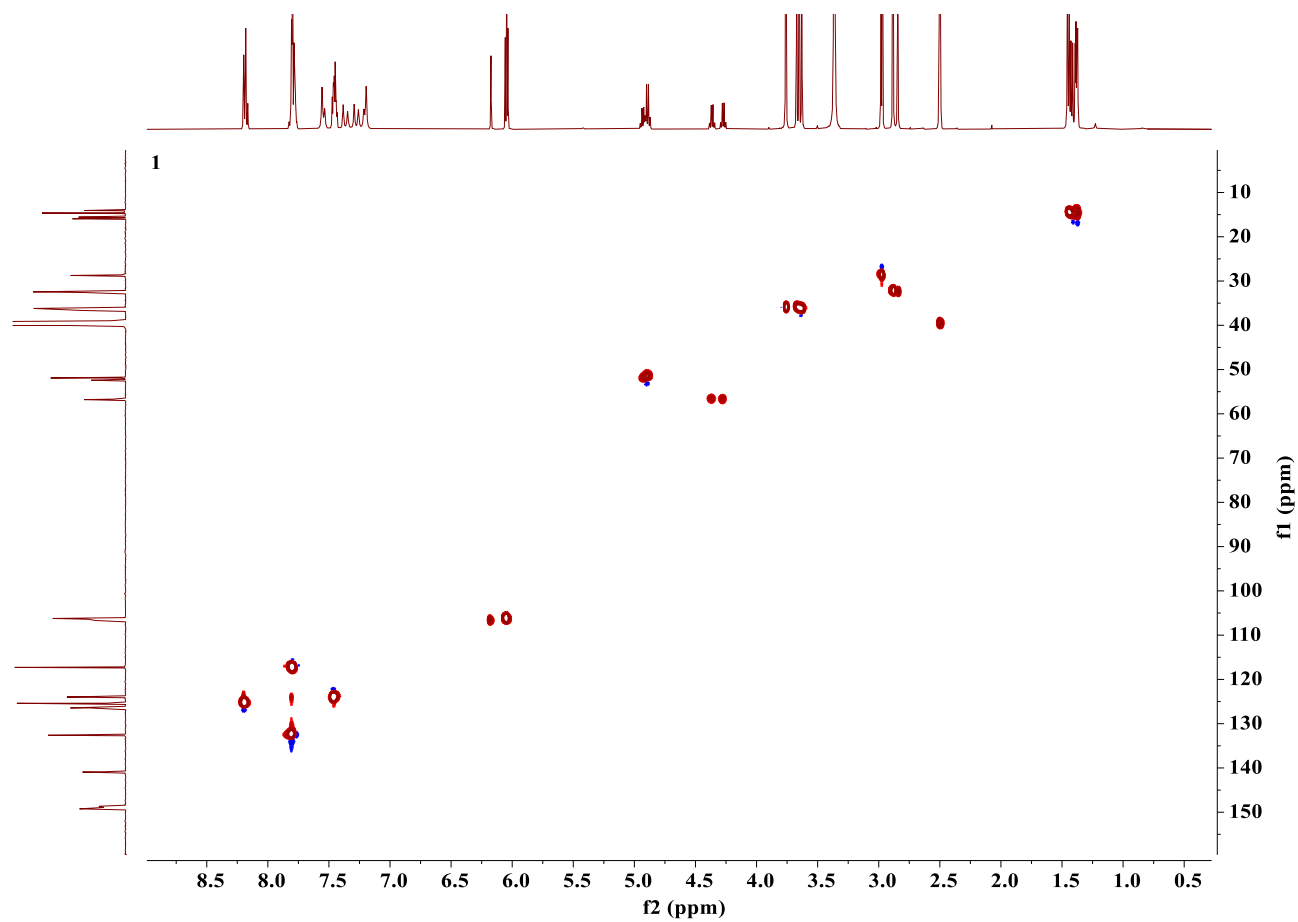
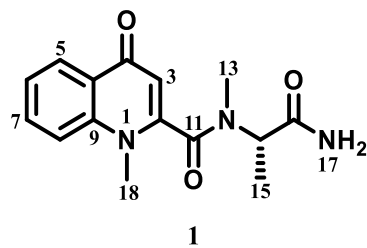


Figure S26: ^1H - ^{13}C HSQC of **1**. DMSO- d_6 , 500 MHz for ^1H and 125 MHz for ^{13}C NMR.

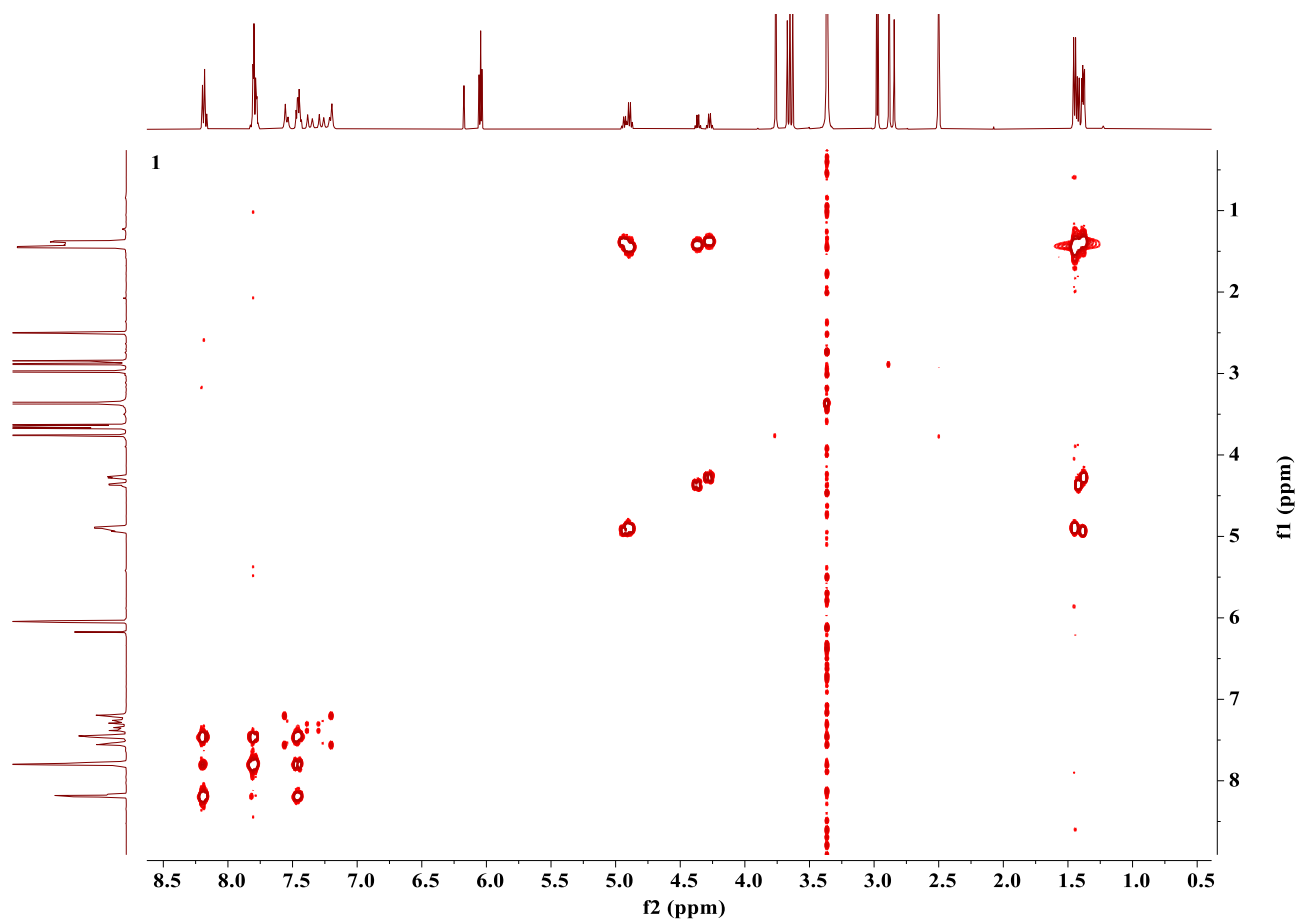
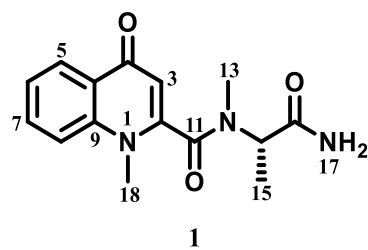


Figure S27: ^1H - ^1H COSY of **1**. DMSO- d_6 , 500 MHz for ^1H NMR.

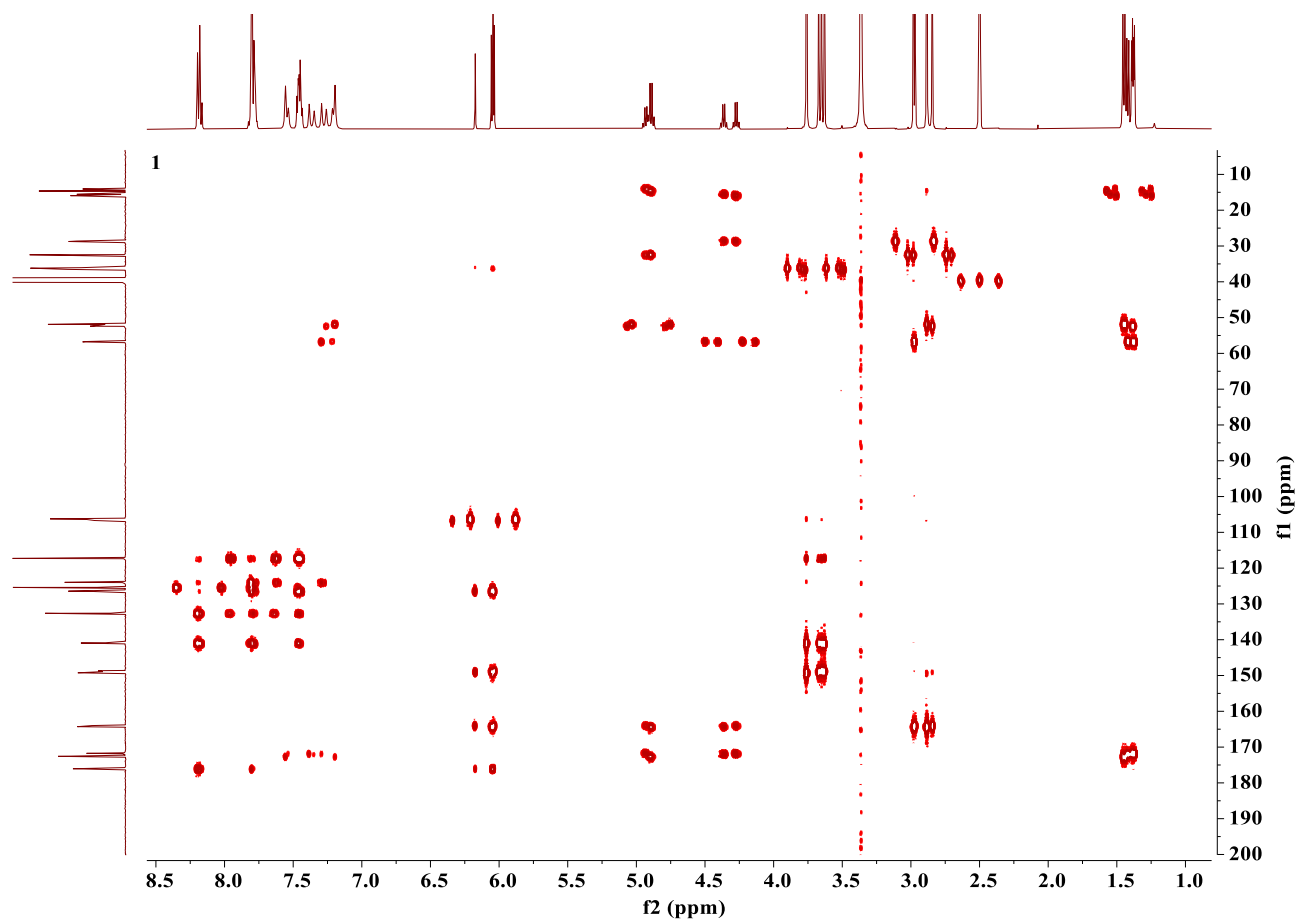
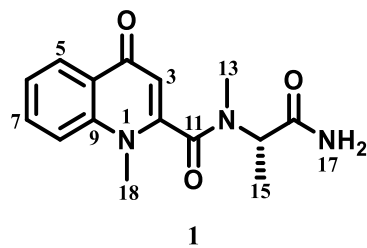


Figure S28: ^1H - ^{13}C HMBC of **1**. DMSO- d_6 , 500 MHz for ^1H and 125 MHz for ^{13}C NMR.

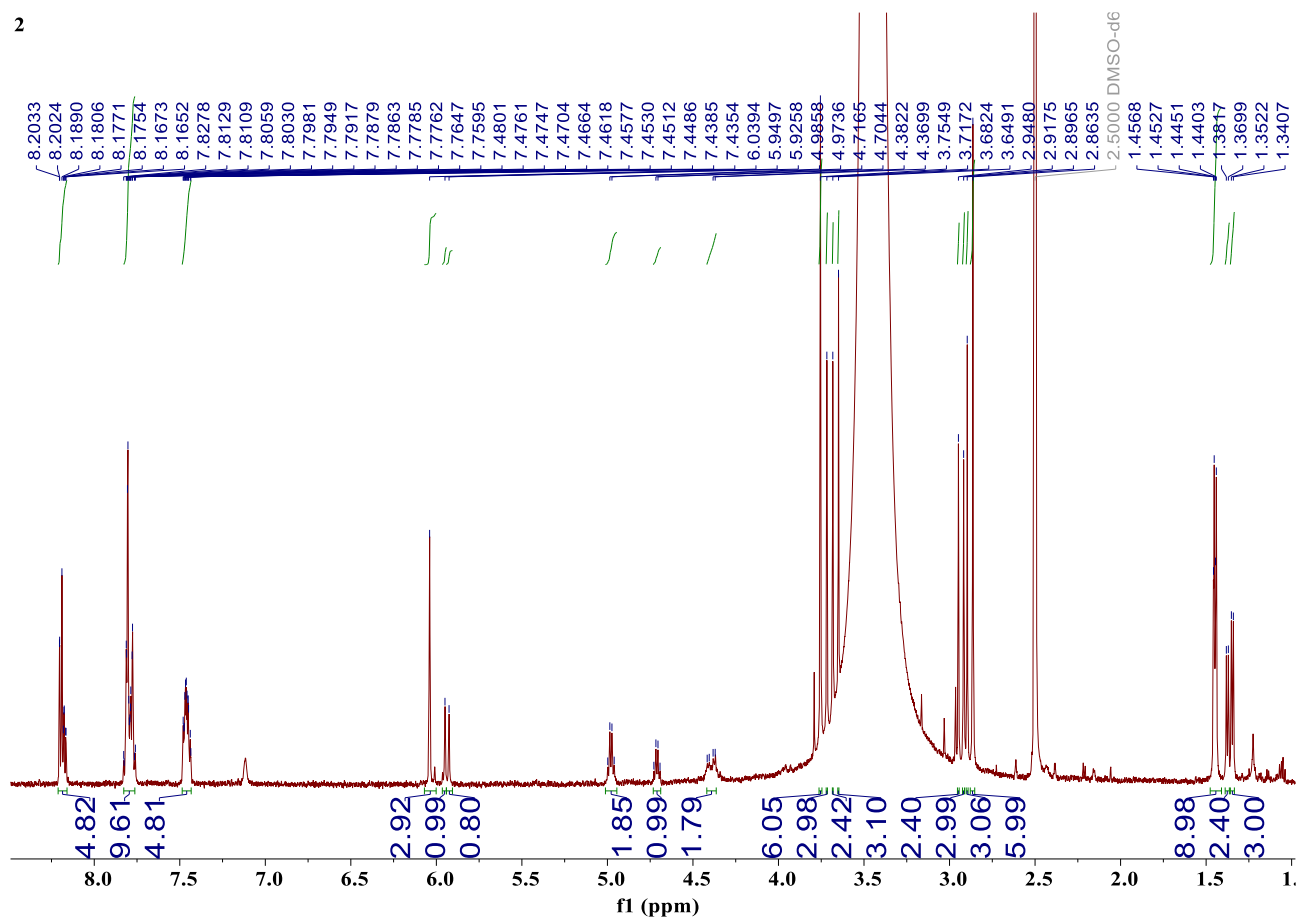
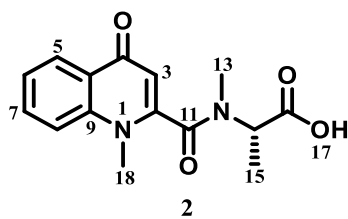


Figure S29: ¹H NMR of **2**. DMSO-*d*₆, 600 MHz for ¹H NMR.

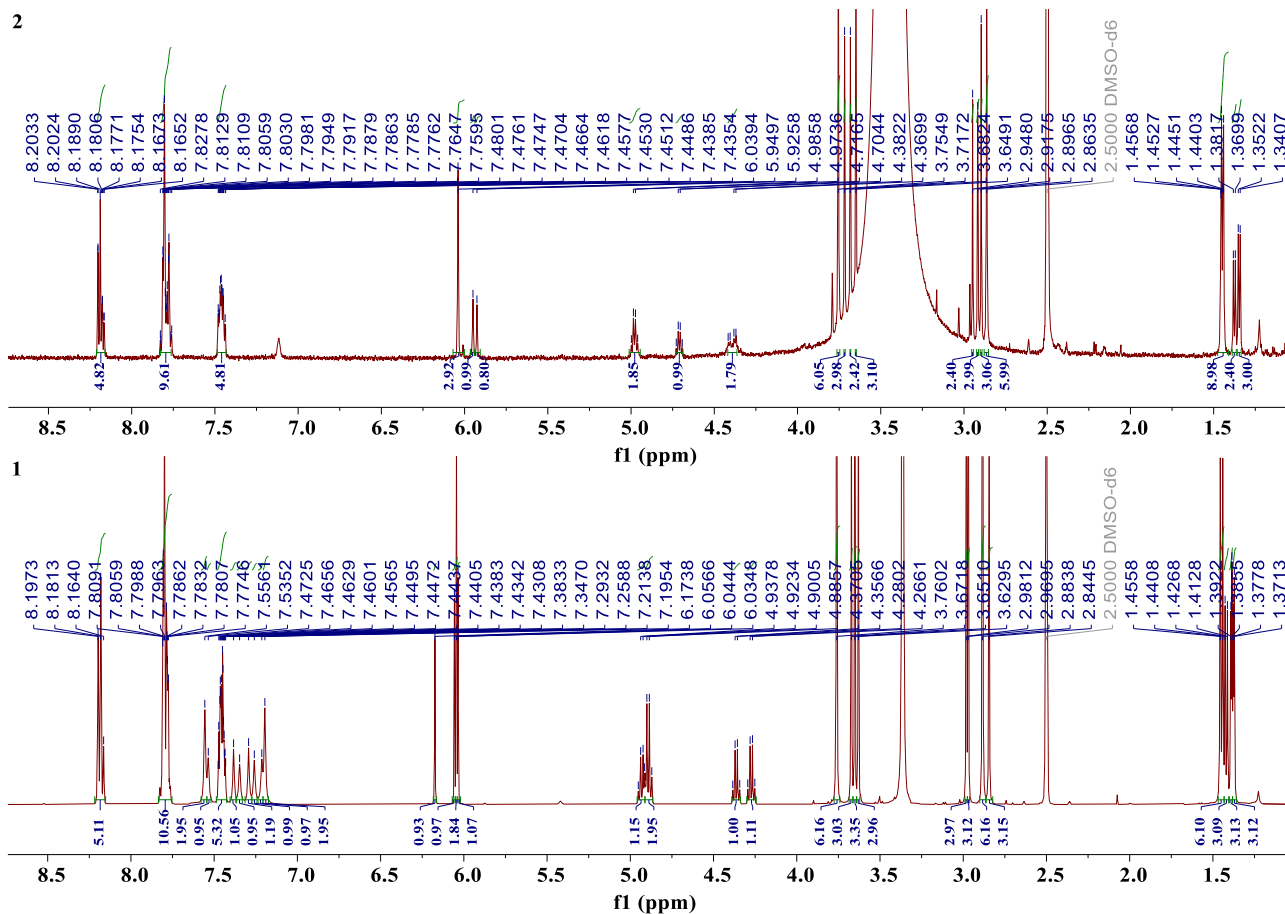
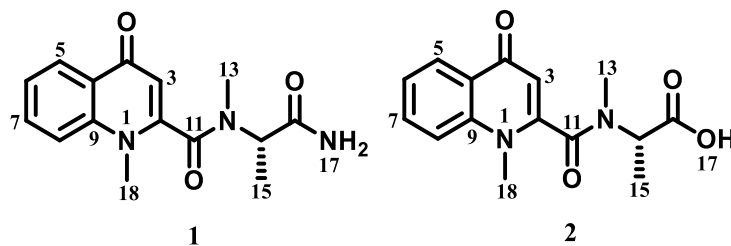
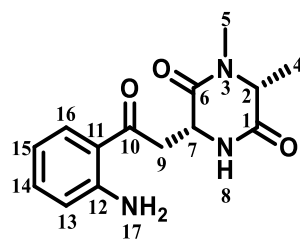


Figure S30: Comparison of ^1H NMR of **1** and **2**. DMSO- d_6 , 500 MHz for ^1H NMR of **1**, 600 MHz for ^1H NMR of **2**.



3

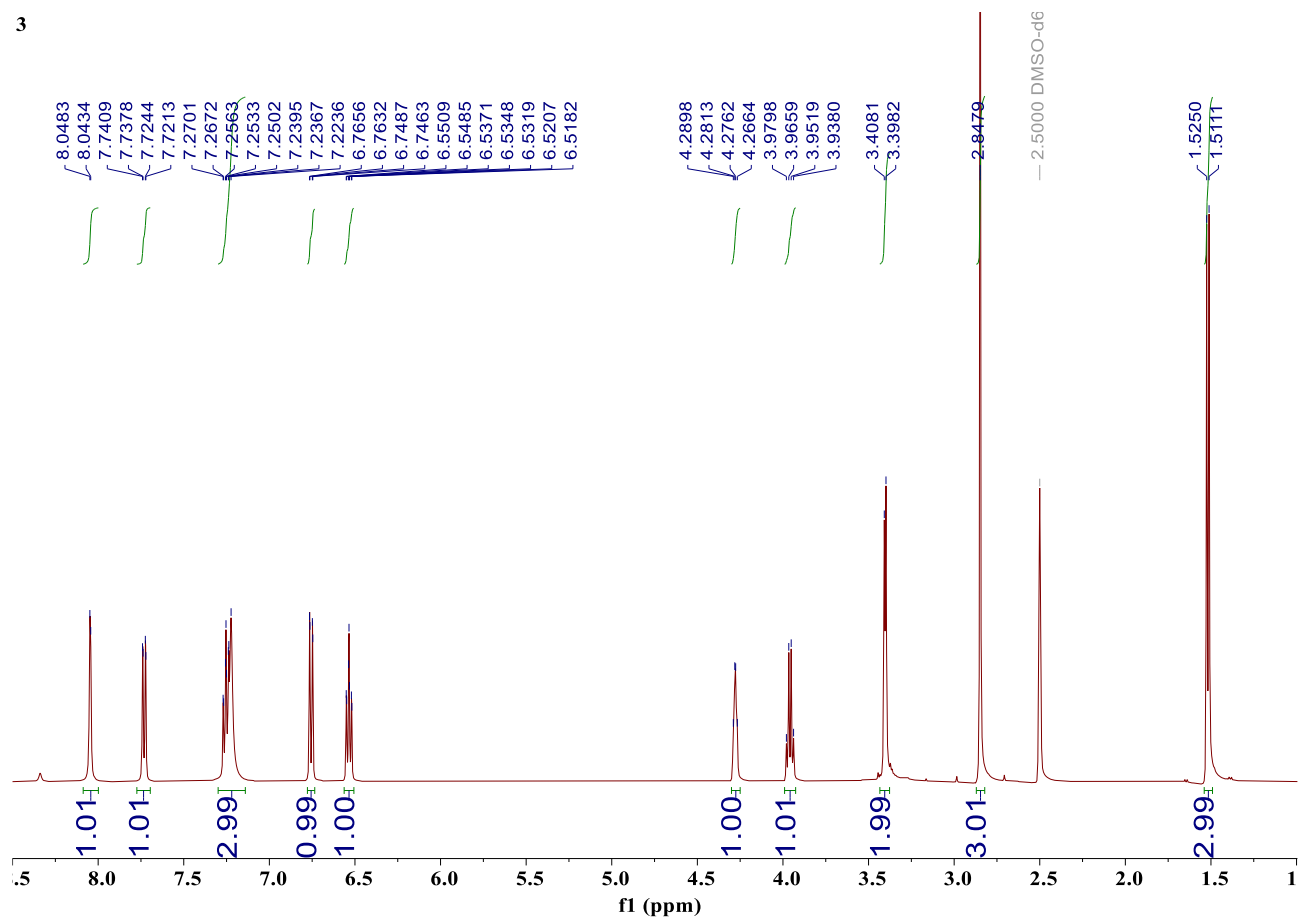


Figure S31: ^1H NMR of **3**. DMSO- d_6 , 500 MHz for ^1H NMR.

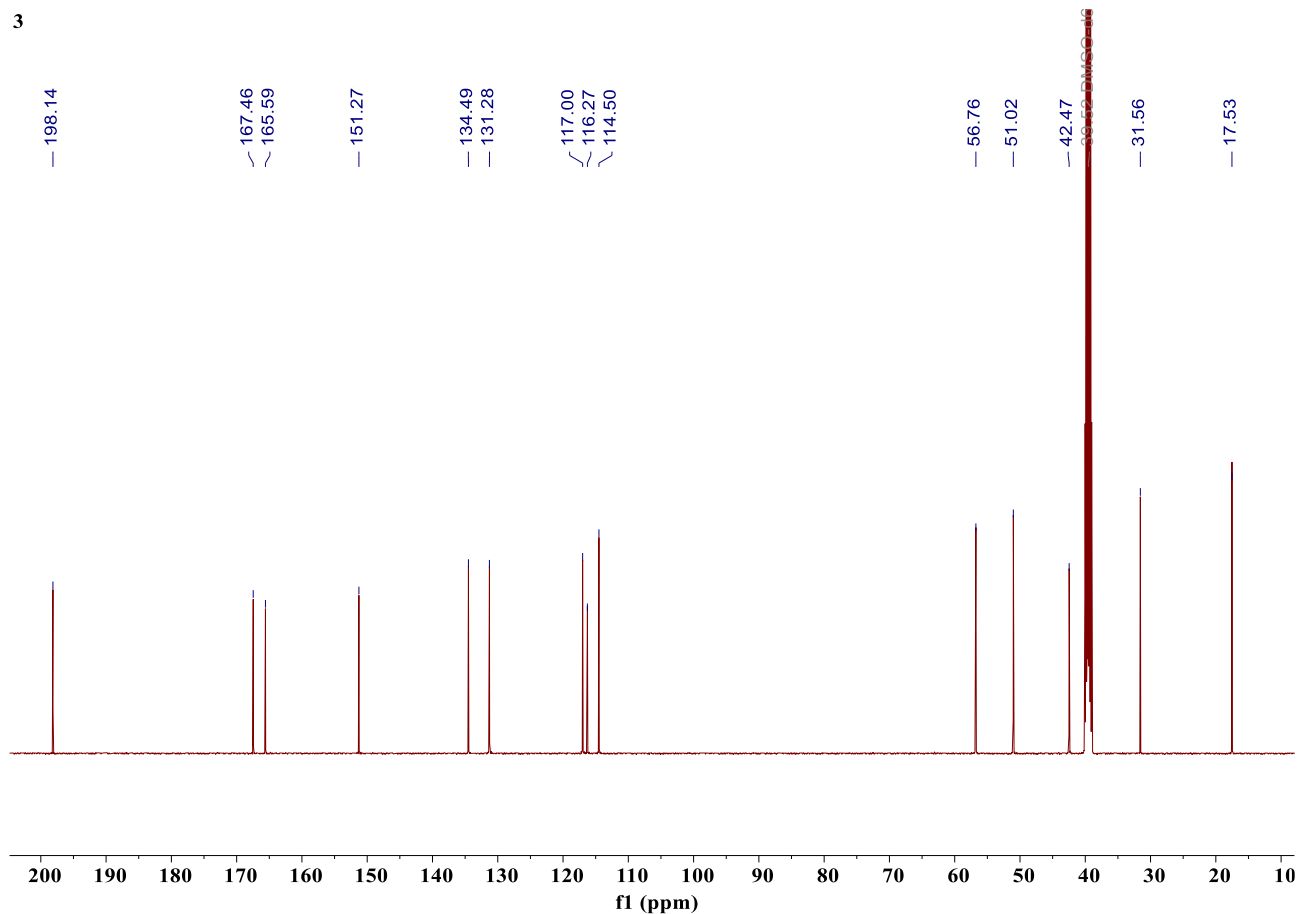
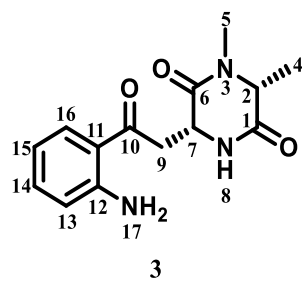


Figure S32: ^{13}C NMR of **3**. DMSO- d_6 , 125 MHz for ^{13}C NMR.

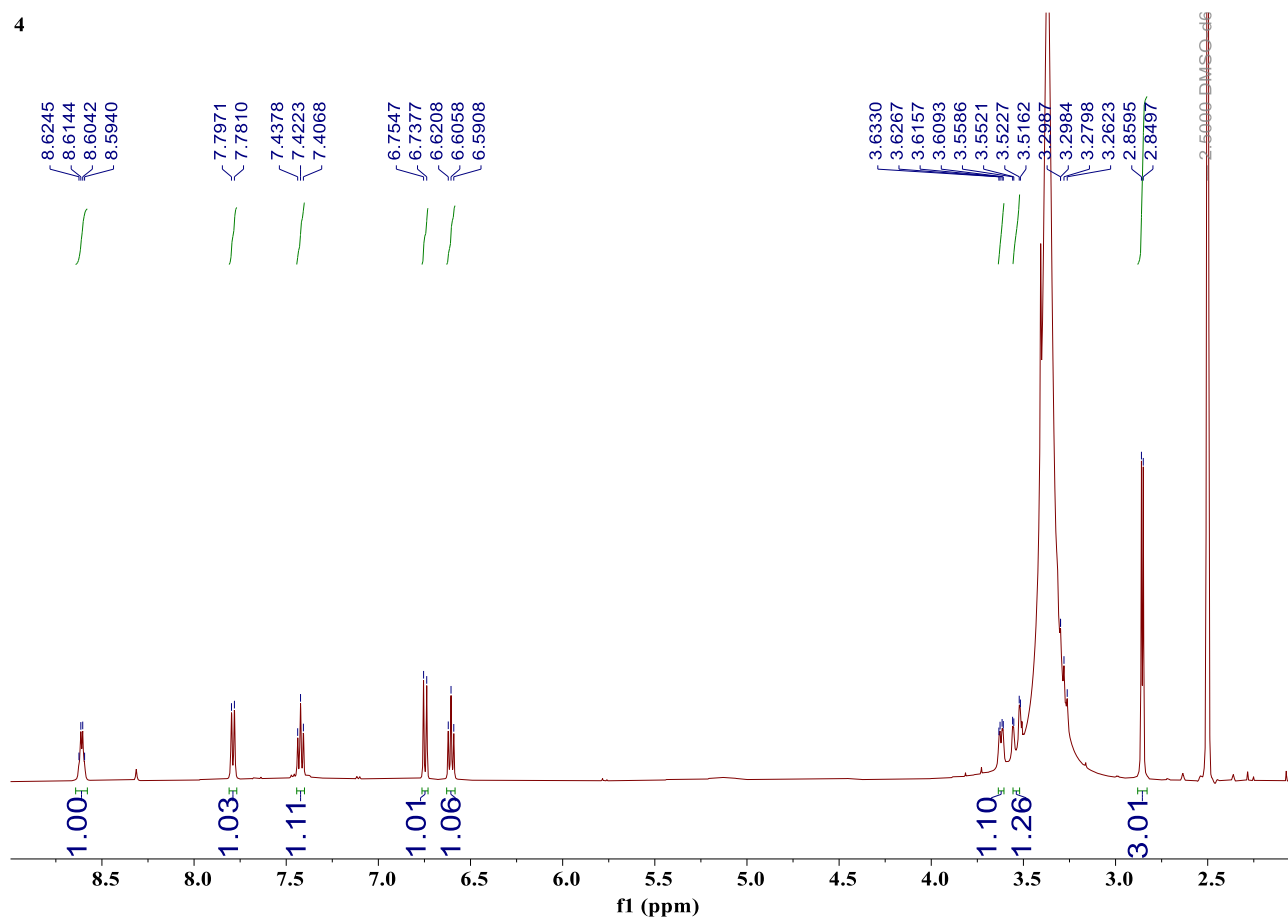
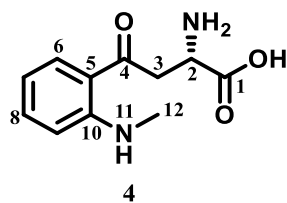


Figure S33: ^1H NMR of **4**. DMSO- d_6 , 500 MHz for ^1H NMR.

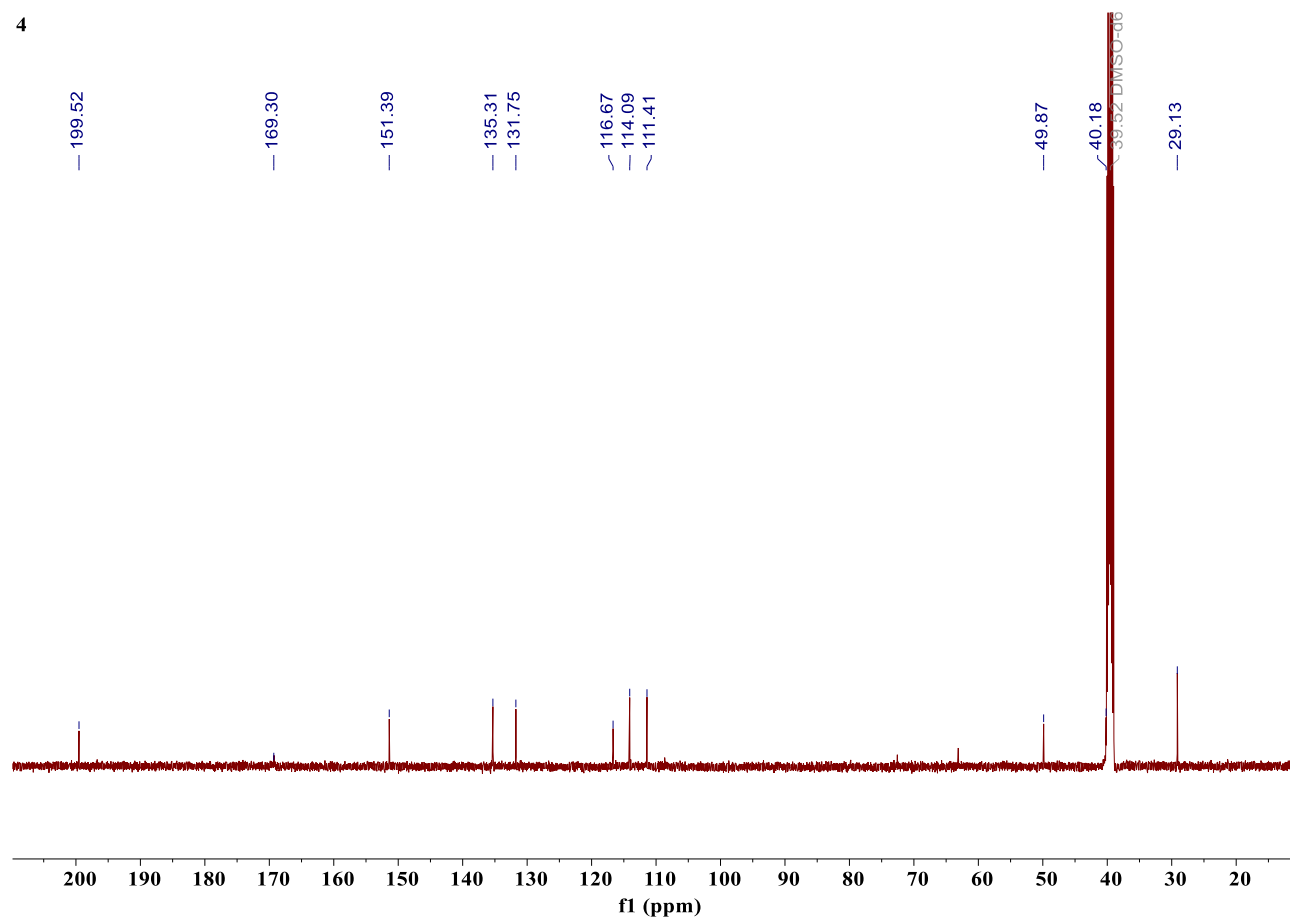
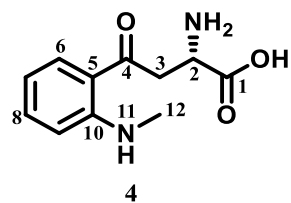


Figure S34: ^{13}C NMR of **4**. DMSO- d_6 , 125 MHz for ^{13}C NMR.

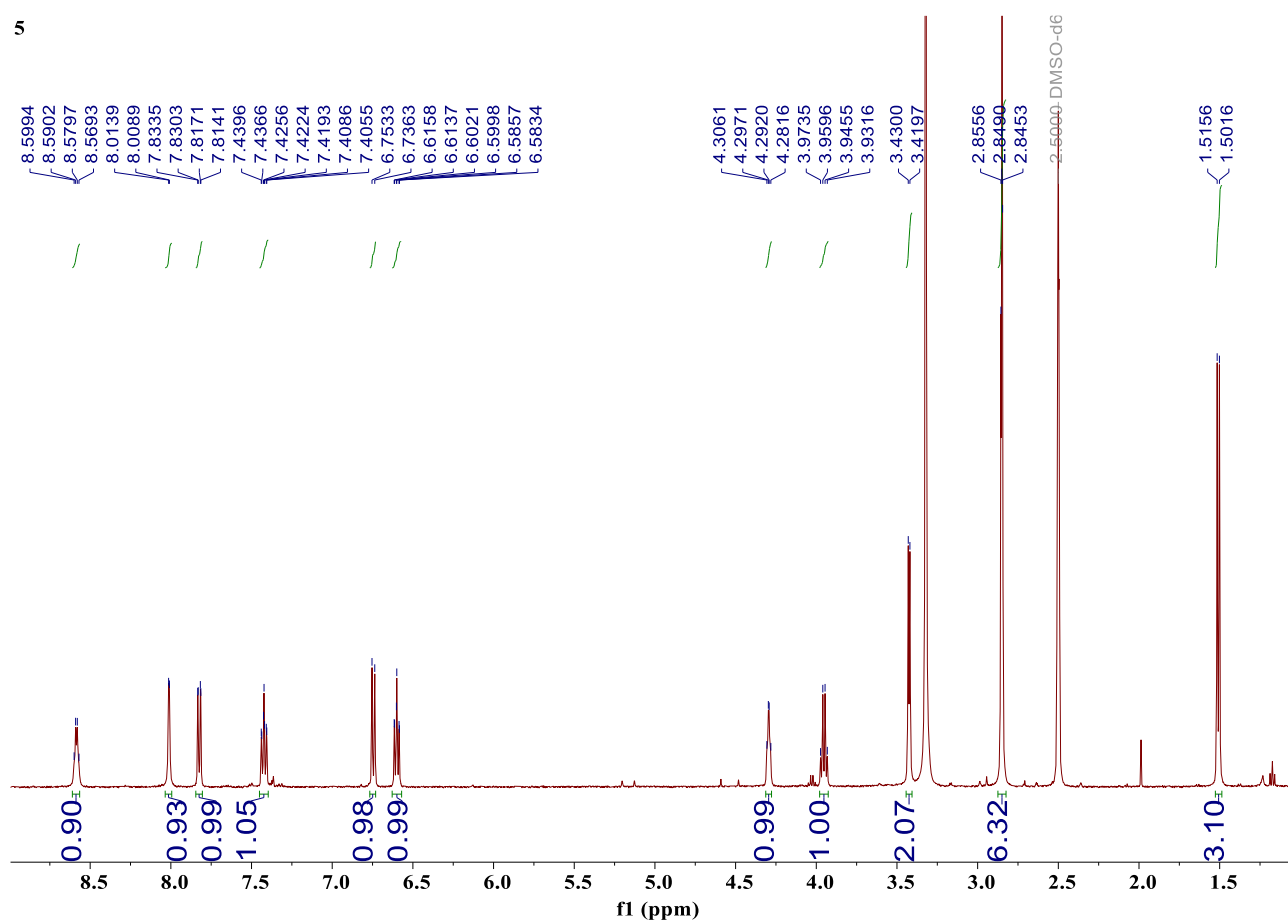
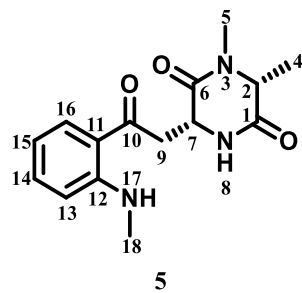


Figure S35: ¹H NMR of **5**. DMSO- *d*₆, 500 MHz for ¹H NMR.

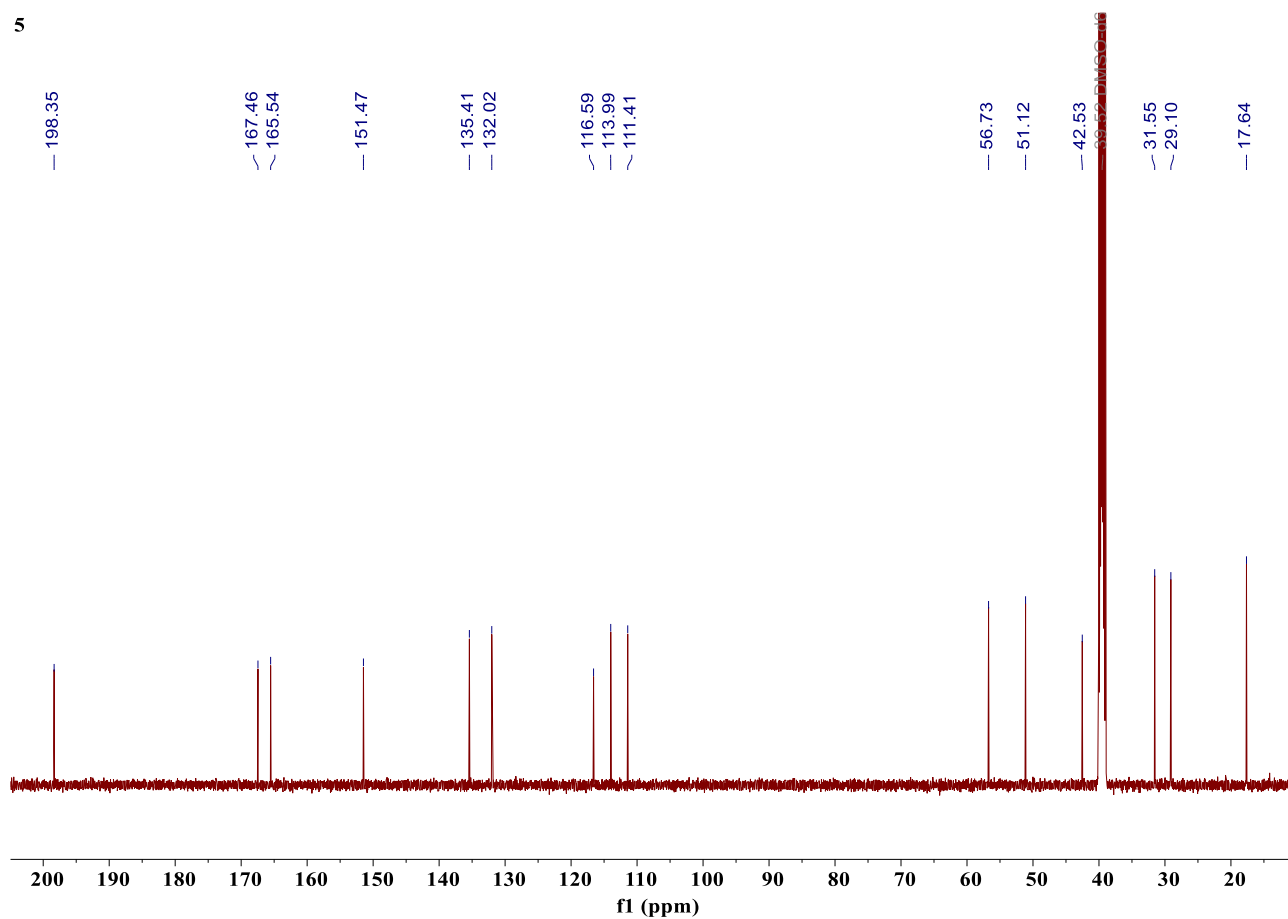
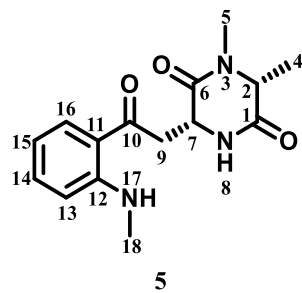


Figure S36: ^{13}C NMR of **5**. DMSO- d_6 , 125 MHz for ^{13}C NMR.

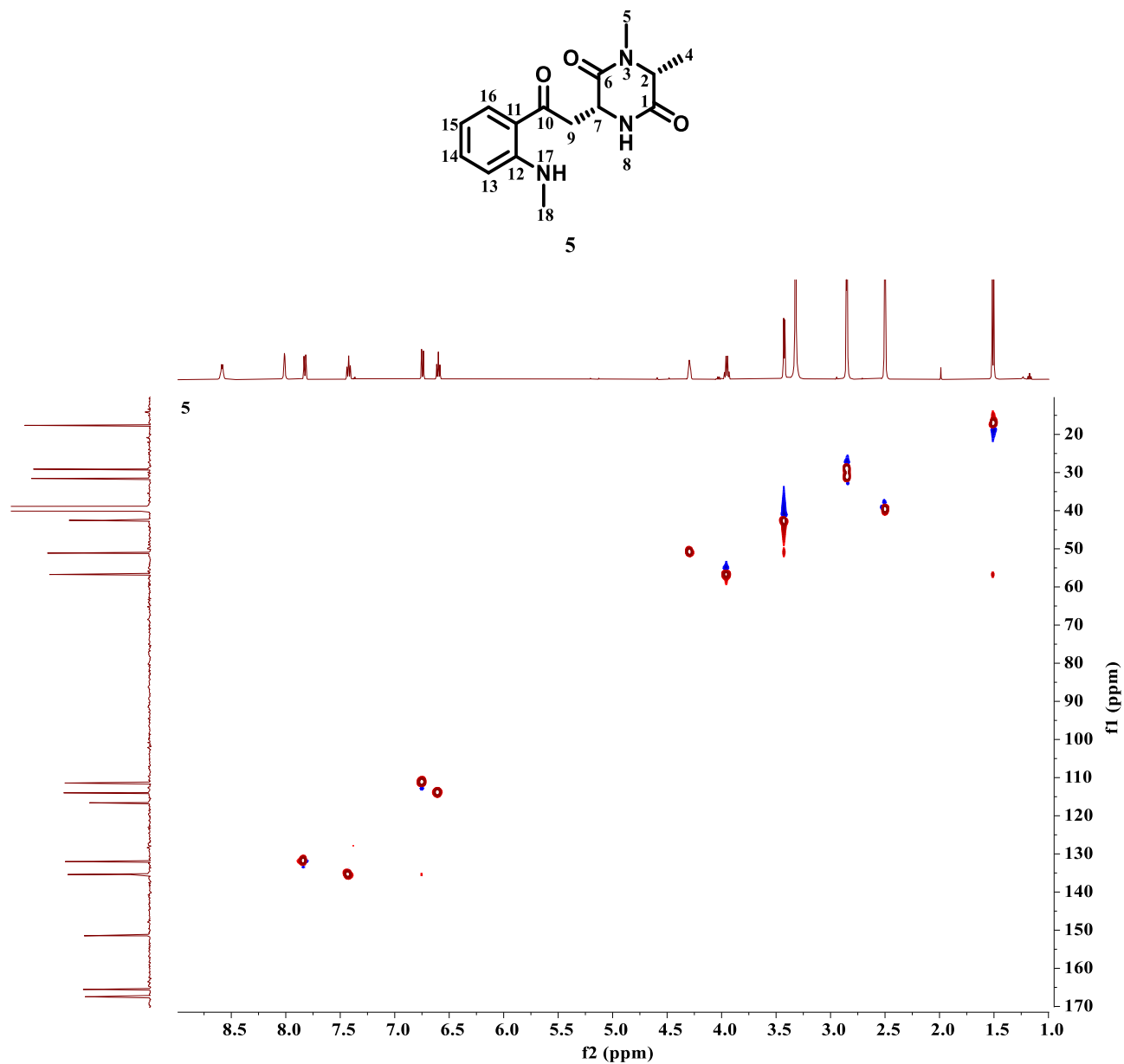


Figure S37: ^1H - ^{13}C HSQC of **5**. DMSO- d_6 , 500 MHz for ^1H and 125 MHz for ^{13}C NMR.

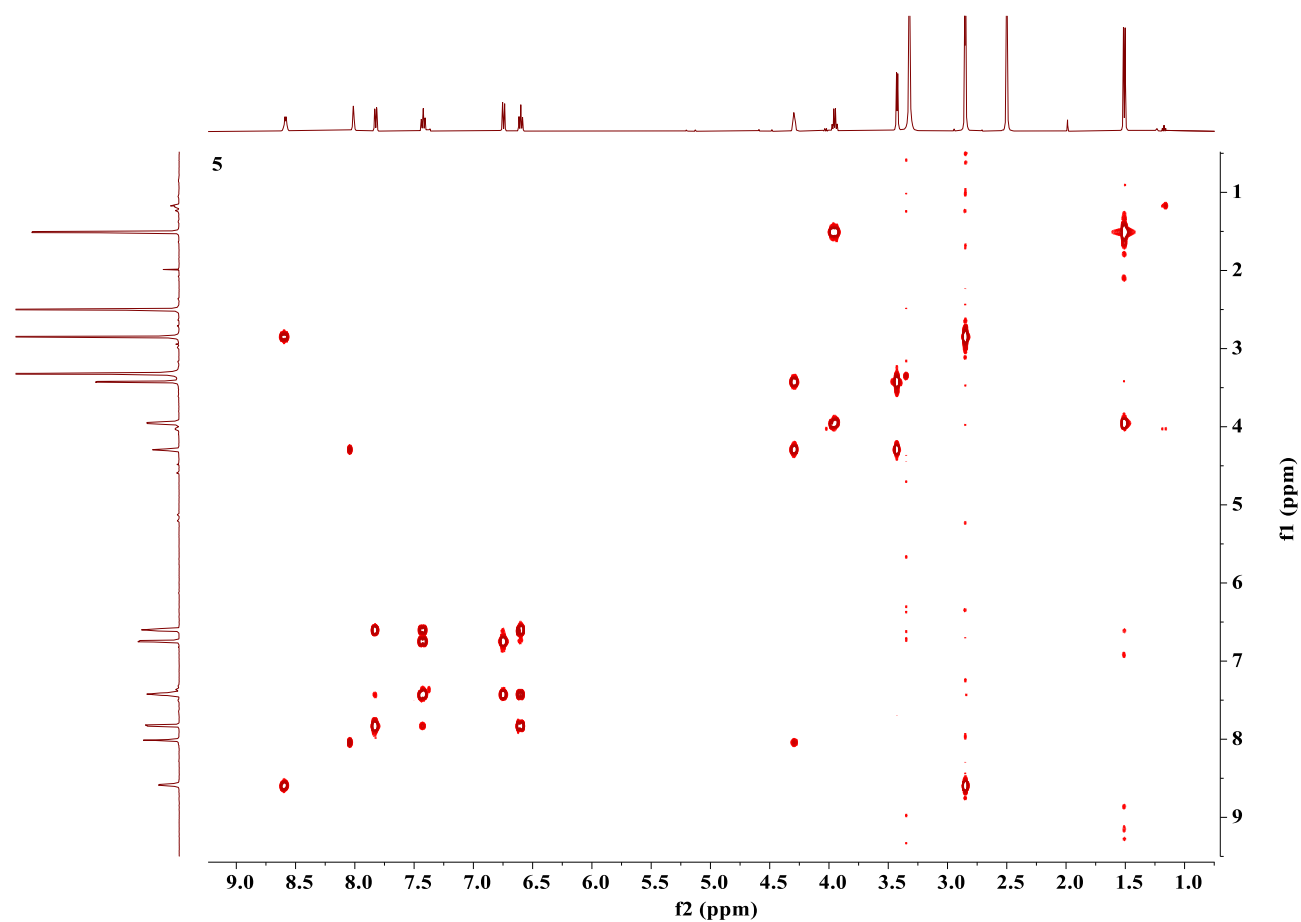
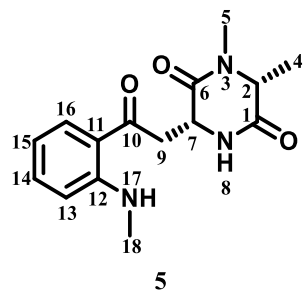


Figure S38: ^1H - ^1H COSY of **5**. DMSO- d_6 , 500 MHz for ^1H NMR.

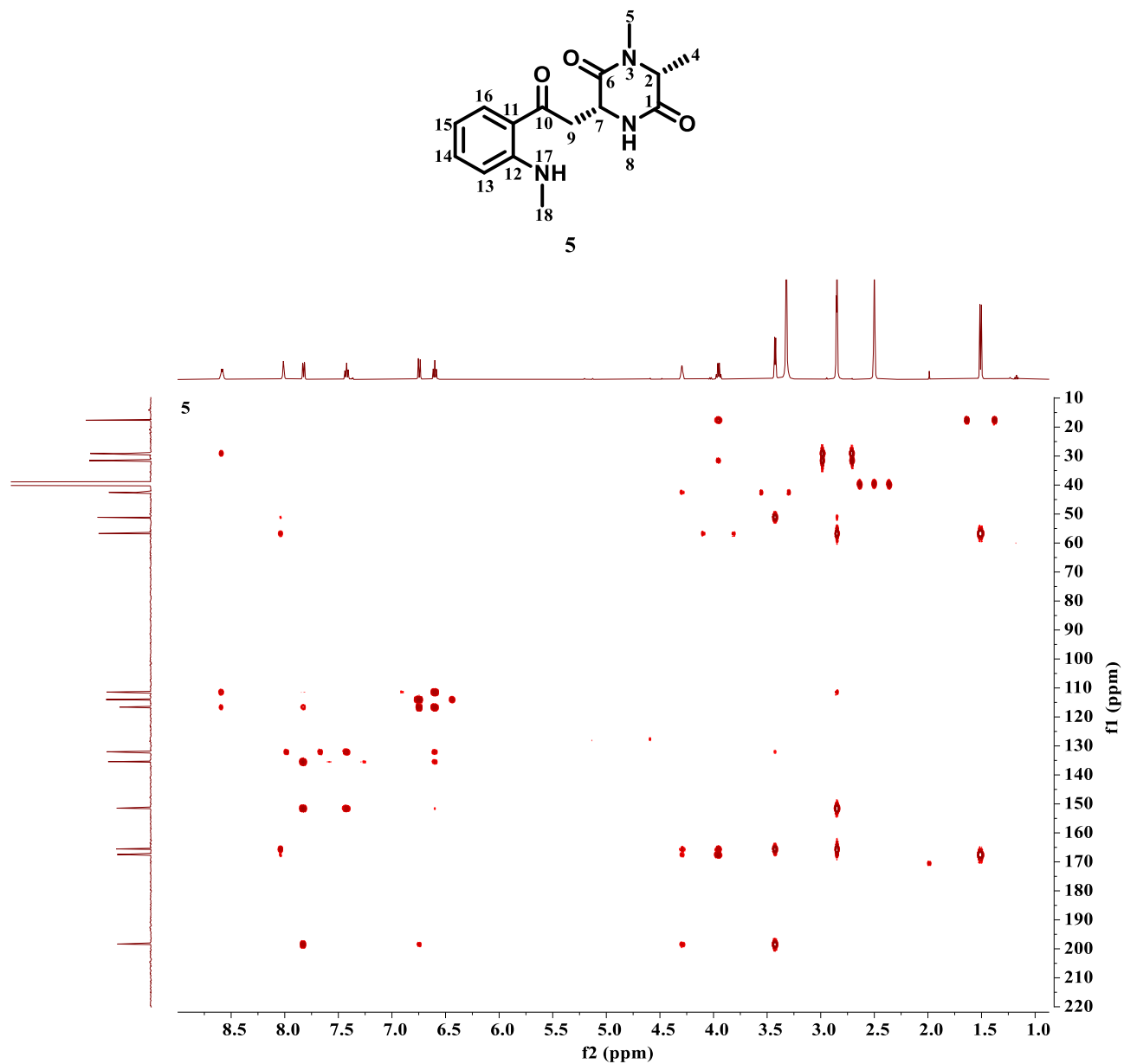


Figure S39: ^1H - ^{13}C HMBC of **5**. DMSO- d_6 , 500 MHz for ^1H and 125 MHz for ^{13}C NMR.

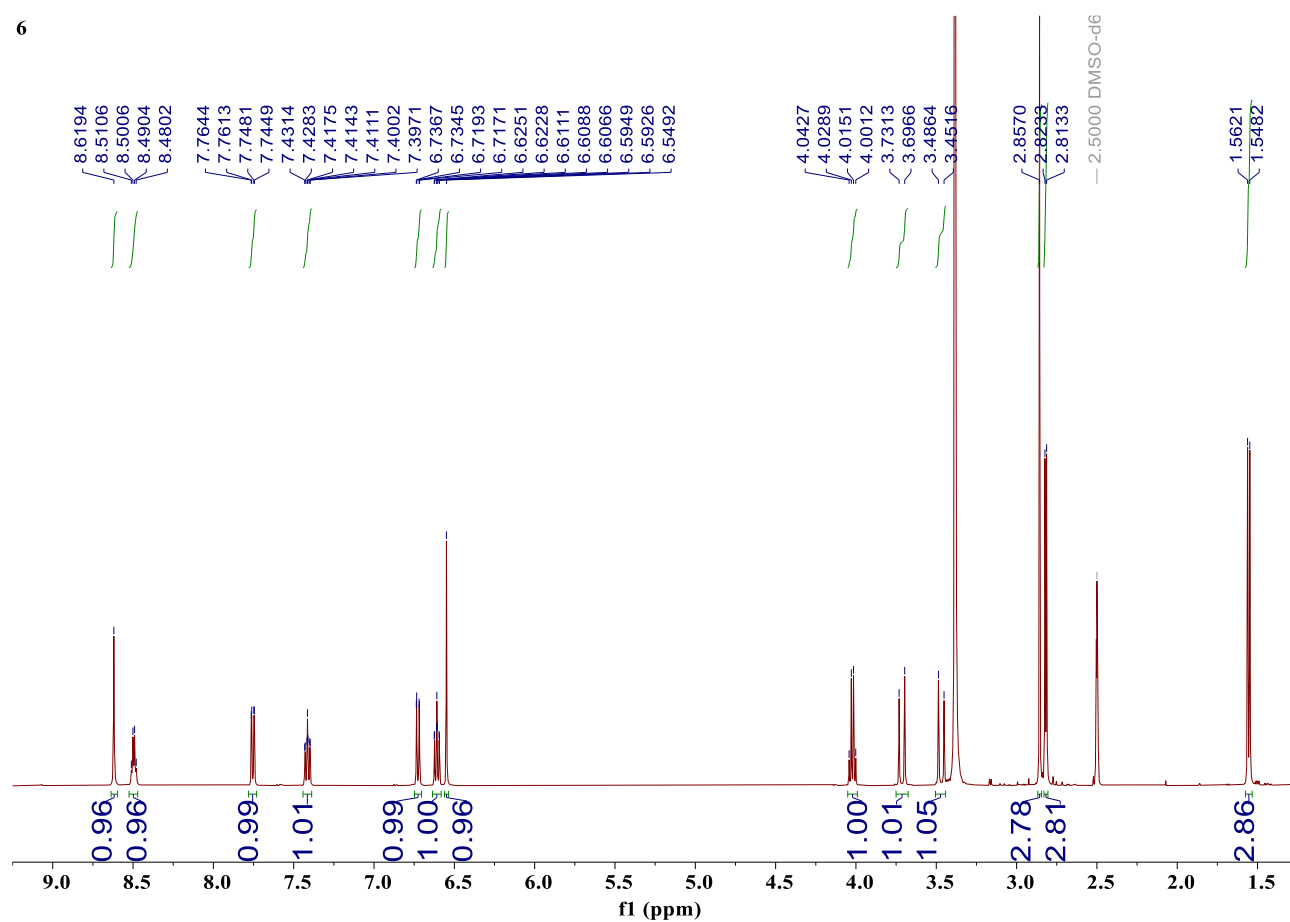
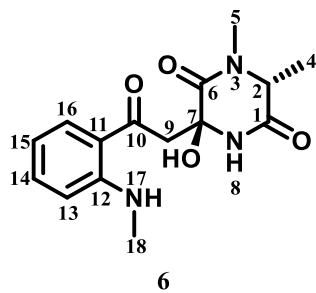


Figure S40: ^1H NMR of **6**. DMSO- d_6 , 600 MHz for ^1H NMR.

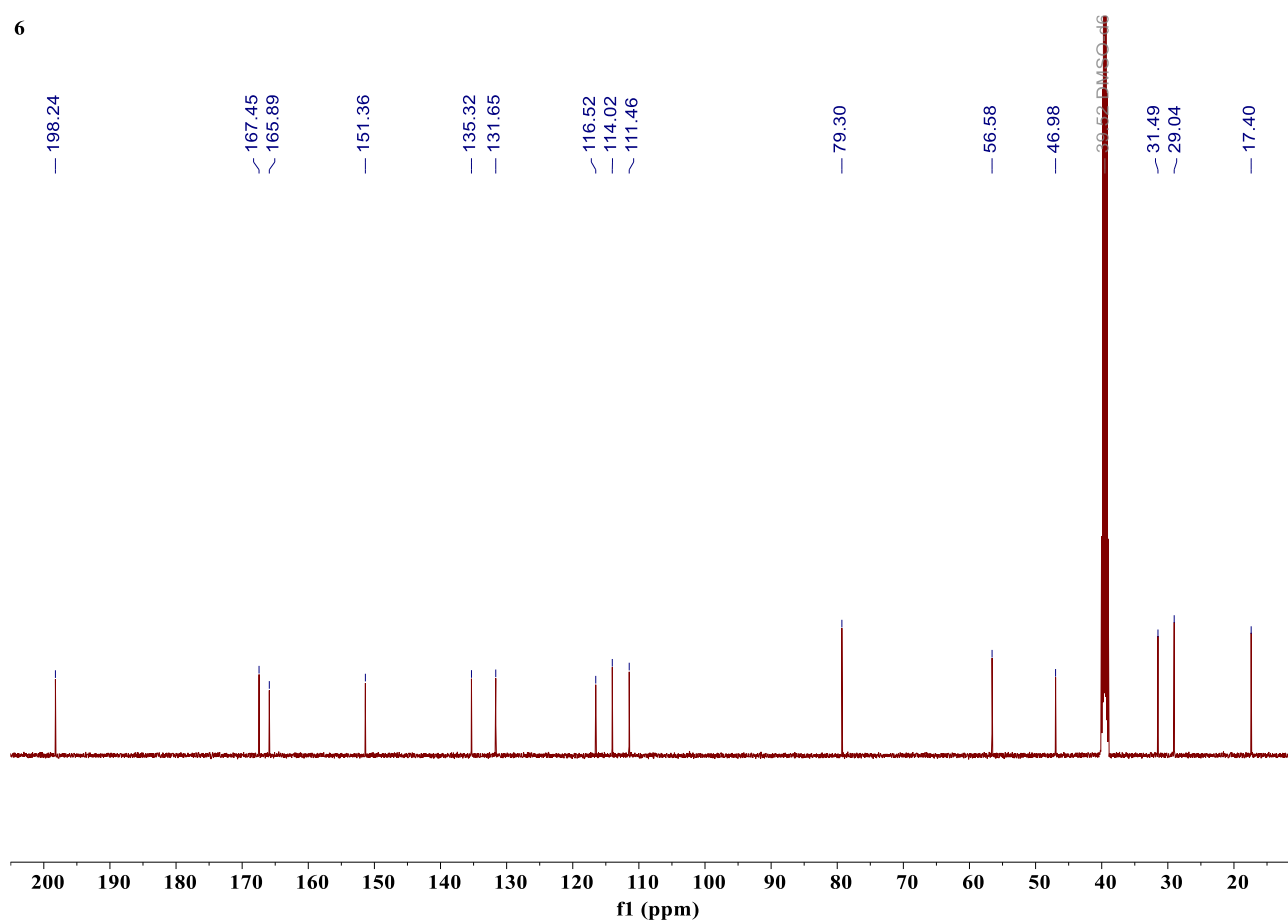
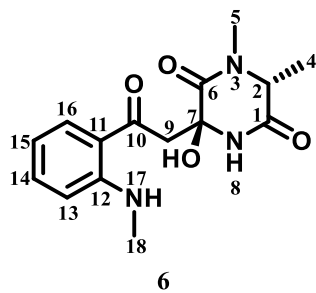


Figure S41: ^{13}C NMR of **6**. DMSO- d_6 , 150 MHz for ^{13}C NMR.

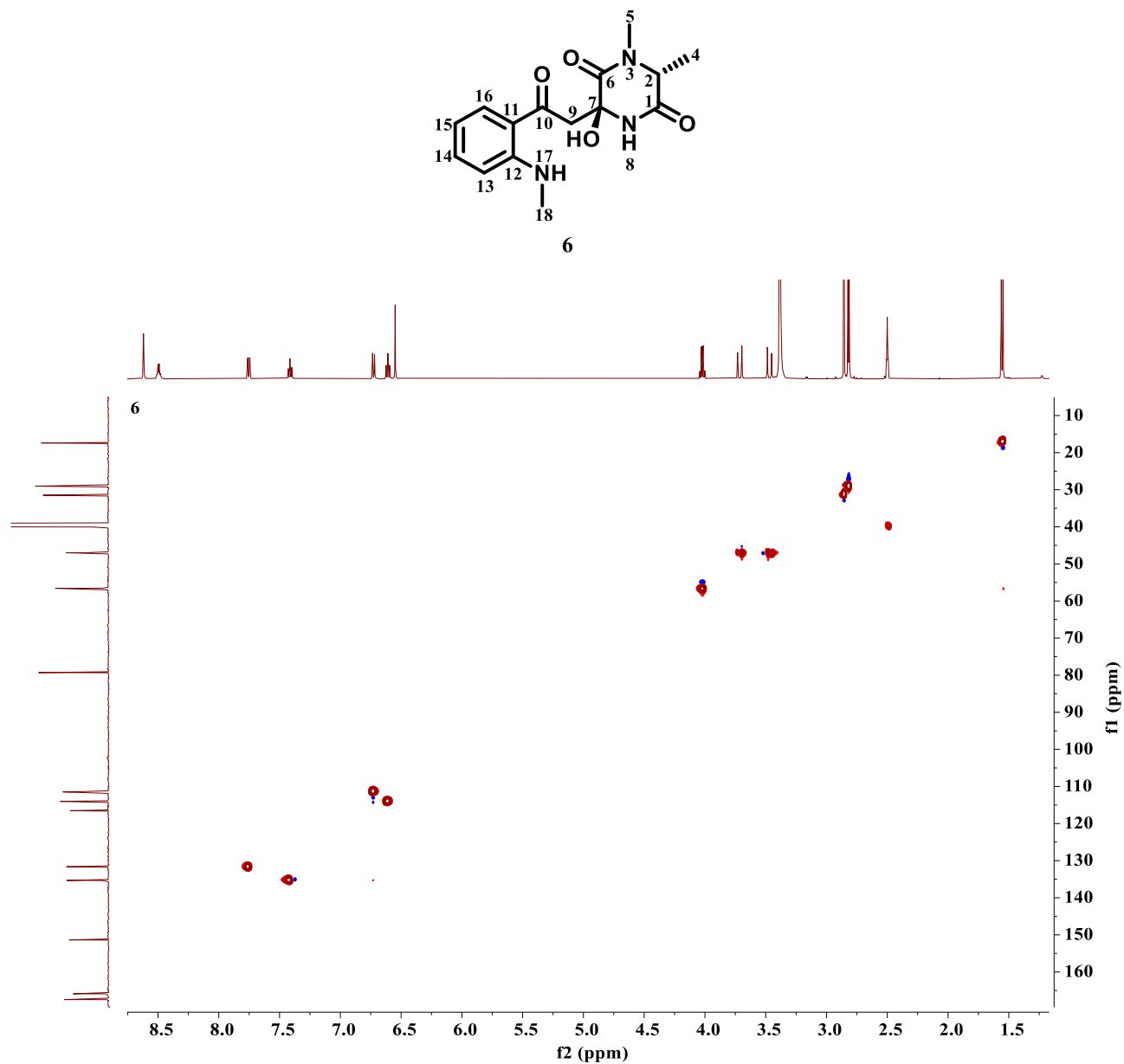


Figure S42: ^1H - ^{13}C HSQC of **6**. DMSO- *d*₆, 600 MHz for ^1H and 150 MHz for ^{13}C NMR.

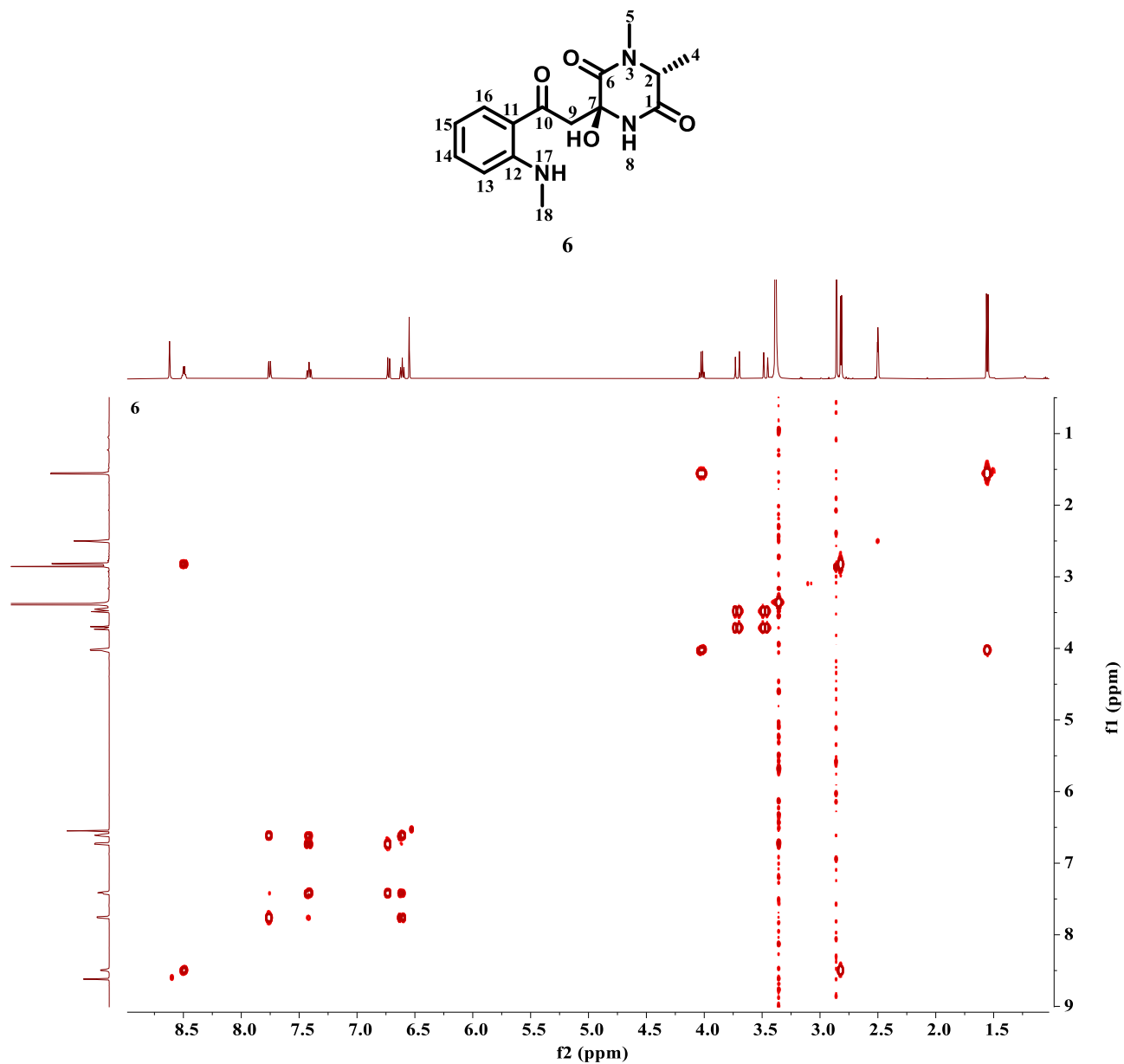


Figure S43: ^1H - ^1H COSY of **6**. DMSO- d_6 , 600 MHz for ^1H NMR.

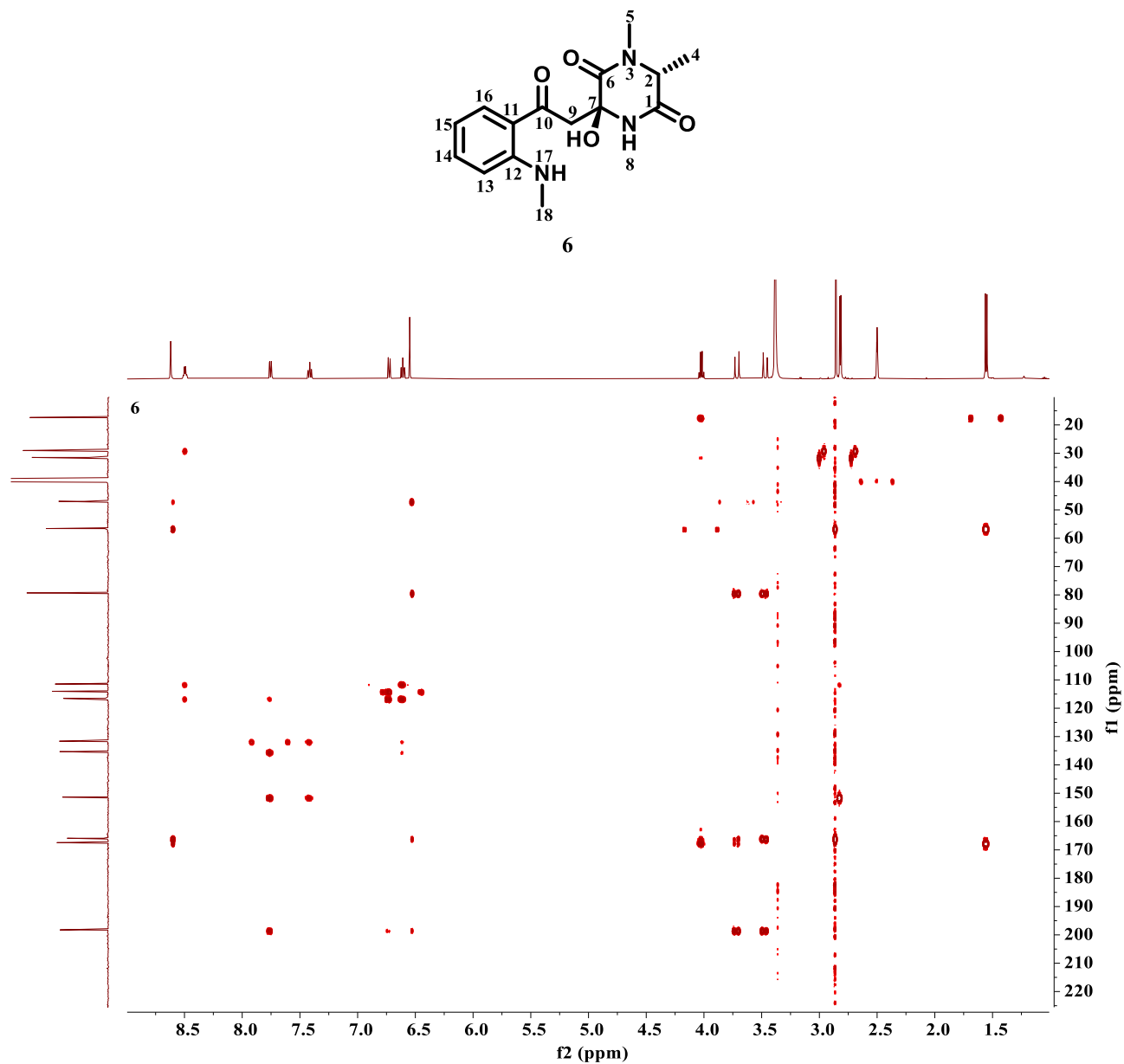


Figure S44: ^1H - ^{13}C HMBC of **6**. DMSO- d_6 , 600 MHz for ^1H and 150 MHz for ^{13}C NMR.

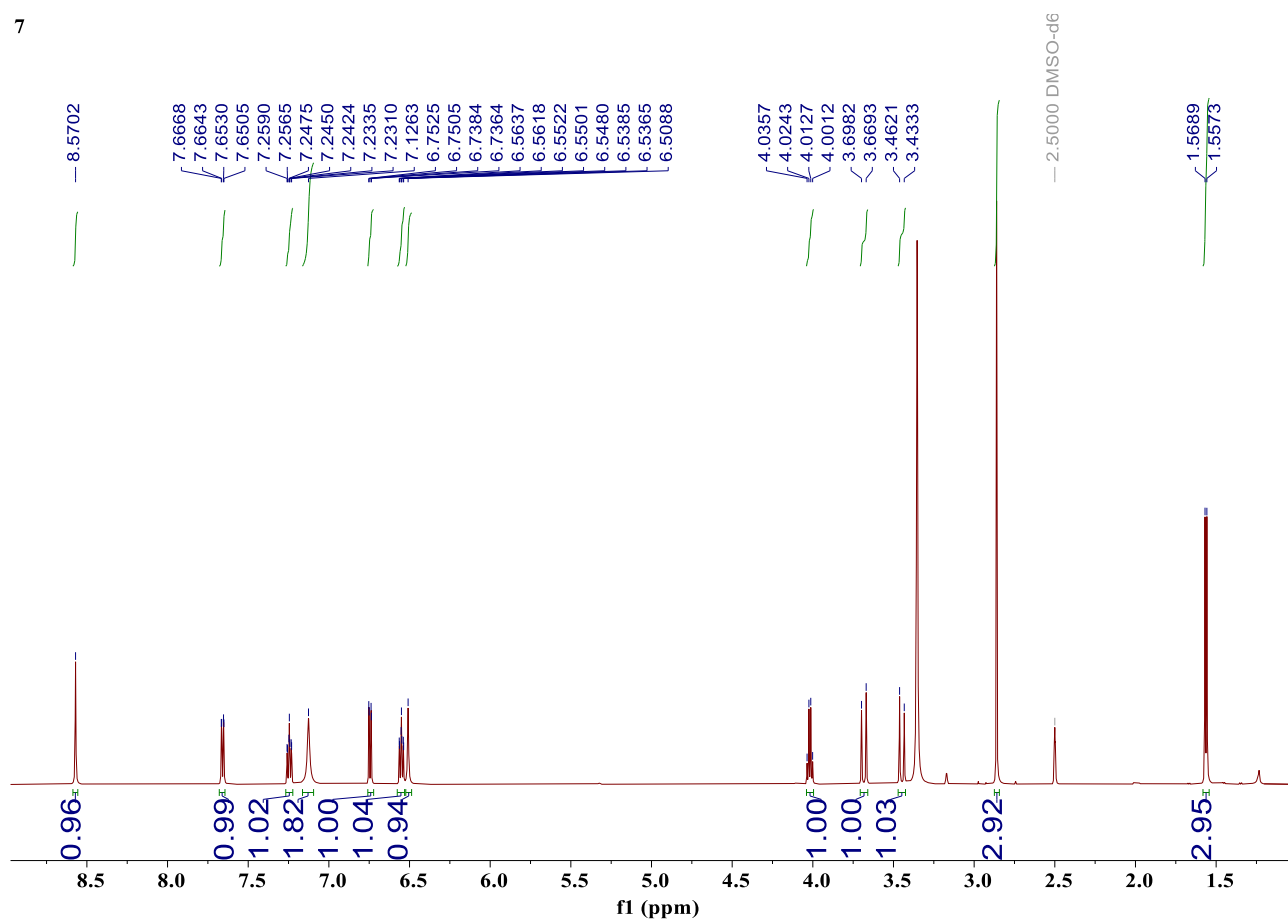
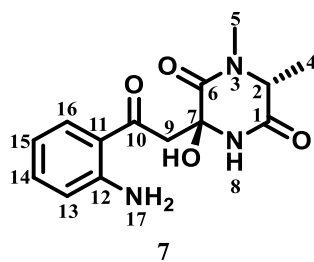


Figure S45: ¹H NMR of 7. DMSO- d₆, 600 MHz for ¹H NMR.

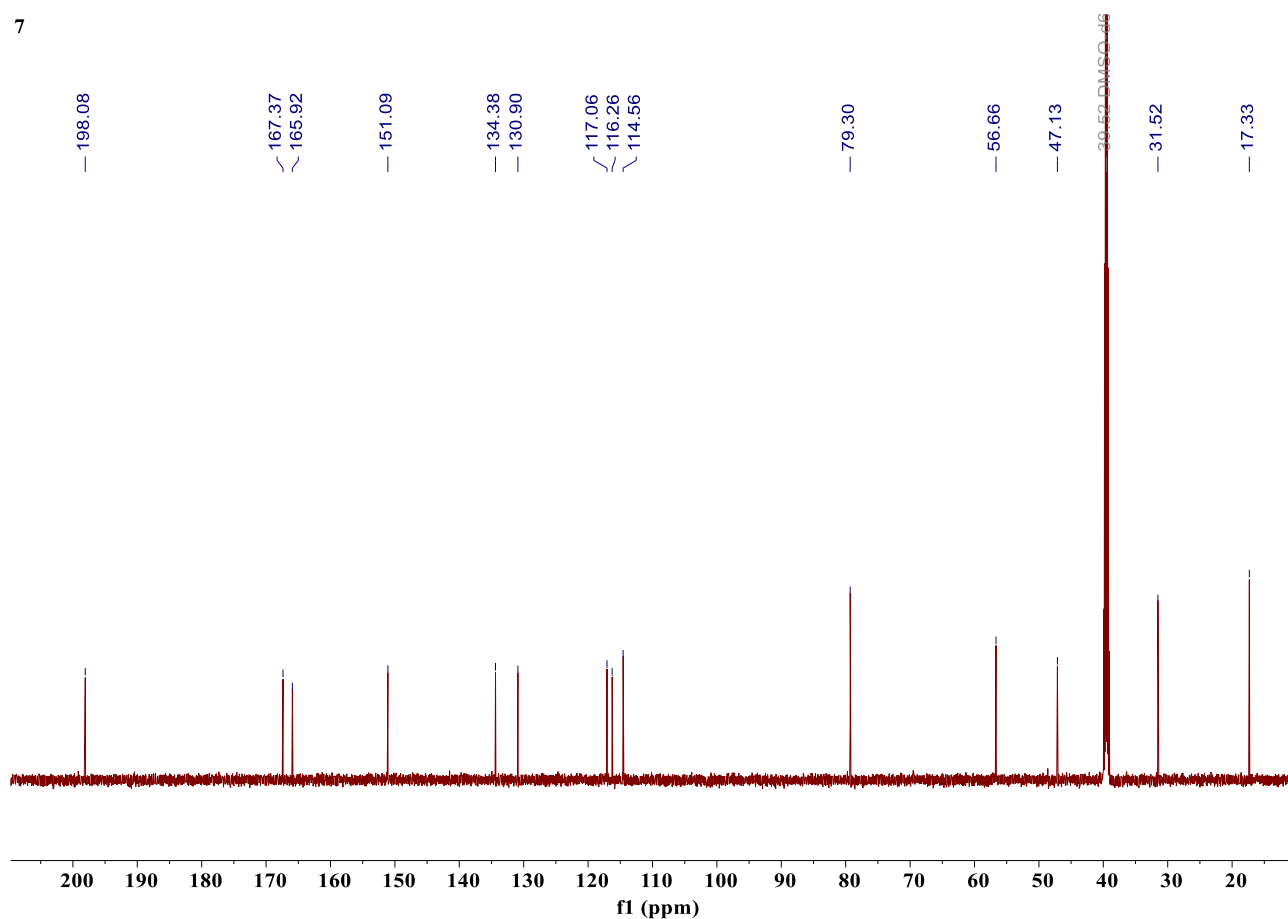
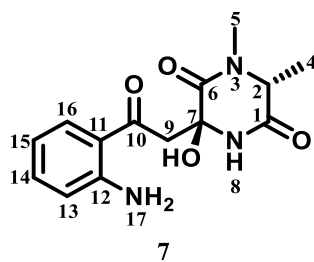


Figure S46: ^{13}C NMR of 7. DMSO-*d*₆, 150 MHz for ^{13}C NMR.

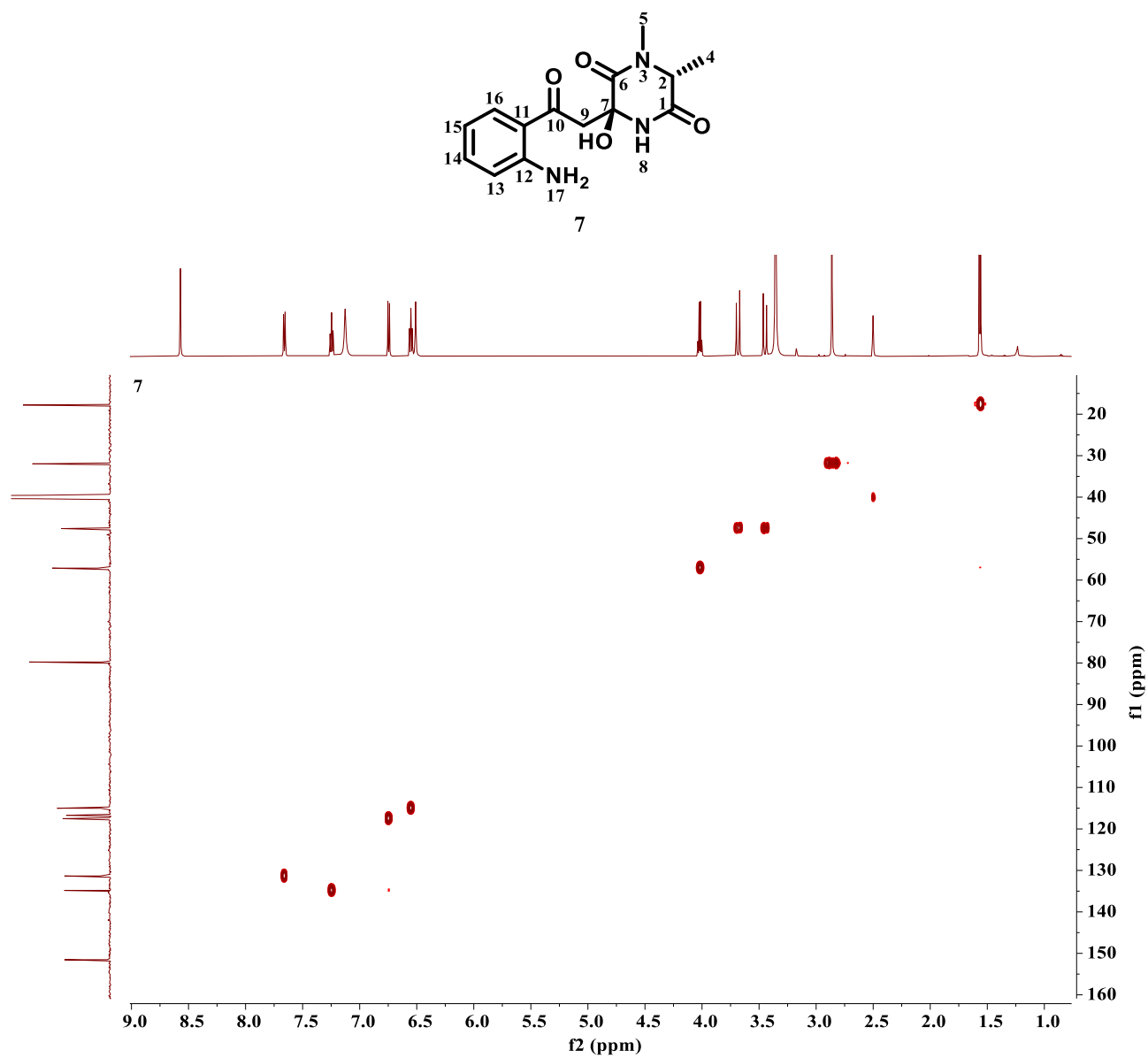


Figure S47: ^1H - ^{13}C HSQC of 7. DMSO- *d*6, 600 MHz for ^1H and 150 MHz for ^{13}C NMR.

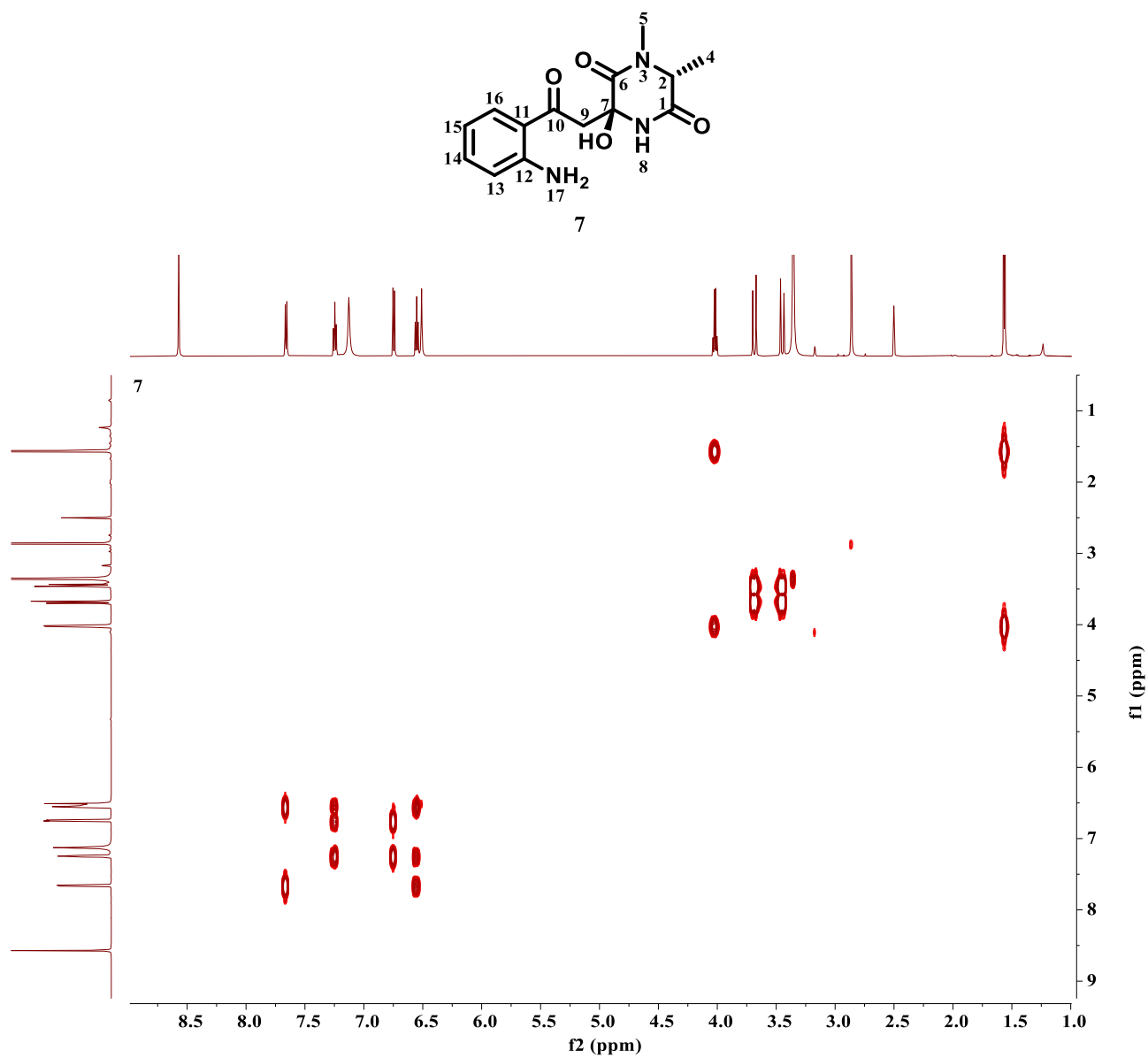


Figure S48: ^1H - ^1H COSY of 7. DMSO- d_6 , 600 MHz for ^1H NMR.

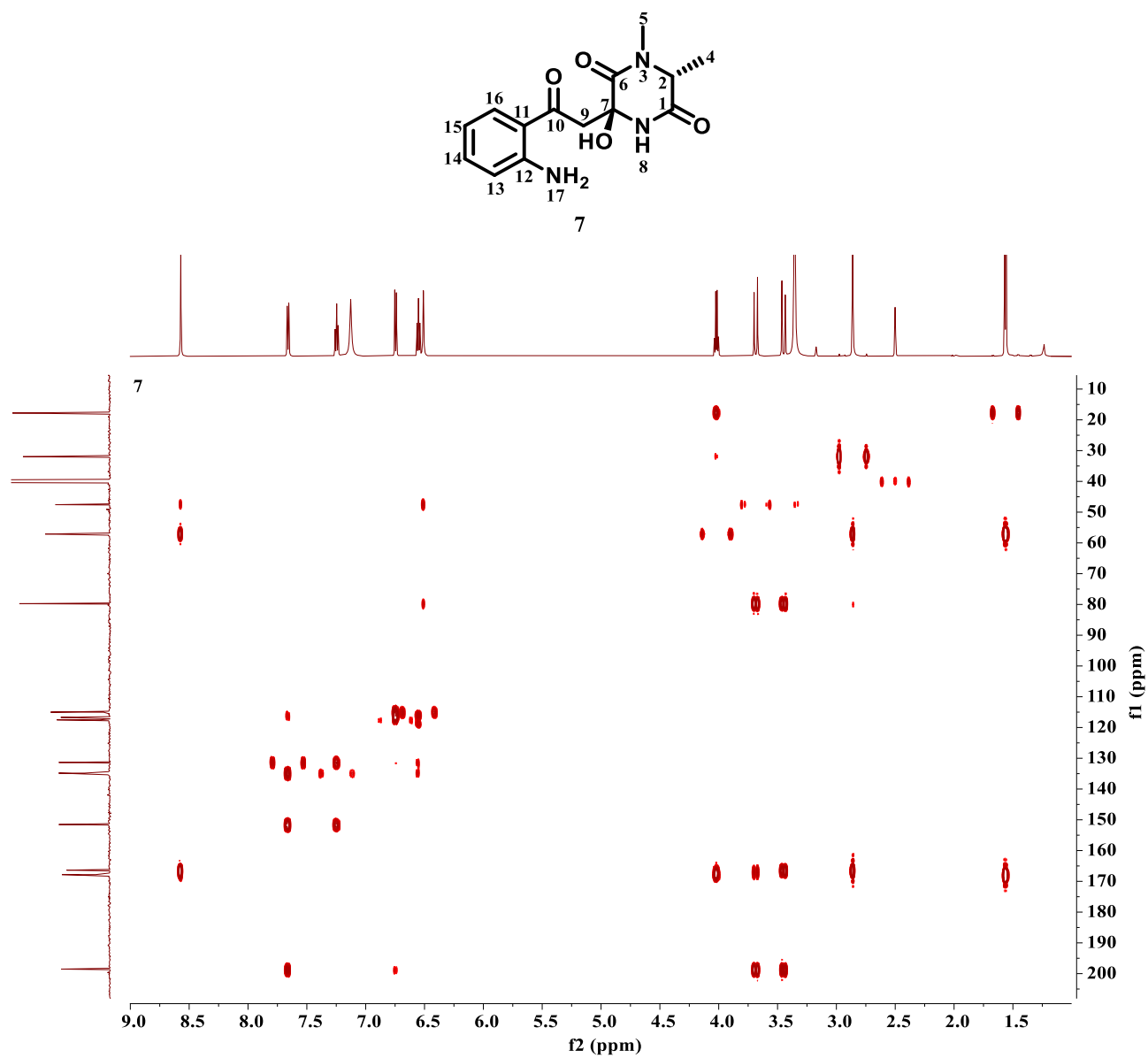
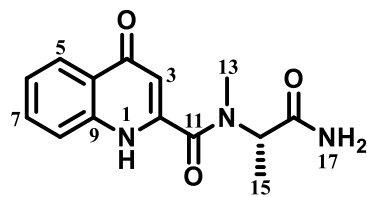


Figure S49: ^1H - ^{13}C HMBC of 7. DMSO- d_6 , 600 MHz for ^1H and 150 MHz for ^{13}C NMR.



8

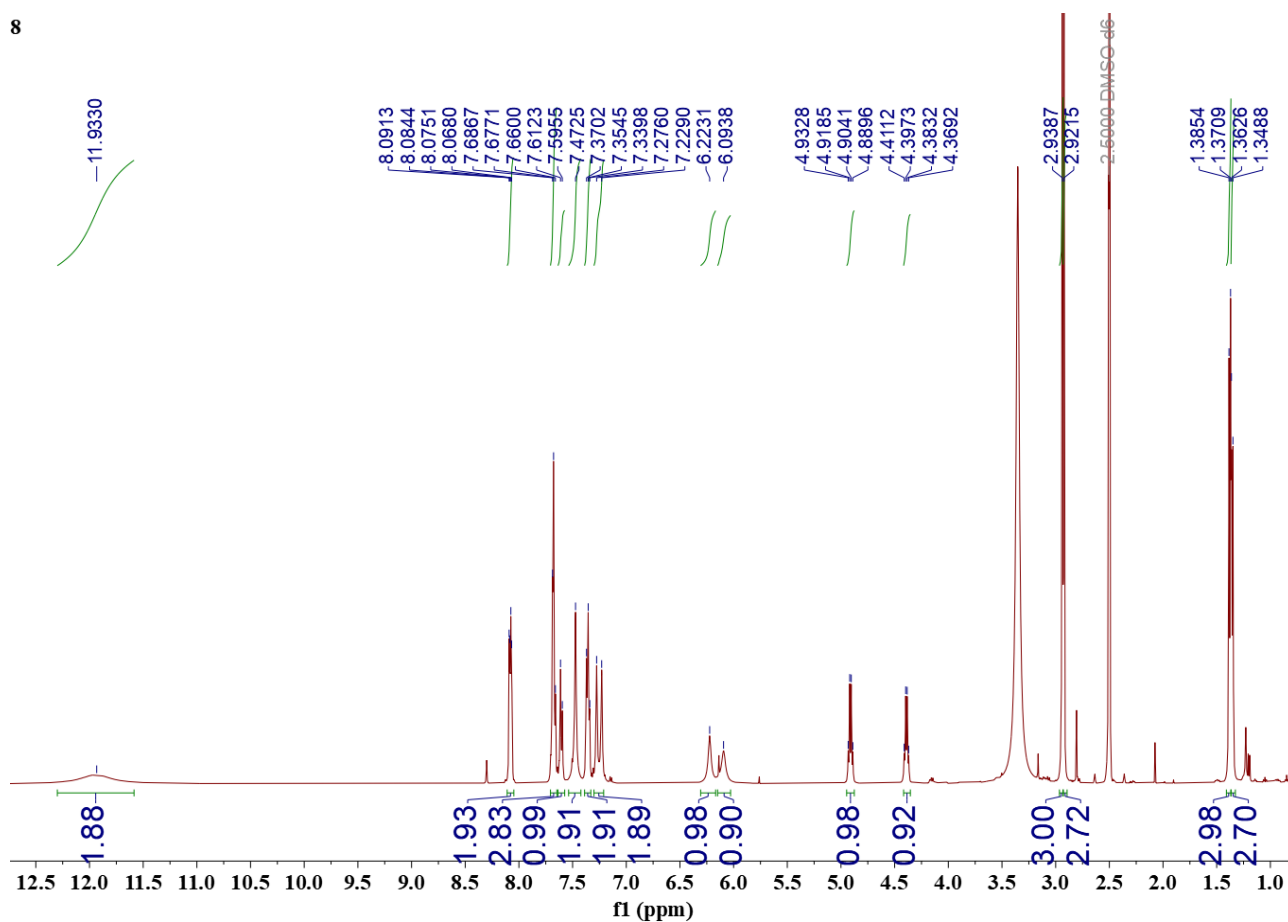
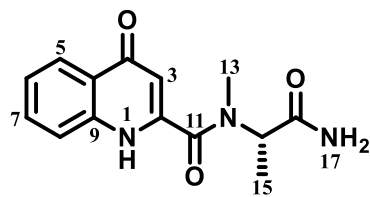


Figure S50: ^1H NMR of **8**. DMSO- d_6 , 500 MHz for ^1H NMR.



8

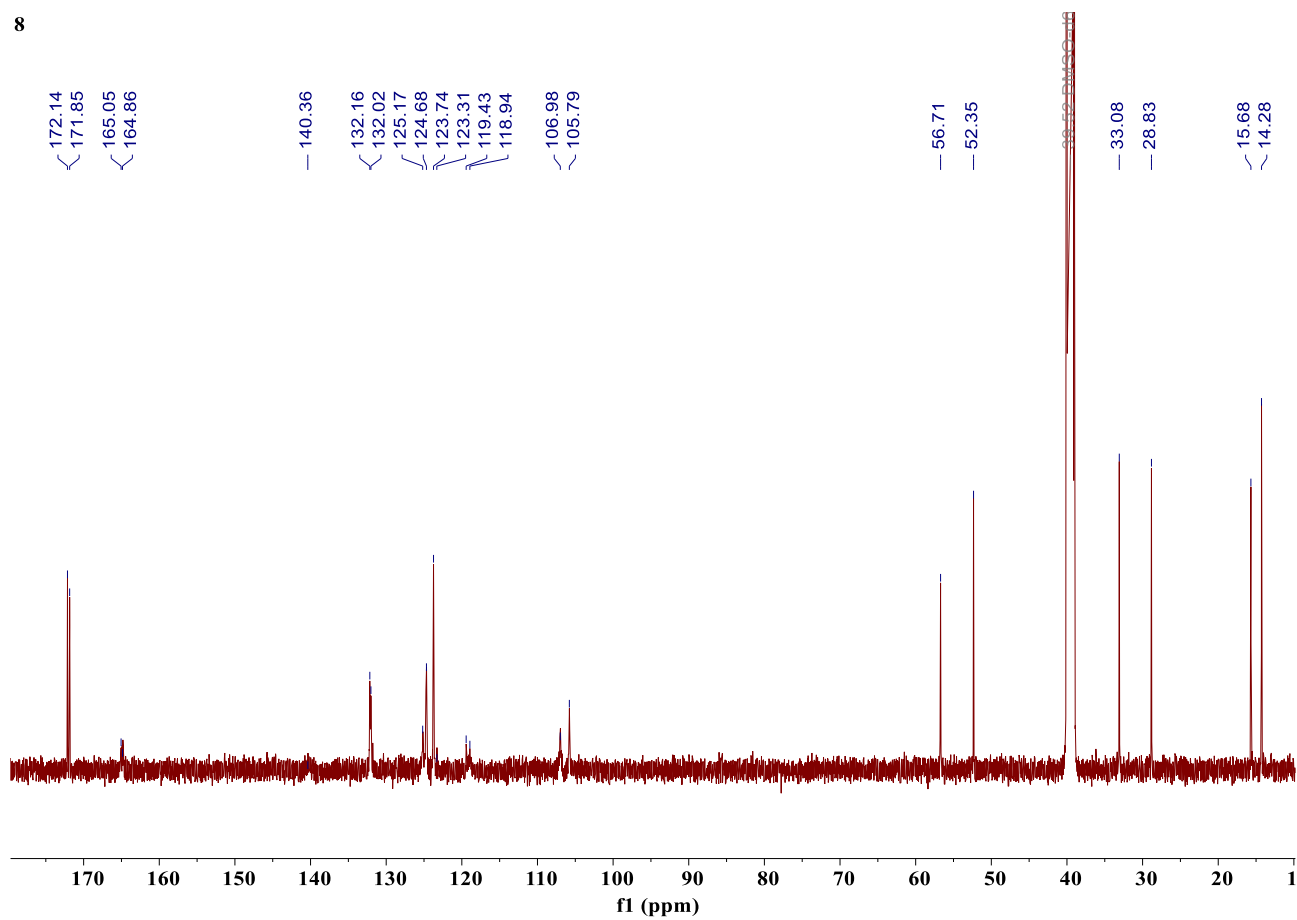


Figure S51: ^{13}C NMR of **8**. DMSO- d_6 , 125 MHz for ^{13}C NMR.

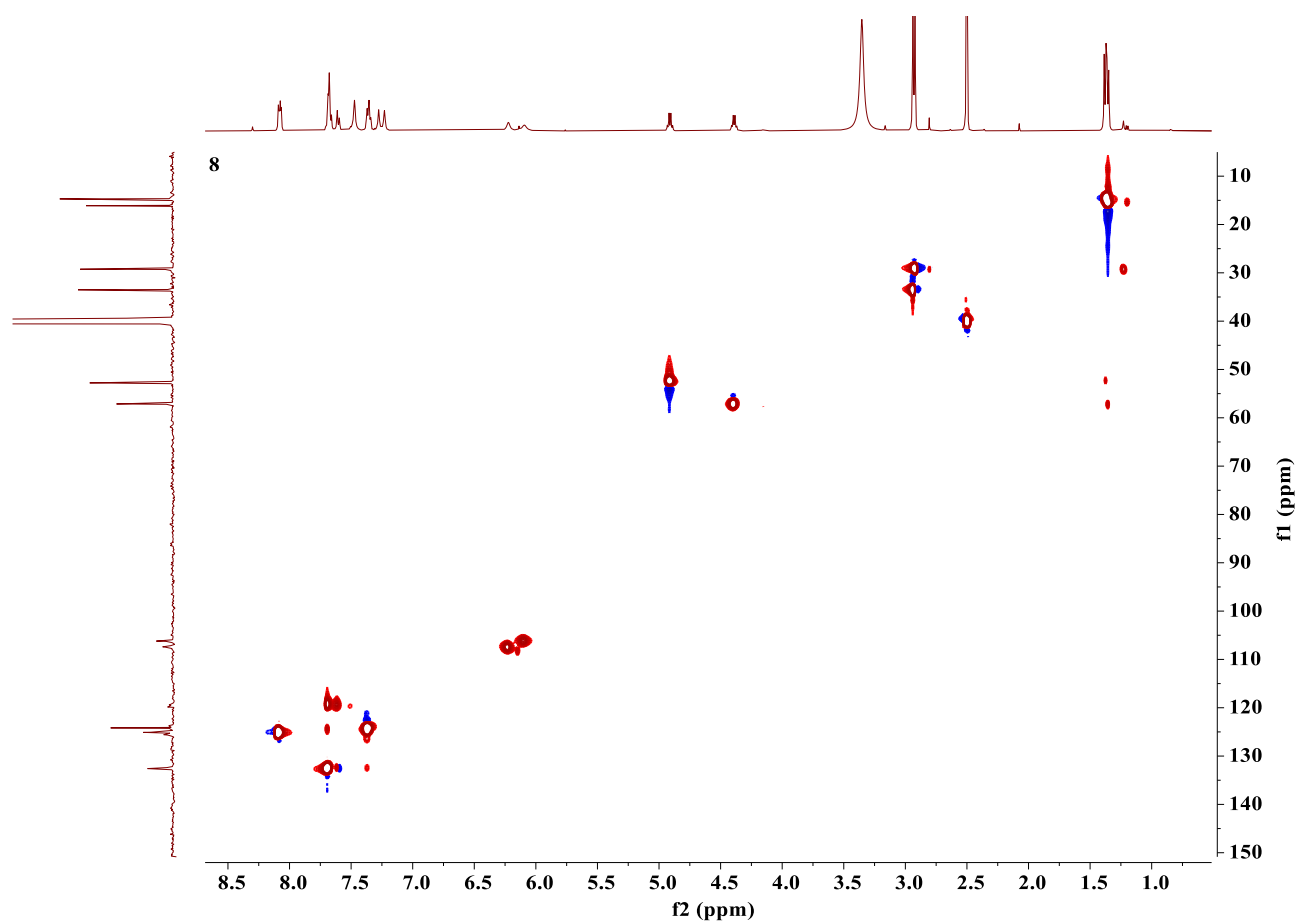
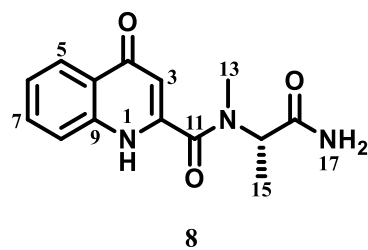


Figure S52: ^1H - ^{13}C HSQC of **8**. DMSO- *d*₆, 500 MHz for ^1H and 125 MHz for ^{13}C NMR.

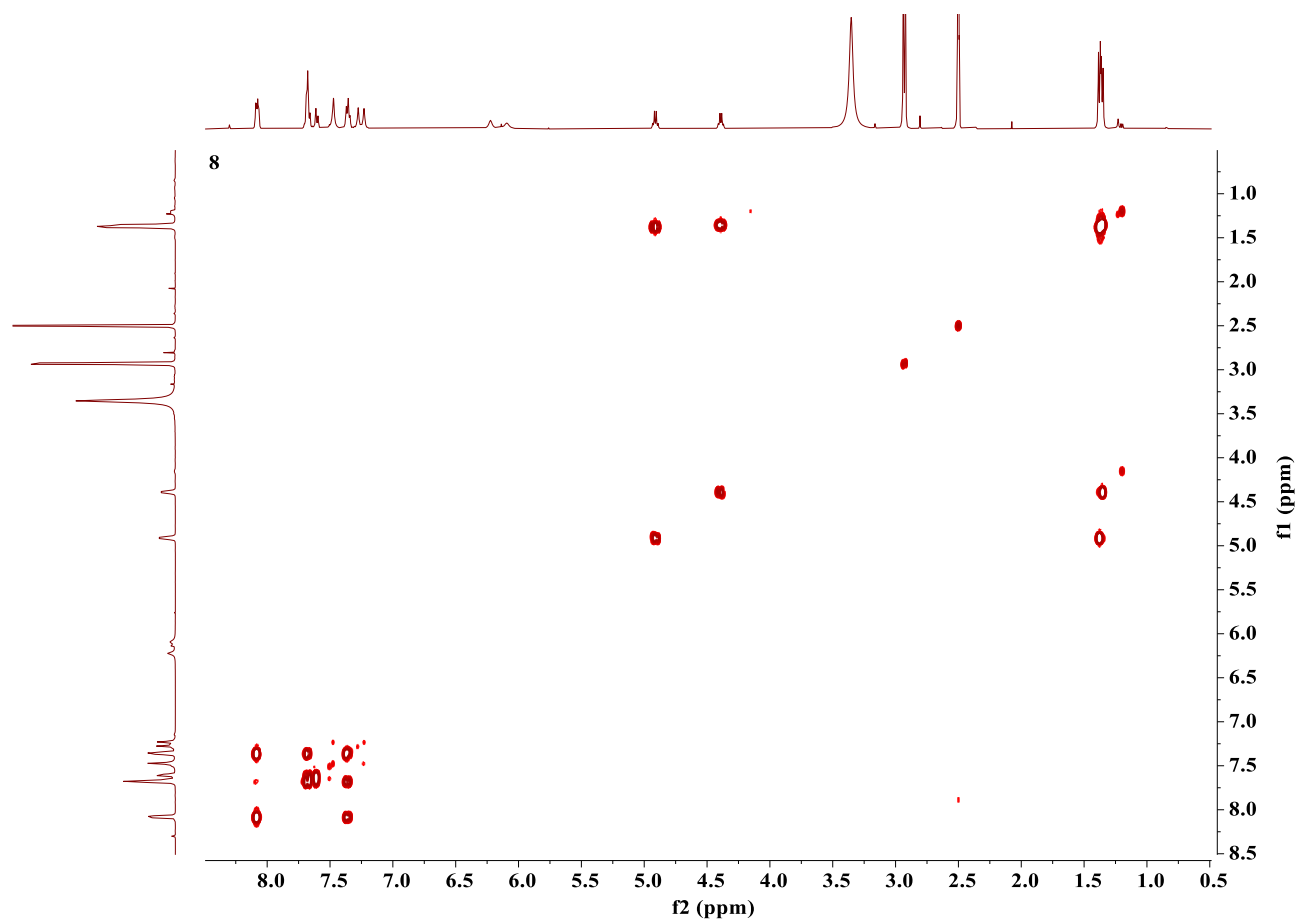
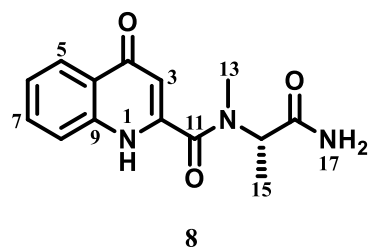


Figure S53: ^1H - ^1H COSY of **8**. DMSO- d_6 , 500 MHz for ^1H NMR.

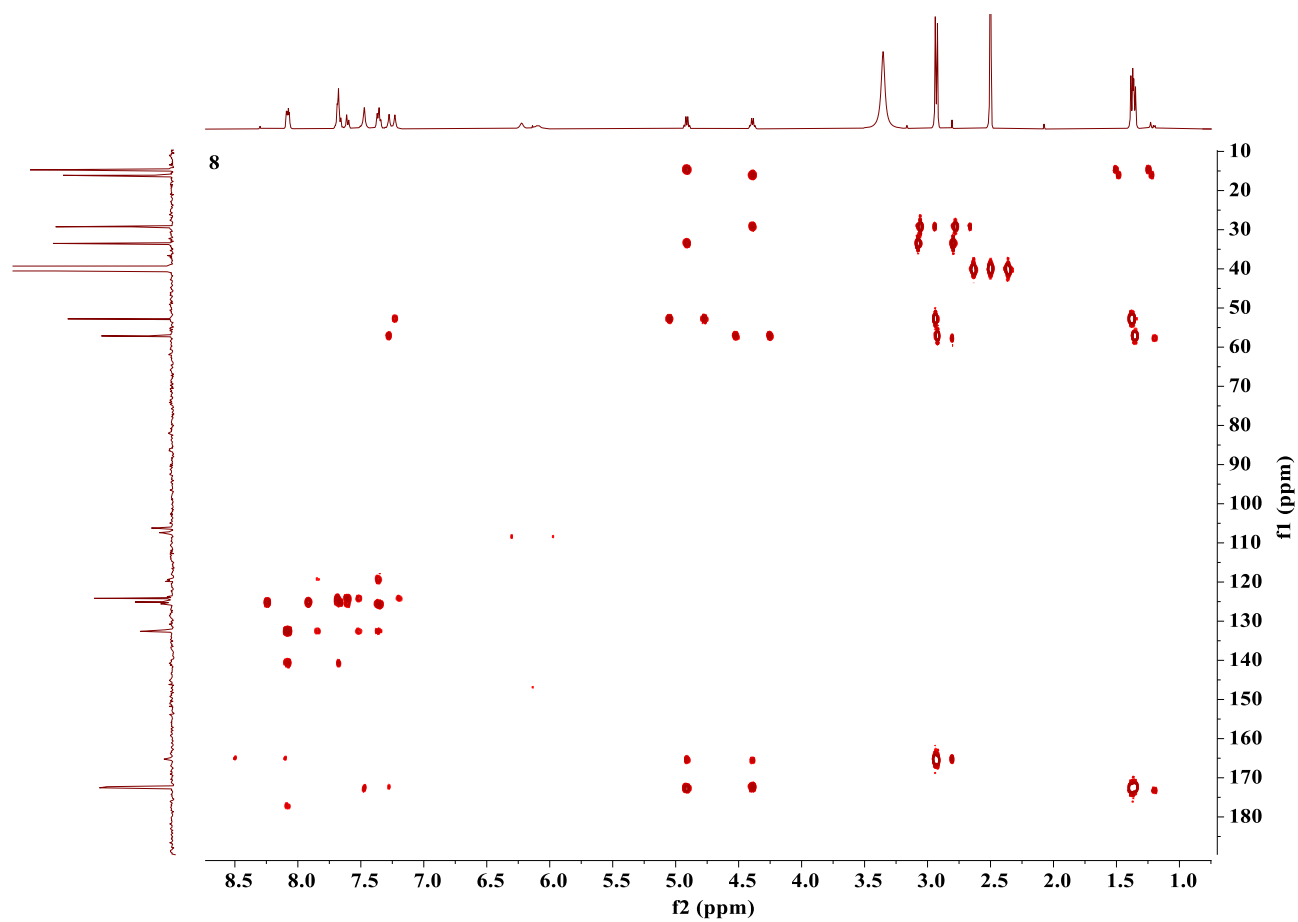
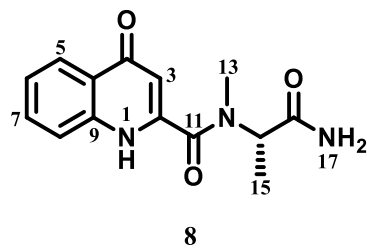


Figure S54: ^1H - ^{13}C HMBC of **8**. DMSO- d_6 , 500 MHz for ^1H and 125 MHz for ^{13}C NMR.

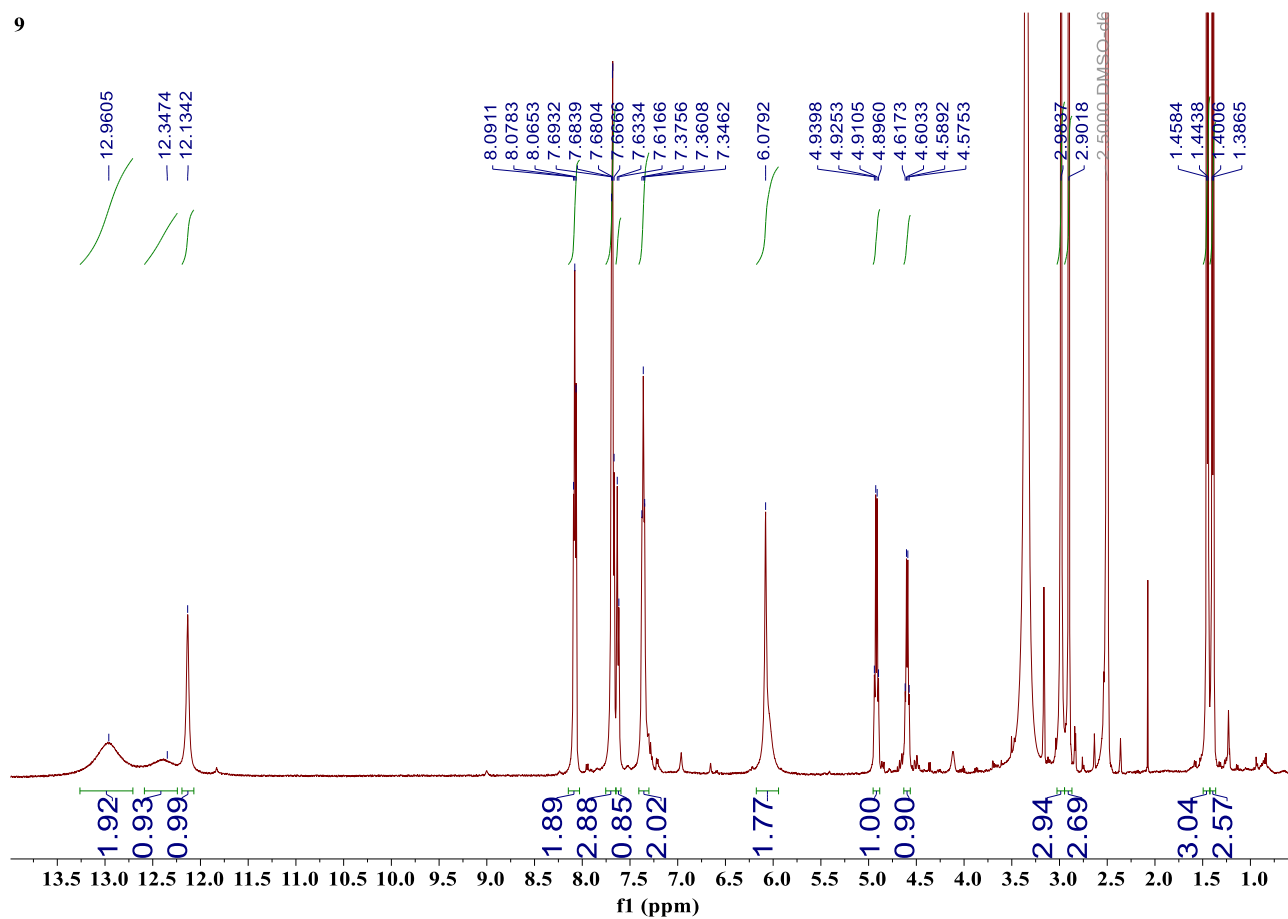
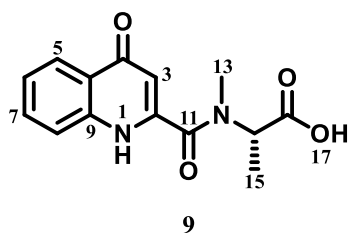


Figure S55: ¹H NMR of **9**. DMSO-*d*₆, 500 MHz for ¹H NMR.

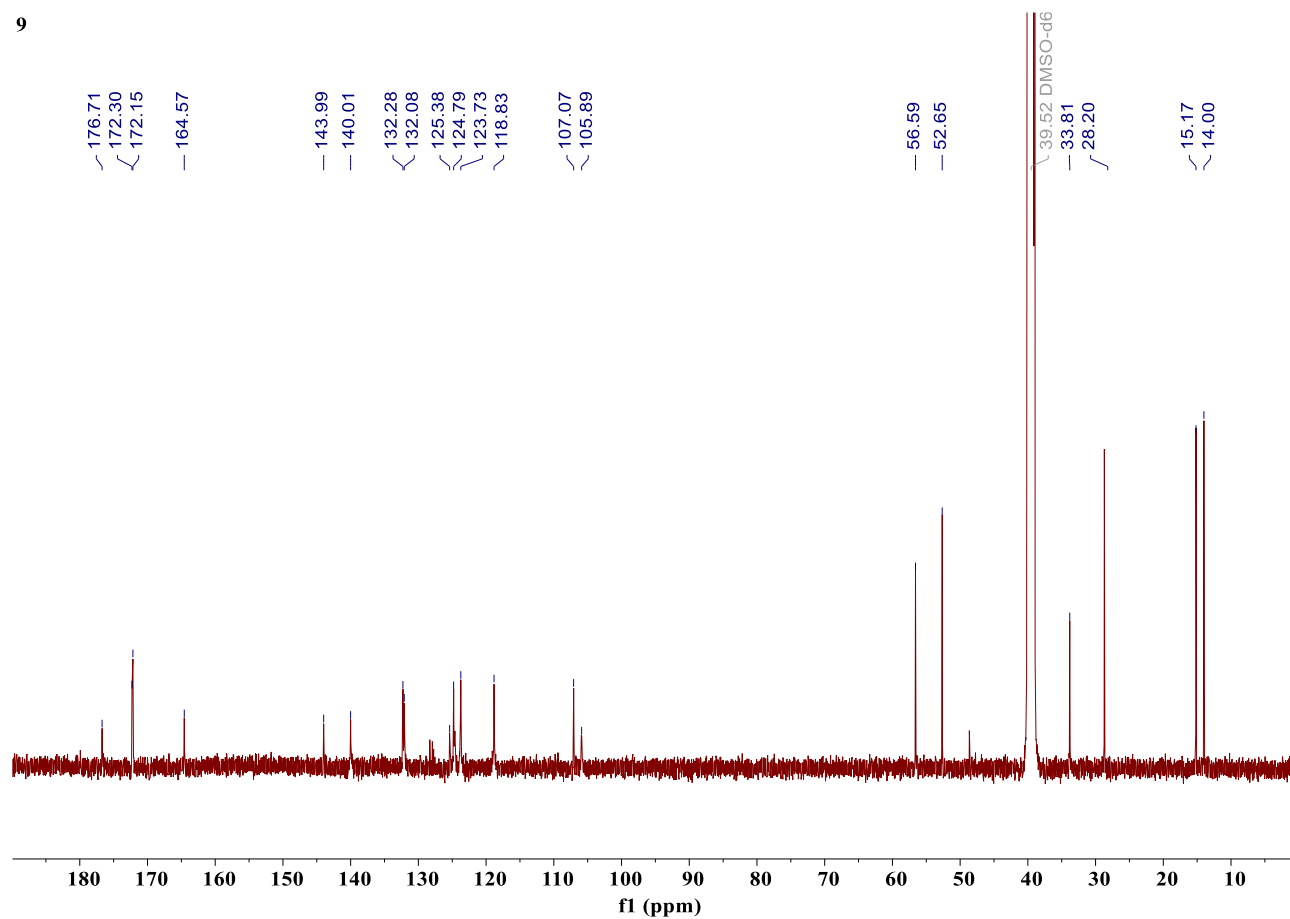
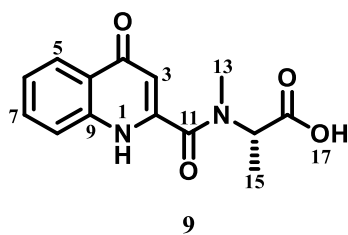


Figure S56: ^{13}C NMR of **9**. DMSO- d_6 , 125 MHz for ^{13}C NMR.

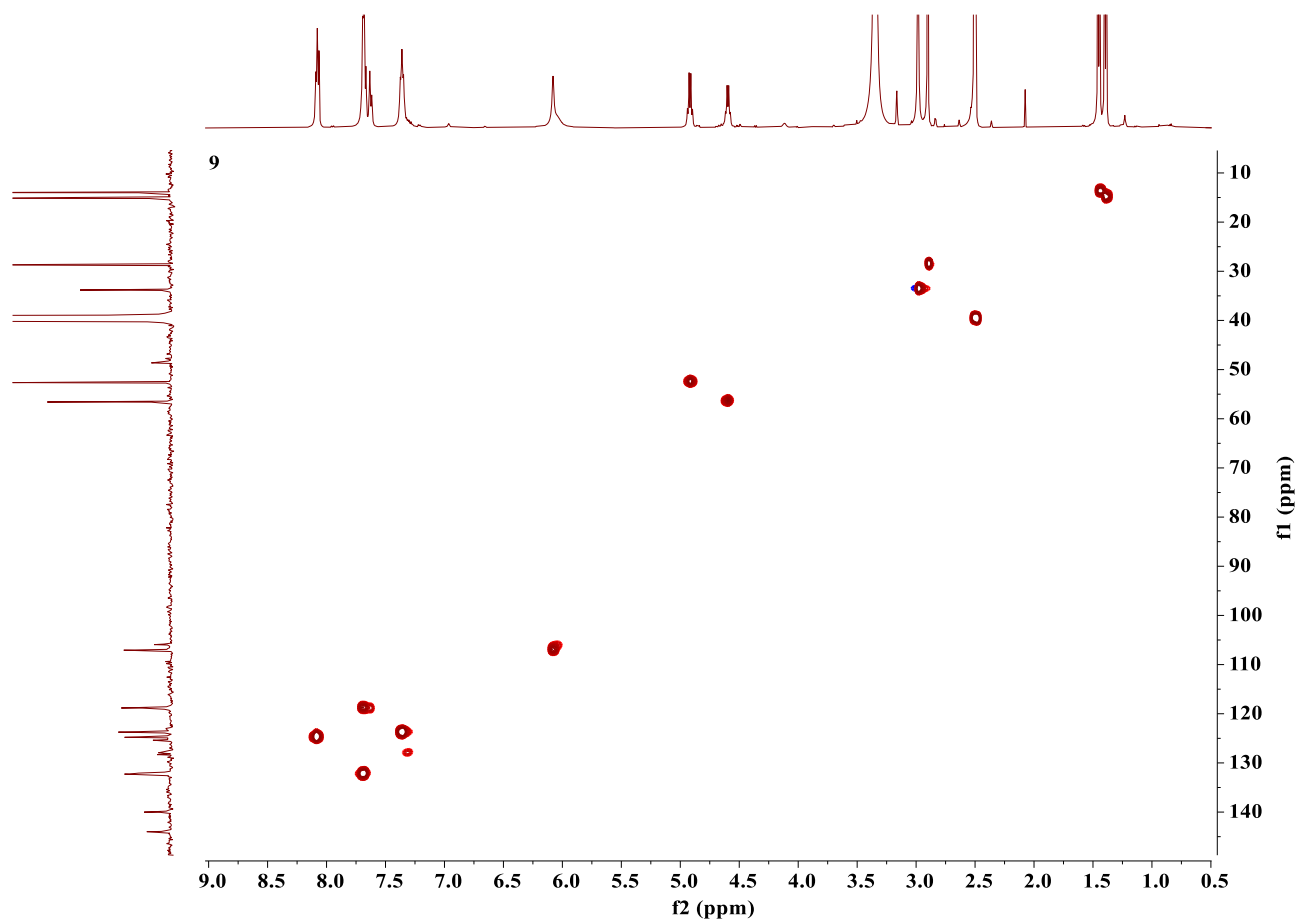
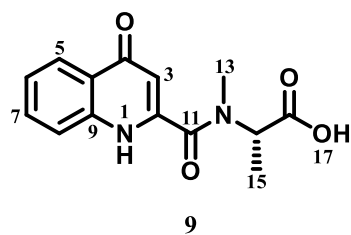


Figure S57: ^1H - ^{13}C HSQC of **9**. DMSO- d_6 , 500 MHz for ^1H and 125 MHz for ^{13}C NMR.

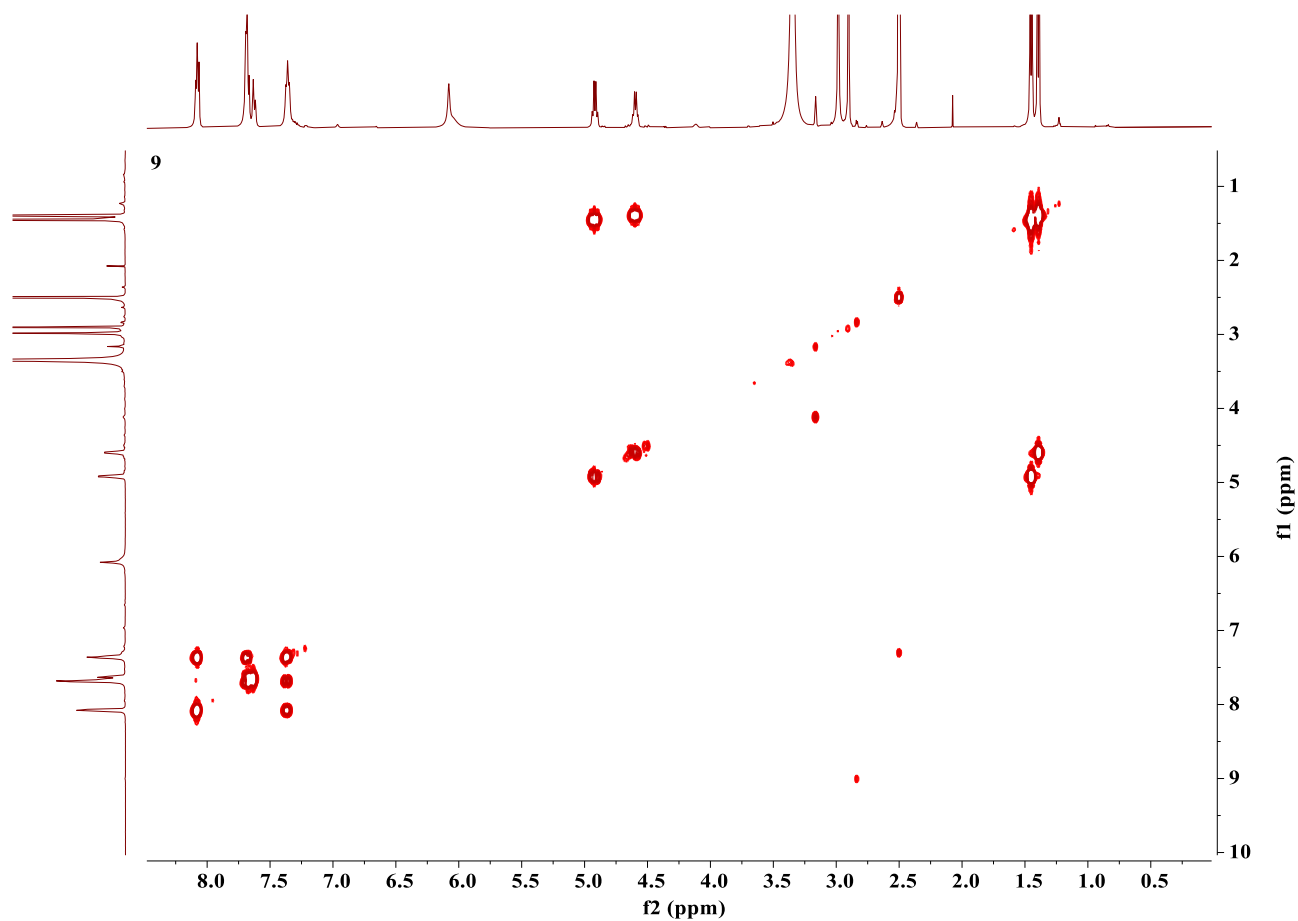
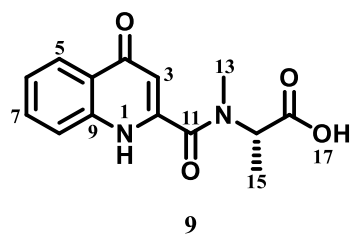


Figure S58: ^1H - ^1H COSY of **9**. DMSO- d_6 , 500 MHz for ^1H NMR.

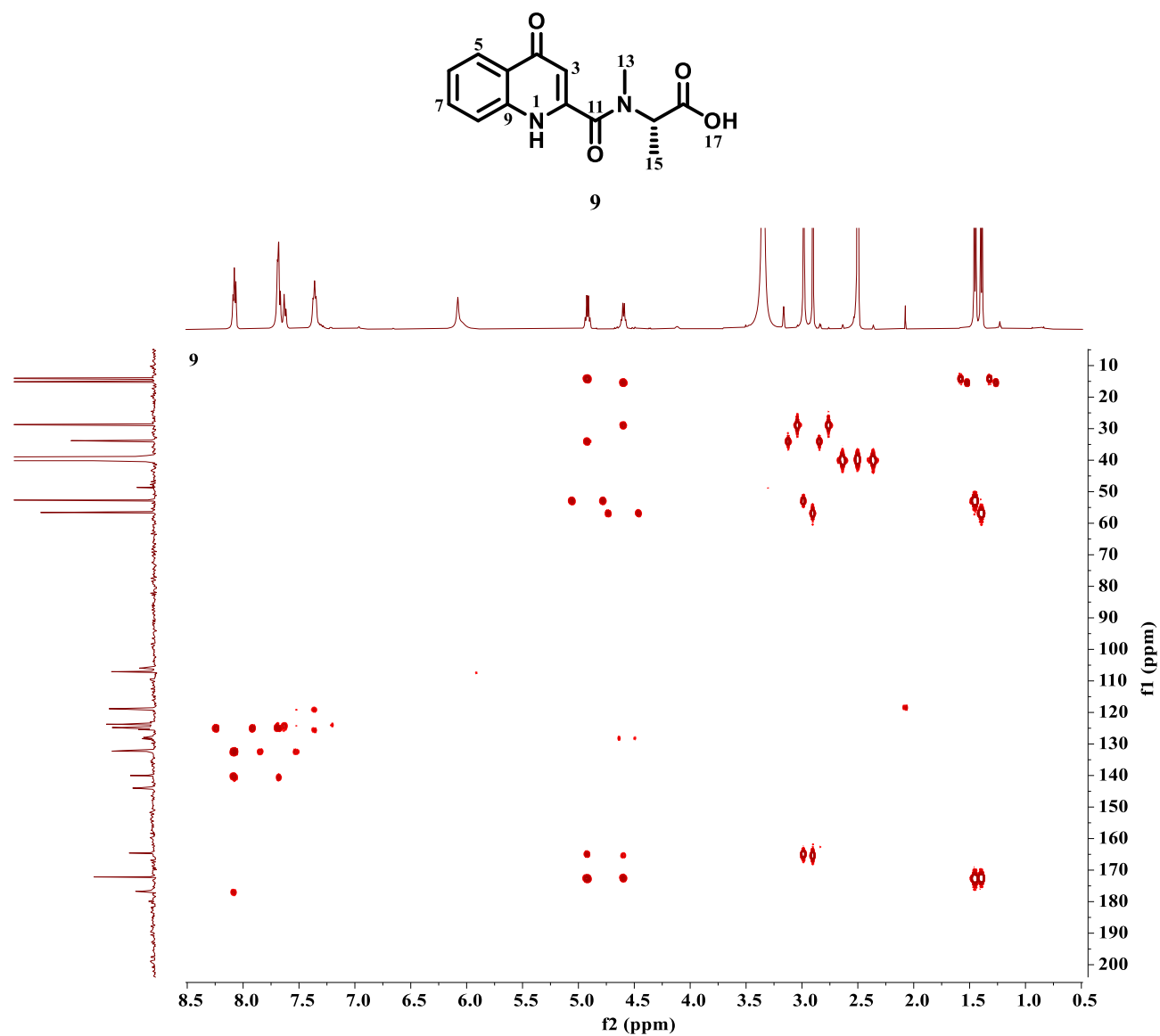


Figure S59: ^1H - ^{13}C HMBC of **9**. DMSO- d_6 , 500 MHz for ^1H and 125 MHz for ^{13}C NMR.

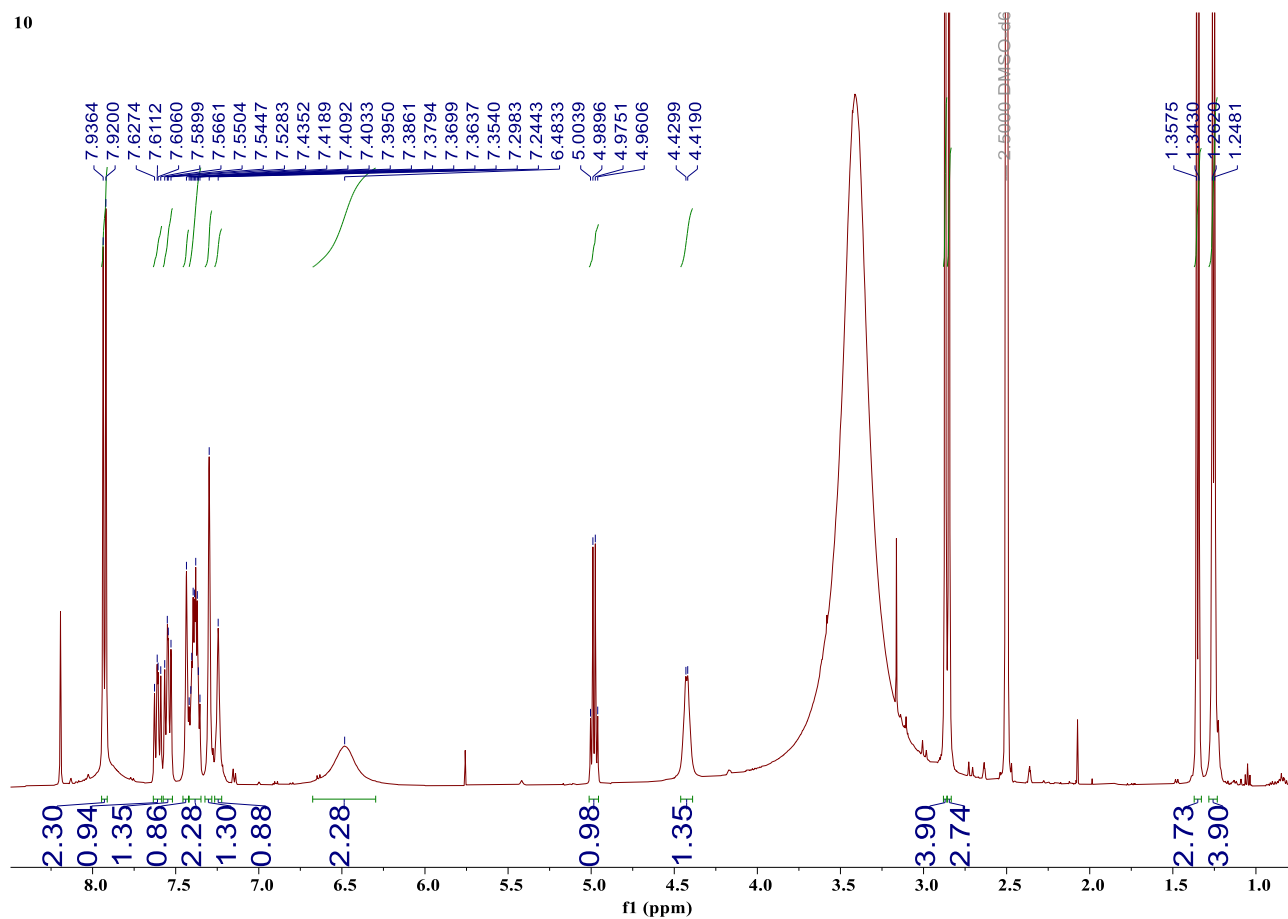
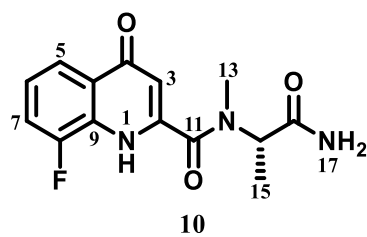
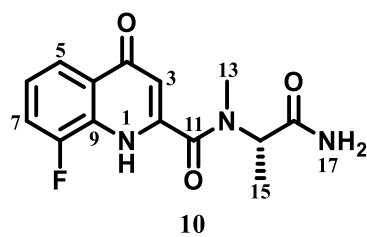


Figure S60: ^1H NMR of **10**. DMSO- d_6 , 500 MHz for ^1H NMR.



10

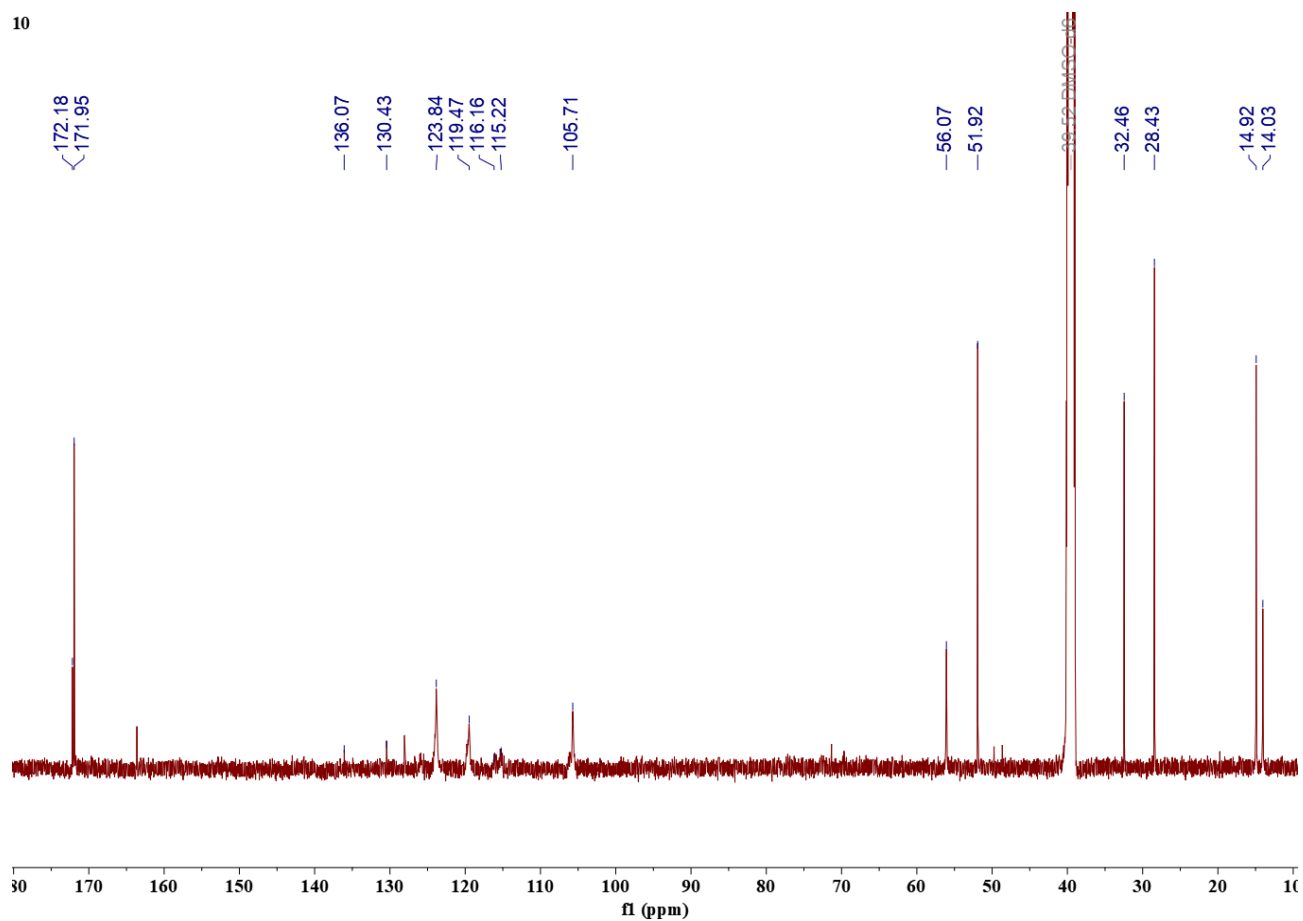


Figure S61: ^{13}C NMR of **10**. DMSO- d_6 , 125 MHz for ^{13}C NMR.

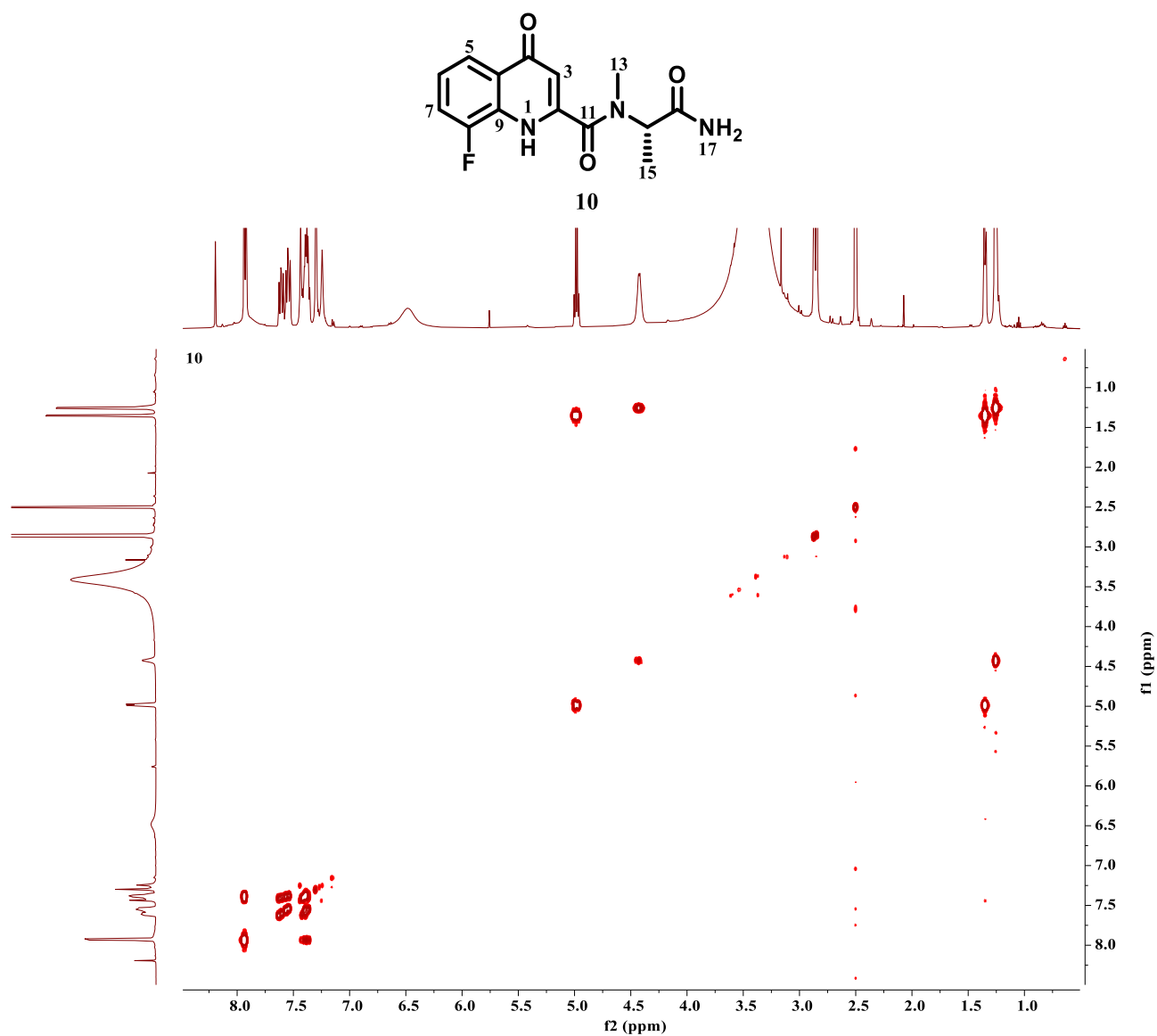


Figure S62: ^1H - ^1H COSY of **10**. DMSO- d_6 , 500 MHz for ^1H NMR.

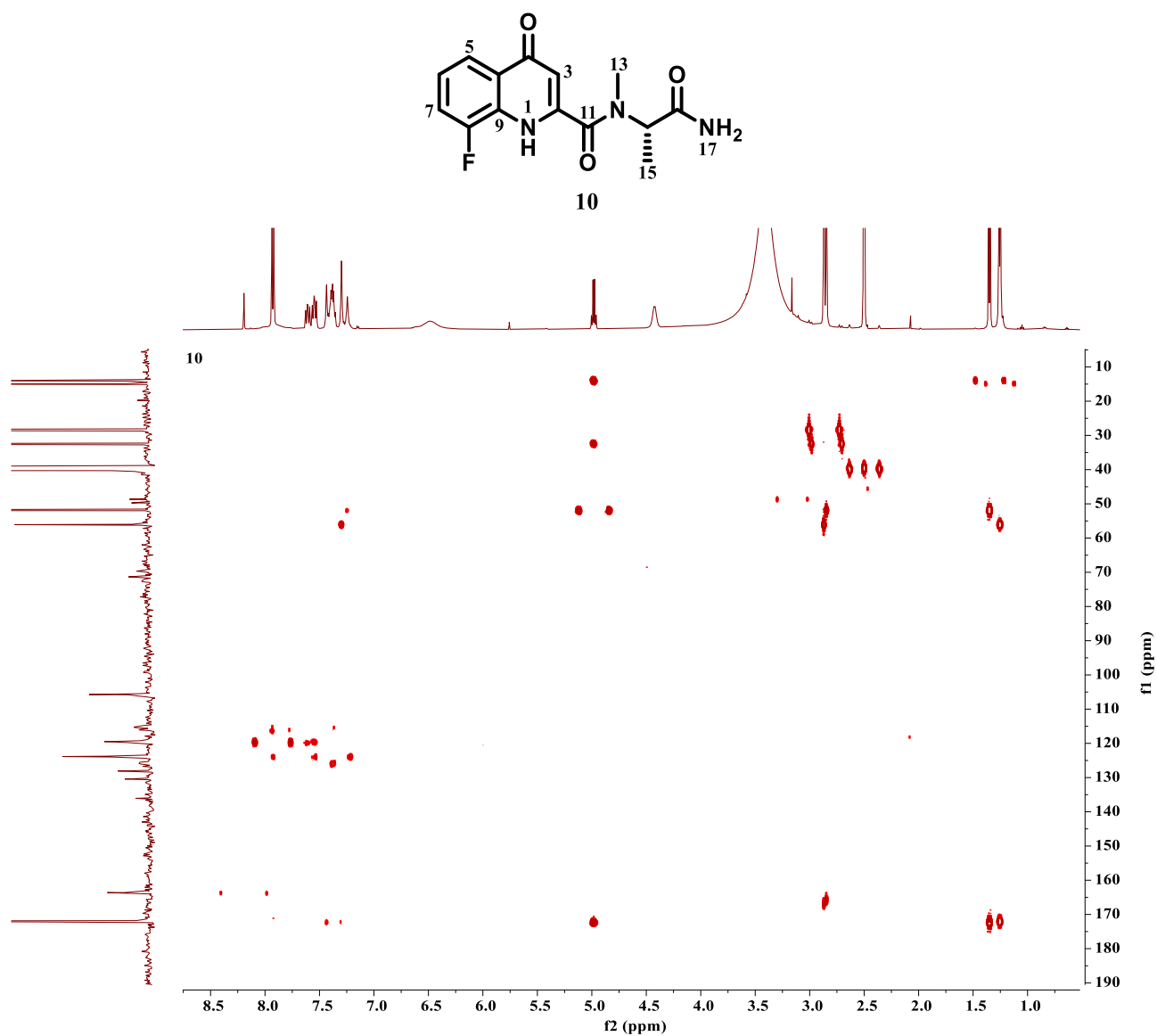


Figure S63: ^1H - ^{13}C HMBC of **10**. DMSO- d_6 , 500 MHz for ^1H and 125 MHz for ^{13}C NMR.

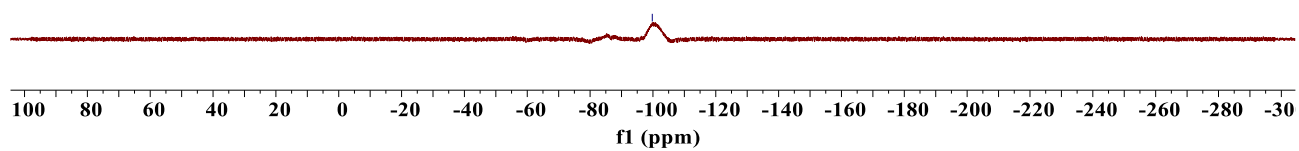
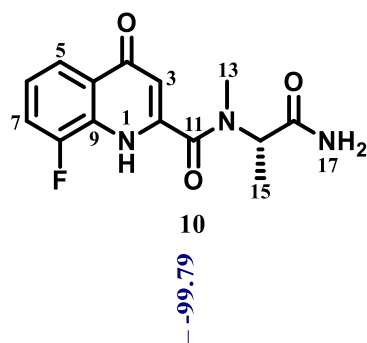


Figure S64: ^{19}F NMR of **10**. DMSO- d_6 , 500 MHz for ^{19}F NMR.

Due to scalar coupling with the quadrupolar ^{14}N nucleus, the corresponding ^{19}F signal is broadened.

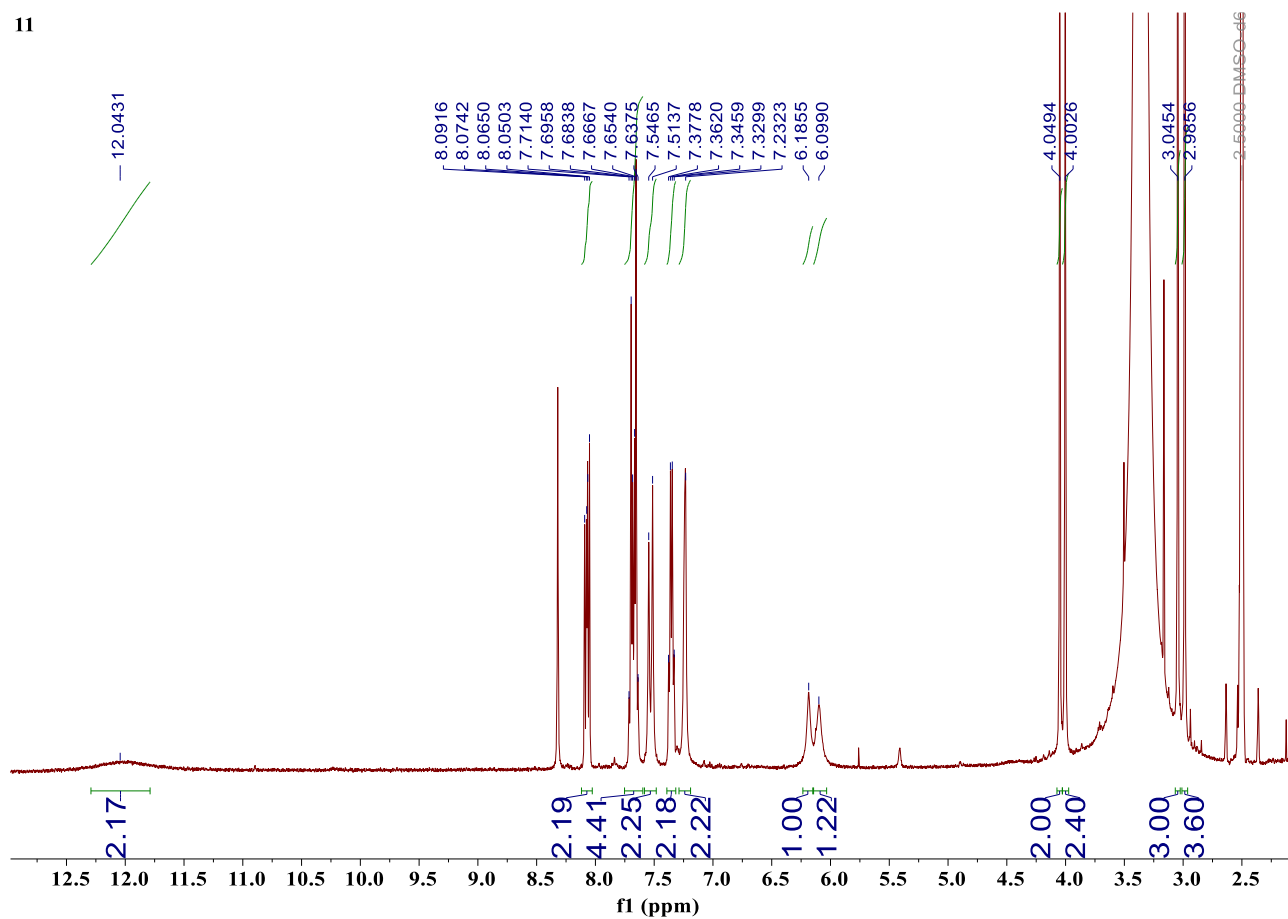
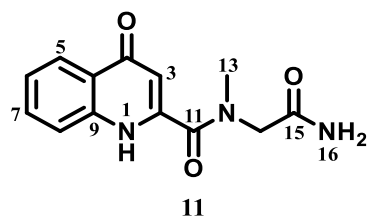


Figure S65: ¹H NMR of **11**. DMSO- *d*₆, 500 MHz for ¹H NMR.

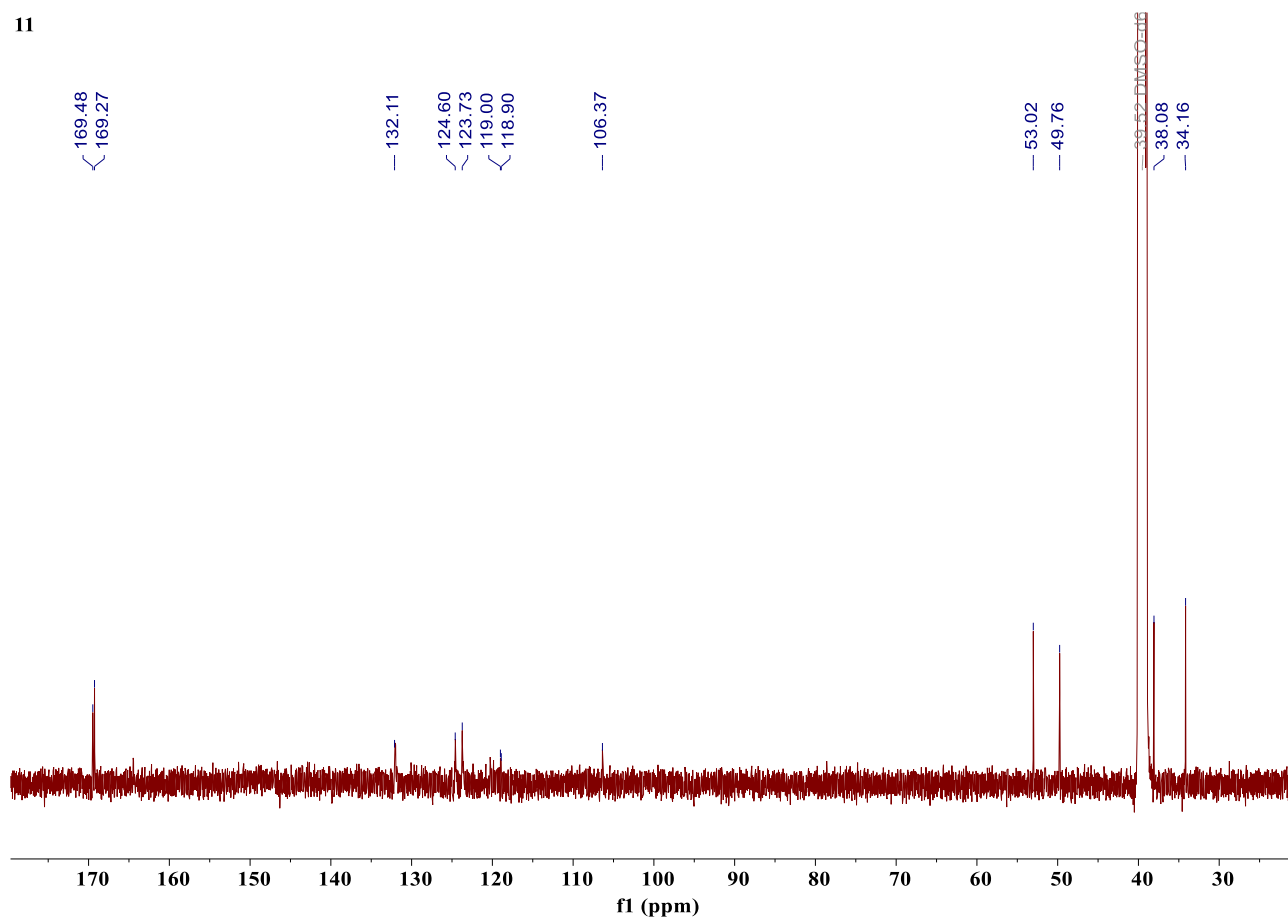
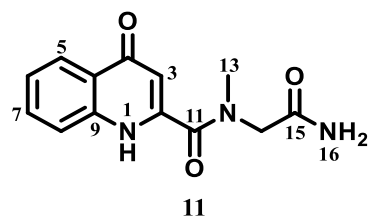


Figure S66: ^{13}C NMR of **11**. DMSO- d_6 , 125 MHz for ^{13}C NMR.

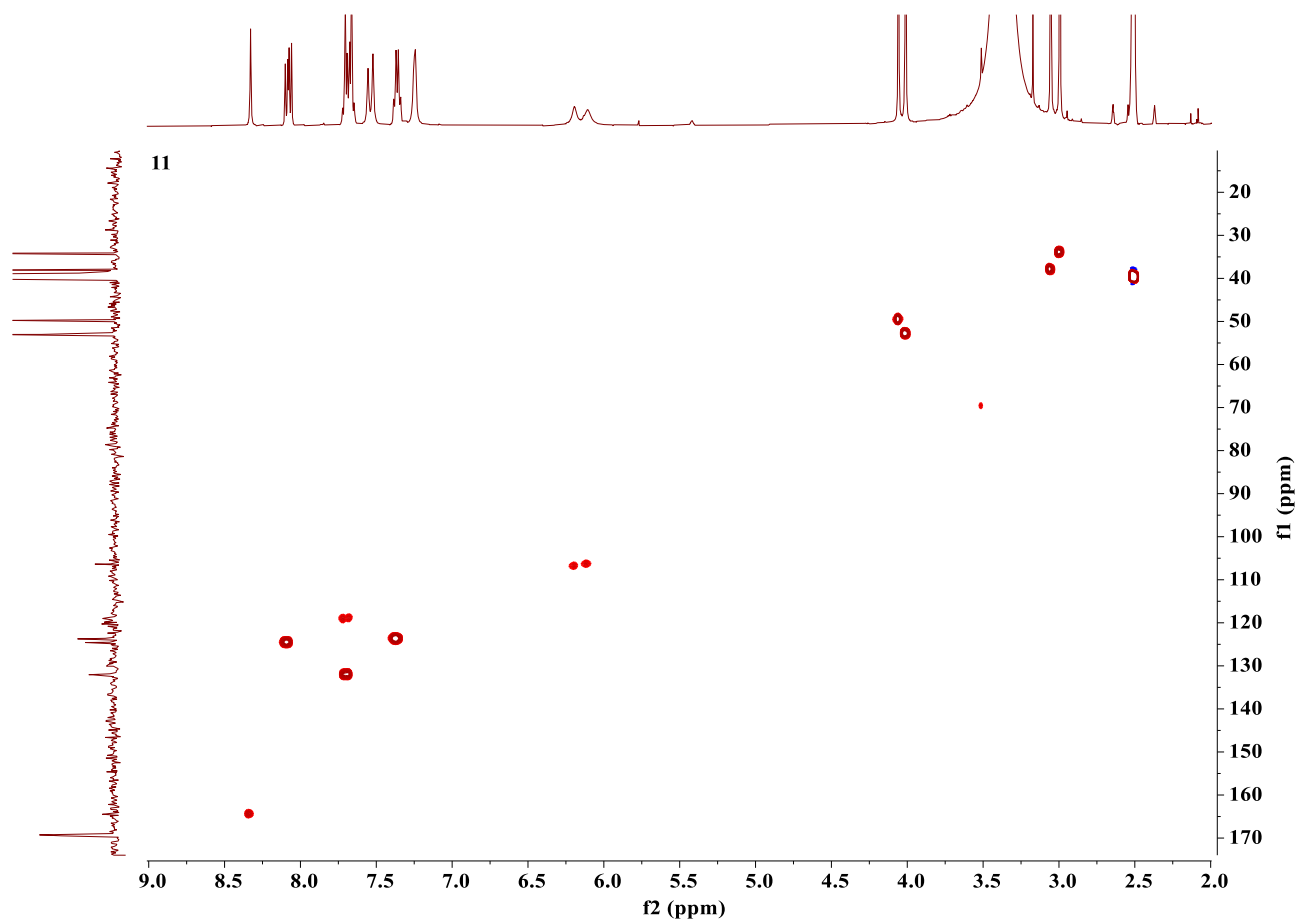
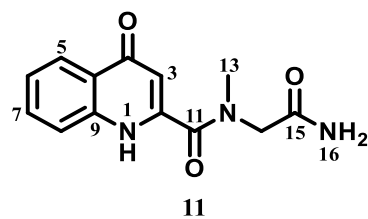


Figure S67: ^1H - ^{13}C HSQC of **11**. DMSO- d_6 , 500 MHz for ^1H and 125 MHz for ^{13}C NMR.

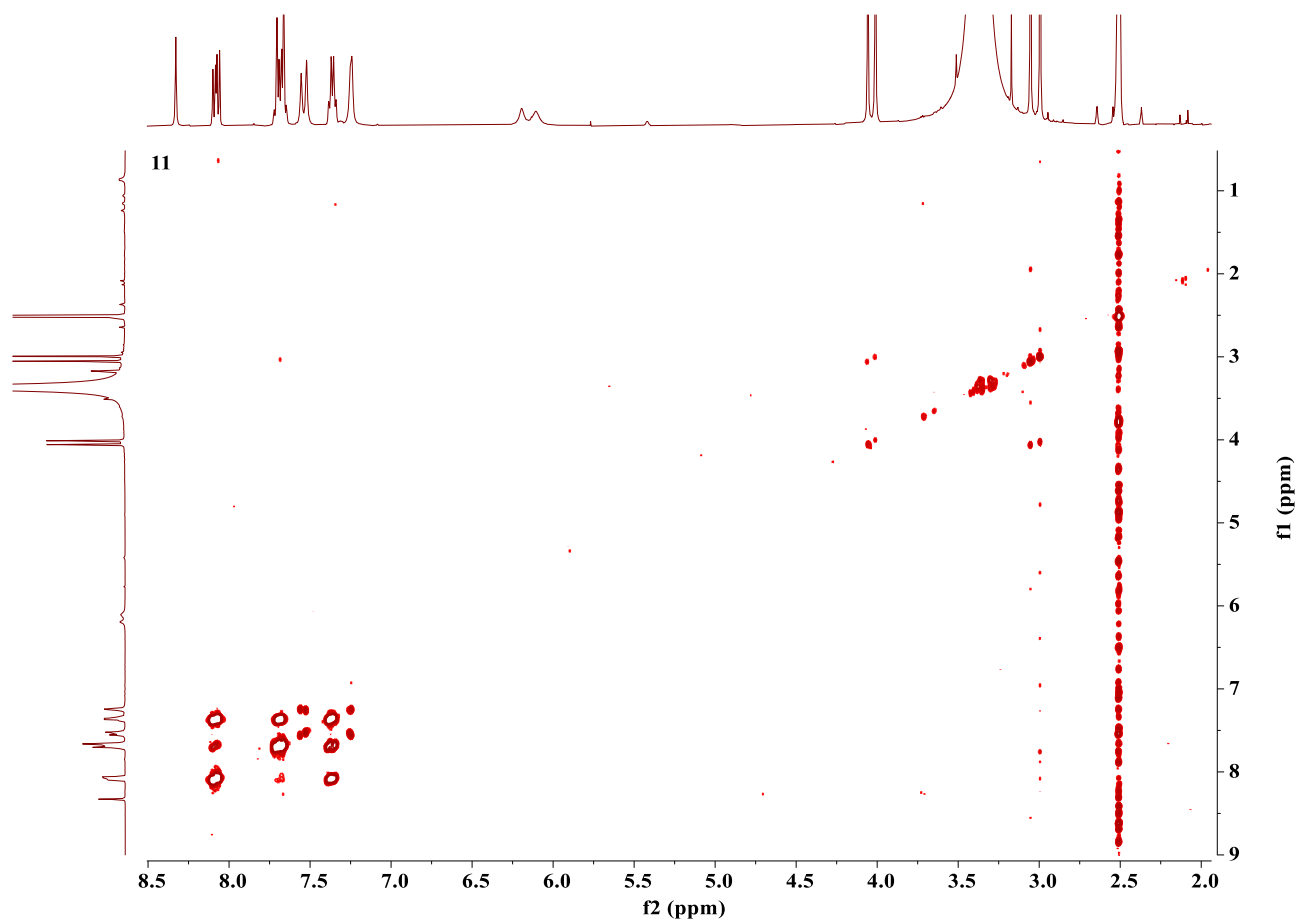
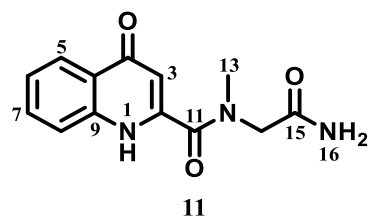


Figure S68: ^1H - ^1H COSY of **11**. DMSO- d_6 , 500 MHz for ^1H NMR.

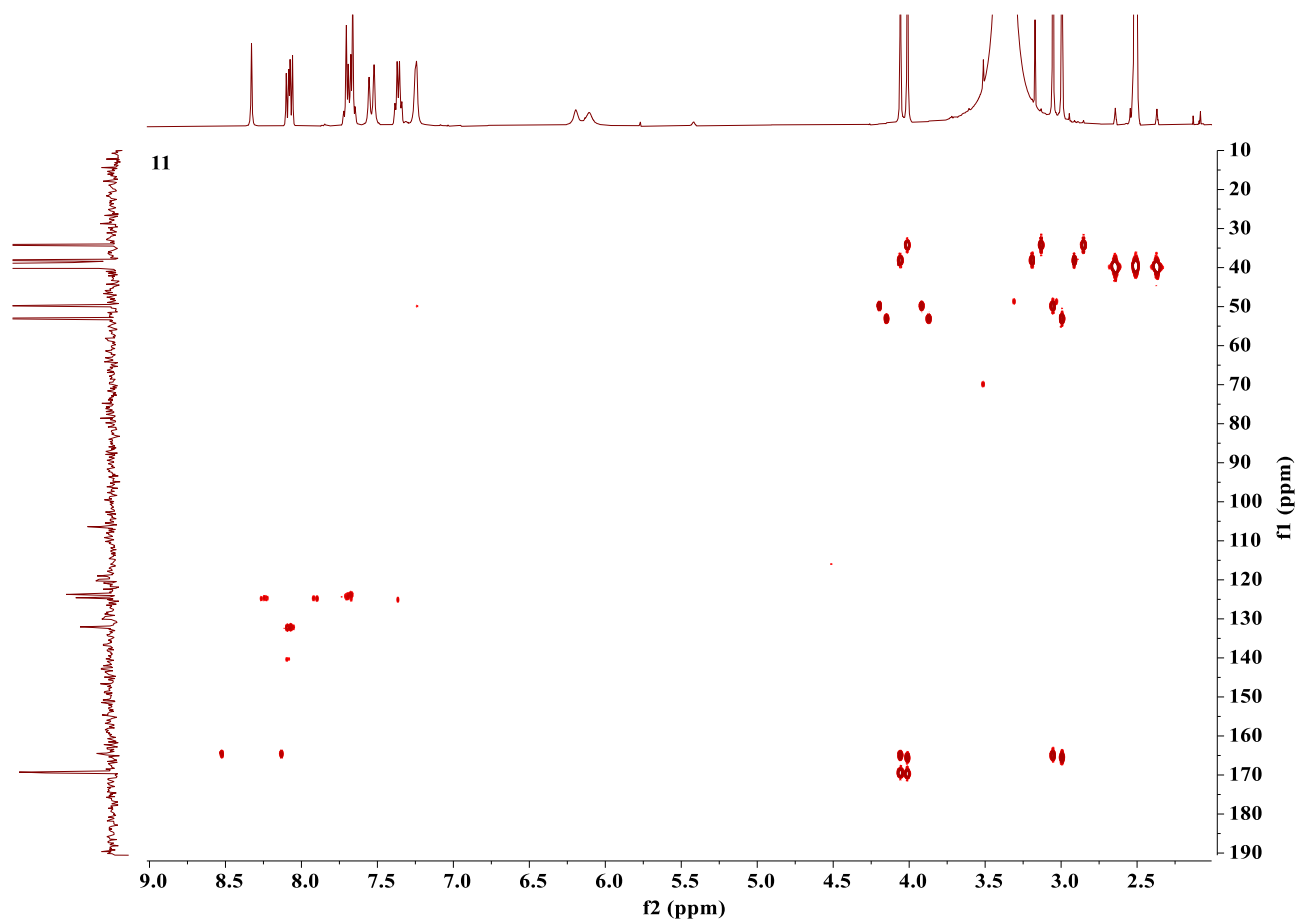
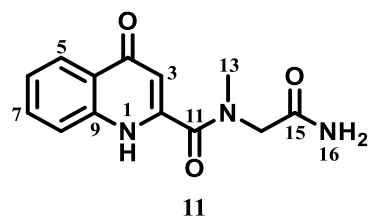


Figure S69: ^1H - ^{13}C HMBC of **11**. DMSO- d_6 , 500 MHz for ^1H and 125 MHz for ^{13}C NMR.



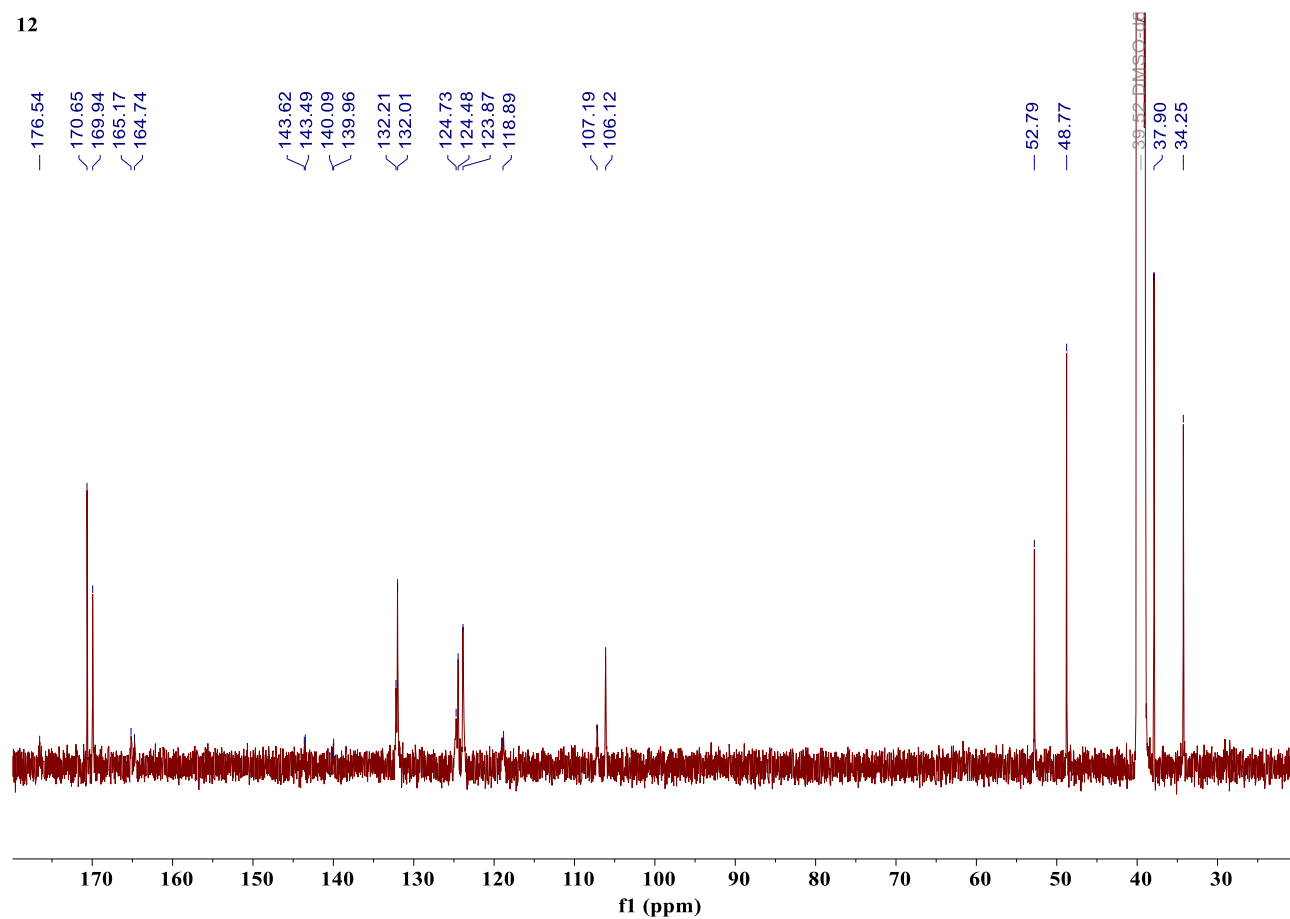
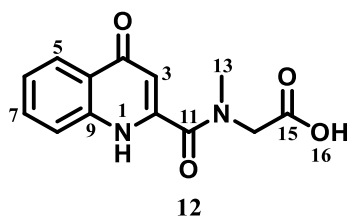


Figure S71: ^{13}C NMR of **12**. DMSO- d_6 , 125 MHz for ^{13}C NMR.

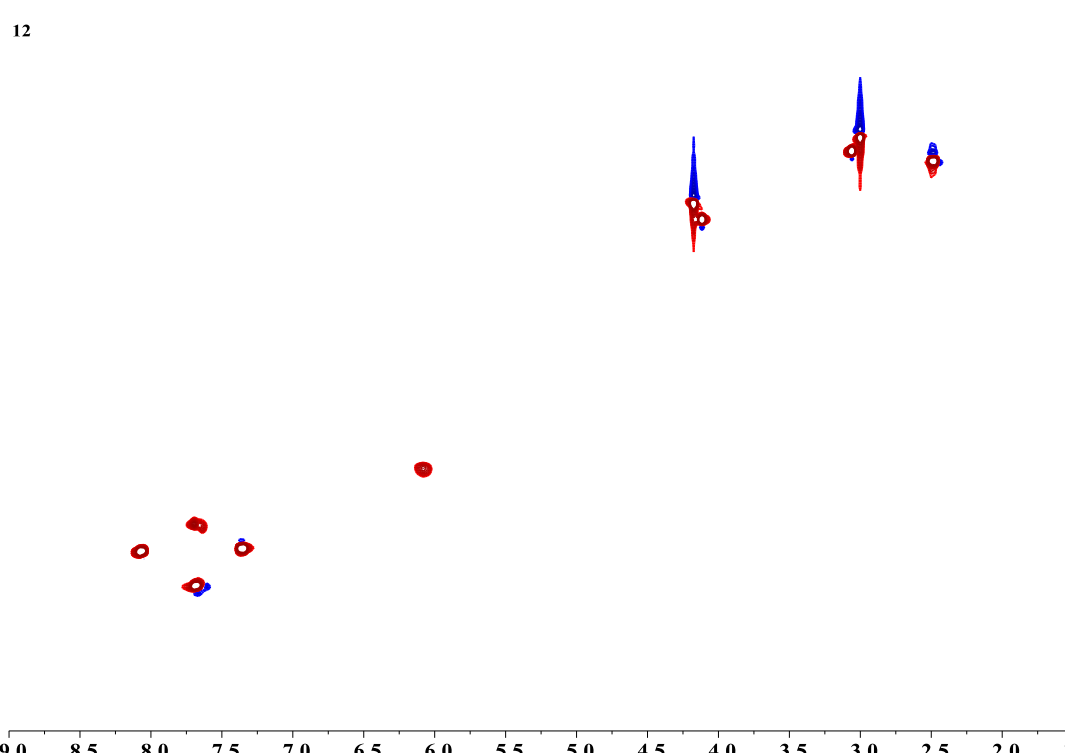


Figure S72: ^1H - ^{13}C HSQC of **12**. DMSO- d_6 , 500 MHz for ^1H and 125 MHz for ^{13}C NMR.

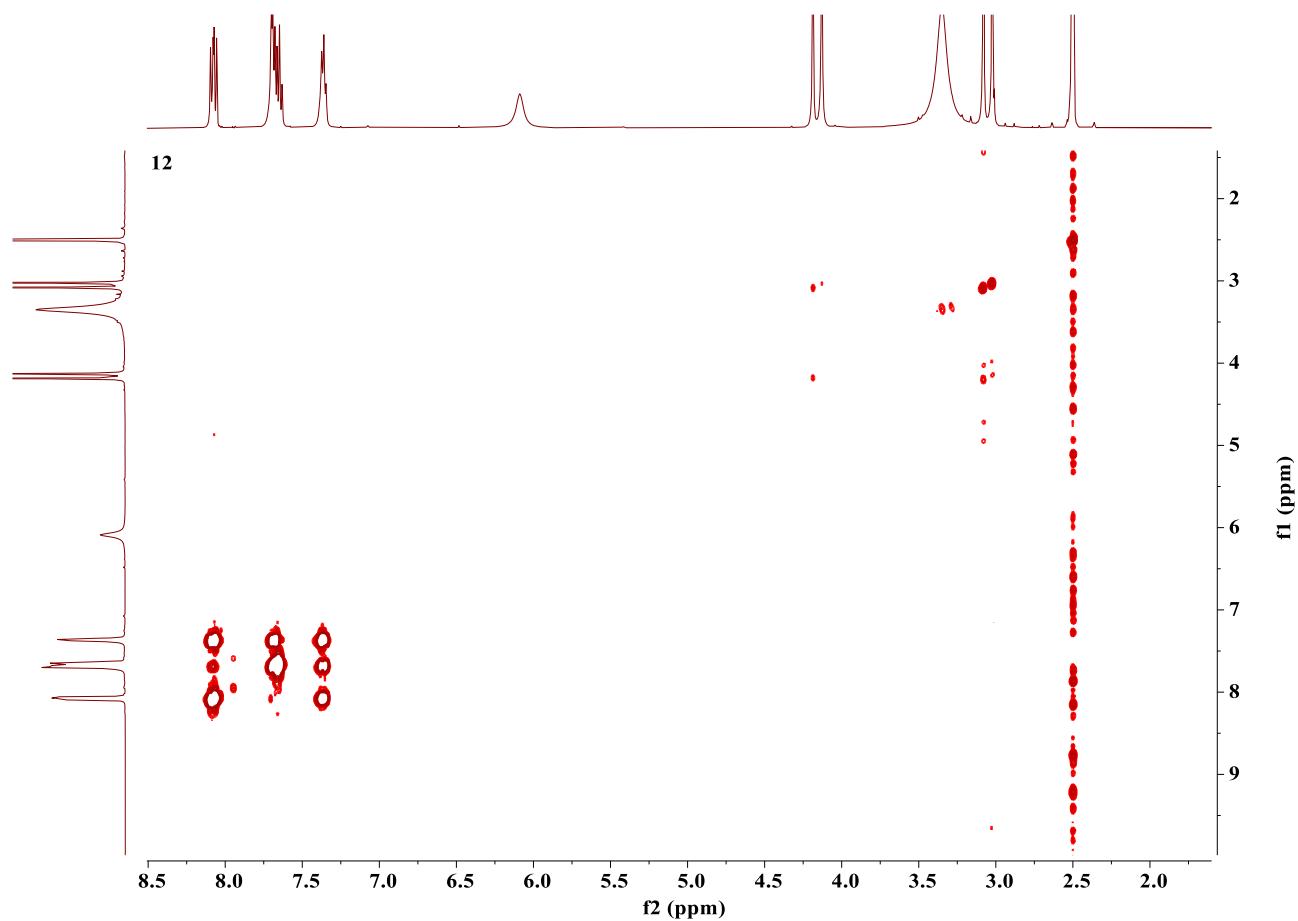
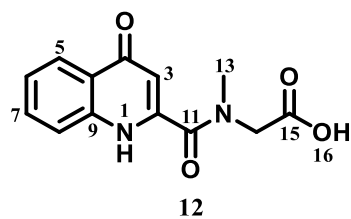


Figure S73: ^1H - ^1H COSY of **12**. DMSO- d_6 , 500 MHz for ^1H NMR.

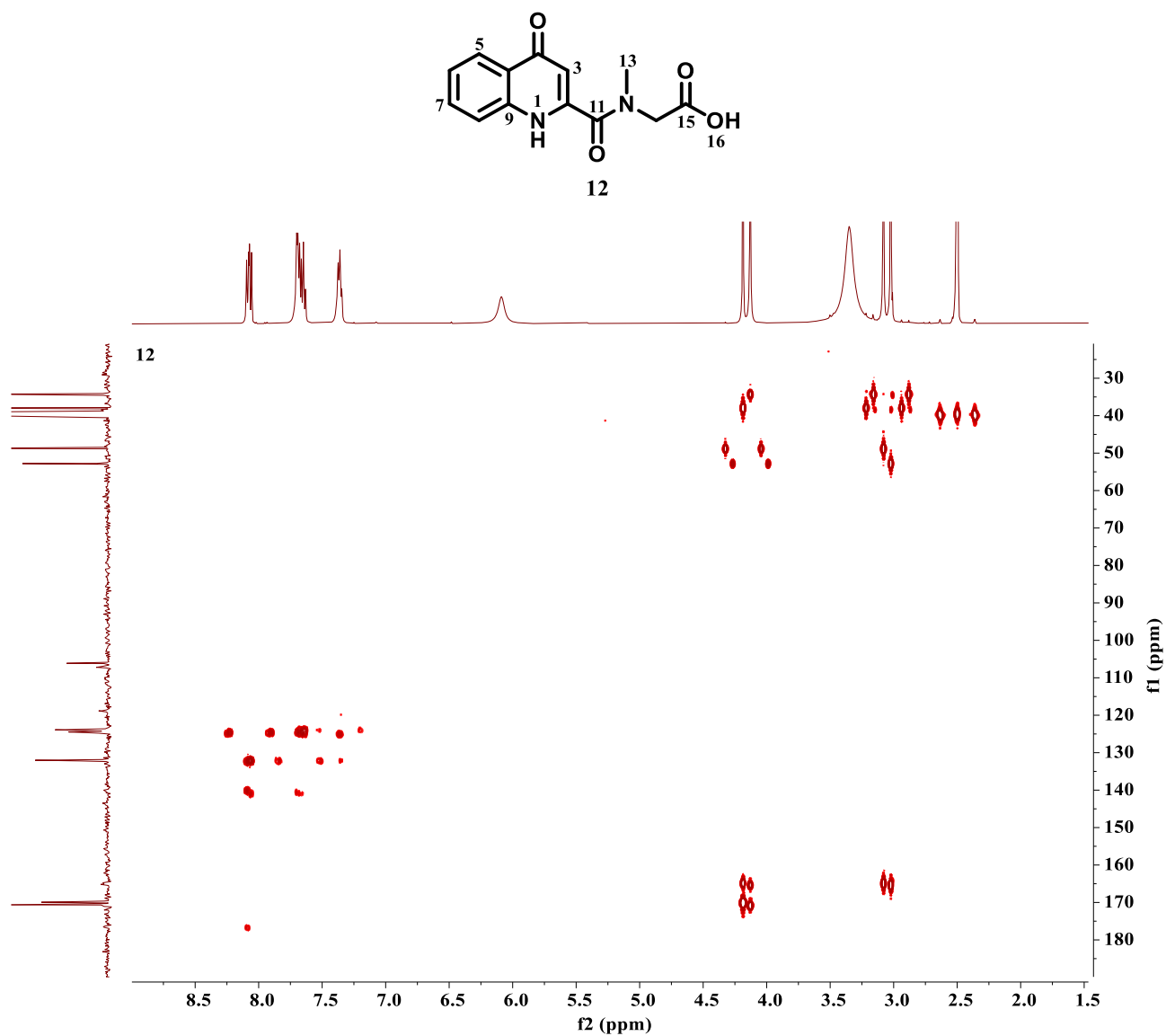


Figure S74: ^1H - ^{13}C HMBC of **12**. DMSO- d_6 , 500 MHz for ^1H and 125 MHz for ^{13}C NMR.

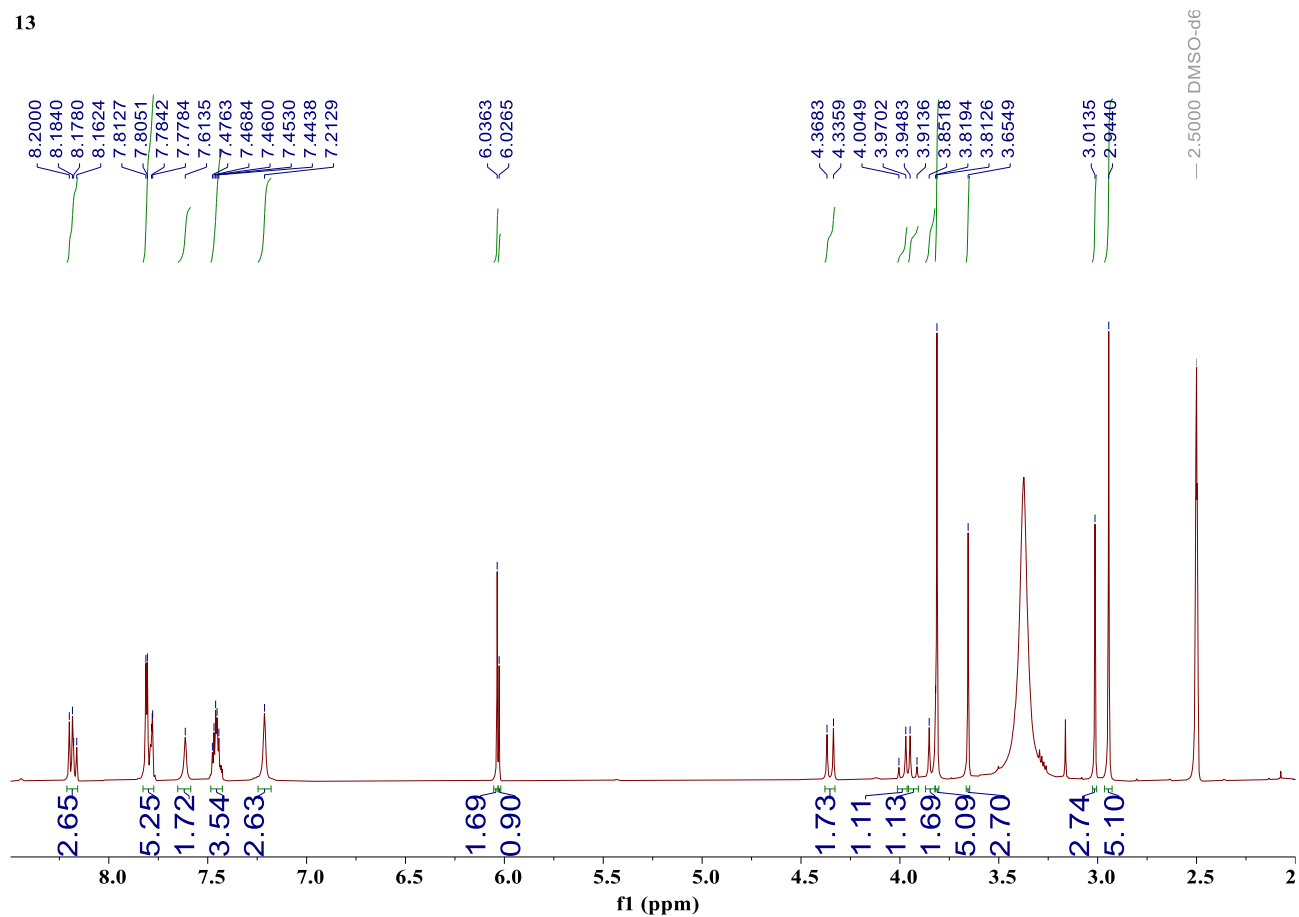
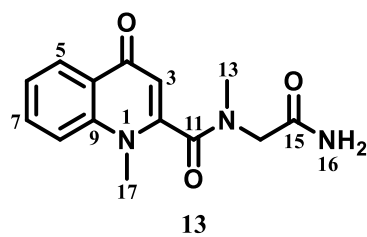
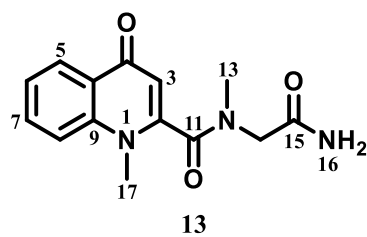


Figure S75: ^1H NMR of **13**. DMSO- d_6 , 500 MHz for ^1H NMR.



13

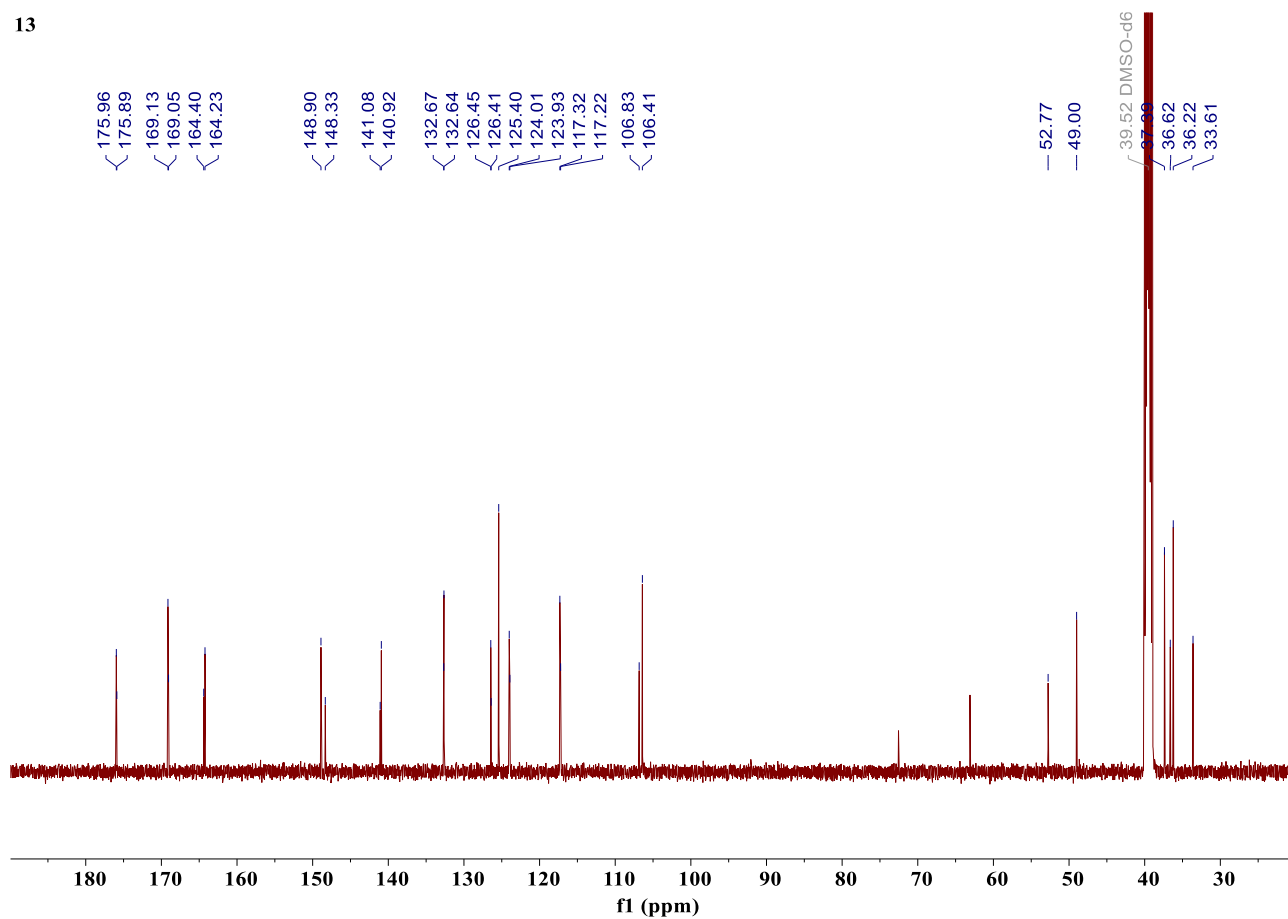


Figure S76: ^{13}C NMR of **13**. DMSO- d_6 , 125 MHz for ^{13}C NMR.

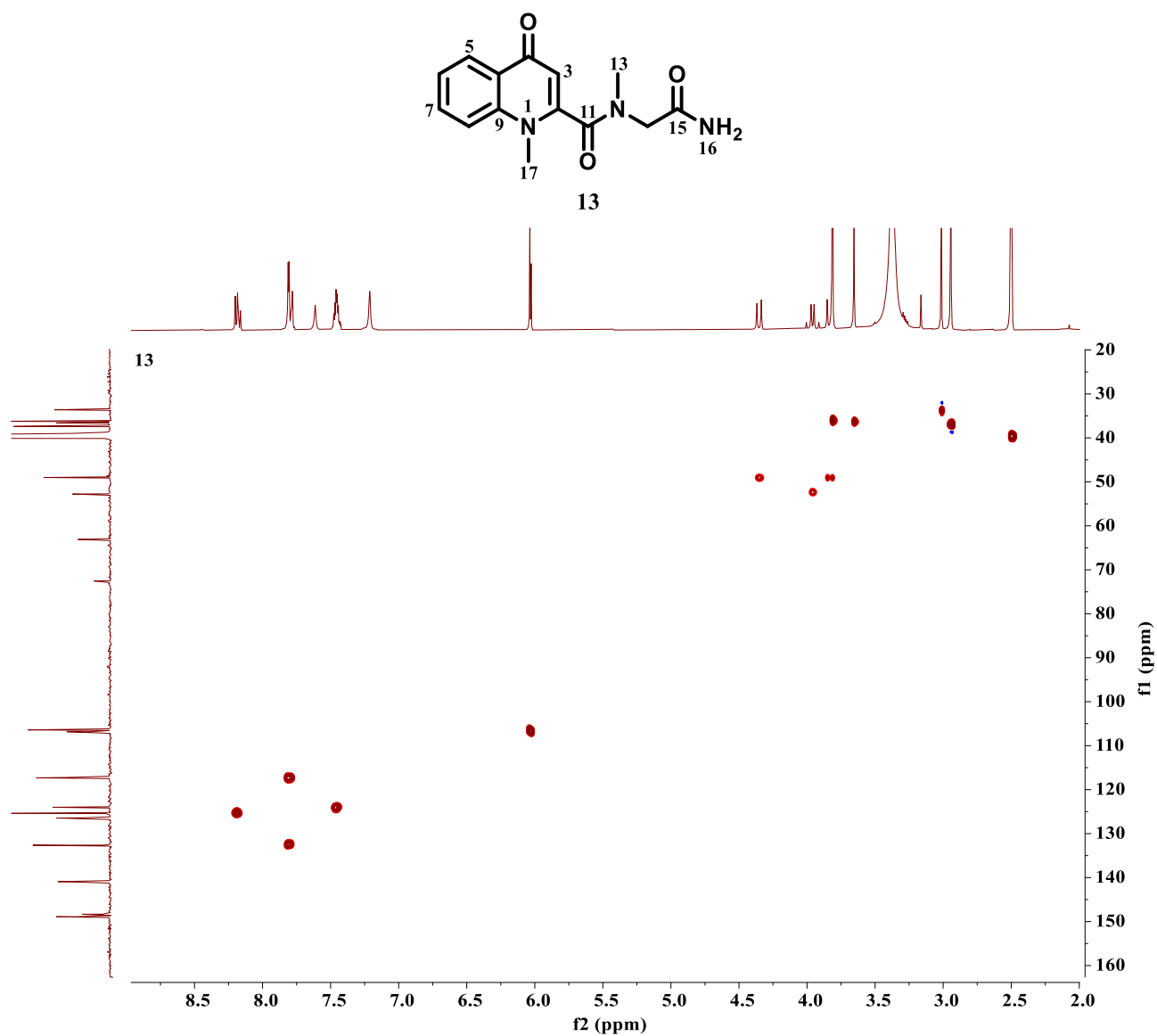


Figure S77: ^1H - ^{13}C HSQC of **13**. DMSO- *d*₆, 500 MHz for ^1H and 125 MHz for ^{13}C NMR.

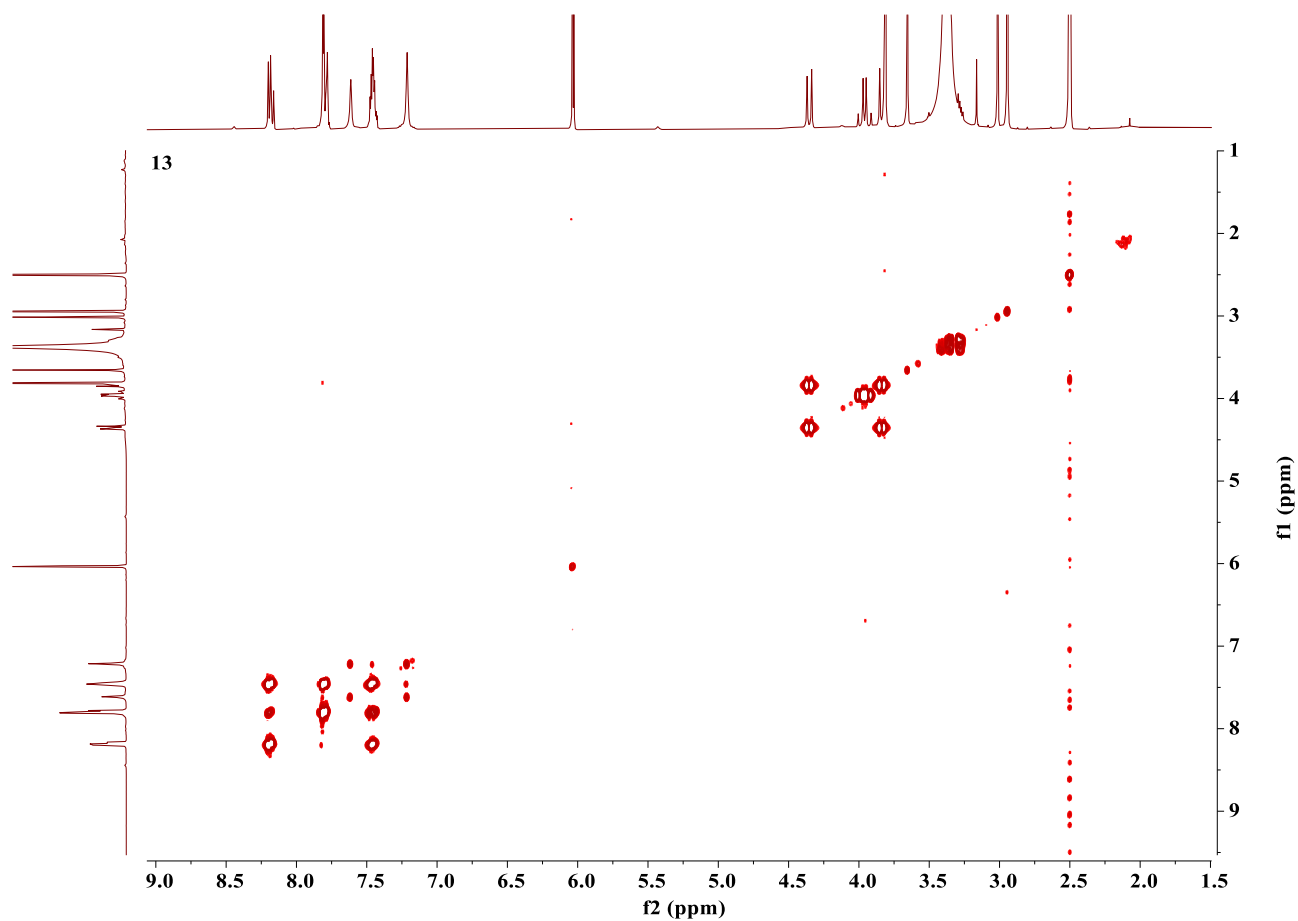
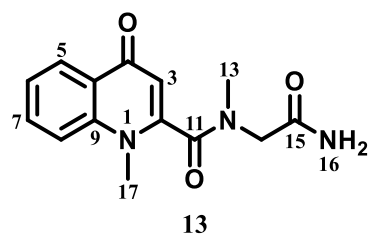


Figure S78: ^1H - ^1H COSY of **13**. DMSO- d_6 , 500 MHz for ^1H NMR.

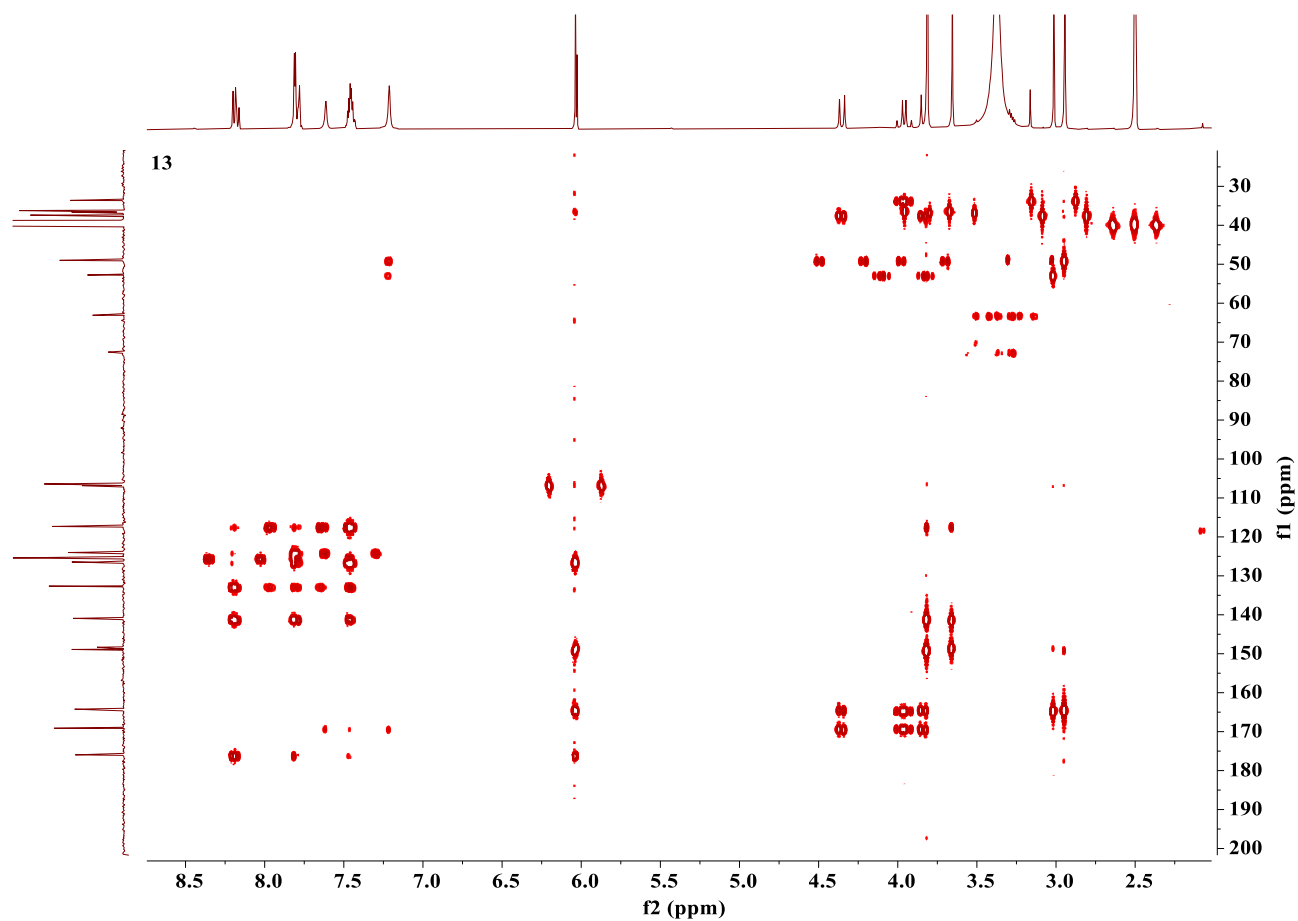
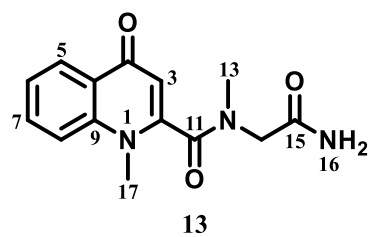


Figure S79: ^1H - ^{13}C HMBC of **13**. DMSO- *d*₆, 500 MHz for ^1H and 125 MHz for ^{13}C NMR.

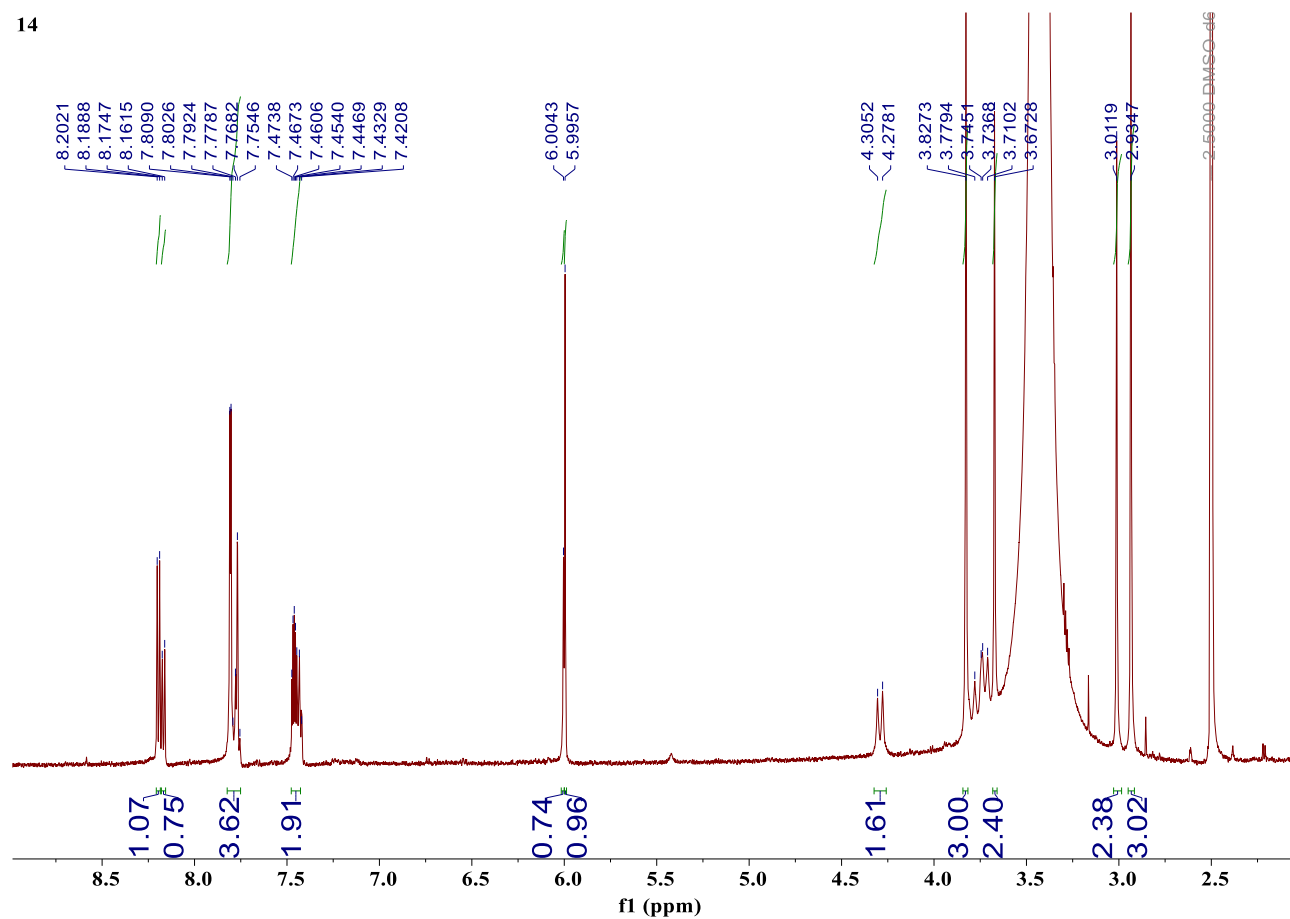
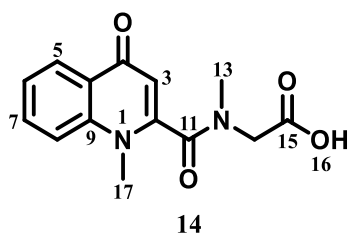


Figure S80: ^1H NMR of **14**. DMSO- d_6 , 600 MHz for ^1H NMR.

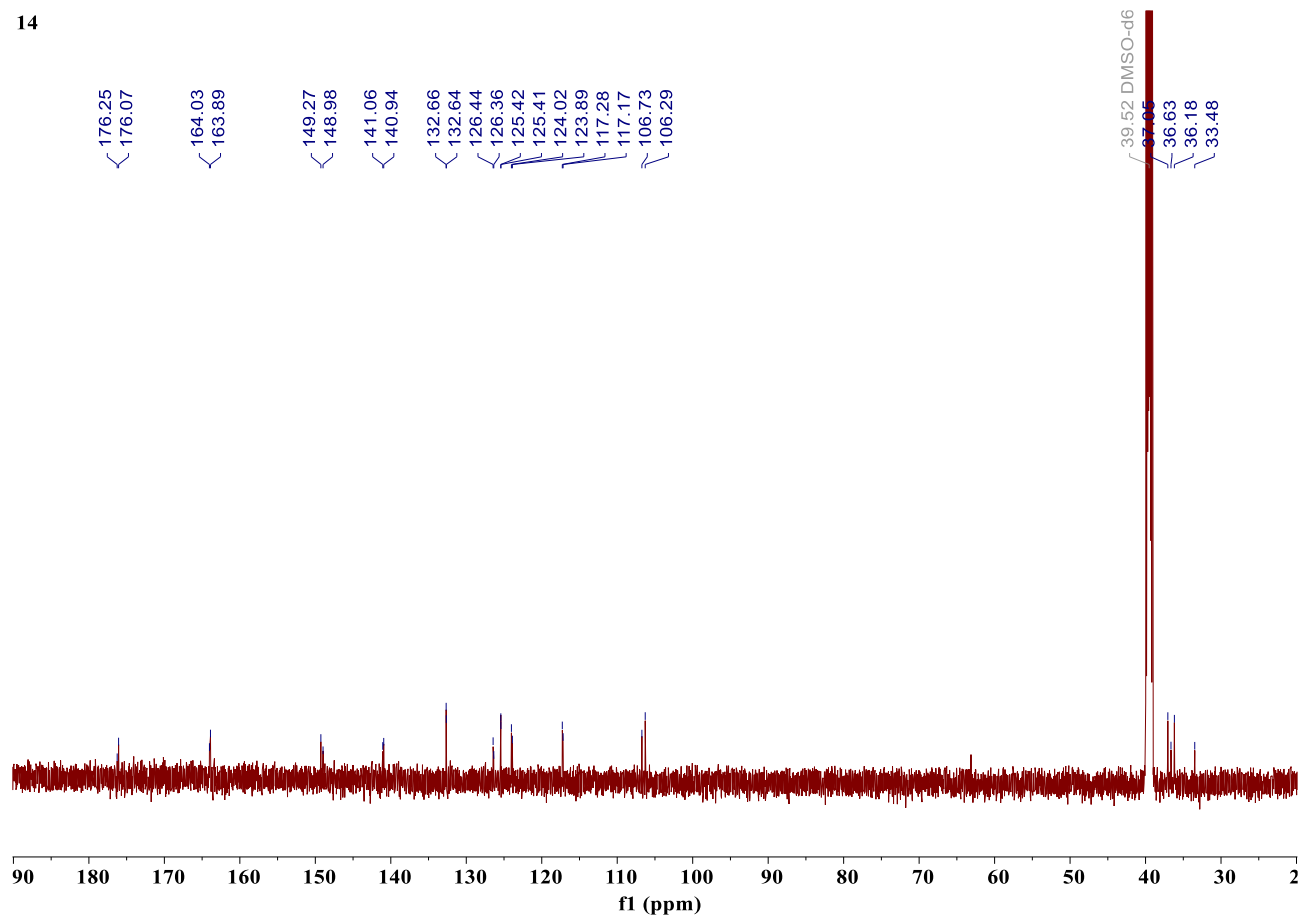
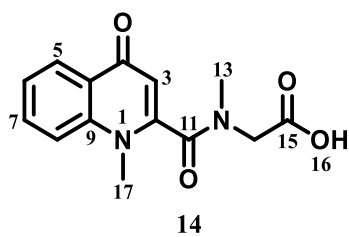


Figure S81: ^{13}C NMR of **14**. DMSO- d_6 , 150 MHz for ^{13}C NMR.

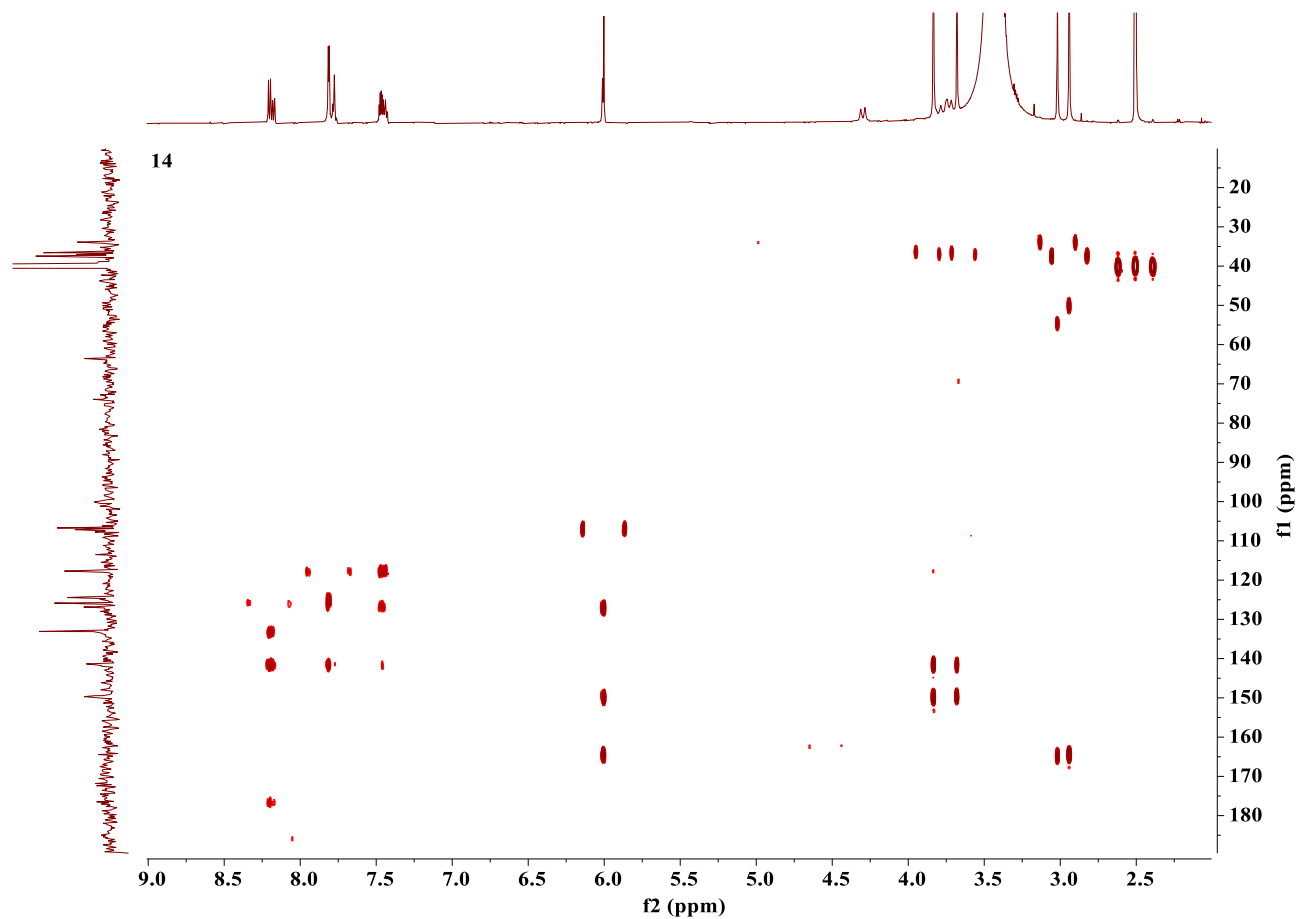
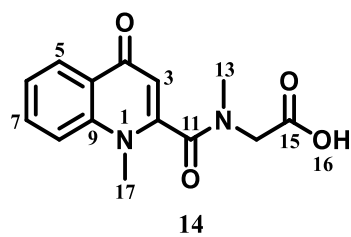


Figure S82: ^1H - ^{13}C HMBC of **14**. DMSO-*d*₆, 600 MHz for ^1H and 150 MHz for ^{13}C NMR.

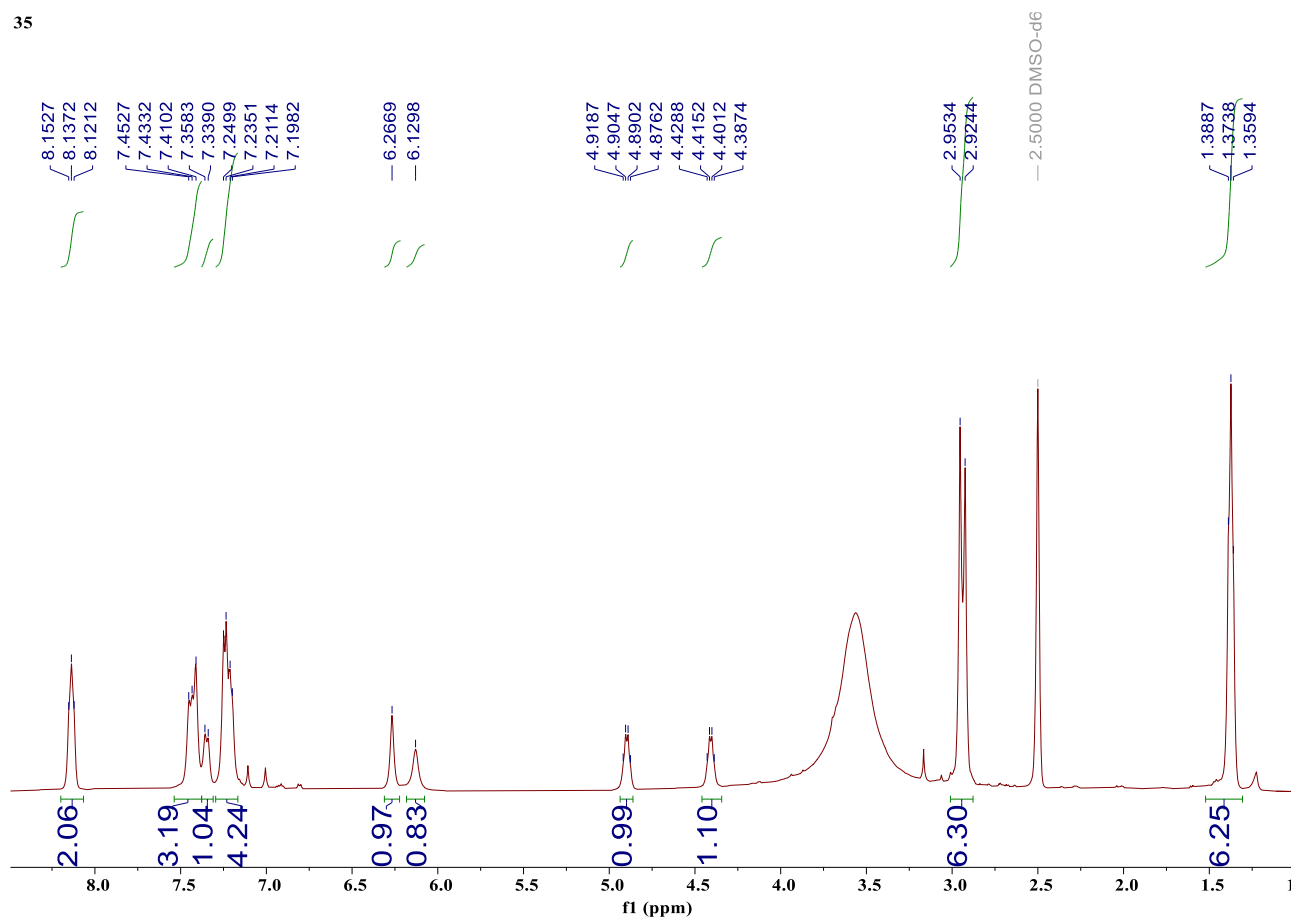
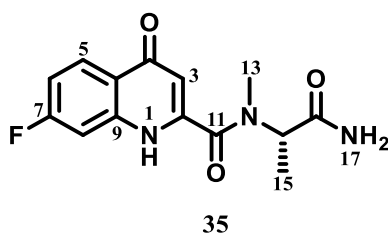
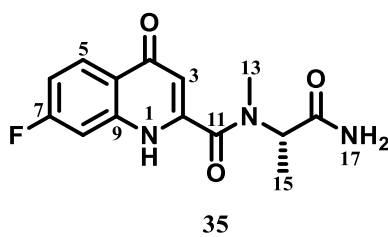


Figure S83: ^1H NMR of **35**. DMSO- d_6 , 500 MHz for ^1H NMR.



35

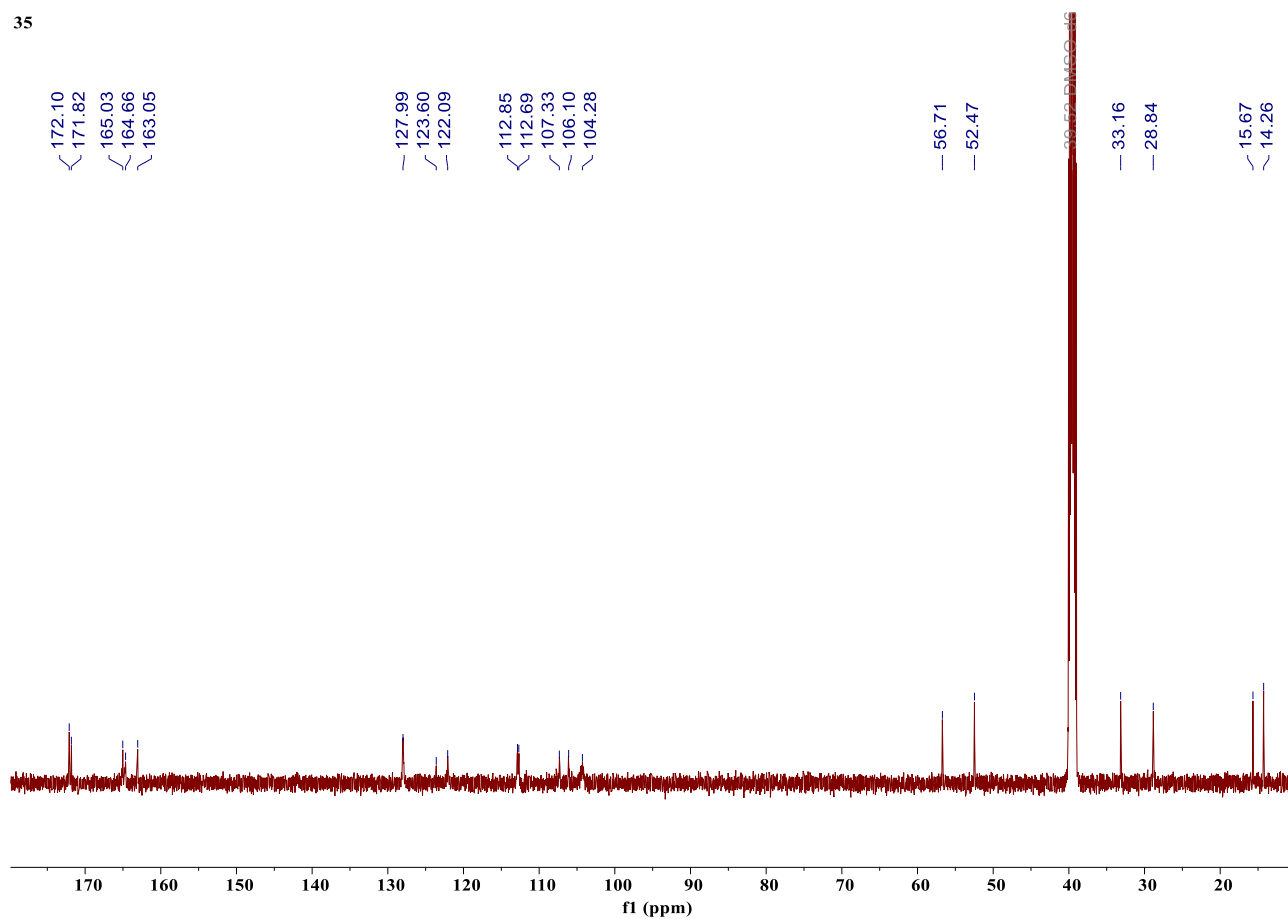
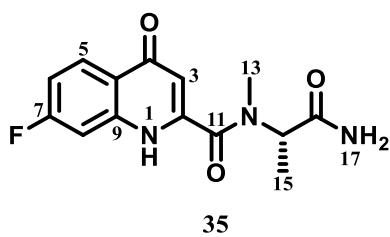


Figure S84: ^{13}C NMR of **35**, DMSO- d_6 , 125 MHz for ^{13}C NMR.



35

-74.02

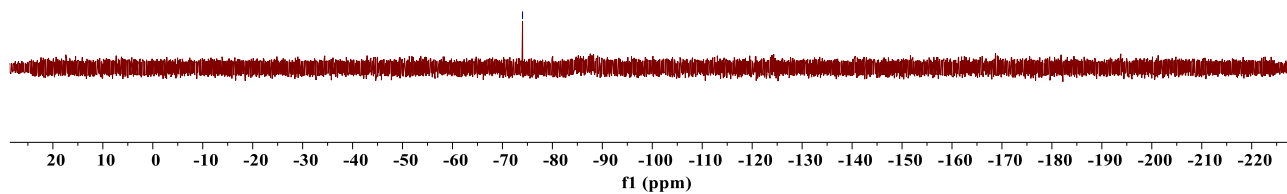
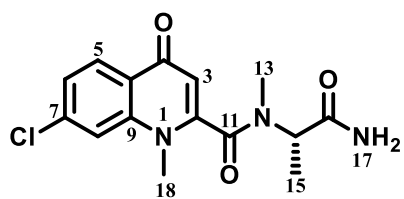


Figure S85: ^{19}F NMR of **35**. DMSO- d_6 , 500 MHz for ^{19}F NMR.



38

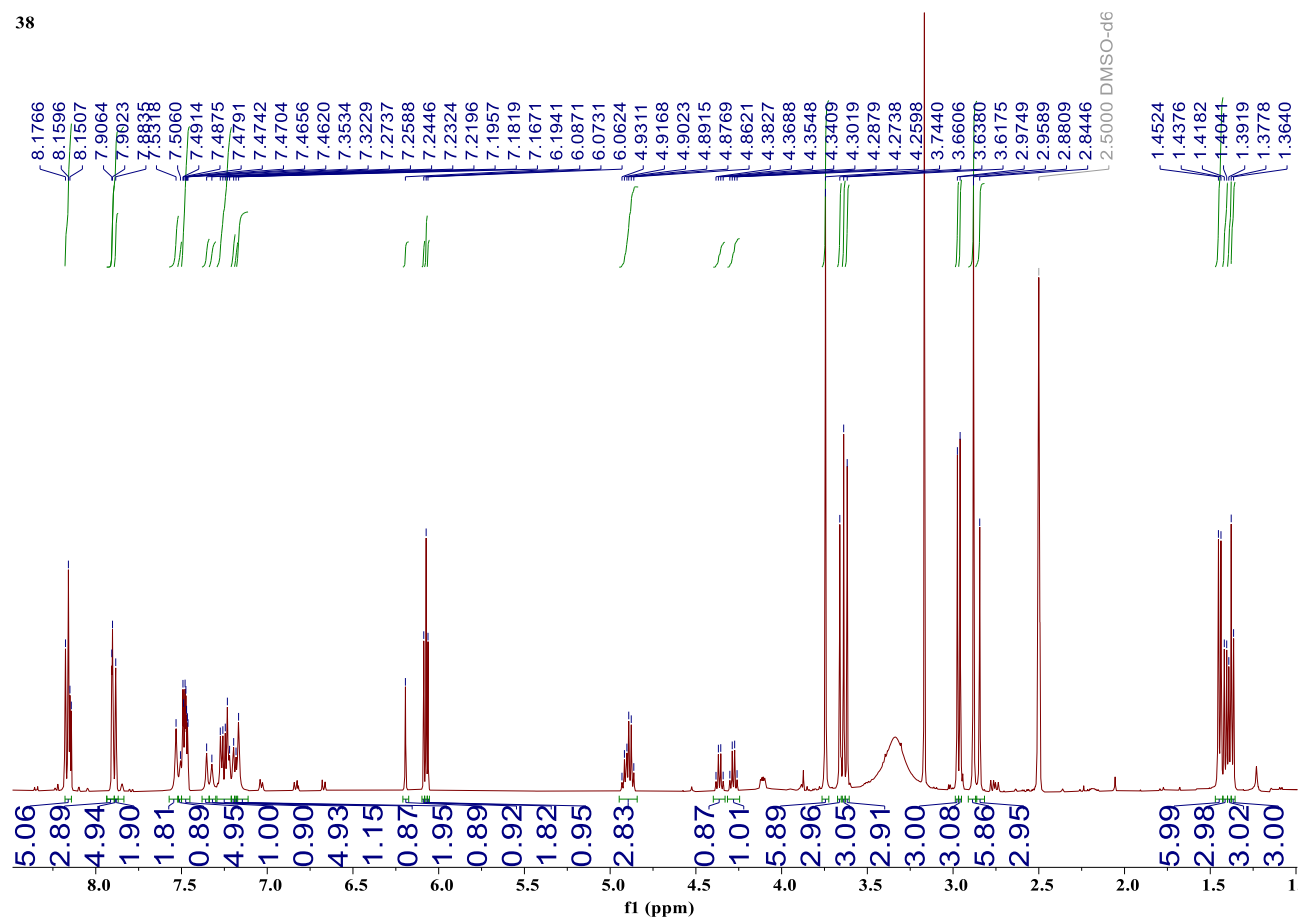
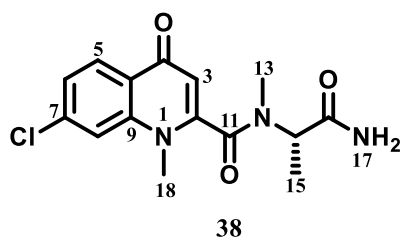
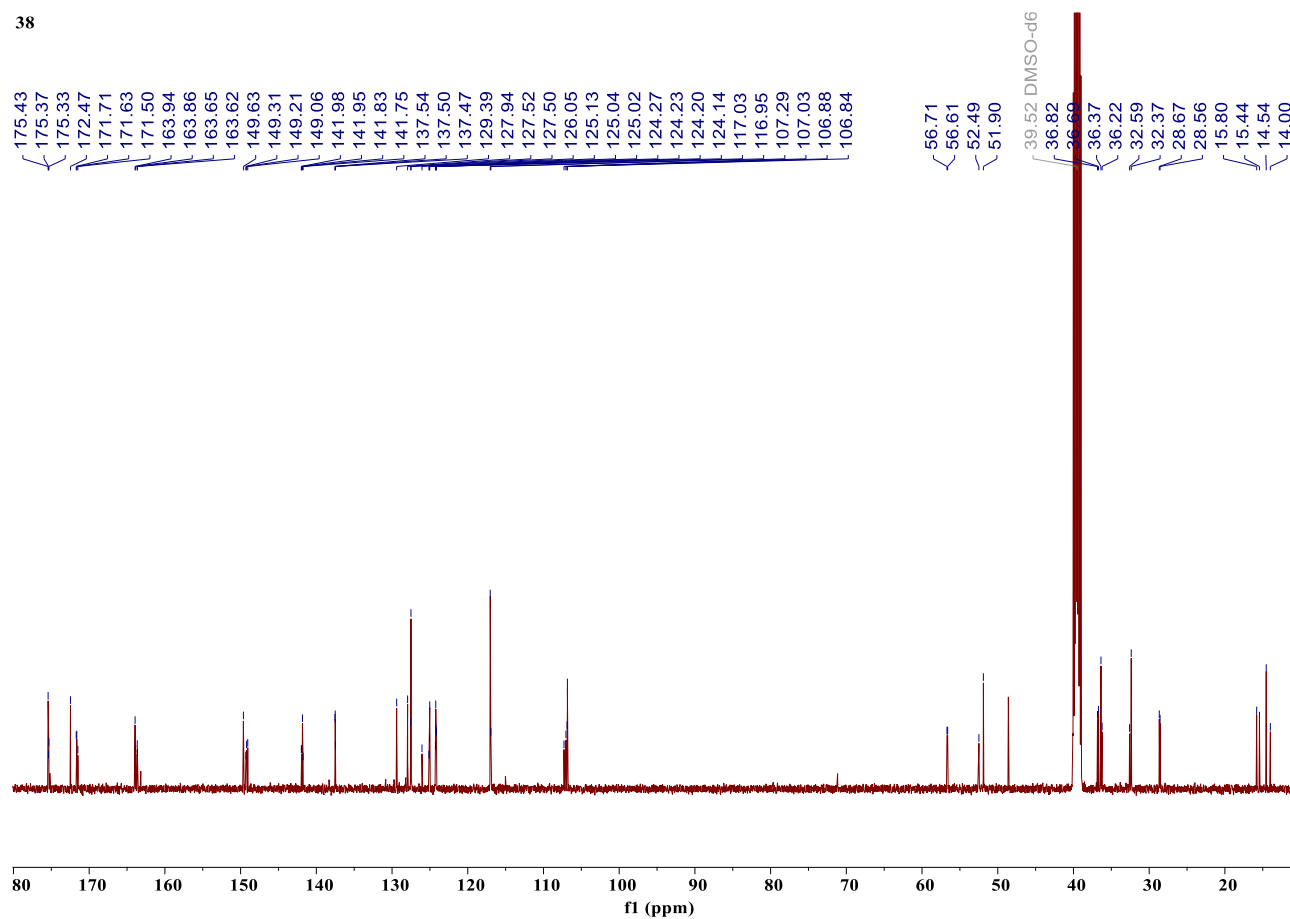


Figure S86: ^1H NMR of 38. DMSO- d_6 , 500 MHz for ^1H NMR.



38



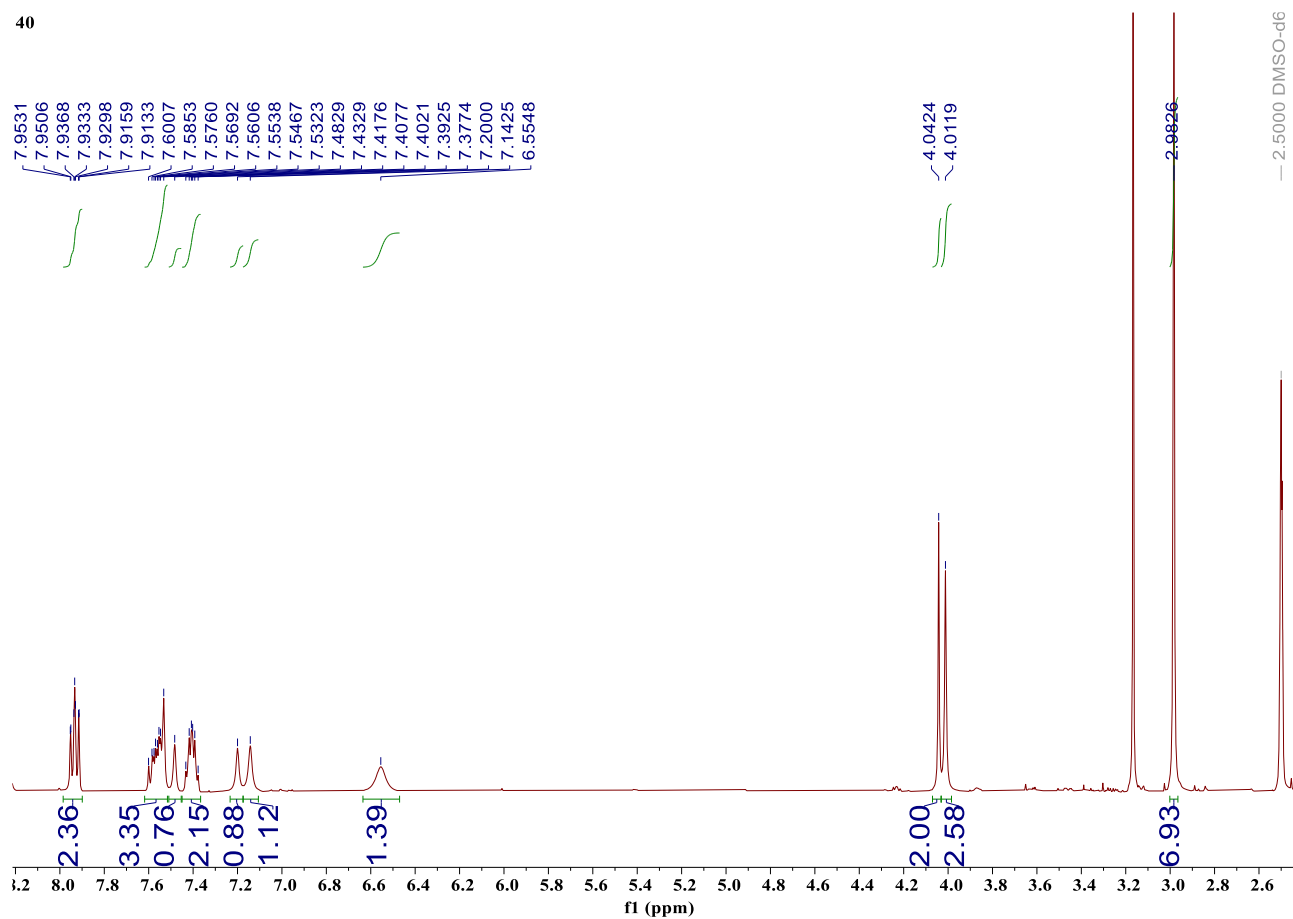
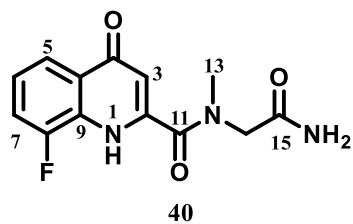
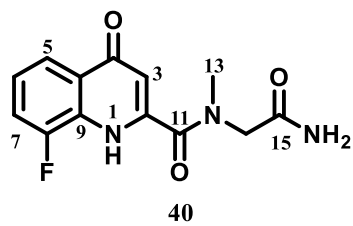
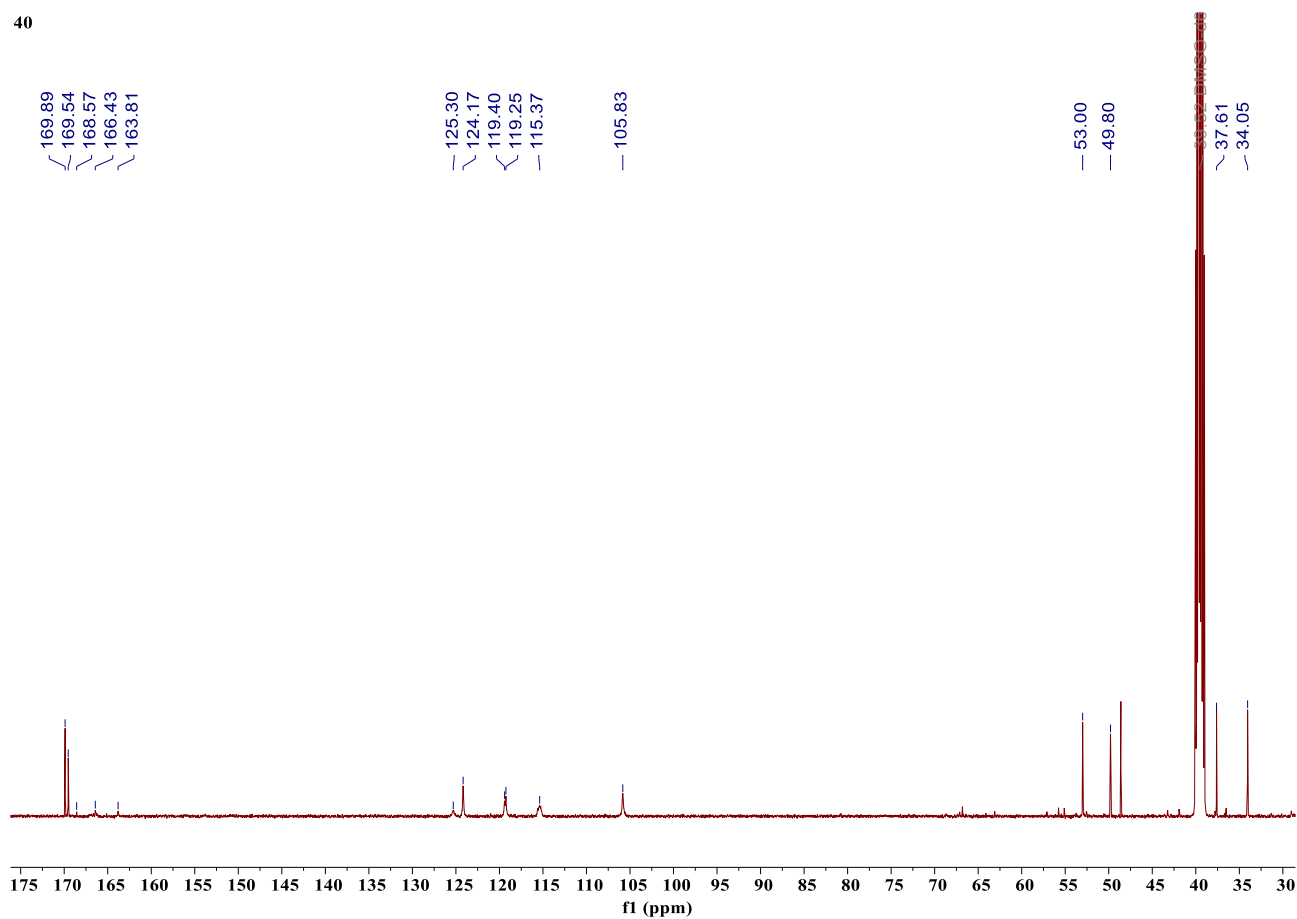
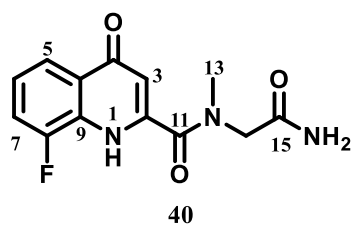


Figure S88: ^1H NMR of **40**. DMSO- d_6 , 500 MHz for ^1H NMR.



40





-97.59

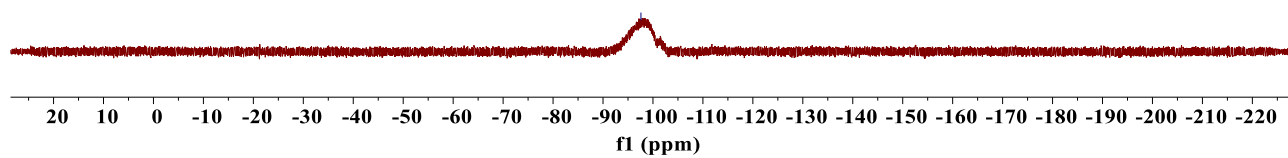


Figure S90: ^{19}F NMR of **40**. DMSO- d_6 , 500 MHz for ^{19}F NMR.

Due to scalar coupling with the quadrupolar ^{14}N nucleus, the corresponding ^{19}F signal is broadened.

SUPPLEMENTARY PROTEIN SEQUENCES

Amidase 1 (NC_066257.1)

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RPWDLDPRCANLPWNESVFRGIQSRPLVIGLILDDGVVKIHPPIERAMRELAAKLEANGHELVTWNTSDHL
KCISLMDRYTVDGGEDVRRDVAVAGEPFIPHVEALVNRGKPISVYEWQMNTERNNLQKKYLDGWNAAR
RSPSGKPVVDVLLSPTMPHVAVPHRSVRWVGTYKIWNLLDYPATFPVDRVRADKDVLPDYPRPNDLDAW
NWSLYDAEAMEGQPVNLQVIGKRLEESLLGAATVIERIWS

Amidase 2 (NC_066259.1)

MPQSWEELAADKRARLAKTIPDEWKVQTLPAEDSVIDFPKKSILSEAEKITEASAADLVSKLAAGELTS
VEVTLAFCKRAAIAQQLTNCAHEFFDAALAQARELDEYYAKHKRPVGPLHGLPISLKDQLRVKGYETSM
GYISWLNKYDEGDSVLTTMLRKAGAVFYVKTSVPQTLVCETVNNIIGRTVNPRNKNWSCGGSSGGEGAI
VGIRGGVIGVGTDIGGSIRVPAAFNGLYGLRPSHGRPLPYAKMANSMEGQETVHSSVGPITHSVEDLRLFTKS
VLGQEPWKYDSKVIPMPWRQSESDIASKIKNGGLNIGYYNFDGNVLPHPILRGVETTVAALAKAGHTVT
PWTPYKHDGFDLISHIYAADGSADVMRDISASGEPAIPNIKDLLNPNIAVNMNELWDTHLQKWNQME
YLEKWREAEEKAGKELDAIIPITPTAAVRHDQFRYYGYASVINLLDFTSVVVPVTFADKNIDKKNESFKAV
SELDALVQEEYDPEAYHGAPVAVQVIGRRLSEERTLAIAEEVGKLLGNVVT

Amidase 3 (NC_066260.1)

MSSTWEAQARKAQYILEQSIPKQWLVPADRLPPADQLNVMDFPKSGMLTDKELAITELSATALVSAMGE
GRFSAAEVAVAFKRSVIGHQLLNFAEFMAEKAILRARELDKYYKETGQLVGPLHGVPIISVKEHIGFRGLT
LNAGYTTWVDNIATDDALILECLEEAGAVFHVRTNQPQSLMHLCCSNNLTGTTLNPNYNRTLSPGGSSGGEG
ASMGFKCAPLGIGTDIGGSIRCPAAFCNAYGFRPTIHRNPLKGSKVPVPGQESILGVIGPLACQCVEDLELFQ
RALLDQEPWDEETSLVPLPWKHVSSTENITVGIMWDDGCVRPHPVTRALSHARERLRALGINVVDWEPY
NHDHGWKIISLYFADAAARNRELIAESGEPTLPLTEWAFQYSRSTPLTVAENWELNCQRDIYRDDYHALM
KSHGVDFILCPAYVGVASVLGESQYWNYTAINWILDQPAAVFSPGLYVDPQLDPVDNSYQPRSAEDEREW
KKYSPERYVGAPIGLQLVGRHFKDEETLAAASLVSKALKNTTNV

Amidase 4 (NC_066261.1)

MNFKYLFYSGQASGNASWEAKASAKRADTLAKIPAEWLLPKEDLDKAANERNLTGPFINQYLSPDDIAI
VEQGSVPLVEKMSAGKLSAVQVTRAFCKTAAIAHQINNCLHEVFFDQALERAQELDNHLSHGSLVGLPLH
GLPISLKDQFHVKGNAATSMGYIGWIDTYEGNPDPNLVFNVNSQVVTPELLAGAVLYCKFGETVNNIIGRTL
NPINQNLSCGGSSGGEGALSALHGSSVGLGTDIGGSVRIPAGFCGIFSLKPTHNRLSYRDVANVIPGQTTYAS
SVGVLGNTIEAIHLVLSSLISTEPWLDRPNVASLPWRQNIADETLSRAGKGGQCTSAGDKQPLKLGVLWTD
GVVTPHPPVARGRLRLTVDAVKKAGHRVSFLKADGGHDINRQLDLSGEPLIPQLRDSFTLRDPMSLLKYQAL
TIEGRDYEAAYS DYWNETANAEDGQIVDAVIMPVAPHAAPVPGKFYHTGRLPTRRPLPPVFLALNPPSGMR
ATVVIPVTRADKNQDLADPDYAPLNELDAKNWEAYDADIYDGAPVGVQIVARKYEEEEKIWAIGKLVYDCL
LQGEAQ

Amidase 5 (NC_066262.1)

MAEETWQAAVAERRRILDLLIPTDWKLSGEFRASLPADGRLLQADPVRKSGILSEDELDTENYSAGQLLQ
RLARGDVSSLAVTTAFCKRAAIAQQLTSCLEHFFAQAIERARYLDEYLRGKVVGPLHGLPISIKDSFQV
KGLHTTVGYVSFLKNGPATENSAMVDLLLDLGAFLYVKTNPQTLMTADSENNIYGRTLNPHNTNLTAGGS
SGGEGALIAFRGSILGVGTDVAGSIRIPALCCGLYGFKPTTDRVPFGGQVSGVAEETPMISPSAGPLAQSFDD

LELFMSRVLAAGPWRYDPAAVAMPWIRHTEPQSLLSVGILPEDKHFPLHPPVKRALESAIENLKRMGHRIR
LPSPSRDIADANRLAFEFFTYGPRLDHITPSGEPVVTSVATLPSPMFSGPPYLSQDLGVFDKIKILQIARKKY
SESWRRTIVENEIDVILAPGAQNTAVPHDSYGWPPYTAIWNLLDYPACIIPYGKASKELDPEPMVVGQQPG
YDPEAVDGAPCALQIVAPRFQDEKCLAAARIIDRDIRV

Amidase 6 (NC_066262.1)

MDWQQAQRKRDELAKIPSGWRISDLLKKADELNDILDIPRQSGILTEREIDITESSDATALISRLKDGEIS
AYEVAVAFCKRAAIAQQLTCCLTEIFFDRAIERAKELDRQYAVTGELVGPLHGIPLSKDSYNVTGVQSTLGY
VSFLDRPALTFNSPMVNILLDAGAVIYVKTHLPQTMMTADSHTNVFGRTRNPYGRNLTAGGSCGGEGALIA
MRGSILGAGTDVGGSLRIPSLCCGTFGFKPSVGRLPFAGQTPPGRIGMAGGIAVSTGPLCTSTRDADLFFRT
VVSSHENLDDNSLGFPYLEPLKLVSSLTIGILPEDPALPLHPCMQRTLITAARKLAAAGHRIVHLPKEELPSL
MDACDLAFRFFNMDPDRTPLRIVKDGGEPIYPSLSMIYNVEGTDPEPTLRQLYDLNVAKAQITAKMRQAW
LKNRVDVVLAPAYQSCAPLHDTYGKNIYTVIWNMVDYPACLIPFGHANKAADADEFIRDVQYSPDYKPDEI
EGAPCHVQLIGRRLKDEVLLQHAKVIEMVLADGK

Amidase 7 (NC_066263.1)

MTEPHWKSIAHQKRLTQLQSIPSPWRLSSSSISSTSPKSPIEIIQTCGILSPQELRWTEVNDITELVCLLASRSV
TSVQLTTAFCKRAAVAQQLTGCLTEIFFDRALQRAKFLDEEFERTGKVTGPLHGVVPSIKDRFDVEGFDTTI
GWVGLANKPAAKSDSVVQLLESMGAVLYVKTNPQSLMMSDSYNHVFGQSVNAFNTKLISGGSSGGEGA
LVGAGGSVLGIGTDIGGSIRVPSNLQGLMHQHYLVPPVAGPMARSVSTIEYFMQSLSDSNPNLDPGCIPI
WRRETAAFPDPTRKLKLGIVYDDGVVRPQPPVMRLMRELARELTNAGHEVVEWDTSLHRTGTTLWTRSIL
ADGGYHCRQLCSLVDEPLIQGMVVGTSADELSSVEKEKLEEEKYAFQESYLAQWVASGIDALLPVTPWL
GYKPKQWVQSNQWLGYTALFNLLNYAAVTAPIGKADGELDHPVAGIDNEWKGYAPRNKADRFNYDQYDI
DLVKGMPVTVQIVGGRYGEEKAVAVAKVVDTLFGR

Amidase 8 (NC_066264.1)

MEYSSIAQRKQDQLWERIPQEWRLAQDKIPGMHSPAESVTNVQYDRNVMDIPRSCGLLSPKEIEITERW
DITGLLRQIHSQSFTAKEVIQAFCKRAAIAHQQLTRIGRPPPPHRLPNWPSSGLPVSVKDSFNIRGVDSTVGIA
ALALHPSEKDAPLVSLQLSGAIIIAKTNVPQTMGALDSANHVFGRTLNPNNRQLTAGGSTGGEGALLALR
GSMIGFGTDIGGSIRVPAMCNGLYGFKPSQGRVPYGGQTDGQPVAKGRISIQAVAGPIAHSVVDINAVMTHL
IPRAQWFGEDCIPGAWPSPLSSFGKRSRPITIGVLPSDGIVEPLPPISKVLNEVAAALRRTPGVNVVQLPVPKA
LPKCQALAGRLMGVDGGIKMLDLVESTGETLIPWLKGRTRGRPLLLSQLAELQATRGDVELELRRELWM
YGSTGHEQTVDAIHPVAPHPVPELDRYNAVGYTSSFVLLDYPAGTVVRNFGRDDLENGKEMEARVIGSW
DKANRELWNEKTVDRTVYLDSPLSVQVITPKQHDIYELYQAMELIDRAIKTDRSKL

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ENERGIES AND MOLECULAR COORDINATES OF CALCULATED STRUCTURES

Molecular coordinates are provided in .xyz format. Geometry optimizations and harmonic frequencies were computed with M06-2X/6-31+G(d,p) using SMD(water) (#p opt freq m062x/6-31+g(d,p) scrf=(smd,solvent=water) int=ultrafine scf=(tight,xqc,maxcycle=512) nosymm). Single-point energies (E_{SP}) were evaluated at M06-2X/aug-cc-pVTZ with the same solvation settings (#p m062x/aug-cc-pVTZ scrf=(smd,solvent=water) int=ultrafine scf=(tight,xqc,maxcycle=512) nosymm). Final Gibbs free energies were obtained as $G(\text{final}) = E_{SP}(\text{aug-cc-pVTZ}) + \Delta G_{\text{corr}}(6-31+g(d,p))$ (298.15 K; energies in Hartree).

NO.	Structure	$E_{SP}(\text{aug-cc-pVTZ})$	$\Delta G_{\text{corr}}(6-31+g(d,p))$	G (final)
1	6	-1048.236893	0.279316	-1047.957577
2	6 (-)	-1047.761915	0.266383	-1047.495532
3	Int II (-)	-1047.735623	0.262760	-1047.472863
4	Int II (-)..H₂O	-1200.637375	0.305387	-1200.331988
5	Int II	-1048.225383	0.275287	-1047.950096
6	Int II..OH (-)	-1124.180698	0.284801	-1123.895897
7	Int III (-)	-1047.758617	0.267131	-1047.491486
8	Int III	-1048.238431	0.280261	-1047.958170
9	1	-971.8120734	0.254377	-971.5576964
10	H₂O	-76.44190091	0.002947	-76.43895391
11	HO (-)	-76.44190091	-0.007698	-75.95376892
12	TS1	-1047.728469	0.264053	-1047.464416
13	TS2	-1124.17728	0.286447	-1123.890833

6				C	2.02147900	-1.27317800	0.55401200
0 1				N	3.30774400	-0.96891300	0.36006500
C	-4.60907800	1.40254600	-0.81153200	C	3.82410400	0.39918300	0.29462900
C	-5.04981900	0.07401200	-0.82787200	C	2.99888700	1.38086800	1.09544500
C	-4.21565000	-0.96290400	-0.44093500	N	1.76977900	1.00070700	1.48904500
C	-2.88449600	-0.71979000	-0.02699800	O	3.46402000	2.49209600	1.36625900
C	-2.41928700	0.62777800	-0.05996400	C	3.95028700	0.88651100	-1.15122500
C	-3.30479200	1.65840500	-0.42206200	C	4.19608500	-2.03052900	-0.11314100
C	-1.02674600	1.05605400	0.20990900	O	1.56819800	-2.41780200	0.42932100
C	0.15059000	0.19405800	-0.22118900	O	0.19383400	-0.63085900	1.98168600
O	-0.79326200	2.16378900	0.69144100	N	-2.10064100	-1.75667500	0.40267600
C	1.05372700	-0.15267200	0.97587500	C	-2.63149600	-3.09707300	0.57352200

H	-5.26601900	2.21173800	-1.11035400	H	-5.87044900	1.87570100	-0.40477300
H	-6.06489700	-0.15739300	-1.13764600	H	-6.37787200	-0.49993200	0.18835700
H	-4.59791800	-1.97665000	-0.43948400	H	-4.57849800	-2.09603400	0.59974200
H	-2.92345900	2.67504300	-0.42586100	H	-3.49644800	2.56746100	-0.64162500
H	-0.17580000	-0.72450700	-0.70872800	H	-0.67406900	-0.66906400	-1.39437800
H	0.73239400	0.78354500	-0.93695900	H	0.09069700	0.81930900	-1.98826800
H	4.81251700	0.39215200	0.76273600	H	3.34380300	-0.32147400	1.70348700
H	1.17743800	1.73347800	1.87621500	H	1.01704700	2.17977700	0.10061800
H	2.96816200	0.91406400	-1.63195200	H	5.37956100	0.81879600	-0.28702700
H	4.38733400	1.88716200	-1.16626700	H	5.32514500	1.12862000	1.46017000
H	4.60234100	0.21340400	-1.71287300	H	5.66815500	-0.49985900	0.87491900
H	5.22328200	-1.67062100	-0.05966500	H	4.39651700	-2.42397100	0.33748900
H	4.08727100	-2.90851300	0.52431400	H	3.60551100	-2.66455300	-1.23667200
H	3.95864300	-2.30620900	-1.14461000	H	5.00576900	-1.58256800	-1.10936100
H	-1.31407100	-1.52753100	0.99863200	H	-1.03539200	-1.20349400	0.73804700
H	-3.48173300	-3.12428800	1.26699500	H	-2.85707200	-2.86431000	1.97528200
H	-2.95139500	-3.51747700	-0.38404200	H	-2.80821100	-3.51501500	0.32320900
H	-1.83477200	-3.72515100	0.97230700	H	-1.31649800	-3.39325600	1.26785300
H	0.70665000	-1.00303100	2.71710100				

6 (-)

-1 1

C	-5.06933200	1.16578600	-0.23142600
C	-5.34738800	-0.16749600	0.09849600
C	-4.32875600	-1.07809300	0.32414100
C	-2.96511700	-0.69866400	0.22534000
C	-2.68298700	0.64590000	-0.16834400
C	-3.74392000	1.54985300	-0.35445900
C	-1.33045300	1.17073200	-0.46648500
C	-0.26499200	0.29430900	-1.09529700
O	-1.08151700	2.36986300	-0.30312200
C	0.90535500	0.05138700	-0.11768400
C	1.91032400	-0.92661200	-0.75339500
N	3.18832500	-0.83742400	-0.32828900
C	3.59233900	0.11385200	0.72345000
C	2.80714900	1.41898300	0.63538200
N	1.59936200	1.34531500	0.06835400
O	3.28127200	2.47807500	1.07436800
C	5.08804500	0.40298500	0.68204800
C	4.10877500	-1.94025800	-0.60124300
O	1.52054300	-1.80140900	-1.54357600
O	0.49103600	-0.49553600	1.06536100
N	-1.97684300	-1.58238900	0.52658500
C	-2.26592100	-2.90194100	1.05059600

Int II (-)

-1 1

C	-2.29625200	-2.56249100	0.00261200
C	-3.63769800	-2.55539100	0.41985700
C	-4.37559300	-1.38893100	0.43974700
C	-3.80394100	-0.15103200	0.04057200
C	-2.43460300	-0.15103000	-0.38516000
C	-1.72156100	-1.37000900	-0.38982000
N	-4.55229300	0.97568700	0.07498100
C	-5.92472400	1.00729300	0.53875400
C	-1.75599300	1.08411400	-0.77702700
O	-2.32593500	2.18143000	-0.81493000
C	-0.27688700	1.04691900	-1.12275400
C	0.57637100	0.85980900	0.10694400
C	2.07024400	0.63529800	-0.15465400
O	0.16218600	0.98715300	1.24150300
N	2.68941200	-0.35282800	0.48456400
O	2.61767800	1.44667800	-0.91997800
C	4.15272500	-0.48669700	0.36037500
C	2.01008900	-1.37953800	1.27350500
C	4.87096000	0.84125100	0.67418000
C	4.51544600	-1.15367000	-0.96056200
N	5.93494800	1.11127400	-0.03140600
O	4.39401300	1.51584200	1.65882900

H	-1.72363700	-3.48280500	-0.01375300
H	-4.11118800	-3.48165500	0.73263700
H	-5.40815000	-1.41645400	0.76662600
H	-0.68760200	-1.38165700	-0.72036500
H	-6.27728100	2.03600900	0.47802600
H	-6.57415500	0.38042400	-0.08133300
H	-6.00772400	0.67116100	1.57797200
H	-0.02509000	0.27608700	-1.85780200
H	-0.00288000	2.01233400	-1.56132800
H	4.43838400	-1.15594400	1.18061700
H	0.94186900	-1.40025100	1.06290500
H	2.42890600	-2.35136500	1.00102400
H	2.17050000	-1.20758700	2.34170700
H	4.26978300	-0.50738400	-1.80715300
H	5.58317600	-1.37494900	-0.99144900
H	3.96406300	-2.09376600	-1.06103600
H	6.33449600	1.97370400	0.34669700
H	-4.10378600	1.83651800	-0.20485000

Int II (-)..H₂O

-1 1

C	-5.25215500	1.01279900	0.16608600
C	-5.59927300	-0.34098900	0.19864100
C	-4.63049200	-1.32864400	0.32215100
C	-3.26258000	-0.99580300	0.40164000
C	-2.90682000	0.37634100	0.30934300
C	-3.90883700	1.35488000	0.22927200
C	-1.50878900	0.86464800	0.22763600
C	-0.48389800	0.12231500	-0.61611500
O	-1.18056500	1.92890600	0.74093600
C	0.62708000	-0.44999400	0.22770500
C	1.87004500	-0.91295000	-0.54023000
N	3.08451200	-0.64673500	-0.05145100
C	3.26649300	0.18551900	1.16099800
C	2.43653400	1.48051900	1.13121200
N	2.10159500	1.95741400	-0.04247800
O	2.17301700	1.98017900	2.27150900
C	4.72831500	0.56903800	1.38709000
C	4.24917300	-1.14796200	-0.78208700
O	1.63624900	-1.58137300	-1.56325000
O	0.50119500	-0.69073800	1.41446600
N	-2.30441300	-1.97225800	0.57609600
C	-2.69313200	-3.34462200	0.86169000

H	-6.01333500	1.78011400	0.07807500
H	-6.64336500	-0.63453700	0.14070900
H	-4.93497600	-2.36710600	0.37723100
H	-3.60366400	2.39580800	0.17539100
H	-0.92823200	-0.69406500	-1.19422200
H	-0.05231400	0.84188100	-1.31987200
H	2.92610500	-0.38927700	2.02625300
H	1.55821000	2.81194000	0.08835800
H	5.10721000	1.20266100	0.58044000
H	4.78253400	1.13181000	2.32136000
H	5.36773100	-0.31056300	1.48472700
H	4.80397100	-1.86462200	-0.17149800
H	3.90585900	-1.64457100	-1.68682800
H	4.91024200	-0.32325800	-1.05461500
H	-1.47917900	-1.67579600	1.08268700
H	-3.34116100	-3.42224500	1.74429500
H	-3.21435800	-3.78316000	0.00692300
H	-1.78639000	-3.92244800	1.04021500
O	2.80686400	1.45002300	-2.54973000
H	2.56981600	1.58613900	-1.56731500
H	3.57249600	0.86057800	-2.56595400
O	0.81776600	0.01131100	-3.83614600
H	1.50402000	0.58132100	-3.42630700
H	0.74712100	-0.71807200	-3.20140600

Int II

0 1

C	-2.27323200	-2.55264400	0.00774600
C	-3.62118600	-2.55027700	0.40408300
C	-4.36288800	-1.38625300	0.41422000
C	-3.78879700	-0.14582900	0.02618600
C	-2.41283700	-0.14090100	-0.37775900
C	-1.69611600	-1.35792400	-0.37361600
N	-4.54110800	0.97830000	0.05008000
C	-5.92168500	1.00457400	0.48958500
C	-1.73256300	1.09687100	-0.75686900
O	-2.30394800	2.19304400	-0.80059400
C	-0.24877000	1.06366200	-1.08491100
C	0.59122100	0.86442100	0.15050700
C	2.08566000	0.63738600	-0.10249000
O	0.17112900	0.97420600	1.28385500
N	2.70033700	-0.35707600	0.54001700
O	2.65489700	1.44003200	-0.85673400
C	4.14845800	-0.50710100	0.36371100

C	2.02708100	-1.39936500	1.31638600
C	4.84348800	0.83083400	0.64049000
C	4.47730300	-1.18425900	-0.96322300
N	5.89743700	1.13991700	-0.12177200
O	4.47676600	1.53898600	1.58750600
H	-1.69776300	-3.47128000	-0.00130900
H	-4.09663100	-3.47851100	0.70793400
H	-5.40040400	-1.41742500	0.72467800
H	-0.65724600	-1.36653000	-0.68868700
H	-6.27681400	2.03205200	0.42330200
H	-6.55770800	0.37585800	-0.14238100
H	-6.02176300	0.66732700	1.52691200
H	0.01114500	0.29980400	-1.82463500
H	0.03052700	2.03320600	-1.51136300
H	4.48191100	-1.15276100	1.18418900
H	0.95825500	-1.41181900	1.10983900
H	2.44243100	-2.36608700	1.02224000
H	2.19456400	-1.24475700	2.38542000
H	4.21845000	-0.54565200	-1.81170500
H	5.53716500	-1.44652000	-1.01512600
H	3.90586200	-2.11298200	-1.03586000
H	6.40310700	1.99520500	0.07157100
H	-4.09100500	1.84109700	-0.22104000
H	6.16820300	0.58800100	-0.92296000

Int II..OH (-)

-1 1

C	-4.38556100	-1.64955200	0.60337900
C	-3.37771400	-2.58922300	0.35969700
C	-2.14459700	-2.19148300	-0.14279800
C	-1.88161600	-0.83734300	-0.40609600
C	-2.87285600	0.11763700	-0.09065000
C	-4.12539200	-0.30455100	0.37968400
N	-0.64354900	-0.44128700	-0.91571700
C	-0.59963000	0.24017000	-2.21477400
C	-2.61118100	1.58024500	-0.11251800
O	-3.52294700	2.38021500	-0.29144600
C	-1.21270000	2.12901600	0.14281300
C	-0.32713800	1.22908900	0.96979200
C	1.14701000	1.15174200	0.55838400
O	-0.72619500	0.75149900	2.01829200
N	1.91841600	0.13609700	0.98013900
O	1.59891200	2.10313700	-0.09988700
C	3.35515600	0.29813400	0.69734100

C	1.40798700	-1.03771800	1.69396300
C	3.66394500	0.37422800	-0.80127900
C	4.23082300	-0.78321600	1.32329500
N	2.99883500	-0.43195800	-1.61997100
O	4.57262800	1.14803100	-1.17449300
H	-5.35153200	-1.96492300	0.98286200
H	-3.56181700	-3.64286300	0.54738700
H	-1.37358900	-2.92479800	-0.36325600
H	-4.87464300	0.44716500	0.60916600
H	-0.49789200	-0.48854100	-3.02858900
H	-1.51239400	0.81420700	-2.39103500
H	0.25359800	0.92508100	-2.25341600
H	-0.75088700	2.38789900	-0.81214100
H	-1.34530400	3.06323500	0.70022100
H	3.65588500	1.27040800	1.09884000
H	2.04342600	-1.89163700	1.47007000
H	1.39831800	-0.87339200	2.77464400
H	0.39837500	-1.27093700	1.35857300
H	5.27475700	-0.50564100	1.15887400
H	4.05967100	-0.85155200	2.40021100
H	4.06199400	-1.76242400	0.86774800
H	3.28125900	-0.38873100	-2.59317400
H	2.29865100	-1.20747000	-1.33513600
H	0.04230000	-1.22494000	-0.92899100
O	1.32022300	-2.40032500	-1.23721400
H	1.42347300	-2.89257200	-0.41336900

Int III (-)

-1 1

C	-5.26250800	-0.90050600	-0.18188800
C	-4.50899300	-1.83603500	-0.90231300
C	-3.13290700	-1.71582800	-1.02998800
C	-2.43775500	-0.62745100	-0.45121600
C	-3.20923300	0.31046900	0.28818900
C	-4.59771500	0.15934100	0.41092900
N	-1.06807000	-0.48463900	-0.59762100
C	-0.33777300	-1.67728000	-1.01461700
C	-2.55069400	1.45922800	0.92721500
O	-3.14433200	2.22236900	1.69289300
C	-1.12112800	1.67848100	0.54581700
C	-0.38413500	0.31701900	0.48368300
C	1.03739900	0.69222800	-0.08603000
O	-0.39680900	-0.29167800	1.68178000
N	2.17221500	0.32231500	0.56318400

O	1.10875500	1.45911900	-1.06163800	C	3.40013800	1.01936800	0.21793300
C	3.39068700	1.05301200	0.20442500	C	2.29033200	-0.92941000	1.35877400
C	2.32395100	-0.87335900	1.40108900	C	4.11450400	0.55792000	-1.05230700
C	4.07347500	0.56674900	-1.07410100	C	4.38630100	1.02746200	1.38740700
C	4.40954000	1.08388200	1.34573400	N	3.74241800	-0.59308100	-1.62066000
N	3.68928700	-0.59742900	-1.60723900	O	5.03272700	1.25765300	-1.50559700
O	4.98012100	1.25542200	-1.56823700	H	-6.30386400	-1.07234300	0.04357600
H	-6.33754100	-1.00740300	-0.08760900	H	-4.98422300	-2.73400000	-1.26747500
H	-5.00303400	-2.68352100	-1.36897300	H	-2.58688400	-2.49163200	-1.55442700
H	-2.59691700	-2.47448900	-1.58525400	H	-5.11015600	0.86950000	1.06240200
H	-5.14306300	0.90370600	0.98362500	H	-0.67534600	-1.92112600	-2.10596900
H	-0.64591100	-1.98902400	-2.01417500	H	0.73332500	-1.39569100	-1.21189600
H	0.72574500	-1.44469600	-1.07424500	H	-0.41797300	-2.51736900	-0.44494100
H	-0.47607500	-2.51943100	-0.32279000	H	-1.06831400	2.22232000	-0.39557000
H	-1.12198700	2.18122200	-0.42840900	H	-0.58759000	2.26395700	1.31344400
H	-0.63405600	2.32559000	1.27968600	H	3.10476500	2.05077100	0.01311600
H	3.09316300	2.08087700	-0.01465700	H	3.24590800	-1.40577100	1.12998800
H	3.24421500	-1.38772100	1.11033300	H	2.25976400	-0.69275800	2.42481500
H	2.38419500	-0.61790000	2.46190200	H	1.50048500	-1.63507500	1.12022800
H	1.48723500	-1.54534700	1.24426600	H	5.18617900	1.73403400	1.15959600
H	5.21409700	1.77065300	1.07696300	H	3.88179300	1.35889000	2.29794000
H	3.93394400	1.44949400	2.25880400	H	4.83468500	0.04686200	1.56196400
H	4.84664300	0.10153400	1.53934800	H	4.22865200	-0.90906200	-2.45059000
H	4.14977700	-0.92962200	-2.44533300	H	3.00075600	-1.16701600	-1.24619400
H	2.94912200	-1.15559400	-1.20668300	H	-0.12529200	0.21928000	2.32436200

Int III

0 1

C	-5.23398600	-0.94945600	-0.08294500
C	-4.48970300	-1.87727100	-0.81944300
C	-3.11806100	-1.73843600	-0.98791600
C	-2.43162400	-0.64270900	-0.42791300
C	-3.18922100	0.29277900	0.31955600
C	-4.57126200	0.12649100	0.48248500
N	-1.05505400	-0.48552900	-0.61557200
C	-0.32109700	-1.65097600	-1.11064200
C	-2.53175900	1.46514400	0.91394900
O	-3.12309500	2.27581800	1.62286300
C	-1.08895700	1.66579200	0.54730000
C	-0.37077200	0.32520800	0.37243400
C	1.06078100	0.68349500	-0.13349600
O	-0.39286900	-0.37359000	1.60493500
N	2.16964200	0.28039100	0.52990600
O	1.13486900	1.46337900	-1.09113200

I

0 1

C	-2.96860200	0.52249300	0.12421400
C	-2.25039400	1.79184700	0.28588700
C	-0.82465400	1.68058500	0.32630600
O	-2.85108900	2.88823700	0.37781000
C	-0.20870800	0.46400100	0.21693900
C	1.30050100	0.42876700	0.32769000
N	2.01043600	0.39015800	-0.80527300
C	3.47306100	0.35088100	-0.70703200
C	3.88561600	-0.80943200	0.20567600
N	4.93985500	-0.61855400	1.00408700
O	3.29300200	-1.89421400	0.12814200
C	1.42728900	0.20670400	-2.13254000
C	-4.37459700	0.51522900	0.06480700
C	-5.07242100	-0.66485200	-0.09011100
C	-4.36457800	-1.87526700	-0.19051300
C	-2.98425000	-1.90182700	-0.13464400

C	-2.26489200	-0.69792500	0.02489600
C	4.05594600	1.72031400	-0.37445600
N	-0.87495500	-0.70850700	0.08364600
C	-0.16328100	-1.98749000	-0.03528200
O	1.81314400	0.47809600	1.45452500
H	-0.21977000	2.57387000	0.43773600
H	3.82994700	0.04910300	-1.69752300
H	5.25595100	-1.38390700	1.58644100
H	5.37930700	0.28433100	1.11001500
H	0.36602400	0.45294100	-2.12324600
H	1.92899800	0.87721700	-2.83344400
H	1.56422100	-0.82848100	-2.46134400
H	-4.89391100	1.46465100	0.14421800
H	-6.15620400	-0.66198500	-0.13509200
H	-4.90429900	-2.80875400	-0.31445500
H	-2.46727400	-2.84929200	-0.21594000
H	3.68111700	2.44815600	-1.09781400
H	3.77367800	2.04470200	0.63003500
H	5.14585600	1.70613800	-0.45872100
H	0.90885200	-1.82857600	0.06440600
H	-0.36863400	-2.43730500	-1.00922400
H	-0.48700000	-2.66288800	0.75849500

H₂O

0 1

O	0.02108300	1.36957900	0.00000000
H	0.98582800	1.40964700	0.00000000
H	-0.26319900	2.29237700	0.00000000

OH (-)

-1 1

O	0.03252300	1.38888900	0.00000000
H	0.99692100	1.38888900	0.00000000

TS1

-1 1

C	-5.01815600	1.24846400	-0.27665200
C	-5.38555800	-0.05367100	0.07751100
C	-4.42741300	-1.01945800	0.35065400
C	-3.04898500	-0.72350800	0.26705800
C	-2.67460500	0.58676700	-0.14164300
C	-3.66826900	1.55135900	-0.37402700
C	-1.27949400	1.02301400	-0.40747400
C	-0.28854100	0.08716600	-1.08365600

O	-0.94521200	2.18553500	-0.19801200
C	0.84072300	-0.32805900	-0.16312900
C	2.03116600	-1.00267100	-0.84401900
N	3.24406600	-0.90206500	-0.28301600
C	3.56487000	0.06754900	0.78207000
C	2.68976300	1.32739000	0.73632100
N	2.07870900	1.58536300	-0.39660100
O	2.63476900	1.98741300	1.81894300
C	5.02426600	0.52465900	0.69753500
C	4.30054100	-1.79883000	-0.74816600
O	1.77865500	-1.70563300	-1.83820500
O	0.64932900	-0.51008300	1.04285400
N	-2.10997800	-1.67442500	0.58708100
C	-2.49994600	-2.93265100	1.20092500
H	-5.76964000	2.00214600	-0.48415800
H	-6.43590900	-0.31993300	0.15322600
H	-4.74481800	-2.01139300	0.64963900
H	-3.35141100	2.54582200	-0.67378900
H	-0.78013400	-0.83039100	-1.42234600
H	0.11360400	0.60560900	-1.95643700
H	3.39584200	-0.40442300	1.75701400
H	1.47434700	2.39898500	-0.26774400
H	5.23101000	0.98102600	-0.27545100
H	5.19422800	1.27125800	1.47630400
H	5.72442500	-0.29613800	0.86157200
H	4.83114700	-2.20853900	0.11377000
H	3.84749000	-2.61705800	-1.30401300
H	5.01295800	-1.27981700	-1.39498300
H	-1.21606400	-1.31782700	0.90533600
H	-3.08007900	-2.78721000	2.12149500
H	-3.09249700	-3.53700500	0.50891200
H	-1.59220400	-3.48538000	1.44317200

TS2

-1 1

C	-4.63824200	-1.62960200	0.17796100
C	-3.67506300	-2.58336800	-0.15814000
C	-2.38186300	-2.19185200	-0.50026800
C	-2.04216900	-0.83874300	-0.50833800
C	-2.99802200	0.12453300	-0.13759800
C	-4.29618300	-0.28193200	0.19068800
N	-0.73308000	-0.41568400	-0.86295600
C	-0.64007700	0.14436400	-2.22615800
C	-2.64156400	1.56498100	-0.00274500

O	-3.51655200	2.42301600	0.03671700
C	-1.18835500	1.98228000	0.10506300
C	-0.29794800	0.87983100	0.65166900
C	1.18399800	0.98970600	0.25843200
O	-0.62122300	0.36153600	1.74162600
N	2.07430800	0.09983100	0.73919100
O	1.53682000	1.94776300	-0.45049900
C	3.48003000	0.39669900	0.43884500
C	1.73834600	-1.13829500	1.44825600
C	3.85404700	0.14552400	-1.02201400
C	4.46557200	-0.33726800	1.34790400
N	3.12757700	-0.70772300	-1.73755800
O	4.86228300	0.72978800	-1.47220600
H	-5.64449000	-1.93794600	0.44125700
H	-3.93230600	-3.63803500	-0.15912900
H	-1.63246000	-2.92756200	-0.77697200
H	-5.02353800	0.46950100	0.48162800
H	-0.72941500	-0.66173200	-2.96319400
H	-1.43378200	0.87060300	-2.41279800
H	0.33105500	0.63093700	-2.34912700
H	-0.85738500	2.34480000	-0.87095200
H	-1.14457900	2.83006200	0.79733500
H	3.61856100	1.47063500	0.58559300
H	2.41134700	-1.92703000	1.11084200
H	1.84424100	-1.01282900	2.52943100
H	0.71983700	-1.43951600	1.22513300
H	5.46208900	0.07130200	1.16887400
H	4.21096200	-0.17261700	2.39751900
H	4.49546800	-1.41105100	1.15004800
H	3.44727200	-0.86395400	-2.68722500
H	2.29415900	-1.27833100	-1.41303000
H	-0.04484600	-1.24034800	-0.87165300
O	1.03706500	-2.33825500	-1.31221800
H	1.18890800	-2.90939700	-0.54915500